

Medical English

MEDICAL ENGLISH

A Textbook of Medical Terminology

JIM HUTCHINS; JUSTIN BURR; TRAVIS PRICE; JORDAN WEST; AND
MADDISON JOHNSTON



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ACCESSIBILITY OF THIS RESOURCE

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This statement was last updated on April 21, 2026.

Medical Terminology is a language class. The language we are learning is Medical English, a form of English used by medical professionals worldwide to communicate with one another. When medical professionals communicate with patients, a translator is sometimes required because not all people speak Medical English.

By the time you finish this course, you will be fluent in Medical English.

Like all language courses, the best way to learn is by immersion. In Unit 1, to teach you word building, we will flood your brain with Medical English words. In Units 2 and 3, we will cover some of the details of Medical English common to all organ systems. In Units 4 through 12, we will devote a Unit to each of the body's organ system and introduce Medical English words that are specific to that organ system.

In each Unit, you will be given a table of root words, a table of prefixes, and a table of suffixes. Then you will be given a word list that combines those root words, prefixes, and suffixes into new Medical English words.

Cover design by Alexia Wade and Jim Hutchins.

PART I

UNIT 1: THE BASICS OF MEDICAL ENGLISH WORD BUILDING

In this Unit, we will learn the basics of word building in Medical English and introduce you to Medical English plurals.

UNIT 1 INTRODUCTION

Medical English is based on combining word parts (prefixes, roots, suffixes) to form medical terms that are understandable to other medical professionals. To speak “medical,” it’s important to learn the meaning of each word part, as well as how to combine parts to make longer medical terms. It is also important to properly spell and pronounce each term. This first unit covers commonly encountered prefixes, roots, and suffixes used across all branches of medicine.

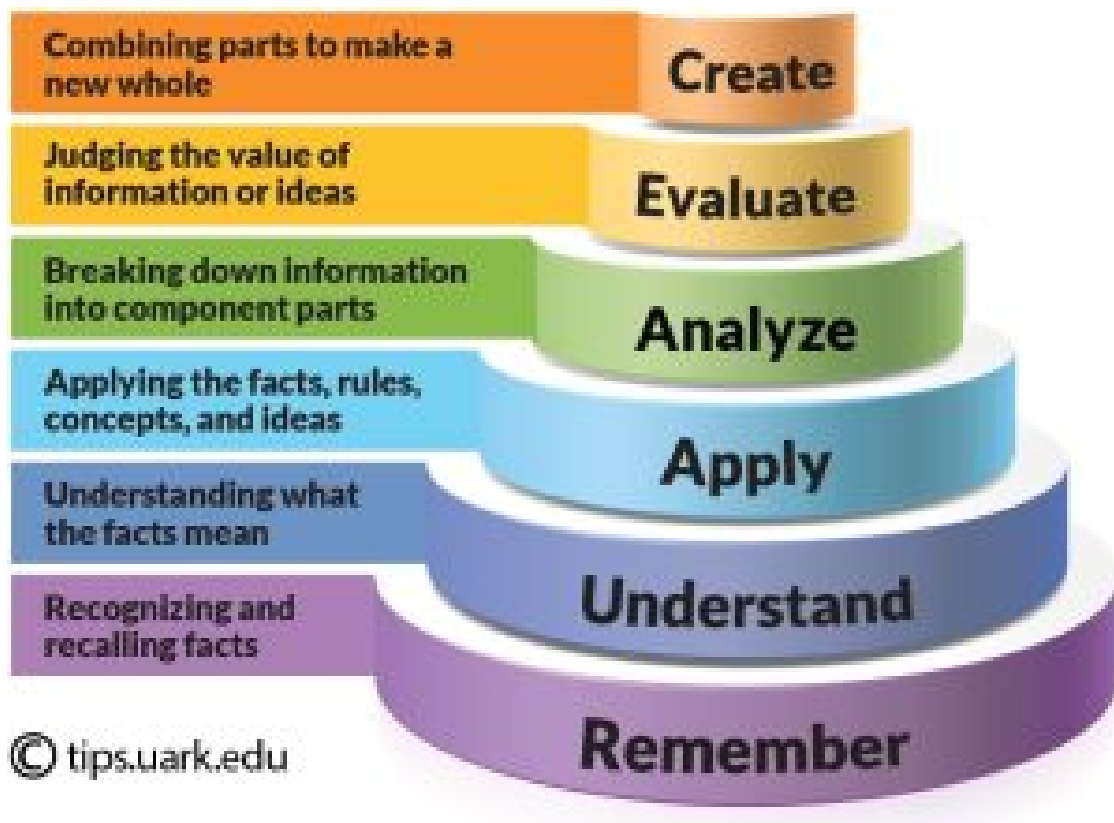
Many of the earliest scientists and physicians who studied the human body and described its parts, functions, and the diagnosis and treatment of disease used the Latin and Greek languages to document their discoveries. As a result, the language used to talk about the human body today is based in these languages. Developing fluency in this medical language can sometimes be intimidating, however once the basics of word building are mastered, the rest is relatively easy. Word building describes the way in which different word parts come together to describe a body part, physiologic process, or medical procedure. These parts include prefixes, word roots, and suffixes.

About Learning Objectives

I will define [*learning objectives*](#) for each unit. Reading, and understanding, these learning objectives is an important part of your education. If you don’t know where you’re headed, then the world’s finest map can’t get you there.

Learning objectives are based on a theory of learning called Bloom’s Taxonomy. Taxonomy is a good Medical English word: its [*etymology*](#) (word origin and breakdown) comes from the Greek word root *taxis* (“arrangement”), as in tactics, and the Greek suffix *-nomy*, “method”. It’s the science of classification. Here, we shorten *taxis* to *tax-*, add a combining vowel (*-o-*) to make it sound nice to our ears, and then the ending *-nomy*.

[*Bloom’s Taxonomy is how Benjamin Bloom and his scientific collaborators*](#) described the science of learning. It goes like this:



Unit 1 figure 1 Bloom's Taxonomy Pyramid

Objectives

Here are the first set of objectives, covering Unit 1. If you complete the activities and self-study in this unit, you will be able to:

- Recognize the origins of Medical English.
- Define the word parts of Medical English words: prefix, root, suffix, and word endings.
- Build hundreds of Medical English words from the list of prefixes, roots, and suffixes provided.
- Recall the singular and plural forms of Medical English words.

Only [Med Terms] in the Building Podcast Episode 1

Our team (Maddison Johnston, Justin Burr, and Jim Hutchins) love words very much. We want to share that love with you. This podcast is made to listen to while you go about your day to day activities and want to grab a little Medical English time while you are doing other things.



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://ucn.pressbooks.pub/medicalenglish/?p=41>

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THE BASICS OF MEDICAL ENGLISH WORD BUILDING

Only [Med Terms] in the Building Episode 1



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Medical English words consist of at least a root word. Root words are usually nouns and usually describe a body part.

A Medical English root word might be preceded by one, two, or (rarely) three prefixes. As the name implies, the prefix comes before the root word.

A Medical English root word might be followed by one, two, or (rarely) three suffixes. Suffixes come after the root word.

Many Medical English words also have a word ending that indicates its meaning. For example, I indicate.

I have given you an indication.

Here, the ending “-tion” changes the meaning of the verb “indicate” ([from the Latin *indicare*, “to point out, show”](#)) to a noun describing that action or state of being. If I indicate something, that is an indication.

For example, in the development of an organism, each of the cells may become *different* from its parent cell (*stem cell*). [The verb *differentiate* means “to become different”](#). The ending -tion turns the verb *differentiate* into the noun *differentiation* (we drop the -te in differentiate to make it flow better), so ***differentiation*** is the process by which each cell becomes different from the stem cell that created it.

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ROOT WORDS

Root words are usually nouns and describe a body component or process.

The word root is the core or central meaning of the term and is very often a reference to a specific body part. Medical terms typically have at least one word root, but may contain many of them strung together. When word roots are combined with other roots or combined with suffixes, a vowel (most often o) is used to build the word. When a root is combined with a vowel we call the new word part a combining form. For the purposes of this text we will use word root and combining form interchangeably.

Rarely, a medical word has two or even three or more roots.

Hyphens and slashes are used to help us identify word parts or when a root has been combined with a vowel to form a combining form.

TABLE LISTING UNIT 1 WORD ROOTS AND THEIR DEFINITIONS

This is the first of the tables you will encounter in this book. Since the book is primarily made up of these tables, it's important that you pay careful attention to them and understand how they work.

Each table has a search panel at the top of the table. You can type any text you want to search into this panel, for example, if you are trying to break down a word to understand its meaning. This is a powerful tool that you should practice using.

The top blue bar has the title of each column. You can click on these titles to sort the table by the contents of that column.

Inside the table are hypertext links to web pages that will give you more information, if you need it.

At the bottom, you will see a listing of the total number of entries. Remember you are responsible for **ALL** of them.

Root	Meaning	Etymology with link
aden/o aden adeno	gland	Greek adēn, "gland"
ang/i/o ang angi angio	vessel	Greek angeion, "case", "capsule", "vessel"
arthr/o arthr arthro	joint	Greek arthron, "joint"
card/i/o card cardi cardio	heart	Greek kardia, "heart"
chol/e chol chole	bile	Greek choler, "bile" Greek khloazein means "green"
cyst/o cyst cysto	bladder; fluid-filled space	Greek kystis, "bladder", "pouch"
cyt/o cyt cyto	cell	Greek kytos, "hollow", "receptacle", "basket"
derm/at/o derm dermat dermato dermo	skin	Greek derm, "skin", "hide", "leather"
gastr/o gastr gastro	stomach	Greek gaster, "stomach", "paunch", "belly"
glyc/o glyc glyco	sweet, sugar	Greek glykys, "sweet"
hem/at/o hem hemat hemato	blood	Greek haima, "blood" Read more about Greek words with "ai" in them
lith/o lith litho	stone	Latin lithus, "stone", "rock"
mast/o mast masto	breast; nipple	Greek mastos, "woman's breast"

Root	Meaning	Etymology with link
my/o/s my myo myos	muscle	Greek mys, "muscle", "mouse"
nephr/o nephr nephro	kidney	Greek nephros, "kidney"
neur/o neur neuro	nervous system	Greek neura, "nerve"
orch/i/o orch orchi orchio	testicles	Greek orkhis, "orchid", "testicles"
os/t/e/o os ost oste osteo	bone	Greek osteon, "bone"
ot/o ot oto	ear	Greek otos, "ear"
scler/o scler sclero	hard	Greek sklēros, "hard"
splen/o splen spleno	spleen	Latin splen, "spleen"
ten/o ten tend teno tendo tendin tendon	tendon	Greek tenon, "tendon", "sinew"
thyroid/o thyroid thyroido	thyroid gland	Greek thyreoeides, "shield-shaped"
ur/o ur uro	urine	Greek ourein Latin urine
vas/o vas vaso	vessel, tube, duct	Latin vasa, "vessel"

PREFIXES

Prefixes, when they are present, are at the beginning of the term and often modify the root of the word.

TABLE LISTING UNIT 1 PREFIXES AND THEIR DEFINITIONS

Prefix	Meaning
a- an-	not
anti-	against (an action)
brady-	slow
di- dia-	through
dys-	bad
endo-	inside
epi-	on top of
hemi-	half
hyper-	too much
hypo-	too little
intra-	inside
macro-	large
meso-	in the middle
micro-	small
olig- oligo-	several; a handful
pan-	all; everything or everyone
para-	beside; near
peri-	around
poly-	many; too much
tachy-	fast
trans-	through

SUFFIXES

A suffix, if it is present, is always found at the end of a medical term. The suffix modifies the meaning of the term, and despite being at the end, it is usually the first word part that we list when we build a word in Medical English.

For example, the first word in the [Unit 1 Word List](#) is adenectomy.

The word is built from

root	aden	gland
suffix	ectomy	surgical removal

When building the word, we start with the suffix

“surgical removal of...”

and end with the only other word part we have, the root:

“surgical removal of a gland”

Note that in a few cases, suffixes are built from two parts, either a prefix and suffix, or a root and a suffix.

Some examples of this are:

ec— a prefix meaning “out of” plus

–**tomy** meaning “cutting” gives us

–**ectomy** surgical removal (literally, “cutting out of”)

hyper— a prefix meaning “too much” plus

–**trophy** meaning “nourishment” or “growth” gives us

–**hypertrophy** meaning “too much growth”

–**malac** a root that is taken from the [Greek word malakos, “softening”](#) plus

–**ia** a suffix meaning “disease”

–**malacia** meaning “softening due to disease”

–**scler** a root meaning “hardening” plus

–**osis** a suffix meaning “process” gives us

–**sclerosis** “the process of hardening”)

Sometimes, the end of a Medical English word is not technically a suffix but really more of a word ending. There is no bright line between a word ending and a suffix. In general, though, a word ending slightly changes the meaning of a root word while a suffix adds a new meaning.

neur root word meaning *nervous system*

–**al** word ending meaning *pertaining to*

neural means *pertaining to the nervous system*

neur root word meaning *nervous system*

–**on** word ending meaning *unit of anatomical structure*

neuron is a microscopic anatomical unit in the brain; the brain and spinal cord consist of about 80 billion neurons

neph root word meaning *kidney*

–**on** word ending meaning *unit of anatomical structure*

nephron is a microscopic anatomical unit in the kidney; the kidney consists of about 1 million nephrons

neph root word meaning *kidney*

–**itis** suffix meaning *inflammation*

nephritis then means *inflammation of the kidney*

neph root word meaning *kidney*

nephro- is the combining form (we add the “o” to make it sound better)

–**lith** suffix (also root word) meaning *stone*

nephrolith then means *stone in the kidney*

TABLE LISTING UNIT 1 SUFFIXES AND THEIR DEFINITIONS

Suffix or ending	Type	Meaning
-a	word ending	no meaning
-al	word ending	pertaining to
-algia	suffix	pain
-ar	word ending	pertaining to
-ary	word ending	pertaining to
-cele	suffix	hernia (something that sticks out where it shouldn't)
-centesis	suffix	puncture
-desis	suffix	binding together
-dynia	suffix	pain
-e	word ending	no meaning
-eal	word ending	pertaining to
-ectomy	suffix	cutting out of
-emia	suffix	present in blood, indicating disease
-esis	word ending	process
-etic	word ending	a drug or condition causing a process
-eum	word ending	anatomical structure word ending
-gram	suffix	a record of a test or procedure
-graphy	suffix	a test or procedure
-hypertrophy	suffix	too much growth
-ia	word ending	disease
-iac	word ending	pertaining to
-ial	word ending	pertaining to
-iasis	suffix	presence of an abnormality
-ic	word ending	pertaining to
-is	word ending	anatomical structure word ending
-itis	suffix	inflammation
-ium	word ending	anatomical structure word ending
-lith	suffix	stone
-logist	suffix	one who studies
-logy	suffix	study of

Suffix or ending	Type	Meaning
-lysis	suffix	breaking down; destroying
-malacia	suffix	softening due to disease
-megaly	suffix	enlargement due to disease
-oid	suffix	looks like; resembles
-oma	suffix	tumor
-on	word ending	unit of anatomical structure
-osis	suffix	process
-pathy	suffix	state of disease
-penia	suffix	low levels of
-pexy	suffix	surgical fixation
-plasty	suffix	surgical revision
-poiesis	suffix	creation process
-ptosis	suffix	drooping
-rrhage	suffix	bursting forth
-rrhaphy	suffix	suturing
-rrhea	suffix	abnormal flow
-rrhexis	suffix	rupture
-scler	suffix	hardening
-scope	suffix	instrument for looking inside
-scopy	suffix	the process of looking inside
-spasm	suffix	involuntary contraction
-stomy	suffix	creating an opening (or "mouth") with surgery
-tomy	suffix	cutting
-trophy	suffix	nourishment, growth
-uria	suffix	present in urine, indicating disease
-y	word ending	disease

SINGULAR AND PLURAL FORMS OF A WORD

In standard English, we often indicate a plural by adding an “s” onto the end of a word (plural | plurals). Sometimes we even change a “-y” to an “-ies” to form a plural (candy | candies). We see this in Medical English as well.

Singular	Plural
candy	candies
biopsy	biopsies
capillary	capillaries
myopathy	myopathies
necropsy	necropsies
ovary	ovaries

In Medical English, we sometimes take the plural ending from the original language (typically, Greek or Latin). Thus one *bacterium* but many *bacteria*.

In the table below, we have grouped the words by word ending. The left column shows the singular form of the word; the right column the plural form. At this point, you don’t need to learn the meanings of the words (but you can if you want); just recognize the pattern of the word ending and its plural form.

Singular	Plural
alveolus	alveoli
bronchus	bronchi
canaliculus	canaliculi
glomerulus	glomeruli
homunculus	homunculi
microvillus	microvilli
nevus	nevi
nucleus	nuclei
nucleolus	nucleoli
plexus	plexi
stimulus	stimuli
thrombus	thrombi
villus	villi
ampulla	ampullae
bursa	bursae
chorda	chordae
concha	conchae
fimbria	fimbriae
lacuna	lacunae
lamella	lamellae
larva	larvae
lunula	lunulae
macula	maculae
papilla	papillae
petechia	petechiae
pleura	plurae
plica	plicae
scapula	scapulae
sequela	sequelae
tendina	tendinae
vertebra	vertebrae
aponeurosis	aponeuroses
cystocentesis	cystocenteses
diagnosis	diagnoses
epididymis	epididymides
iris	irides
mycosis	mycoses
necrosis	necroses
neurosis	neuroses
testis	testes
xanthosis	xanthoses
appendix	appendices
cicatrix	cicatrices
cortex	cortices*
thorax	thoraces

atrium	atria
bacterium	bacteria
brachium	brachia
cilium	cilia
flagellum	flagella
haustum	haustra
ovum	ova
pericardium	pericardia
periostium	periosteae
proprium	propria
ganglion	ganglia
mitochondrion	mitochondria
spermatozoon	spermatozoa
carcinoma	carcinomata
sarcoma	sarcomata

*this does not follow the pattern. I don't know why.

UNIT 1 WORD LIST

Word	Definition
adenectomy	surgical removal of a gland
adenitis	inflammation of a gland
adenoma	tumor derived from a gland
adenomegaly	enlargement of a gland
adenopathy	disease of a gland
adenosis	abnormal change in a gland
amastia	condition of no breast tissue
amyotrophic	condition of no muscle nourishment
angiocarditis	inflammation of the blood vessels and heart
angiogram	record of the blood vessels
angiography	visualizing [or recording from] the blood vessels
angiolith	stone in the blood vessels
angioma	tumor derived from blood vessels
angioplasty	surgical alteration of a blood vessel
angiopoiesis	creation of blood vessels
angiorrhaphy	surgical repair of blood vessels
angiosclerosis	abnormal hardening of blood vessels
angioscope	device for visualizing blood vessels
anorchidism	condition of no testicles
antiarthritic	drug that opposes arthritis
anuria	condition of no urine
arteriosclerosis	abnormal hardening of arteries
arthralgia	pain in the joints
arthrectomy	surgical removal of a joint
arthritis	inflammation of a joint
arthrocele	herniation in a joint
arthrocentesis	puncture of a joint
arthrodesis	binding together of a joint
arthrodynia	pain in the joints
arthrogram	record of a joint

Word	Definition
arthrography	visualizing [or recording from] a joint
arthrolysis	breakdown of a joint
arthropathy	disease of a joint
arthroscopic	pertaining to a procedure for visualizing a joint
arthroscopy	procedure for visualizing a joint
arthrotomy	cutting into a joint
asplenia	condition of no spleen
bradycardia	condition of slow heartbeat
cardia	upper opening of the stomach, which resembles a heart
cardiac	pertaining to the heart
cardiogram	record of the heart
cardiologist	person who studies the heart
cardiology	study of the heart
cardiomegaly	enlargement of the heart
cardiomyopathy	disease of the heart muscle
cardiomyotomy	cutting into the heart muscle
cardiovascular	pertaining to the heart and blood vessels
carditis	inflammation of the heart
cholangiogastrostomy	making an opening between the gallbladder and stomach
cholangiogram	record of the gall bladder
cholangiography	visualizing [or recording from] the gall bladder
cholangioma	tumor derived from the gall bladder
cholangitis	inflammation of the bile vessels
cholecystalgia	pain in the gall bladder
cholecystectomy	surgical removal of the gall bladder
cholecystitis	inflammation of the gall bladder
cholecystogram	record of the gall bladder
cholelith	stone in the gall bladder
cholelithiasis	an abnormal stone in the bile
cholelithotomy	cutting out a stone from the gall bladder

Word	Definition
cystalgia	pain in the urinary bladder
cystectomy	surgical removal of the urinary bladder
cystic	pertaining to the urinary bladder or pertaining to fluid-filled sacs
cystitis	inflammation of the urinary bladder
cystocele	herniation of the urinary bladder
cystodynia	pain in the urinary bladder
cystogram	record of the urinary bladder
cystography	visualizing [or recording from] the urinary bladder
cystolith	stone in the urinary bladder
cystolithectomy	surgical removal of a stone in the urinary bladder
cystoma	tumor derived from the urinary bladder
cystoptosis	drooping of the urinary bladder
cystorrhexis	bursting of the urinary bladder
cystoscopy	procedure for visualizing the urinary bladder
cystospasm	involuntary contraction of the urinary bladder
cystostomy	making an opening in the urinary bladder
cystotomy	cutting into the urinary bladder
cystoureteritis	inflammation of the urinary bladder and the tubes leading from the kidney to the urinary bladder
cystourethrocele	herniation of the urinary bladder and the tube leading out of it
cystourethrogram	record of the urinary bladder and the tube leading out of it
dermatalgia	pain in the skin
dermatitis	inflammation of the skin
dermatodynia	pain in the skin
dermatology	study of the skin
dermatolysis	breakdown of the skin
dermatoscope	device for visualizing the skin
dermatoscopy	procedure for visualizing the skin
dermatosis	abnormal change in the skin
dermis	a layer of the skin

Word	Definition
dermoid	resembling the skin
dermopathy	disease of the skin
dermoscopy	procedure for visualizing the skin
diuresis	separation of urine from the body
diuretic	drug promoting the separation of urine from the body
dysuria	condition of abnormal or painful urination
endocarditis	inflammation of the inner lining of the heart
endocardium	the inner lining of the heart
endomysium	an inner lining surrounding muscle
endoneurium	an inner lining surrounding nerve
epicardium	an outside layer of the heart
epidermal	pertaining to an outside layer of the skin
epidermis	an outside layer of the skin
epigastric	the upper surface of the stomach
epimysium	an outside layer of muscle
epineurium	an outside layer of nerve
gastralgia	pain in the stomach
gastrectomy	surgical removal of the stomach
gastric	pertaining to the stomach
gastritis	inflammation of the stomach
gastrodynia	pain in the stomach
gastromalacia	softening of the stomach
gastropexy	surgical fixation of the stomach
gastroplasty	surgical alteration of the stomach
gastroscope	device for visualizing the stomach
glycemia	condition of sugar in the blood
glycemic	pertaining to sugar in the blood
glycolysis	breakdown of sugar
glycopenia	condition of low sugar
glycosuria	condition of sugar in the urine

Word	Definition
hemarthrosis	condition of blood in a joint
hematology	study of the blood
hematoma	blood collecting in a closed space (literally, "tumor derived from blood")
hematopoiesis	creation of the blood
hematuria	condition of blood in the urine
heminephrectomy	surgical removal of half of the kidneys (i.e. one kidney)
heminephroureterectomy	surgical removal of one kidney and the tube leading from the kidney to the bladder
hemolysis	breakdown of blood cells
hemorrhage	abnormal flow of blood
hemorrhagic	pertaining to abnormal flow of blood
hyperglycemia	condition of high sugar in the blood
hypermastia	condition of enlarged breasts
hypernephroma	enlarged tumor derived from kidney tissue
hypersplenism	condition of enlarged spleen
hyperthyroidism	condition of enlarged or overactive thyroid
hypodermic	pertaining to a layer beneath the skin
hypodermis	a layer beneath the skin
hypoglycemia	a condition of low sugar in the blood
hypoglycemic	pertaining to a condition of low sugar in the blood
hypomastia	condition of small breasts
hypothyroidism	condition of low thyroid function
intra dermal	pertaining to a layer within the skin
lithectomy	surgical removal of a stone
lithiasis	an abnormal stone
lithocystotomy	cutting into the bladder to remove a stone
lithonephritis	inflammation of the kidney due to a stone
lithonephrotomy	cutting into the kidney to remove a stone
macrocytosis	abnormal change in cells [of the blood]
macromastia	condition of enlarged breasts
macrotia	condition of enlarged ears

Word	Definition
mastalgia	pain in the breasts
mastectomy	surgical removal of the breasts
mastitis	inflammation of the breasts
mastoidalgia	pain in a structure resembling the breasts [the mastoid process of the temporal bone]
mastoidectomy	surgical removal of a structure resembling the breasts [the mastoid process of the temporal bone]
mastoiditis	inflammation of a structure resembling the breasts [the mastoid process of the temporal bone]
mastoidocentesis	puncture of a structure resembling the breasts [the mastoid process of the temporal bone]
mastopexy	surgical fixation of the breasts
mesonephric	pertaining to a structure in the middle of the kidney
microcytosis	abnormal change resulting in small cells [in the blood]
micromastia	condition of small breasts
microtia	condition of small ears
myocardial	pertaining to the muscle layer of the heart
myocarditis	inflammation of the muscle layer of the heart
myocardium	the muscle layer of the heart
myocele	herniation in a muscle
myodesis	binding together of a muscle
myodynia	pain in a muscle
myography	[visualizing or] recording from the muscles
myolysis	breakdown of the muscles
myoma	tumor derived from muscle
myomalacia	softening of the muscles
myomectomy	surgical removal of a tumor derived from [uterine] muscles
myometritis	inflammation of the muscle layer of the uterus
myometrium	the muscle layer of the uterus
myopathy	disease of the muscles
myoplasty	surgical alteration of the muscles
myorrhaphy	surgical repair of the muscles
myosclerosis	abnormal hardening of the muscles

Word	Definition
myositis	inflammation of the muscles
myospasm	involuntary contraction of a muscle
myotomy	cutting into a muscle
nephralgia	pain in the kidney
nephrectomy	surgical removal of the kidney
nephrocele	herniation of the kidney
nephrogram	record of the kidney
nephrography	visualization of [or recording from] the kidney
nephrohypertrophy	enlargement of the kidney
nephrolithiasis	an abnormal stone in the kidney
nephrolithotomy	cutting into the kidney to remove a stone
nephrologist	person who studies the kidney
nephrology	study of the kidney
nephroma	tumor derived from the kidney
nephromalacia	softening of the kidney
nephromegaly	enlargement of the kidney
nephron	anatomical unit of the kidney
nephropathy	disease of the kidney
nephropexy	surgical fixation of the kidney
nephroptosis	drooping of the kidney
nephrorrhaphy	surgical repair of the kidney
nephrosclerosis	abnormal hardening of the kidney
nephroscopy	procedure for visualizing the kidney
nephrosis	abnormal change in the kidney
nephrosonography	visualizing the kidney using sound waves
nephrosplenopexy	surgical fixation of the kidney and spleen
nephrostomy	making an opening in the kidney
nephrotomy	cutting into the kidney
nephroureterectomy	surgical removal of the kidney and ureters [tubes leading from the kidney to bladder]
neuralgia	pain in the nervous system

Word	Definition
neurectomy	surgical removal of a nerve
neuritis	inflammation in a nerve
neuroarthropathy	disease of the nerves leading to destruction of a joint
neurodynia	pain in the nervous system
neuroglycopenia	condition of low sugar in the nervous system
neurologist	person who studies the nervous system
neurology	study of the nervous system
neurolysis	breakdown of nervous system tissue
neuroma	tumor derived from the nervous system
neuron	anatomical unit of the nervous system
neuropathy	disease of the nervous system
neuroplasty	surgical alteration of a nerve
neurorrhaphy	surgical repair of a nerve
neurosclerosis	abnormal hardening of the nerves
neurosis	abnormal change in the nervous system
neurotomy	cutting into a nerve
oligocythemia	a handful of cells in the blood [i.e. abnormally low numbers of red blood cells]
oliguria	a handful of urine is produced [i.e. abnormally low amount of urine]
orchidalgia	pain in the testicles
orchidectomy	surgical removal of a testicle
orchiditis	inflammation of the testicles
orchidopexy	surgical fixation of the testicles
orchidotomy	cutting into the testicles
orchiectomy	surgical removal of a testicle
orchiodynia	pain in the testicles
orchioopathy	disease of the testicles
orchiopexy	surgical fixation of the testicles
orchioplasty	surgical alteration of the testicles
orchitis	inflammation of the testicles
ostalgia	pain in a bone

Word	Definition
ostectomy	surgical removal of a bone
osteitis	inflammation in a bone
osteoarthritis	inflammation in the bone and joints
osteodynia	pain in a bone
osteolysis	breakdown of a bone
osteomalacia	softening of the bones
osteopathy	disease of the bones
osteopenia	condition of low bone density
osteoplasty	surgical alteration of a bone
osteosclerosis	abnormal hardening of the bones
osteotomy	cutting into a bone
otalgia	pain in the ear
otitis	inflammation of the [inside structures of the] ear
otodynia	pain in the ear
otolith	stone in the ear [these are normal microscopic structures and are needed for hearing]
otoneurologist	person who studies the ear and nervous system
otoplasty	surgical alteration of the ear
otorrhea	abnormal flow from the ear
otosclerosis	abnormal hardening of the [bones of the middle] ear
otoscope	device for visualizing the [inside structures of the] ear
otosteal	pertaining to the bones of the [middle] ear
pancytopenia	condition of low blood cells of all types [i.e. abnormally low numbers of red and white blood cells]
paramesonephric	pertaining to structures near the middle of the kidney
paraurethral	pertaining to structures near the tube leading from the bladder to the outside world
pericardial	pertaining to the membrane surrounding the heart
pericardiocentesis	puncture of the membrane surrounding the heart
pericardiotomy	cutting into the membrane surrounding the heart
pericarditis	inflammation of the membrane surrounding the heart
pericardium	the membrane surrounding the heart

Word	Definition
perimysium	a membrane surrounding muscle
perineurium	a membrane surrounding nerve
periosteum	a membrane surrounding bone
polyadenopathy	disease of many glands [lymph nodes]
polycystic	pertaining to many fluid-filled sacs
polycythemia	too many cells in the blood [i.e. abnormally high numbers of red blood cells]
polymyositis	inflammation of many muscles
polyneuritis	inflammation of many nerves
polyneuropathy	disease of many nerves
polyuria	condition of too much urine
sclera	hardened organ [i.e., the white of the eye]
scleritis	inflammation of the white of the eye
sclerodermatitis	inflammation of the skin with hardening
scleromalacia	softening of the white of the eye
sclerotomy	cutting into the white of the eye
splenalgia	pain in the spleen
splenectomy	surgical removal of the spleen
splenitis	inflammation of the spleen
splenodynia	pain in the spleen
splenolysis	breakdown of the spleen
splenomalacia	softening of the spleen
splenomegaly	enlargement of the spleen
splenopathy	disease of the spleen
splenorrhexis	bursting of the spleen
tachycardia	condition of fast heart rate
tenalgia	pain in a tendon
tendinitis	inflammation of a tendon [alternative spelling of tendonitis]
tendodesis	binding together of a tendon
tendolysis	breakdown of a tendon
tendon	anatomical unit of connective tissue

Word	Definition
tendonectomy	surgical removal of a tendon
tendonitis	inflammation of a tendon
tendoplasty	surgical alteration of a tendon
tenodesis	binding together of a tendon
tenodynia	pain in a tendon
tenolysis	breakdown of a tendon
tenoplasty	surgical alteration of a tendon
tenorrhaphy	surgical repair of a tendon
tenotomy	cutting into a tendon
thyrocele	herniation in the thyroid [i.e. swelling of the thyroid gland]
thyroid	resembling a shield [a gland in the neck that resembles a shield]
thyroidectomy	surgical removal of the thyroid gland
thyroiditis	inflammation of the thyroid gland
thyroidotomy	cutting into the thyroid gland
thyromegaly	enlargement of the thyroid gland
transdermal	through the skin
transurethral	through the urethra [a surgical approach]
uremia	condition of urine in the blood [i.e. urea and other chemicals which should be in the urine]
ureteralgia	pain in the tubes leading from the kidney to the bladder
ureterectomy	surgical removal of the tubes leading from the kidney to the bladder
ureteritis	inflammation of the tubes leading from the kidney to the bladder
ureterocele	herniation [ballooning] of the tubes leading from the kidney to the bladder
ureterolithiasis	condition of having a stone in the tubes leading from the kidney to the bladder
ureteronephrectomy	surgical removal of the kidney and the tubes leading from the kidney to the bladder
ureteroplasty	surgical alteration of the tubes leading from the kidney to the bladder
ureterorrhaphy	surgical repair of the tubes leading from the kidney to the bladder
ureteroscopy	procedure for visualizing the tubes leading from the kidney to the bladder
urethra	the tube leading from the bladder to the outside world
urethrectomy	surgical removal of the tube leading from the bladder to the outside world
urethreoplasty	surgical alteration of the tube leading from the bladder to the outside world

Word	Definition
urethritis	inflammation of the tube leading from the bladder to the outside world
urethrocele	herniation [ballooning] of the tube leading from the bladder to the outside world
urethrocystitis	inflammation of the bladder and the tube leading from the bladder to the outside world
urethrodynia	pain in the tube leading from the bladder to the outside world
urethrogram	record of the tube leading from the bladder to the outside world
urethropexy	surgical fixation of the tube leading from the bladder to the outside world
urethrorrhea	abnormal flow from the tube leading from the bladder to the outside world
urethroscope	device for visualizing the tube leading from the bladder to the outside world
urethroscopy	procedure for visualizing the tube leading from the bladder to the outside world
urethrospasm	involuntary contraction of the tube leading from the bladder to the outside world
urethrotomy	cutting into the tube leading from the bladder to the outside world
urinary	pertaining to urine
urodynia	pain with urination
urologist	person who studies the urinary system
urology	study of the urinary system
uropathy	disease of the urinary system
uropoiesis	creation of urine
urostomy	making an opening in the abdominal wall to release urine
vasospasm	involuntary contraction of the blood vessels
vasovasostomy	making an opening in the vas deferens [the vessel leading from the testes to the urethra; vasectomy reversal]

HOW THE UNIT 1 WORD LIST WAS BUILT

Your responsibility is to learn the [21 prefixes](#), [25 roots](#), and [56 suffixes](#) in the linked tables. With these 102 parts, you can build the 346 words you will learn in this Unit. If you want to see how those words are built, this optional table will show you how it was done.

“Missing” letters (i.e. letters that have been dropped to make the word sound “better”) are indicated in square brackets.

[h]em, for example, is the root *hem* (blood) with the h dropped.

Combining forms (i.e. word parts that have a letter, usually an “o”, to make the word sound “better”) are indicated with slashes.

ang

angi

angio

are combined to make “ang/i/o” in its entry in the root word table.

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
Greek ourein Latin urine					ur ur/o urin				
Greek nephros , "kidney"					neph neph/o				
Greek mys , "muscle", "mouse"					my my/o my/s				
Greek kardia , "heart"					card card/i card/i/o				
Greek kytos , "hollow", "receptacle", "basket"					cyt cyt/o				
Greek gaster , "stomach", "paunch", "belly"					gastr gastr/o				
Latin lithus , "stone", "rock"					lith lith/o				
Greek neura , "nerve"					neur neur/o				
Greek derm , "skin", "hide", "leather"					derm derm/at derm/ at/o derm/o				
Greek angeion , "case", "capsule", "vessel"					ang ang/i/o				
Greek arthron , "joint"					arthr arthr/o				
Greek sklēros , "hard"					scler scler/o				
Greek adēn , "gland"					aden aden/o				
Greek glykys , "sweet"					glyc glyc/o				
Greek osteon , "bone"					os os/t os/t/e osteo				

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
Greek haima, "blood"					hem hem/o hemat hemat/o				
Latin splen, "spleen"					splen splen/o				
Greek choler, "bile" (Greek khloazein means "green")					chol chol/e				
Greek tenon, "tendon", "sinew"					ten tend tendin tendon				
Greek thyreoeides, "shield-shaped"					thyroid thyroid/o				
Greek mastos, "woman's breast"					mast mast/o				
Greek orkhis, "orchid", "testicles"					orch orch/i orch/i/o orchid orchid/o				
Greek otos, "ear"					ot ot/o				
Latin vasa, "vessel"					vas vas/o				
this government website has a list of word parts									
					aden		ec	tomy	adenector
					aden		itis		adenitis
					aden		oma		adenoma
					aden		megaly		adenomeg
					aden		pathy		adenopath
					aden		osis		adenosis
	a				mast		ia		amastia
					my/o		trophic		amyotrop
				ang/i/o	card		itis		angiocard

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
					ang/i/o	card	itis		angiocard
					ang/i/o		gram		angiogram
					ang/i/o		graphy		angiograp
					ang/i/o		lith		angiolith
					ang/i		oma		angioma
					ang/i/o		plasty		angioplast
					ang/i/o		poiesis		angiopoie
					ang/i/o		rrhaphy		angiorrha
					ang/i/o		scler	osis	angioscler
					ang/i/o		scope		angioscop
	an		orchid		ism				anorchidi
	anti		arthr		it[is]	ic			antiarthri
	an		ur		ia				anuria
	arter/i/o						scler	osis	arterioscle
	arthr						algia		arthralgia
	arthr						ec	tomy	arthrector
	arthr						itis		arthritis
	arthr/o						cele		arthrocele
	arthr/o						centesis		arthrocen
	arthr/o						desis		arthrodesi
	arthr/o						dynia		arthrodyn
	arthr/o						gram		arthrogran
	arthr/o						graphy		arthrograp
	arthr/o						lysis		arthrolysis
	arthr/o						pathy		arthropath
	arthr/o						plasty		arthroplas
	arthr/o						scopic		arthroscop
	arthr/o						scopy		arthroscop

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
					arthr/o		tomy		arthroton
	a				splen		ia		asplenia
	brady				card		ia		bradycard
					card		ia		cardia
					card		iac		cardiac
					card/i/o		gram		cardiogram
					card/i/o		logist		cardiologist
					card/i/o		logy		cardiology
					card/i/o		megaly		cardiomegaly
					card/i/o	my/o	pathy		cardiomyopathy
					card/i/o	my/o	tomy		cardiomyectomy
					card/i/o	vascul	ar		cardiovascular
					card		itis		carditis
				cholangio	gastr/o		stomy		cholangio
			chol		ang/i/o		gram		cholangiogram
			chol		ang/i/o		graphy		cholangiography
			chol		ang/i/o		oma		cholangioma
			chol		ang		itis		cholangitis
					chol/e	cyst	algia		cholecystalgia
					chol/e	cyst	ec	tomy	cholecystectomy
					chol/e	cyst	itis		cholecystitis
					chol/e	cyst/o	gram		cholecystogram
					chol/e	lith			cholelith
					chol/e				cholelithiasis
					chol/e	lith/o	tomy		cholelithotomy
Greek kystis, "bladder", "pouch"					cyst				
					cyst/o				cyst/o
					cyst		algia		cystalgia
					cyst		ec	tomy	cystectomy

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
inflammation (–itis) of the bladder (–cyst)					cyst		ic		cystic
					cyst		itis		cystitis
					cyst/o		cele		cystocele
					cyst/o		dynia		cystodynia
					cyst/o		gram		cystogram
					cyst/o		graphy		cystography
					cyst/o		lith		cystolith
					cyst/o	lith	ec	tomy	cystolithec
					cyst		oma		cystoma
					cyst/o		ptosis		cystoptosis
					cyst/o		rrhexis		cystorrhexis
					cyst/o		scopy		cystoscopy
					cyst/o		spasm		cystospasm
					cyst/o		stomy		cystostomy
					cyst/o		tomy		cystotomy
the name <i>ureter</i> is from ancient Greece and means "the tube leading from the kidney to the bladder"				cyst/o	ureter		itis		cystoureteritis
				cyst/o	urethr/o		cele		cystourethral
				cyst/o	urethr/o		gram		cystourethrogram
					derm/at		algia		dermatalgia
					derm/at		itis		dermatitis
					derm/ at/o		dynia		dermatodynia
					derm/ at/o		logy		dermatology
					derm/ at/o		lysis		dermatolysis
					derm/ at/o		scope		dermatoscopy

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
					derm/ at/o		scopy		dermatoscopy
					derm/at		osis		dermatosis
					derm		is		dermis
					derm		oid		dermoid
					derm/o		pathy		dermopathy
					derm/o		scopy		dermoscopy
pertaining to (–esis) throughput (dia–) of urine (ur)	di[a]				ur		esis		diuresis
inducing or promoting urination	di[a]				ur		etic		diuretic
	dys				ur		ia		dysuria
	endo				card		itis		endocarditis
	endo				card		ium		endocardium
	endo				my/s		ium		endomyocardium
	endo				neur		ium		endoneurium
	epi				card		ium		epicardium
	epi				derm		al		epidermal
	epi				derm		is		epidermis
	epi						ic		epigastric
	epi				my/s		ium		epimysium
	epi				neur		ium		epineurium
							algia		gastralgia
							ec	tomy	gastrectomy
					gastr		ic		gastric
							itis		gastritis
					gastr/o		dynia		gastrodynia
					gastr/o		malac	ia	gastromalacia
					gastr/o		pexy		gastropexy
					gastr/o		plasty		gastroplasty

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
-ectomy (Greek) means "cutting out of"					gastr/o		scope		gastroscope
					glyc		emia		glycemia
					glyc		emic		glycemic
					glyc/o		lysis		glycolysis
					glyc/o		penia		glycopenia
				glyc/o/s	ur		ia		glycosuria
				hem	arthr		osis		hemarthrosis
					hemato		logy		hematology
					hemat		oma		hematoma
					hemato		poiesis		hematopoiesis
				hem/at	ur		ia		hematuria
	hemi				neph		ec	tomy	heminephrectomy
	hemi				neph	ur	ec	tomy	heminephrectomy
					hem/o		lysis		hemolysis
					hem/o		rrhage		hemorrhage
					hem/o		rrhagic		hemorrhagic
	hyper				glyc		emia		hyperglycemia
	hyper				mast		ia		hypermastia
	hyper				neph		oma		hypernephroma
	hyper				splen		ism		hypersplenism
	hyper				thyroid		ism		hyperthyroidism
	hypo				derm		ic		hypodermic
	hypo				derm		is		hypodermis
	hypo				glyc		emia		hypoglycemia
	hypo				glyc		emic		hypoglycemic
	hypo				mast		ia		hypomastia
	hypo				thyroid		ism		hypothyroidism
	intra				derm		al		intradermic

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
					lith		ec	tomy	lithectomy
					lith		iasis		lithiasis
				lith/o	cyst/o		tomy		lithocysto
				lith/o	nephr		itis		lithoneph
				lith/o	nephr/o		tomy		lithoneph
	macro				cyt		osis		macrocyt
	macro				mast		ia		macromas
	macro				ot		ia		macrotia
					mast		algia		mastalgia
					mast		ec	tomy	mastecto
					mast		itis		mastitis
					mast		oid	algia	mastoidal
					mast		oid	ectomy	mastoid
					mast		oid	itis	mastoidit
					mast		oid/o	centesis	mastoido
					mast/o		pexy		mastopexy
middle, intermediate, halfway	meso				nephr		ic		mesoneph
	micro				cyt		osis		microcyt
	micro				mast		ia		micromas
	micro				ot		ia		microtia
				my/o	card		ial		myocardia
				my/o	card		itis		myocardit
				my/o	card		ium		myocardiu
					my/o		cele		myocele
					my/o		des	is	myodesis
					my/o		dynia		myodyn
					my/o		graphy		myograph
					my/o		lysis		myolysis

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
					my		oma		myoma
					my/o		malac	ia	myomalacia
					my		om[a]	ectomy	myomectomy
				my/o	metr		itis		myometritis
				my/o	metr		ium		myometrium
					my/o		pathy		myopathy
					my/o		plasty		myoplasty
					my/o		rrhaphy		myorrhaphy
					my/o		scler	osis	myosclerosis
					my/o/s		itis		myositis
					my/o		spasm		myospasm
					my/o		tomy		myotomy
					nephr		algia		nephralgia
					nephr		ec	tomy	nephrectomy
					nephr/o		cele		nephrocele
					nephr/o		gram		nephrogram
					nephr/o		graphy		nephrography
					nephr/o		hyper	trophy	nephrohypertrophy
					nephr/o	lith	iasis		nephrolithiasis
					nephr/o	lith/o	tomy		nephrolithotomy
					nephr/o		logist		nephrologist
					nephr/o		logy		nephrology
					nephr		oma		nephroma
					nephr/o		malac	ia	nephromalacia
					nephr/o		megaly		nephromegaly
					nephr		on		nephron
					nephr/o		pathy		nephropathy
					nephr/o		pexy		nephropexy

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
					nephr/o		ptosis		nephroptosis
					nephr/o		rrhaphy		nephrorrhaphy
					nephr/o		scler	osis	nephrosclerosis
					nephr/o		scopy		nephroscopy
					nephr		osis		nephrosis
					nephr/o	son/o	graphy		nephrosomography
					nephr/o	splen/o	pexy		nephrosplenopexy
					nephr/o		stomy		nephrostomy
					nephr/o		tomy		nephrotomy
					nephr/o	ureter	ec	tomy	nephroureterectomy
					neur		algia		neuralgia
					neur		ec	tomy	neurectomy
					neur		itis		neuritis
				neur/o	arthr/o		pathy		neuroarthropathy
					neur/o		dynia		neurodynia
				neur/o	glyc/o		penia		neuroglycopenia
					neur/o		logist		neurologist
					neur/o		logy		neurology
					neur/o		lysis		neurolysis
					neur		oma		neuroma
					neur		on		neuron
oligo					neur/o		pathy		neuropathy
					neur/o		plasty		neuroplasty
					neur/o		rrhaphy		neurorrhaphy
					neur/o		scler	osis	neurosclerosis
					neur		osis		neurosis
					neur/o		tomy		neurotomy
					cyt		[h]emia		oligocythemia

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
	olig[o]				ur		ia		oliguria
					orchid		algia		orchidalgia
					orchid		ec	tomy	orchidectomy
					orchid		itis		orchiditis
					orchid/o		pexy		orchidopexy
					orchid/o		tomy		orchidotomy
					orch/i		ec	tomy	orchiectomy
					orch/i/o		dynia		orchiodynia
					orch/i/o		pathy		orchiopathy
					orch/i/o		pexy		orchiopexy
					orch/i/o		plasty		orchioplasty
					orch		itis		orchitis
					os/t		algia		ostalgia
					os/t		ec	tomy	ostectomy
					os/t/e		itis		osteitis
	os/t/e/o				arthr		itis		osteoarthritis
					osteo		dynia		osteodynia
					osteo		lysis		osteolysis
					osteo		malac	ia	osteomalacia
					osteo		pathy		osteopathy
					osteo		penia		osteopenia
					osteo		plasty		osteoplasty
					os/t/e/o		scler	osis	osteosclerosis
					osteo		tomy		osteotomy
					ot		algia		otalgia
					ot		itis		otitis
					ot/o		dynia		otodynia
					ot/o		lith		otolith

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word
data-url="https://www.etymonline.com/word/para- ">Greek "beside, near"				ot/o	neur/o		logist		otoneurologist
					ot/o		plasty		otoplasty
					ot/o		rrhea		otorrhea
					ot/o		scler	osis	otosclerosis
					ot/o		scope		otoscope
					ot	os/t/e	al		otosteal
	pan				cyt/o		penia		pancytopenia
	para	meso			nephro		ic		paramesonephric
	para				urethra		al		paraurethral
	peri				card		ial		pericardial
	peri				card/i/o		centesis		pericardiocentesis
	peri				card/i/o		tomy		pericardiectomy
	peri				card		itis		pericarditis
	peri				card		ium		pericardium
	peri				myo		ium		perimysium
	peri				neur		ium		perineurium
	peri				ost		eum		periosteum
	poly				aden/o		pathy		polyadenopathy
	poly				cyst		ic		polycystic
	poly				cyt		[h]emia		polycythemia
	poly				myo/o/s		itis		polymyositis
	poly				neur		itis		polyneuritis
	poly				neur/o		pathy		polyneuropathy
	poly				ur		ia		polyuria
					scler		a		sclera
					scler		itis		scleritis
				scler/o	derm/at		itis		sclerodermitis

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word	
declawing a cat is a tendonectomy							scler/o	malac	ia	scleromalacia
							scler/o	tomy		sclerotomy
							splen	algia		splenalgia
							splen	ec	tomy	splenectomy
							splen	itis		splenitis
							splen/o	dynia		splenodynia
							splen/o	lysis		splenolysis
							splen/o	malac	ia	splenomalacia
							splen/o	megaly		splenomegaly
							splen/o	pathy		splenopathy
							splen/o	rrhexis		splenorrhexis
		tachy					card	ia		tachycardia
		ten					algia		tenalgia	
		tend					ec	tomy	tendectomy	
		tendin					itis		tendinitis	
		tend/o					desis		tendodesis	
		tend/o					lysis		tendolysis	
		tend					on		tendon	
		tendon					ectomy		tendonectomy	
		tendon					itis		tendonitis	
		tend/o					plasty		tendoplasty	
		ten/o					dynia		tenodynia	
		ten/o					lysis		tenolysis	
		ten/o					plasty		tenoplasty	
		ten/o					rrhaphy		tenorrhaphy	
		ten/o					tomy		tenotomy	
		thyr/o					cele		thyrocele	
		thyroid					ec	tomy	thyroidectomy	

[declawing a cat is a tendonecctomy](#)

Etymology	Prefix	Prefix	Root	Root	"Main" Root	Root	Suffix or Word Ending	Suffix or Word Ending	Word	
the word <i>urethra</i>, coined by Hippocrates, means "the tube leading from the bladder to the outside world"	trans						thyroid	itis	thyroiditis	
							thyroid/o	tomy	thyroidotomy	
							thyr/o	megaly	thyromegaly	
							derm	al	transdermal	
							urethr	al	transurethral	
							ur	[h]em	ia	uremia
							ureter	algia	ureteralgia	
							ureter	ec	tomy	ureterectomy
							ureter	itis	ureteritis	
							ureter/o	cele	ureterocoele	
	ureter/o					lith	ureter/o	iasis	ureterolithiasis	
							neph	ec	tomy	ureteronephrectomy
							ureter/o	plasty	ureteroplasty	
							ureter/o	rrhaphy	ureterorrhaphy	
							ureter/o	scopy	ureteroscopy	
							urethr	a	urethra	
							urethr	ec	tomy	urethrectomy
							urethr/o	plasty	urethreoplasty	
							urethr	itis	urethritis	
							urethr/o	cele	urethrocele	
	urethr/o						cyst	itis	urethrocystitis	
							urethr/o	dynia	urethrodynia	
							urethr/o	gram	urethrography	
							urethr/o	pexy	urethropexy	
							urethr/o	rrhea	urethrorrhea	
urethr/o	scope	urethroscopy								
urethr/o	scopy	urethroscopy								

PART II

UNIT 2: TALKING ABOUT THE HUMAN BODY

UNIT 2 INTRODUCTION

Medical English is the language spoken in every area of healthcare in the United States and even in many foreign countries. Having a clear and uniform way of talking about the body is critical to ensure patients receive accurate care and to help minimize errors. To avoid confusion, healthcare workers use a set of standardized terms when talking about the human body.

Objectives

After completing this unit, you will be able to:

- Name the directional terms used to describe the human body in Medical English.
- Identify the components of a medical record which uses the Subjective, Objective, Assessment and Plan (SOAP) note format.

TALKING ABOUT THE HUMAN BODY

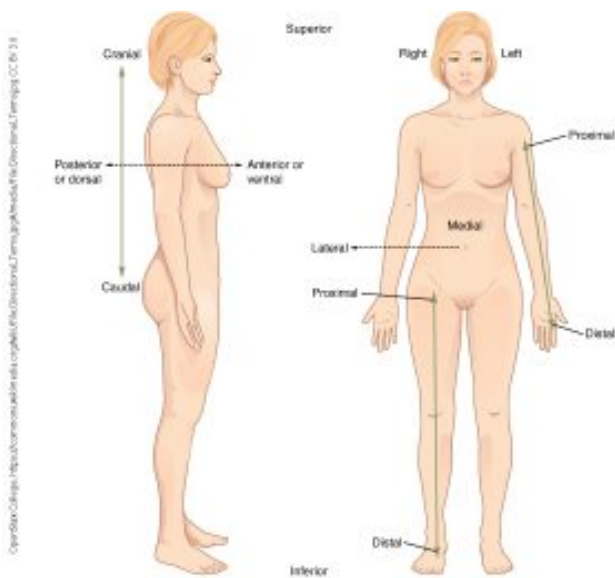
Only [Med Terms] in the Building Episode 2



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<https://ucn.pressbooks.pub/medicalenglish/?p=176>

Anatomical Position



Unit 2 figure 1 Anatomical Position

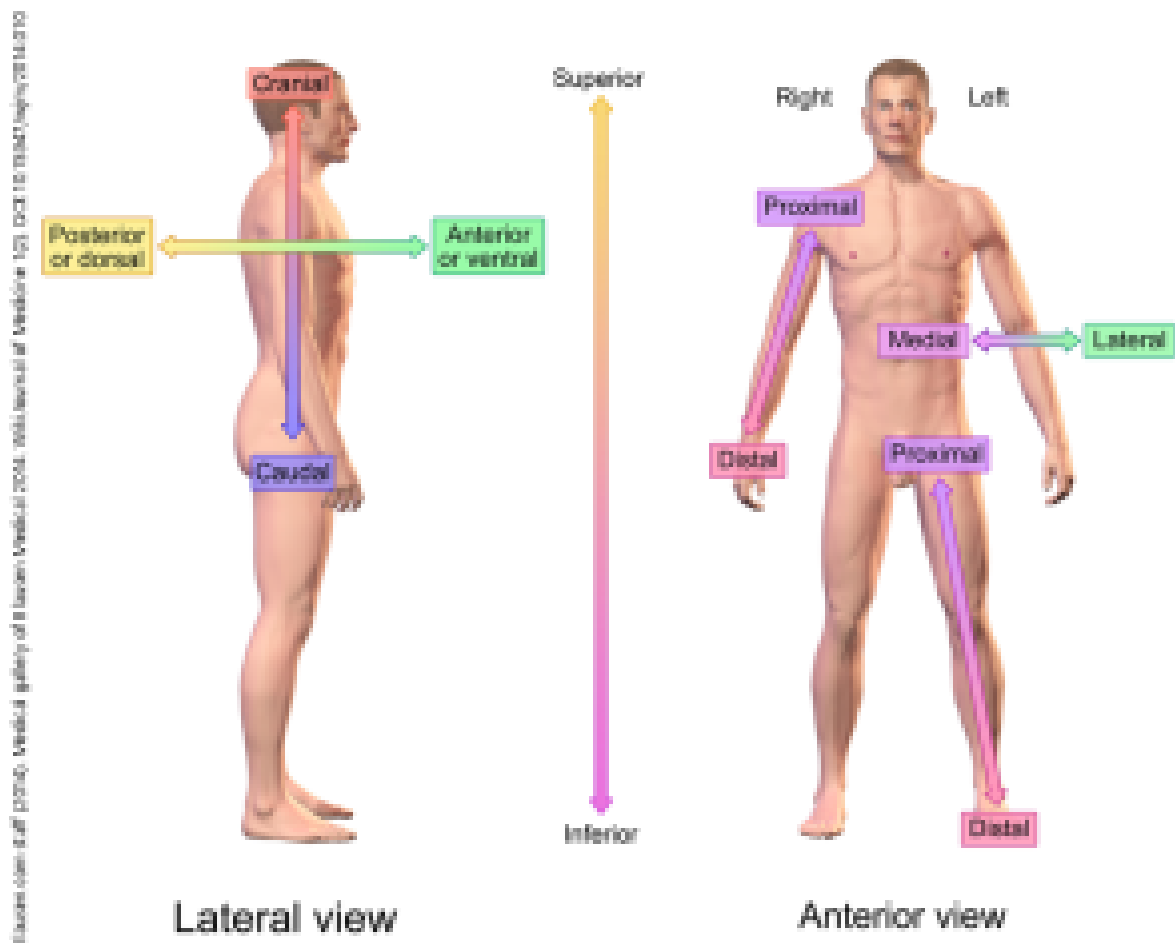
In real life, the human body doesn't spend a lot of time in any single position, however, when we are talking about the body it is important to have a sort of default position to refer to. Without this, you couldn't describe the arm as being to the side of the stomach, because it would always depend on how the person is positioned.

This default position is referred to as *anatomical position* and is demonstrated in the image. This position is useful for describing different parts of the body as well as for defining various body regions.

Similar to the standard position of the body, we use a set of terms to describe directions so everyone is clear about what is being said. If we are simply referencing a part of the body you will find the root

for that body part combined with a suffix that means pertaining to, like -al or -ic. Examples include the term **spinal**, which means *pertaining to the spine*, or **pelvic**, which means *pertaining to the pelvis*.

Directional Terms

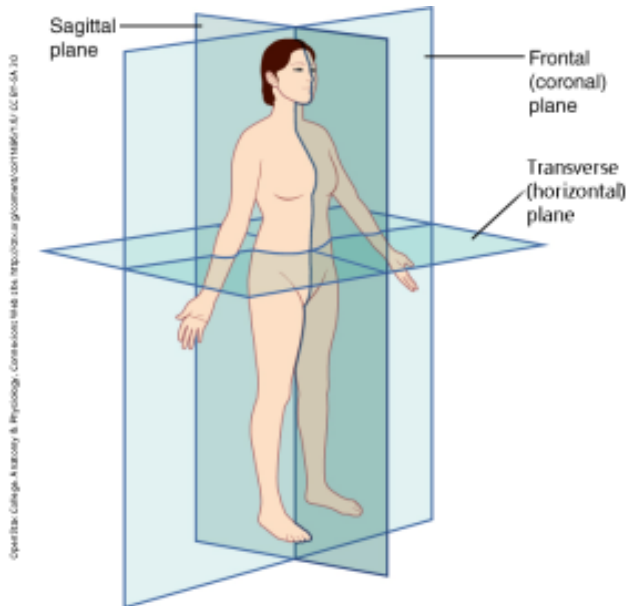


Unit 2 figure 2 Directional References

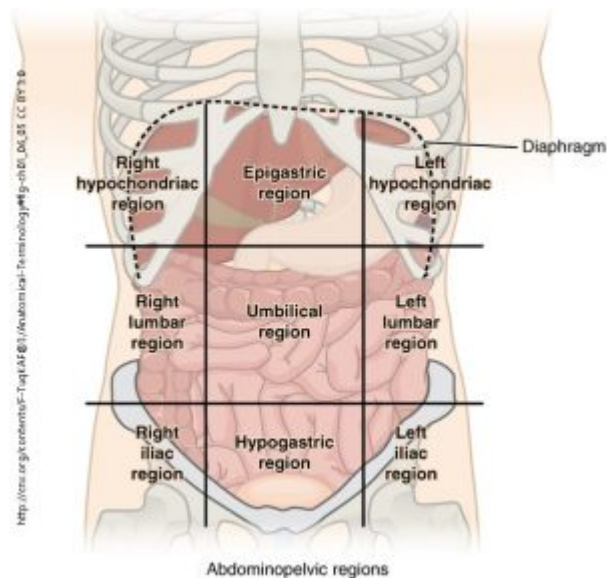
If we are describing the location of a part of the body with reference to another, we use specific terms to avoid confusion. These terms may also apply to individual organs as well. For example, we could say that the heart is superior to the stomach, but we could also say that the right atrium of the heart is superior to the right ventricle. Next time you find yourself in front of a mirror, practice using these directional terms. Can you describe your nose with reference to your eyes? How about your ears? How would you describe your hand with reference to your shoulder? Don't forget to use anatomical position and remember that sometimes more than one term may be used to describe body part locations. For example, your feet are inferior to your knees but your feet are distal to your knees as well.

Planes of Section

One more important way in which we talk about the body is by referring to sections of it, or what we commonly refer to as planes. It's necessary to know the reference plane of what we are looking at when we examine organs, whether for learning purposes or in clinical settings like when looking at an X-ray. This image illustrates the three cardinal planes of the human body. A fourth plane, not shown in the image, is an oblique plane and refers to any sections not described by one of the three cardinal planes.



Unit 2 figure 3 Planes of Section



Unit 2 figure 4 Abdominal Regions

The human abdomen is divided up into 9 abdominal regions, tic-tac-toe board style. The regions are centered on the **umbilicus**, the navel or belly-button.

The names of the regions are shown in the image at left. These names are used by clinicians to describe the location of pain, for example. The location of pain can help clinicians narrow down their **differential diagnosis**, the (hopefully short) list of possible diagnoses that are consistent with the patient's signs and symptoms. For example, pain near the border of the **right iliac region** and the **hypogastric region** is a common symptom of appendicitis, an inflammation of the appendix which can become life-threatening. (Even

though appendicitis is a new word to us at this point in the course, we can recognize what it means: the root appendix plus the suffix -itis, meaning "inflammation of". Appendicitis is inflammation of the appendix.)

Standardized Medical Records: The SOAP Note

The clarity afforded us by common terminology extends to what we do in clinical settings. It is important for anyone entering healthcare to be able to recognize and understand the terms that are used to describe medical

procedures and processes. The acronym SOAP is used to categorize the different components of patient care and is often the foundation for how medical records are created.

- The S stands for subjective and is used to describe the patient's own experiences, views, and feelings, including the reasons why they have sought medical care.
- The O stands for objective and describes information that can be gathered about the patient without requiring the patient's input. These findings include things like vital signs, physical exam findings, and medical laboratory results.
- The A is for assessment and is the result of an analysis of the subjective and objective information about the patient. Assessment is another way of describing what the healthcare provider believes is wrong with the patient.
- The P stands for plan and covers what will be done to further assess the patient and treat their condition.

Video Attribution

- Justin Burr, Jim Hutchins & Maddison Johnston. (2024, December 18). *Only [Med Terms] in the Building Episode 2*. [Video]. Kaltura.

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UNIT 2 PREFIXES

This is a complete list of prefixes that are used in the words we are building for Unit 2. Some of these prefixes appeared in the Unit 1 prefix table. Those previously-used prefixes are displayed in **gray** text so that you can easily spot the new prefixes you need to learn for this unit, but you still have access to the definitions of the prefixes from previous units.

Prefix	Definition
a-	not
ab-	off, from, away
ana-	up, throughout, again
beni-	for good
bi-	two, double
com- con-	together, together with, in combination
contra-	against, opposite
de-	from, down, out
dia-	between
dis-	separate, apart
dys-	bad
epi-	on top of
ex-	out, off, from, beyond
hyper-	too much
hypo- (before process)	too little
hypo- (before anatomical structure)	below
idio-	one's own, peculiar
im- in-	not
ipsi-	same (side)
mal- mali-	badly
meta-	between, with, after; change
micro-	small
non-	not
pro-	in front of; in favor of
radio-	X-rays or other radiation
re-	again
sub-	an underlayer; support

Prefix	Definition
sym– syn– sys–	with, together
trans–	through
un–	not
un– uni–	one

UNIT 2 ROOTS

Word roots introduced in Unit 1 are colored gray.

Root Word	Definition
abdomin	belly
acerb[o]/a	harsh, bitter
acute	sharp
aise ease	state of rest or well-being
alert	aware of surroundings; vigilant
anter/i	front
antrum	cave
assur	secure
ausculta	listen attentively to
benign	well-mannered
calor	heat
candida	fungus; usually used to designate the specific fungus <i>Candida albicans</i>
cardio	heart
caud	tail
charge	obligation
chondr	cartilage (as in the ribs)
chronic	pertaining to time
coron	crown
cran/i	head; skull
crino	secretion
dem/o dem/i/o	people, population
different	set apart
dila	increase in size
dist	farther away
dolor	pain
dors dorsum	back
drome	route, road
endo	inside

Root Word	Definition
etio	cause
febr	fever
front	front
gastr	stomach
genera	ancestry
genet	birth
gn[us]	birth
gnos	knowing
graph	device used for a test or procedure
hereditar	transmitted in a family line
homeo	same
horizon[t]	parallel to the ground
iatro	physician, healer
il[eum]	flank
infer	underneath
inguin	groin
later	to the side
lesion	damage, injury
lethargic	drowsy
lipid	fat
local	place
lumb[us]	lower back
marked	making an impression
med/i	towards the middle
morbid	sick
mortal	dead
observa	what is seen
occult	what is unseen
orient	towards the east (from using a compass)
ox	shortening of oxygen

Root Word	Definition
palliat	covering
palm	front surface of the hand
palpa	feeling with the fingers
path/o	disease
pend	hanging
percuss	tapping
phylax	protection
physi/o	function (literally "nature")
plant	bottom surface of the foot
port	carry
posi	place
poster/i	behind
press	pressure; dent
produc	giving rise to
progress	moving forward
prone	face down, bending forward
prox/im	near
ptom	something that falls on you
recurr	happening again
remark	making a dent again
remiss	diminishing; becoming less
rostr	nose
rubor	redness
rupt	breaking, bursting
sagitt	arrow
scope	instrument for looking inside
sequel	what comes next
skelet	bones
stas stasis	standing still

Root Word	Definition
statim	immediately
stem	standing
sterile	not producing offspring
super/i	toward the top
supine	face up, bending backward
tens	stretching, pressure (cf. tendon in Unit 1)
tom (Unit 1 suffix table)	cutting
tribut	to pay
turgor	swelling
umbilic[us]	navel, belly-button
vascul	little vessel (vas + cule)
vaso	vessel
ventr	belly
vers	across

UNIT 2 SUFFIXES

Suffixes introduced in Unit 1 are colored gray.

Suffix	Definition
–able	capable of
–al	pertaining to
–ance	to give
–ant	state of being
–ar	pertaining to
–ary	pertaining to
–ed	word ending creating past tense
–emia	present in blood, indicating disease
–ent	state of being
–gen	arising from
–ia	disease
–iac	pertaining to
–ial	pertaining to
–iasis	presence of an abnormality
–ic	pertaining to
–ical	pertaining to
–ile	pertaining to
–ing	verb ending that makes the continuous tense (ongoing process)
–ion	word ending that turns a verb into a noun
–ior	having a state
–is	anatomical structure
–ity	turns a noun into a state of being
–ive	to go
–ized	turns a noun into a verb
–logist	one who studies
–logy	study of
–olus	diminutive suffix
–or	pertaining to
–ory	pertaining to
–ous	word ending that turns a noun into an adjective

Suffix	Definition
--------	------------

-tion	word ending that turns a verb into a noun
-------	---

-tive	to go
-------	-------

-y	disease
----	---------

UNIT 2 ABBREVIATIONS

Starting with this unit, each unit will include some commonly used Medical English abbreviations for you to learn.

Abbreviation	Meaning
BP	blood pressure
c or c/ or c with a line over it	with (Latin <i>cum</i> , "with")
Dx	diagnosis
HR	heart rate
Hx	history
NPO	<i>nil per os</i> (Latin, "nothing by mouth")
PRN	<i>pro re nata</i> (Latin, "as the thing is needed")
Pt	patient
RR	respiratory rate
Rx	prescription (Latin <i>recipe</i>)
s or s/ or s with a line over it	without (Latin <i>sans</i> , "without")
Sx	sign and/or symptom
Sz	seizure
Tx	treatment
wt	weight

UNIT 2 WORD LIST

Word Usage	Word	Definition
SOAP note words	abdominal	pertaining to the abdomen (belly)
SOAP note words	abrupt	sudden
SOAP note words	acute	occurring suddenly (cf. chronic or insidious)
SOAP note words	afebrile	without fever
SOAP note words	alert	in a state of vigilance
general	anatomical	pertaining to anatomy
general	anatomy	body structure
directional term	anterior	toward the front (in humans, same as ventral)
general	antrum	body cavity
SOAP note words	auscultation	the act of listening through a stethoscope
SOAP note words	benign	not harmful
directional term	bilateral	on two sides (cf. unilateral)
general	cancer	disease characterized by abnormal, rapidly dividing cells
general	candidiasis	the presence of a fungal infection (typically, with the common fungus <i>Candida albicans</i>)
SOAP note words	cardiovascular	pertaining to the heart and blood vessels
directional terms	caudal	towards the tail (buttocks)
SOAP note words	chronic	occurring over a long time (cf. acute)
directional term	contralateral	on the opposite side
planes of section	coronal	anatomical plane that divides the body into front and back (same as frontal)
directional term	cranial	toward or pertaining to the head
SOAP note words	degeneration	the process of falling apart

Word Usage	Word	Definition
SOAP note words	diagnosis	the recognition of a disease from its signs and symptoms
SOAP note words	differential	separating one thing from another
SOAP note words	discharge	releasing from the hospital
SOAP note words	disease	condition of being unwell (literally, "without ease")
SOAP note words	disposition	frame of mind; outcome
directional term	distal	further away from the point of attachment (generally used only for extremities; cf. proximal)
directional term	dorsal	pertaining to the back
directional term	dorsum	the back
SOAP note words	dyslipidemia	condition of abnormal fats in the blood
SOAP note words	endocrinology	study of internal secretions (i.e. hormones)
SOAP note words	epidemiology	study of a disease in a population
abdominal regions	epigastric	on top of the stomach
SOAP note words	etiology	study of the origins of an illness
SOAP note words	exacerbation	the process of getting worse
SOAP note words	febrile	with a fever
planes of section	frontal	anatomical plane that divides the body into front and back (same as coronal)
general	genetic	pertaining to birth; a condition you are born with
general	hereditary	passed down through the genes
general	homeostasis	balance in body systems (literally, "keeping things the same")
planes of section	horizontal	anatomical plane that divides the body into top and bottom (same as transverse)

Word Usage	Word	Definition
SOAP note words	hypertension	abnormal condition of blood pressure that is too high
abdominal regions	hypochondriac	below the rib cartilage
abdominal regions	hypogastric	below the stomach
SOAP note words	hypotension	abnormal condition of blood pressure that is too low
SOAP note words	hypoxia	abnormal condition of oxygen concentration in the blood that is too low
SOAP note words	iatrogenic	condition caused by a medical procedure
SOAP note words	ictal	occurring suddenly
SOAP note words	idiopathic	without obvious cause; a disease that seems to come out of nowhere
abdominal regions	iliac	pertaining to the groin region
SOAP note words	impression	initial observations made by a clinician
SOAP note words	indolent	not causing pain; low priority for treatment
directional term	inferior	toward the bottom
abdominal regions	inguinal	pertaining to the groin
SOAP note words	insidious	slow and "sneaky" onset of a disease
directional term	ipsilateral	on the same side
directional term	lateral	toward the side (away from the midline, cf. medial)
SOAP note words	lesion	damage or injury
SOAP note words	lethargic	appearing to be drowsy
SOAP note words	localized	in one place, not widespread

Word Usage	Word	Definition
abdominal regions	lumbar	pertaining to the back
SOAP note words	malaise	disease, sickness
SOAP note words	malignant	harmful
SOAP note words	marked	especially noticeable (pronounced "mark-ud")
directional term	medial	toward the midline (away from the side, cf. lateral)
SOAP note words	metastasis	movement from one place to another
general	microscope	device for seeing small things
SOAP note words	morbidity	pertaining to disease
SOAP note words	mortality	pertaining to death
SOAP note words	noncontributory	not relevant to the patient's illness
SOAP note words	observation	watching the patient to assess further development of signs or symptoms
SOAP note words	occult	hidden; in an unexpected place
SOAP note words	oriented	a patient knows where, when, and who they are
SOAP note words	palliative	care not designed to cure, but to make the patient more comfortable
directional term	palmar	towards or pertaining to the palm
SOAP note words	palpation	examining by touching
SOAP note words	pathogen	an organism that gives rise to disease
SOAP note words	pathology	study of disease
general	pathophysiology	study of body dysfunction in disease
SOAP note words	pending	waiting for results

Word Usage	Word	Definition
SOAP note words	percussion	examining by tapping
general	physiology	study of body function
directional term	plantar	towards or pertaining to the underside of the foot
directional term	posterior	toward the back (in humans, same as dorsal)
SOAP note words	prognosis	predicting the future of a disease
SOAP note words	progressive	getting worse over time
directional term	prone	face down (cf. supine)
SOAP note words	prophylaxis	treatment to prevent or discourage disease
directional term	proximal	closer to the point of attachment (generally used only for extremities; cf. distal)
SOAP note words	radiographic	making an image using X-rays or other radiation
SOAP note words	reassurance	to give encouragement to help a patient's progress
SOAP note words	recurrent	happening over and over again
SOAP note words	remission	process of a disease going away
SOAP note words	reproductive	to bring into existence again; pertaining to the process of creating children
directional term	rostral	toward the nose
planes of section	sagittal	anatomical plane that divides the body into left and right
SOAP note words	sequelae	things that come out one after another; result of an illness (pronounced "sequel-ay")
SOAP note words	skeletal	pertaining to the bones of the body
general	stat	immediately
SOAP note words	sterile	unable to reproduce (of individuals or of micro-organisms)

Word Usage	Word	Definition
directional term	superior	toward the top
directional term	supine	face up (cf. prone)
SOAP note words	supportive	lifting up
SOAP note words	symptom	abnormal feelings reported by the patient (cf. signs, which are the result of objective clinical tests)
SOAP note words	syndrome	symptoms which occur together
SOAP note words	systemic	pertaining to an entire organ system or the entire body
planes of section	transverse	anatomical plane that divides the body into top and bottom (same as horizontal)
abdominal regions	umbilical	pertaining to the navel ("belly-button")
directional term	unilateral	on one side (cf. bilateral)
SOAP note words	unremarkable	not prominent; not important
SOAP note words	vasodilation	increase in the diameter of a blood vessel
directional term	ventral	pertaining to the abdomen (belly)

HOW THE UNIT 2 WORD LIST WAS BUILT

You are not required to learn how the Unit 2 words were built; you are responsible for the prefixes, roots, suffixes, and word list tables for each Unit. But if you want to know how to build the Unit 2 words, this table can serve as your guide.

Etymology	Prefix1	Prefix2	Root Root	"Post-Root"	Suffix Word End	Word
			abdomin		al	abdominal
	ab		rupt			abrupt
Latin acutus, "sharp, pointed"			acute			acute
	a		febr		ile	afebrile
Italian all'erte "to the high tower!"			alert			alert
	ana		tom		ical	anatomical
Greek ana + temnein, "to cut up"	ana		tom		y	anatomy
Latin ante (as in prefix), "more in front; earlier"			anter/i		or	anterior
Greek antron, "cave"			antrum			antrum
Latin auscultare, "listen attentively to"			ausculta		tion	auscultation
Latin benignus, "well-born" (cf. malignant)	beni		gn[us]			benign
	bi		later		al	bilateral
Calor is literally the Latin word for "heat", as in calorie.			calor			calor
">Latin cancer, from Greek karkinos, which has three meanings like the English word "cancer": a crab, the constellation which was thought to resemble a crab, or a tumor which penetrates and grabs onto normal tissue like a crab			cancer			
">Latin, candida, "shining white" from which we get the English words incandescent or candle; Latin albicans is "white" so it literally means "shining white-white"			candid[a]		iasis	candidiasis
			cardio	vascul	ar	cardiovascular
Latin cauda, tail. Also seen as cauda equina ("horse's tail") at the inferior end of the spinal cord			caud		al	caudal
Greek khronos, "time"			chronic			chronic
	contra		later		al	contralateral
			coron		al	coronal
			cran/i		al	cranial

Etymology	Prefix1	Prefix2	Root Root	"Post-Root"	Suffix Word End	Word
	de		genera		tion	degeneration
Greek diagignōskein, "to know one thing apart from another"	dia		gnos		is	diagnosis
			different		ial	differential
	dis		charge			discharge
	dis		ease			disease
Latin dis- "apart" + ponere "to put, place"	dis		posi		tion	disposition
			dist		al	distal
The Latin word <i>dolor</i> is pain or sadness. The name Dolores derives from St. Mary of the Sorrows (Spanish, <i>Maria de los Dolores</i>)			dolor			dolor
It is also featured in a hilarious episode of Seinfeld.						
Latin dorsum, "back"			dors		al	dorsal
			dorsum			dorsum
	dys		lipid		emia	dyslipidemia
			endo	crino	logy	endocrinology
Greek epi-, "among, upon" + dêmos, "a people" + -logy. Compare to democracy, demo + Greek - kratos, "rule, strength". Democracy is literally "rule by the people" and epidemic is literally "[disease] on top of the people".	epi		dem/i/o		logy	epidemiology
	epi		gastr		ic	epigastric
Greek aitia, "cause, responsibility"; also spelled ætiology			etio		logy	etiology
	ex		acerb[o]/a		tion	exacerbation
			febr		ile	febrile
			front		al	frontal
			genet		ic	genetic
			hereditar		y	hereditary
			homeo	stasis		homeostasis

Etymology	Prefix1	Prefix2	Root Root	"Post-Root"	Suffix Word End	Word
			horizon[t]		al	horizontal
	hyper		tens		ion	hypertension
	hypo		chondr		iac	hypochondriac
	hypo		gastr		ic	hypogastric
	hypo		tens		ion	hypotension
	hyp[o]		ox		ia	hypoxia
literally, "caused by a healer"; a condition caused by a medical procedure			iatro		gen + ic	iatrogenic
blow or strike; smite; describes a disease that "hits you upside the head"			ict[us]		al	ictal
	idio		path		ic	idiopathic
Latin ileum, singular form of ilia "flank, groin"			il[eum]		iac	iliac
Latin impressionem, "making a mark on through pressure"	im		press		ion	impression
First used to mean "causing no pain" (in-, "not" + -dol[or], "pain")	in		dol[or]		ent	indolent
			infer		ior	inferior
Latin inguinalis, "groin"			inguin		al	inguinal
From Latin insidiosus, "lying in wait", as an enemy crouched to ambush	in		sidi[osus]		ous	insidious
	ipsi		later		al	ipsilateral
Latin lateris, "the side"			later		al	lateral
Latin læsionem, "damage, injury"			lesion			lesion
Greek lethargikos, "drowsy"			lethargic			lethargic
			local		ized	localized
Latin lumbus "hip, loin"			lumb[us]		ar	lumbar
French word meaning "bad state of being" cf. disease	mal		aise [ease]			malaise
"badly born" mal- + Latin -gnus ("born")	mal/i		gn[us]		ant	malignant

Etymology	Prefix1	Prefix2	Root Root	"Post-Root"	Suffix Word End	Word
Clearly defined; easy to distinguish			marked			marked
			med/i		al	medial
Greek meta-, "change" + -histanai, "to stand or place"	meta		stasis			metastasis
	micro		scope			microscope
French morbidité, "in a state of disease"			morbid		ity	morbidity
			mortal		ity	mortality
literally, "join together to pay tribute to". Con- substitutes for com- before a t.	non	con	tribut		ory	noncontributory
			observa		tion	observation
			occult			occult
			orient		ed	oriented
Latin palliatavus, "under a cloak; concealed"			palliat		ive	palliative
			palm		ar	palmar
			palpa		tion	palpation
			path/o		gen	pathogen
			patho		logy	pathology
			patho	physi/o	logy	pathophysiology
			pend		ing	pending
			percuss		ion	percussion
			physio		logy	physiology
			plant		ar	plantar
Latin posterus (initial meaning coming after in time; here, means coming after anatomically)			poster/i		or	posterior
	pro		gnos		is	prognosis
			progress		ive	progressive
Latin pronus, "bent forward"			prone			prone
Greek prophylassein, "to keep watch, guard"	pro		phylax		is	prophylaxis

Etymology	Prefix1	Prefix2	Root Root	"Post-Root"	Suffix Word End	Word
			prox/im		al	proximal
	radio		graph		ic	radiographic
re, "again" + Latin assecurar, "to secure" + ance, "to give"	re		assur		ance	reassurance
			recurr		ent	recurrent
Latin remissionem, "diminishing; going away"			remiss		ion	remission
French reproduire	re		produc		tive	reproductive
Latin rostrum, "beak, muzzle, snout"			rostr		al	rostral
"Calor, dolor, rubor, turgor" is an old Latin mnemonic for the four cardinal signs of inflammation (in English: heat, pain, redness, swelling).			rubor			rubor
Latin sagittalis, "like an arrow"			sagitt		al	sagittal
			sequel			sequelae
			skelet		al	skeletal
Stat is a word beloved of TV shows and movies with a medical theme. It is a shortening of <i>statim</i>, Latin for "immediately, right away".			stat[im]			stat
			sterile			sterile
			super/i		or	superior
			supine			supine
	sup [sub]		port		ive	supportive
Greek píptō, fall	sym		ptom			symptom
Greek, "running together"	syn		drome			syndrome
	sy/s		stem		ic	systemic
Latin transvertere, "turning across"	trans		vers[us]			transverse
Latin turgere, "to swell".			turgor			turgor
Latin umbilicus "navel"			umbilic[us]		al	umbilical
	uni		later		al	unilateral
	un		remark		able	unremarkable

Etymology	Prefix1	Prefix2	Root Root	"Post-Root"	Suffix Word End	Word
			vaso	dila	tion	vasodilation
Latin venter, "belly"			ventr		al	ventral

PART III

UNIT 3: TALKING ABOUT MOLECULES, CELLS, AND TISSUES

UNIT 3 INTRODUCTION

The body is made up of trillions of cells that come together to form tissues, which work together to give organs their specific functions. How cells develop, multiply, and function is the basis for everything that happens in the body. The body expends a great amount of energy trying to keep everything working right and in balance. This process is called homeostasis and is central to the field of physiology, or the study of how the body works.

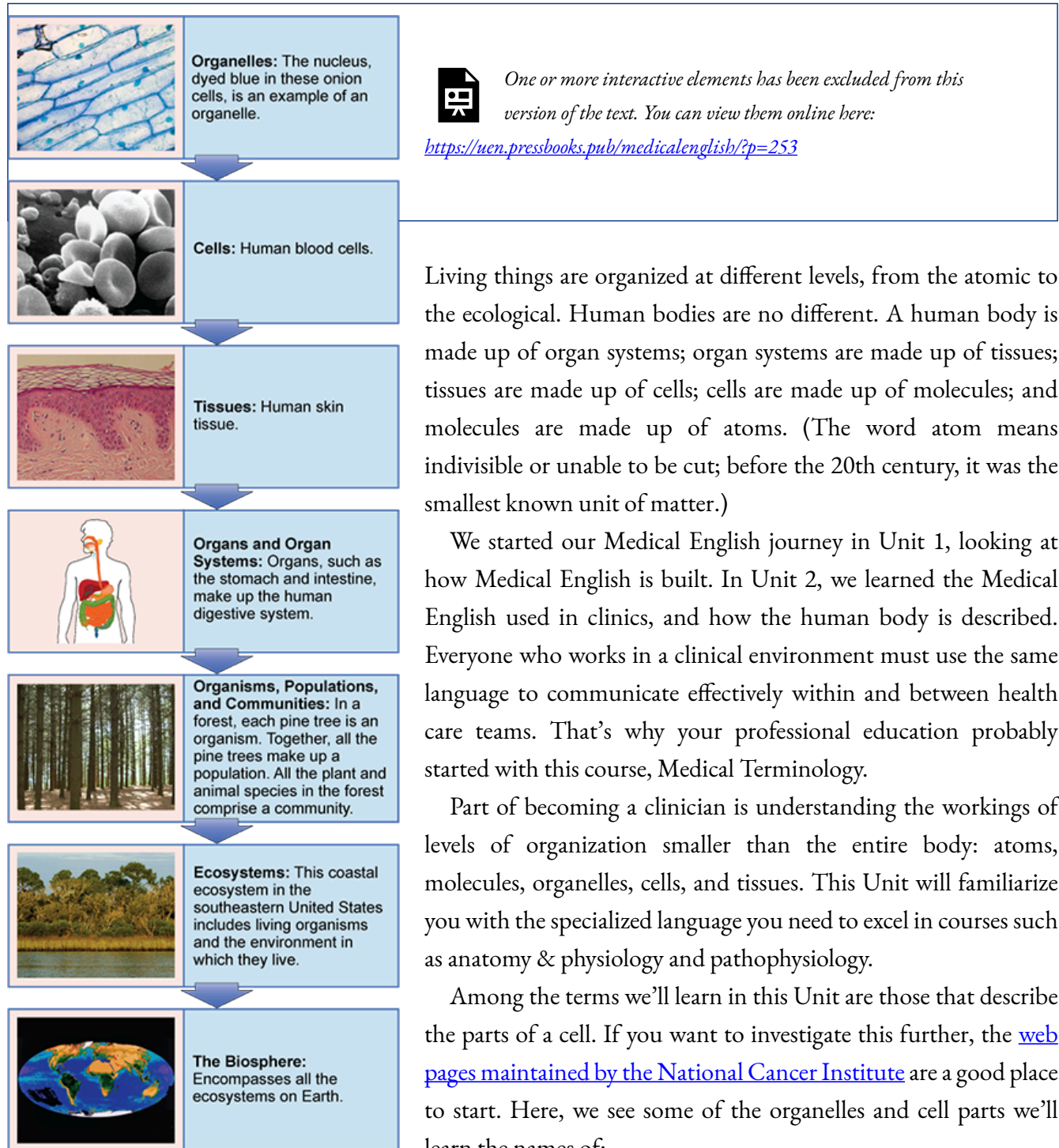
Objectives

After completing this unit, you will be able to:

- Name common terms used to describe molecules, cells, and tissues.
- Describe how these terms are built by combining prefixes, roots, and suffixes in combinations which give them meaning.

TALKING ABOUT MOLECULES, CELLS, AND TISSUES

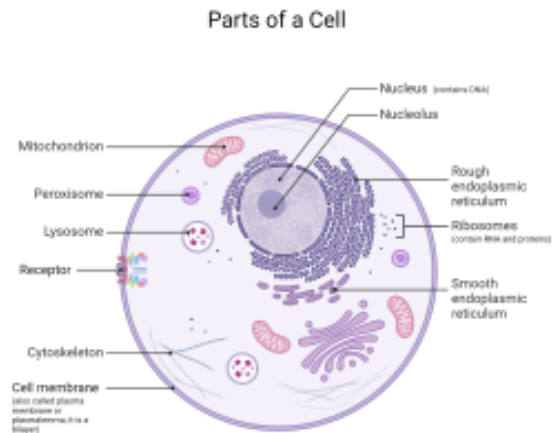
Only [Med Terms] in the Building Episode 3



Unit 3 figure 1 Levels of Organization

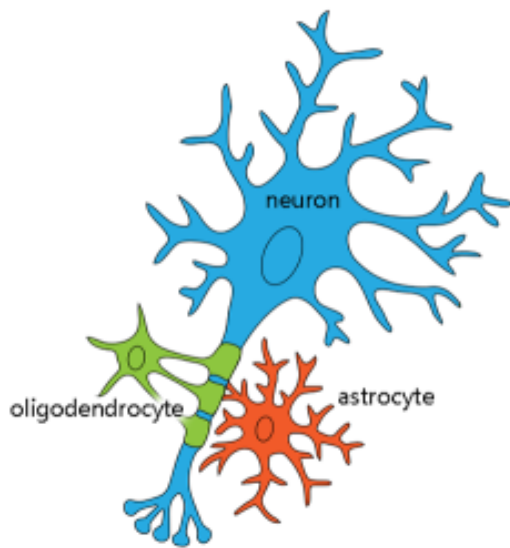
- the cell membrane, a **bilayer** made of **phospholipids** which surrounds the entire cell and keeps the insides in and the outsides out;
- membrane **receptors**, which allow the cell to interact with its environment, and react to external chemical signals like **epinephrine**, which are then turned by the receptor into electrical or chemical changes in the processes of the cell;
- the **nucleus**, which contains the genetic material, and the **nucleolus**, where the genetic material (**DNA**) is copied into **RNA**;
- the **ribosome** (not shown here), where RNA is turned into proteins, the workhorse molecule of the cell (ribosomes are often associated with the **endoplasmic reticulum**);
- mitochondria** (singular **mitochondrion**), which makes the energy needed for proper cell function.

Adapted from "Structural Overview of an Animal Cell" by Bluffdendron (2021).
Retrieved from <https://www.bioninja.com/resources/animal-cell>



Unit 3 figure 2 Parts of a Cell

Akiyao, University of Michigan Medical School, CC SA 3.0
(https://commons.wikimedia.org/wiki/File:Oligodendrocyte_Nb6_Astrocyte.svg)



Unit 3 figure 3 Cell Types

We'll also look at different cell types which make up tissues. Some diseases involve changes in the number and distribution of cells in the body. For example, **leukocytes** are white blood cells, critical for the body's immune defenses. If there are too many leukocytes of the wrong kind, then the disease **leukemia** results. **Astrocytes** (**astroglia**) are star-shaped cells which play an important role in maintaining brain function; if they become cancerous and their numbers increase uncontrollably, an **astrocytoma** may grow in the brain. **Oligodendrocytes** (**oligodendroglia**) have several arms that wrap parts of the nerve cell (**neuron**) in a protective layer. When they become cancerous, an **oligodendroglioma** may result.

Video Attribution

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UNIT 3 PREFIXES

Prefixes introduced in previous units are colored gray.

Prefix	Definition
a-	not
amphi-	around; on both sides
an-	not
ana-	up, throughout, again
anti-	against (an action)
apo-	from, off, away
auto-	self
bi-	two, double
cata-	down, downward
con-	normally, com- and con- mean "with, together". Here, it is an "intensive prefix" which strengthens the word part after.
de-	from, down, out
di-	separate, apart, double, two
en-	inside
endo-	inside
epi-	on top of
eu-	good; well-
exo-	out, outside, without
extra-	outside, more, beyond
haplo-	single
hetero-	other, different
holo-	whole
hyper-	too much
hypo-	too little
inter-	between, among
intra-	inside
iso-	equal
macro-	large
mero-	a part
meta-	between, with, after; change

Prefix	Definition
micro-	small
mono-	one, single
oligo-	Scanty; few - or- several; a handful
par-	beside; near
para-	
poly-	many, much
pro-	in front of; in favor of
pseudo-	false, lying
re-	again
sub-	an underlayer; support
sym-	with, together
telo-	an end; complete
trans-	through
tri-	three

UNIT 3 ROOTS

Root words introduced in previous units are colored gray.

Root Word	Definition
acetyl	the name of a chemical group
acid	a property of some chemicals
aden	gland
adip/o	fat
aero	air
albumen albus	(egg) white
alcoh[ol]	the name of a chemical group; alcohols end in the suffix "-ol"
alkali	a property of some chemicals
allele	one form of a gene (unit of inheritance)
andr/o	male
apex	tip; the highest point
astro	star
bios	life
base	bottom; the lowest point
blast/o	a rapidly dividing cell
body	large molecule
bol	throw (forward movement)
carbo	carbon, an element found in all living things
cell	the unit of life; a mixture of chemicals and organelles bounded by a membrane
centri centro	center
chol/e	bile
chondr/o	cartilage
chondrion	grain
chrom/at/o	color
cilia	eyelash
code	letters that contain a (sometimes hidden) meaning
coll	glue
column	a shape higher than it is wide
crine	secretion; release from cells or from the body

Root Word	Definition
cub[e]	a shape of equal height and width
cyst	fluid-filled space
cyt cyte cyto	cell
dendr/o	tree
diffus	pouring out
dominant	showing a quality if there is only one copy (of a gene)
dynam	power, energy
elast[ic]	capable of being stretched and snaps back into place when released
electr/o	related to electricity
ergon	energy
erythr/o	the color red
ester	the name of a chemical group
estrus	reproductive cycle in female non-human mammals
ethane	the name of a chemical group
expressed	being shown
fibro	fiber
filament	a thread
fruct	fruit
gamete	a cell needed for reproduction
gastr	stomach
gen geno genet	birth (dealing with inheritance)
gland	gland
glia	glue
glob	glob, wad (pertaining to the shape of a protein)
gluco glyco	sweet, sugar; the name of a chemical group that makes a chemical sweet
glycerine	a sweet, sticky chemical (glycer- + -ine)
goblet	goblet, chalice; a drinking vessel which holds a large amount of liquid

Root Word	Definition
guano	bat poop
hem/o	blood
hist/o	tissue
hydra hydro	water
immuno	body defenses against micro-organisms and other invaders
insula	island
ion	particle which carries an electrical charge
kalium	Latin name for the element potassium (found in the vegetable kale)
kary/o	the cell nucleus (from the Greek for nut)
keto	the name of a chemical group
kin/e	movement; also used to describe the addition of a phosphate chemical group
lact	milk
lamina	thin plate, sheet, layer
lariat	a loop
layer	layer
leuk/o	the color white
ligand	a molecule that attaches to a receptor
lip/o	fat
lymph/o	vessels that drain fluid from around the cells
lyse lyso	breakdown
lyte	product of a breakdown
lyze	process of breaking down
magnet	substance which attracts other magnetic substances
malt	grain that has been steeped, germinated, and dried
mega	large (cf. the suffix megaly , "a condition of being large")
mei	less
mer mere	a part
methyl	the name of a chemical group

Root Word	Definition
mito	thread
mole	word describing a unit in chemistry
myc/o	fungus
natr[ium]	Latin word for the element sodium
natur/e	the state something was born in
neo	new
nephr	kidney
neutral	having no electrical charge
nucleus nucle nucleo nucleot	the center of an atom or the center of a cell
organ	instrument
osmos	pushing, thrusting
osteo	bone
oxi oxy	shortened form of the element oxygen
parallel	side-by-side
path	feeling, suffering (used to describe molecules)
path/o	disease (used to describe a medical condition, in the sense of "suffering")
pen[dere]	hang
pept peptido	protein
per	large amount of
phage phago	eat
phase	stage; appearance
pheno	shine; appearance
phil	love
phob	fear
phospho/r	the element phosphorus and related chemical groupings
phot	light
pino	drink

Root Word	Definition
plasm plasma	something molded; plastic or dough
plate	plate
ploid	fold; numbered
pol[e]	one end of a round object, as in North Pole & South Pole
port	to carry; a door
prim	first
prosta[te]	gland in male that surrounds the urethra
prot prote proteo	first, original; class of molecules that are essential for life
ptosis	drooping
receptor	a protein on the surface of a cell that turns a chemical on the outside to a signal on the inside of the cell
recess	showing a quality if there are two copies (of a gene)
regula	rule; ruler or straight piece of wood
ret[e]	net, network
ribose	the name of a chemical group found in cells
script	writing
secre	set apart; release
skelet	bones
sol solut	dissolve
soma somat somato some	body
son	sound
squam	a shape wider than it is high (literally, "fish-scale")
ster	solid; a chemical that is solid at room temperature
stat stit	to set
stratum	layer
sucr	sugar

Root Word Definition

surf[ace]	outside part
test[is]	male testicle
thel[e]	nipple
therm/o	temperature
thromb/o	clot
thym[us]	an organ of the upper chest
tom	cutting
ton	firmness (for example, of tissues or muscle)
top[os]	place
translation	carry across
trop	turning towards
tub[e]	pipe
type	kind
ur	urine
valence	strength
vascul	little vessel
vesic	bladder
villus	hair
zyme	yeast; ferment

UNIT 3 SUFFIXES

Suffixes introduced in previous units are colored gray.

Suffix	Definition
-al	pertaining to
-an	chemical name ending
-ance	to give
-ant	state of being
-ar	pertaining to
-ase	chemical name ending for an enzyme (a protein that promotes the breaking or building of chemical bonds)
-ate	-ate word ending that turns a noun into an adjective that means "showing"; also a chemical name ending
-cule	diminutive suffix (describes a smaller version of the word in front of it)
-ein	chemical name ending
-elle	diminutive suffix
-emia	present in blood, indicating disease
-er	word ending that turns a verb into a noun
-gen	arising from
-ic	pertaining to
-ical	pertaining to
-icle	diminutive suffix
-icul	diminutive suffix
-id	chemical name ending
-ide	chemical name ending
-ified	word ending that turns a noun into an adjective
-il	chemical name ending
-in	chemical name ending for a protein
-ine	chemical name ending
-ion	word ending that turns a verb into a noun
-is	process word ending
-ism	belief; process word ending
-ity	turns a noun into a state of being
-ium	anatomical structure word ending
-ive	to go

Suffix	Definition
-let	diminutive suffix
-logy	study of
-lysis	breakdown process
-oid	resembling
-ol	chemical name ending for an alcohol
-ole	diminutive suffix
-olus	diminutive suffix
-on	unit of anatomical or functional structure
-one	chemical name ending
-ose	chemical name ending for a sugar
-osis	process
-ote	word ending indicating membership in a group
-ous	word ending that turns a noun into an adjective
-penia	low levels of; scarcity
-sion	word ending that turns a verb into a noun
-sis	process word ending
-some	body; structure
-tic	pertaining to; t added to make it sound better
-tion	word ending that turns a verb into a noun
-tive	to go
-type	impression made by
-ula	diminutive suffix
-ule	
-um	anatomical structure word ending

SUFFIXES WITH SPECIAL MEANINGS IN CHEMISTRY

Chemical Name Ending	Meaning
–ase	name for an enzyme, a protein or RNA molecule which promotes the forming or breaking of chemical bonds
–in	name for a protein, a large molecule made up of many amino acids
–ol	name for an alcohol
–ose	name for a sugar

UNIT 3 ABBREVIATIONS

Abbreviation	Meaning
DNA	deoxyribonucleic acid, the carrier of genetic information for the cell
EtOH	ethanol
MTOC	microtubule organizing center, where microtubules are made
RNA	ribonucleic acid, the "go-between" linking DNA to protein synthesis

UNIT 3 WORD LIST

Word	Definition
acetylation	addition of an acetyl group
acetylcholine	chemical signaling molecule
acetylcholinesterase	enzyme that breaks down acetylcholine
acid	chemical that gives off hydrogen ions
acidic	quality of being acid
adenine	molecule found in DNA and RNA; pairs with thymine (in DNA) or uracil (in RNA)
adipocyte	fat cell
aerobic	activity that occurs when oxygen is plentiful
albumin	protein found in egg white; major blood protein
alcohol	chemical group that includes grain alcohol, ethanol (EtOH)
alkali	chemical that absorbs hydrogen ions
alkaline	quality of being alkali
allele	unit of inheritance
amphipathic	a molecule that partially prefers water and partially repels water
anabolic	chemical process that makes larger chemicals out of smaller ones
anaerobic	activity that occurs when oxygen is scarce
anaphase	stage of cell division where chromosomes are separating
androgen	lipid signaling molecule that promotes development of male sexual characteristics
antibody	protein that defends the body against invaders; immunoglobulin
antiparallel	parallel, but running in opposite directions, like a two-lane road
antiport	transport process that involves molecules moving in opposite directions; exchange
apical	toward the top
apocrine	cellular secretion that involves breaking off a piece of the cell
apoptosis	programmed or preplanned cell death
astrocyte	star-shaped cell of the brain; a type of glial cell (same as astroglia)
astroglia	star-shaped cell of the brain; a type of glial cell (same as astrocyte)
atom	the smallest unit of matter; indivisible quantity (name means "can't be cut")
autosome	chromosome that is not involved in sex determination
avascular	without blood vessels
basal	toward the bottom

Word	Definition
base pair	adenine-thymine or cytosine-guanine pairs that form the "rungs" of the twisted ladder structure of DNA
bilayer	two-layered covering; cell membrane
blastocyst	stage of a human embryo where it is a hollow ball of rapidly-dividing cells
carbohydrate	molecule consisting of equal parts carbon and water
catabolic	chemical process that makes smaller chemicals out of larger ones
catalyze	to speed up a chemical reaction
centriole	organelle that acts as a microtubule organizing center (MTOC)
centromere	part of a chromosome in the center, where the two DNA molecules touch
centrosome	organelle that acts as a microtubule organizing center (MTOC)
chimera	animal that is a mixture of two genetic types
cholesterol	important lipid, essential for cell processes
chondrocyte	cartilage cell
chromatin	DNA plus RNA plus associated proteins, visible when cells are dyed and viewed through the microscope
chromosome	chromatin visible through the microscope in X-shaped packages
cilia	eyelash-like extension of the cell, used to move substances on the cell surface
codon	unit of genetic information; thousands of codons make up a gene
colloid	microscopic particles (larger than molecules) surrounded by water
columnar	cells that are higher than they are wide
constitutive	ongoing, continuous
cuboidal	cells that are about as high as they are wide ("cube-like")
cytokinesis	the movement of cells in cell division
cytoplasm	gel-like substance, including organelles, that fills cells and is surrounded by a membrane
cytosine	molecule found in DNA and RNA; pairs with guanine
cytoskeleton	organelle composed of structural support proteins
cytosol	cytoplasm minus organelles
dehydration	chemical reaction which removes water and forms a new chemical bond
denaturation	using heat or chemicals to alter the normal (native) structure of proteins, DNA, or RNA
deoxyribose	a sugar that, along with phosphate groups, forms the backbone of the DNA molecule

Word	Definition
diffusion	process by which molecules move from areas of high concentration to areas of low concentration
diploid	containing two copies of each DNA molecule; the normal number of DNA molecules in most human cells
dominant	in Mendelian genetics, one copy of a dominant allele is expressed as an observable characteristic
elastin	a protein; snaps back into shape when released after being stretched
electrolysis	process of breaking molecules (or hairs) through electricity
electrolyte	charge-carrying ion in the blood
electromagnetic	radiation that depending on energy level can be radio waves, microwaves, visible light, X-rays, or gamma rays
electron	subatomic particle with a negative charge; found on the outside of the atom
endergonic	chemical reaction which requires energy
endocytosis	intake of substances by the cell
endogenous	coming from inside
endoplasmic reticulum	organelle with net-like structure found in cytoplasm
endothelium	tissue lining blood vessels
endothermic	chemical reaction which absorbs heat energy; same thing as endergonic
entropy	the tendency of matter to become disorganized
enzyme	protein that speeds up chemical reactions in the cell
epigenetic	changes in DNA that are outside of the normal molecular genetic processes
epinephrine	chemical signaling molecule
epithelium	tissue lining body cavities and outside surfaces of the body
erythrocyte	red blood cell
erythrocytosis	abnormally high levels of red blood cells
estrogen	lipid signaling molecule that promotes development of female sexual characteristics
ethanol	alcohol that forms part of liquor, wine, and beer; grain alcohol
euchromatin	unspooled chromatin; chromatin in the active process of being used
eukaryotic	cells with a true nucleus
exergonic	chemical reaction which releases energy
exocytosis	release of substances from a cell
exogenous	coming from outside

Word	Definition
exon	the part of a gene that is turned into protein
exothermic	chemical reaction which releases heat energy; same thing as exergonic
extracellular	outside of cells
fibroblast	connective tissue cell
fructose	a sugar; converted by liver to glucose
gamete	eggs in females and sperm in males; the cells specialized for reproduction
gastrulation	process of embryonic development that forms a "little stomach" or invagination
genotype	silent characteristic of DNA that results from the sequence of DNA base pairs
glia	non-excitabile cells of the brain; perform housekeeping duties for the brain
gluconeogenesis	process by which new glucose molecules are made inside the body
glucose	a sugar; main energy carrier in blood
glycerol	alcohol derived from sugar
glycogen	storage form of glucose, mostly found in liver and muscle
glycogenolysis	breakdown of glycogen
glycolysis	breakdown of glucose
glycoprotein	protein + sugar; more protein than sugar
goblet cell	mucus-making cell
Gram stain	a staining technique used to classify bacteria by their microscopic appearance
guanine	molecule found in DNA and RNA; pairs with cytosine
haploid	containing one copy of each DNA molecule; the normal number of DNA molecules in gametes
hemoglobin	protein that carries oxygen and CO ₂ in the blood
heterochromatin	spooled chromatin; chromatin in storage
histological	pertaining to the study of tissues
histology	study of tissues
histone	substance found in tissues that binds to DNA
histopathology	study of how tissues change during disease
histone	protein spool around which DNA is wound
holocrine	cellular secretion that involves the cell exploding
hydrolysis	chemical reaction which adds water and breaks a chemical bond
hydrophilic	a molecule that prefers water

Word	Definition
hydrophobic	a molecule that repels water
hypercholesterolemia	high cholesterol levels in the blood
hyperkalemia	high potassium levels in the blood
hybernatemia	high sodium levels in the blood
hypertonic	solution with more particles per volume than the inside of a cell
hypokalemia	low potassium levels in the blood
hyponatremia	low sodium levels in the blood
hypotonic	solution with fewer particles per volume than the inside of a cell
immunoglobulin	protein that defends the body against invaders; antibody
insulin	protein that helps cells manage blood sugar
interphase	stage of cell life cycle that does not involve active division
intron	the part of a gene that is discarded in protein synthesis
ionic	pertaining to charged particles (atoms or molecules)
isomer	different arrangements of the same atoms in the same quantities
isotonic	solution with the same number of particles per volume as the inside of a cell
isotope	different forms of the same element where the number of neutrons varies but the number of protons stays the same
ketoacidosis	high ketone and acid levels in the blood
ketone	chemical byproduct of lipid breakdown
kinase	enzyme that adds a phosphorus-containing chemical group
lactose	a sugar; found in milk
lamina	layer; "plate"
lariat	loop of RNA that is discarded as an intron
leukocyte	white blood cell
leukocytosis	abnormally high levels of white blood cells
ligand	a signaling molecule or drug that binds a receptor
lipoprotein	a protein that carries fat molecules in the blood
liposome	microscopic balls of fat molecules
lymphocyte	white blood cell (a subtype)
lysosome	organelle that breaks down substances that the cell no longer needs

Word	Definition
macrophage	cell that eats large particles, including bacterial invaders
maltose	a sugar; derived from malt
megakaryocyte	literally, "cell with a large nucleus"; breaks into fragments which help in blood clotting (platelets, also called thrombocytes)
meiosis	process of forming gametes by reorganizing and reducing the amount of DNA in each cell
merocrine	cellular secretion that involves the cell releasing substances using vesicles
metabolism	anabolism + catabolism
metaphase	stage of cell division where chromosomes are lined up
methylation	addition of a methyl ($-CH_3$) group
microfilament	thread-like part of the cytoskeleton
microglia	defensive cell in the brain
microtubule	tube-like part of the cytoskeleton
microvillus	microscopic shaggy hair-like structures on the surface of intestinal cells and other absorptive epithelia
mitochondrion	organelle that creates cellular energy
mitosis	process of cell division that results in diploid cells (non-gamete cells; somatic cells)
mole	unit describing number of molecules in a solution
molecule	arrangements of two or more atoms into larger structures (typically, about one-thousandth of a millimeter)
monomer	single molecular unit
mycoplasma	type of bacterium that causes "walking pneumonia" and genital infections
mycosis	process involving a fungus; fungal disease
neutron	subatomic particle with no charge; found in the nucleus of the atom
norepinephrine	chemical signaling molecule
nucleolus	"little nucleus": RNA-making factory in the nucleus of a cell
nucleoside	part of DNA and RNA: base (adenine, cytosine, guanine, thymine, or uracil) plus sugar
nucleotide	monomer of DNA or RNA; nucleoside plus 1 phosphorus and 4 oxygen atoms
nucleus	organelle where genetic material is stored and manipulated
oligodendrocyte	a cell in the nervous system that forms insulating sheaths; same as oligodendroglia
oligodendroglia	a cell in the nervous system that forms insulating sheaths; same as oligodendrocyte
organelle	microscopic part of a cell where specific cell functions are carried out

Word	Definition
osmosis	process of diffusion through a membrane that only allows water to pass
osteocyte	bone cell
parenchyma	the proper tissue of an organ (as distinguished from connective tissue, blood vessels, etc.)
peptidoglycan	sugar + protein; more sugar than protein
peroxide	water with an extra oxygen atom; H ₂ O ₂
peroxisome	organelle that uses peroxide to break down substances in the cell
phagocytosis	process of cell eating large particles
phenotype	observable characteristic that results from genetic plus epigenetic processes
phospholipid	phosphorus-containing fat molecule
phosphorylation	process of removing a phosphorus-containing chemical group
photon	subatomic particle that carries electromagnetic (light) energy
pinocytosis	process of cell drinking
platelet	cell fragment responsible for blood clotting; thrombocyte
ploidy	number of DNA molecules in the cell
polar	molecule with an unequal distribution of electrons; this makes it water-loving (hydrophilic)
polymer	combination of two to 250 million molecular units
polypeptide	small protein; string of many amino acids (generally, between 2 and 100ish)
polyploid	containing more than two copies of each DNA molecule
primer	a DNA or RNA fragment that starts a chemical process in the cell
prokaryotic	cells without a true nucleus
prophase	stage of cell division that begins the division process
prostaglandin	lipid signaling molecule
proteasome	organelle that breaks down proteins that the cell no longer needs
protein	a string of amino acids which is responsible for most cell functions (generally, between 100 and 34,350 amino acids)
proteoglycan	sugar + protein; more sugar than protein
proton	subatomic particle with a positive charge; found in the nucleus of the atom
pseudostratified	appearing to be layered when there is actually a single layer of cells
receptor	a protein in the cell membrane; when it binds a signaling molecule it causes a change in the cell
recessive	in Mendelian genetics, two copies of a recessive allele are needed to create an observable characteristic

Word	Definition
regulate	to control by rule or direct
renaturation	restoring the normal (native) structure of proteins, DNA, or RNA
resonance	property of molecules where there is extensive sharing of electrons between atoms
reticular	net-like
ribose	a sugar that, along with phosphate groups, forms the backbone of the RNA molecule
ribosome	organelle that is a protein-making machine; consists of RNA plus protein
secretion	release of substances from a cell or from the body
solution	molecular-size particles surrounded by water
somatic	most of the body cells; cells that have no reproductive function (non-gametes)
squamous	cells that are wider than they are high
steroid	lipid signaling molecule based on cholesterol
stratified	layered
stratum	layer
substrate	molecule that an enzyme acts upon
sucrose	a sugar; cane sugar or table sugar
surfactant	chemical substance that helps lipids dissolve in water; detergent
suspension	cells or other large microscopic particles surrounded by water
symport	transport process that involves molecules moving in the same direction
telomere	part of a chromosome at the ends
telophase	stage of cell division that ends the division process
testosterone	lipid signaling molecule that promotes development of male sexual characteristics
thermodynamics	branch of physics that studies how energy is managed by systems like cells
thrombocyte	cell fragment responsible for blood clotting; platelet ("small plate")
thrombocytopenia	abnormally low levels of thrombocytes
thrombocytosis	abnormally high levels of thrombocytes
thymine	molecule found in DNA; pairs with adenine
tonicity	number of particles per unit volume (i.e. concentration of particles)
transcription	process by which RNA is made from DNA
translation	process by which protein is made from RNA
triglyceride	form of lipid found in the blood

Word	Definition
uracil	molecule found in RNA; pairs with adenine
valence	property of atoms important to understand formation of bonds by sharing or trading of electrons
vesicle	organelle that contains molecules to be secreted or absorbed by the cell
villus	shaggy hair-like structures lining the intestines

HOW THE UNIT 3 WORD LIST WAS BUILT

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Acetylation can affect how DNA and proteins act in the body.			acetyl		ation		acetylation
choline isolated from pig and ox bile		acetyl	chol		ine		acetylcholine
			chol	in[e] ester	ase		acetylcholinesterase
Latin acidus, "sour, sharp, tart"			acid				acid
			acid		ic		acidic
derived from the pancreas of an ox			aden		ine		adenine
		adip/o	cyt/e				adipocyte
able to live only if oxygen is present; exercise that uses oxygen as a fuel source		aero	b[ios]		ic		aerobic
">Latin albumen means "the white of an egg"; it is the neuter of Latin albus, "white"			album[en]		in		albumin
Arabic al-kuhul			alcoh[ol]		ol		alcohol
From Arabic al-qaliy, "ashes, burnt ashes" (of a plant called saltwort, which grows in salty soil). Ultimately from Arabic qala, "to roast in a pan".			alkali				alkali
			alkal[i]		ine		alkaline
German allelomorph, from Greek allel+morphē "other form"			allele				allele
	amphi		path		ic		amphipathic
Greek ana- "upward" + Greek -ballein "throw"	ana		bol		ic		anabolic

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
able to live if oxygen is absent (e.g. inside a closed wound); exercise when oxygen is unavailable as a fuel source	an	aero	b[ios]		ic		anaerobic
	ana		phase				anaphase
			andr/o		gen		androgen
	anti		body				antibody
	anti		parallel				antiparallel
Latin portare, "carry"	anti		port				antiport
			ap[ex]		ical		apical
Greek krīnō, "to separate"	apo		crine				apocrine
Greek apo- "separation" + Greek -ptosis "falling off", the shedding of leaves in the autumn or flower petals	apo		ptosis				apoptosis
		astro	cyte				astrocyte
		astro	glia				astroglia
Greek "indivisible"	a		tom				atom
	auto		some				autosome
	a		vascul[um]				avascular
			bas[e]		al		basal
adenine, cytosine, guanine, thymine and uracil are collectively called "bases". They form pairs that hold the two "rungs" of the "twisted ladder" (double helix) of DNA together			base	pair			base pair
	bi		layer				bilayer
		blasto	cyst				blastocyst

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
		carbo	hydr		ate		carbohydrate
Greek kata- "down" + Greek ballein "throw"	cata		bol				catabolic
Greek kata- "down, completely" + Greek lyein "loosen"	cata		lyze				catalyze
			centri		ole		centriole
		centr/o	mere				centromere
		centr/o	some				centrosome
Greek khimaira, an animal that is a mixture of two genetic types, named for the creature (with a lion's head, a goat's body, and a dragon's tail) from Greek mythology that was slain by Bellerophon			chimera				chimera
Greek khole, "bile" + steros, "stiff" (found in gallstones)		chole	ster[os]		ol		cholesterol
		chondr/o	cyte				chondrocyte
			chromat		in		chromatin
		chrom/o	some				chromosome
			cilia				cilia
			cod[e]		on		codon
			coll		oid		colloid
			column		ar		columnar

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
The Latin <i>constituere</i> means "to set up, fix, establish, set in order". Normally, the prefix <i>con-</i> means "together" but here etymologists think it's acting as an emphasis, what they call "an intensive prefix". In politics, this gives us the word "constitution". In cell and molecular biology, "constitutive" means "ongoing, required to operate".	con		stit/u		tive		constitutive
			cub[e]		oid	al	cuboidal
		cyto	kine		sis		cytokinesis
			cyt/o	plasm			cytoplasm
DNA and RNA base isolated from cells			cyt		os[e]	ine	cytosine
		cyto	skelet		on		cytoskeleton
		cyto	sol[ution]				cytosol
	de		hydr/a		tion		dehydration
	de		natur[e]	a	tion		denaturation
	de	oxy	ribose				deoxyribose
			diffus		ion		diffusion
	di		ploid				diploid
Latin <i>dominari</i> "to rule, govern"			dominant				dominant
			elast[ic]		in		elastin
		electr/o	lys[e]		is		electrolysis
electro- + Greek <i>lytos</i> "loosed" (past of <i>lyein</i> "loosen")		electr/o	lyte				electrolyte

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
electro- + Latin magnetum, magnetic stone mined in Magnesia, a province in Greece (Thessaly). The name of the element magnesium, which is not magnetic, comes from the same geographic source.		electr/o	magnet				electromagnetic
Latin electrum, "amber", because when you rub amber it gives off electrons			electr		on		electron
	end[o]		ergon		ic		endergonic
	endo		cyt		osis		endocytosis
	endo		gen		ous		endogenous
	endo		plasm		ic		endoplasmic
Greek endo- + Greek thēlē, "nipple". Formed backwards from "internal epithelium".	endo		thel[e]		ium		endothelium
	endo		therm[o]		ic		endothermic
	en		trop		y		entropy
	en		zyme				enzyme
The root word genetic was coined by Carlyle 1831 as fake-Greek genetikos from Greek root genesis "origin"	epi		genet		ic		epigenetic
Also called adrenaline. Named this because it was first isolated from the adrenal medulla, a gland on top of the kidney.	epi		nephtr		ine		epinephrine
	epi		thel[e]		ium		epithelium
		erythr/o	cyt/e				erythrocyte
		erythr/o	cyt		osis		erythrocytosis

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Estrus (the reproductive cycle in most mammals) + -gen. Humans menstruate monthly; most mammals go through estrus twice a year.			estr[us]/o		gen		estrogen
			ethan[e]		ol		ethanol
	eu		chromat		in		euchromatin
Greek eu- + Greek karyon "nut"	eu		karyo		tic		eukaryotic
	ex[o]		ergon		ic		exergonic
	exo		cyt		osis		exocytosis
	exo		gen		ous		exogenous
From ex[pressed] + [regi]on			ex[pressed]	[regi]on			exon
	exo		therm[o]				exothermic
	extra		cell		ul[e]	ar	extracellular
		fibro	blast				fibroblast
			fruct		ose		fructose
			gamete				gamete
			gastr		ula	tion	gastrulation
The root word genetic was coined by Carlyle 1831 as fake-Greek genetikos from Greek root genesis "origin"		geno	type				genotype
Short for neuroglia: neuro + Greek glia, "glue"			glia				glia
		gluco	neo	genes	is		gluconeogenesis
Greek gleukos, "sweet wine"; it gave its ending to sugars of all sorts			gluc		ose		glucose
Greek glukeros, "sweet"			glycer		ol		glycerol

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
			glyc/o		gen		glycogen
			glyc/o		gen/o	lysis	glycogenolysis
			glyc/o		lysis		glycolysis
		glyc/o	prot[o]		ein		glycoprotein
French gobelet, "cup", in this case a cup of mucus, which is kinda disgusting.			gob[e]let				goblet
">A staining method for bacteria developed by the Danish physician Hans Gram in 1883			Gram	stain			Gram stain
Originally isolated from bat poop			guan[o]		ine		guanine
Greek haplo, "single" + ploos, "fold"	ha[plo]		ploid				haploid
			hem/o	glob	in		hemoglobin
	hetero		chromat		in		heterochromatin
The Greek word histos means "warp" (as in warp and weft) or "web". In the 1800s, when the medical writers of the time were looking for a Greek word that meant "tissue" (even though ancient Greeks did not have microscopes), they decided "web" or "vertical part of woven fabric" was a pretty good analogy.			hist/o		log[y]	ical	histological
			hist/o		logy		histology
The Greek word ιστός, histós, "loom" is the ultimate origin of hist- and histo- words.			hist[o]		one		histone
			hist/o	path/o	logy		histopathology

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
	holo		crine				holocrine
			hydro		lysis		hydrolysis
		hydro	phil		ic		hydrophilic
		hydro	phob		ic		hydrophobic
			chol/e	ster[oid]	ol	emia	hypercholesterolemia
	hyper		kal[ium]		emia		hyperkalemia
	hyper		natr[ium]		emia		hypernatremia
	hyper		ton		ic		hypertonic
	hypo		kal[ium]		emia		hypokalemia
	hypo		natr[ium]		emia		hyponatremia
	hypo		ton		ic		hypotonic
			immuno	glob	ul[e]	in	immunoglobulin
Latin insula, "island", because it comes from the islets of Langerhans in the pancreas			insul[a]		in		insulin
	inter		phase				interphase
intr[agenic] + [regi]on	intr		[genic]	[regi]on			intron
			ion		ic		ionic
Greek iso- "equal" + Greek meros, "part"	iso		mer				isomer
	iso		ton		ic		isotonic
Greek isos "equal" + topos "place". Having the same place in the periodic table.	iso		top[os]				isotope
		keto	acid		osis		ketoacidosis
Gmelin coined in 1851 by dropping the "a" from German Aketon, derived from Latin acetum "vinegar, acid" + Greek-based chemical suffix -one			ket[o]		one		ketone
			kin		ase		kinase

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin lac, "milk" + -ose, chemical ending for sugars from mid-1800s French chemists			lact		ose		lactose
			lamina				lamina
Spanish la reata "the rope"			lariat				lariat
		leuk/o	cyt/e				leukocyte
		leuk/o	cyt		osis		leukocytosis
Latin ligandus, "to bind"			ligand				ligand
		lip/o	prot[o]		ein		lipoprotein
		lip/o	some				liposome
		lymph/o	cyt/e				lymphocyte
		lys/o	some				lysosome
	macro		phage				macrophage
			malt		ose		maltose
	mega		kary/o	cyte			megakaryocyte
Greek meiosis, "lessening"			mei		osis		meiosis
Greek mero-, "partial" + Greek krīnō, "to separate"	mero		crine				merocrine
Greek metabole, "a change" (meta-, "change" + ballein, "to throw")	meta		bol		ism		metabolism
	meta		phase				metaphase
Methylation is used to regulate cellular processes, such as gene expression and cell division.			methyl		ation		methylation
Latin filamentum, "fine untwisted thread"	micro		filament				microfilament

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
	micro		glia				microglia
	micro		tub[e]		ule		microtubule
Latin villus, "tuft of hair, wool"	micro		villus				microvillus
Greek mitos, "thread" + khondros, "granule" + -ion diminutive suffix		mito	chondrion				mitochondrion
			mit[o]		osis		mitosis
German, coined in 1900 by Ostwald, short for Molekül			mole				mole
Coined 1811 by Avogadro from Latin mole, "mass" + diminutive suffix -cule			mole		cule		molecule
Greek monos, "single" + Greek meros, "part"	mono		mer				monomer
">An infection which was first thought to be a mixture of a fungus (myco-) and the cytoplasm of host cells; now known to be caused by a bacterium			myc/o	plasma			mycoplasma
			myc		osis		mycosis
			neutr[al]		on		neutron
another substance, related to epinephrine/ adrenaline, that is isolated from the adrenal medulla, a gland above the kidney	nor[mal]	epi	nephr		ine		norepinephrine
			nucle		olus		nucleolus
			nucle		os[e]	ide	nucleoside
Latin nucleus + -t- for the sound of it + -ide			nucleot		ide		nucleotide
Latin nucleus, "little nut"			nucleus				nucleus

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
	oligo	dendr/o	cyte				oligodendrocyte
Spanish, coined by Río Hortega, oligodendroglía, "many" + "tree branches" + "glue"	oligo	dendr/o	glia				oligodendroglia
			organ		elle		organelle
Greek ōsmos, "thrusting, pushing"			osmos		is		osmosis
		osteo	cyte				osteocyte
			ox	id[e] + a	tion		oxidation
para- + -enkhyma "infusion" (en- + -khein "pour"). The ancients thought that the parenchyma formed from the infusion and clotting of blood into the space occupied by the organ.	par		enchyma				parenchyma
		peptid/o	glyc		an		peptidoglycan
Coined by Thomson 1804 from per-, "large amount" + oxide	per		ox		ide		peroxide
	per		oxi	some			peroxisome
	phag/o		cyt		osis		phagocytosis
Latin phænomenon, "that which appears or is seen"			pheno		type		phenotype
	phospho		lip		id		phospholipid

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin phosphorus, "light-bringing". Phosphorus named by chemists because of its use in matches. Phosphate groups, added in phosphorylation, consist of 1 phosphorus and 4 oxygen atoms. Phosphorylation can change how proteins act in the body.			phosphor		yl	ation	phosphorylation
Greek photos, "light"			phot[os]		on		photon
Greek pinein, "drink"		pino	cyt/o		osis		pinocytosis
Literally, "a little plate".			plate		let		platelet
Greek ploos, "fold"; haploid and diploid came first, then "ploidy" was used to describe the number of chromosomes			ploid		y		ploidy
			pol[e]		ar		polar
Greek polys, "many" + Greek meros, "part"	poly		mer				polymer
	poly		pept[one]		ide		polypeptide
Greek ploos, "fold"; haploid and diploid came first, then "ploidy" was used to describe the number of chromosomes	poly		ploid				polyploid
Latin primus, "first"			prim		er		primer
	pro		karyo		tic		prokaryotic
	pro		phase				prophase
			prosta[te]	gland	in		prostaglandin
			prote[in]a		some		proteasome

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Coined 1838 by Mulder; Greek prôteios "the first quality" + French ending -ine, "derived substance"			prot[o]		ein		protein
		prot/e/o	glyc		an		proteoglycan
Greek prōton, "first"			prot		on		proton
Greek pseudos, "lying" + Latin stratum, "thing spread out, layer"	pseudo		strat[um]		ified		pseudostratified
Latin, receptor "to hold, contain" (a ligand)			receptor				receptor
Latin recess, "to go backward" + -ive which makes an adjective out of a verb			recess		ive		recessive
The Latin word <i>regula</i> means "a straight piece of wood" from which we get "ruler". When you apply a ruler to something, we say it is regulated.			regul[a]		ate		regulate
Latin resonantia, "an echo"; in chemistry, delocalized electrons. The image is of electrons ping-ponging back and forth without being attached to a specific atomic nucleus.	re		son		ance		resonance
Latin rete, "net"			ret[e]		icul	ar	reticular
Reticulum is a combination of Latin <i>rete</i>, "net", and the diminutive suffix "-iculum"			ret[e]		icul	um	reticulum

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
German Ribonsäure, from English gum arabic with letters scrambled for no reason. This is what makes etymology fun!			ribose				ribose
			ribo[se]	some			ribosome
Latin secretionem, "a dividing, separation"			secre		tion		secretion
			solut[e]		ion		solution
			somat[o]		ic		somatic
Latin squamosus, "covered in scales"			squam		ous		squamous
Sterol is a shortening of cholesterol			ster[ol]		oid		steroid
Latin stratum, "thing spread out, layer"			strat[um]		ified		stratified
Latin stratum, "thing spread out, layer"			stratum				stratum
Latin substratum, "spread out underneath"	sub		strat[um]		e		substrate
French sucre, "sugar" + -ose			sucr[e]		ose		sucrose
French surface, "outermost boundary", sur- + face			surf[ace]		ant		surfactant
Latin sub-, "beneath" + pendere, "hanging"	su[b]		pen[dere]		sion		suspension
Latin portare, "carry"	sym		port				symport
Greek telos, "the end, completion" + mer	telo		mer		e		telomere
Greek telos, "the end, completion"	telo		phase				telophase
German Testosteron, coined 1935			test[is]	ster[ol]	one		testosterone
Thermo- + Greek dynamis, "power"		thermo	dynam[is]		ics		thermodynamics

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
		thromb/o	cyt/e				thrombocyte
Latin penuria, "poverty; scarcity"		thromb/o	cyt/o		penia		thrombocytopenia
		thromb/o	cyt		osis		thrombocytosis
Obtained from the thymus gland			thym[us]		ine		thymine
Greek tonikos, "stretching", but since the 1680s used to describe the healthy firmness of tissues			ton		ic	ity	tonicity
	tran[s]		script		ion		transcription
Latin translationem, "transfer of meaning"	trans		translation				translation
	tri		glycer		ide		triglyceride
		ur	ac[etyl]		il		uracil
Latin valentia, "power, capacity"			valence				valence
Latin vesicula, "little blister"			ves		icle		vesicle
Latin villus, "tuft of hair; shaggy hair; wool"			villus				villus

PART IV

UNIT 4: THE INTEGUMENTARY (SKIN) SYSTEM

UNIT 4 INTRODUCTION

With this Unit, we turn from the basics of Medical English to the specific words you need to describe various organ systems. Even though the skin (integument) is the largest organ of the human body, at 12 to 15 kg weight (about 16% of the total weight of the human body) and 1.5 to 2 m² of surface area, it is also the simplest of the organ systems. That's where we'll start this part of the journey.

Objectives

After you complete this unit, you will be able to:

- Recognize prefixes, root words, and suffixes related to the diagnosis and treatment of skin diseases.
- Recall abbreviations related to the diagnosis and treatment of skin diseases.

THE INTEGUMENTARY (SKIN) SYSTEM

Only [Med Terms] in the Building Episode 4

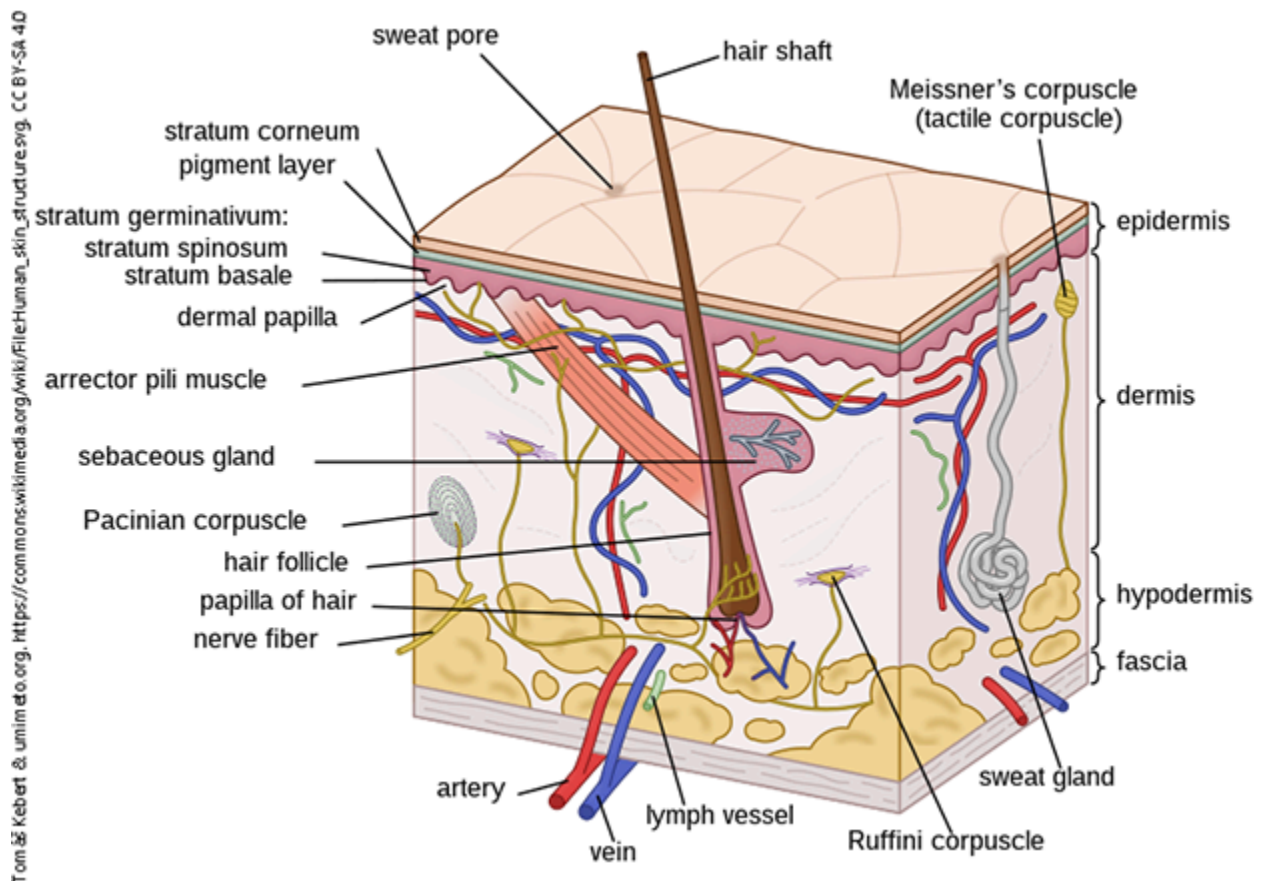


One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://uen.pressbooks.pub/medicalenglish/?p=281>

Corrections: The majority of leather comes from cow skin. The majority of gelatin comes from pig skin.

A [comprehensive list of Acme Corporation products](#) is found here.



Unit 4 figure 1 Skin Diagram

The skin is composed of distinct layers, with the **epidermis** being the most superficial, the **dermis** in the middle, and the subcutaneous layer being the deepest of the three. The subcutaneous layer is also called the **hypodermis**, as shown here. “**Hypodermic**” is generally used to refer to the needle and “subcutaneous” (abbreviated **SC**, **SQ**, or **subQ**, depending on the clinic) to the layer it is inserted into. Injecting substances with a **hypodermic** needle is called administering a drug through the **subcutaneous** (**subQ**) route. Hair and gland structures are surrounded by a layer of epidermis (red and purple in the diagram) although they appear to be located in the dermis. Blood vessels, lymphatic vessels, and nerves are most abundant in the dermis, but may extend into the subcutaneous layer. The epidermis is made up of many layers of cells and is capable of withstanding the friction our skin encounters on a daily basis.

Erythroderma



<https://dermnetnz.org/topics/erythroderma>, CC BY-NC-ND 3.0 NZ

Unit 4 figure 2 Erythroderma

As the body’s first contact point with the outside world, the skin is exposed to a variety of potentially infectious agents as well as harsh conditions, caustic agents, and other substances that can cause irritation, damage, or disease. The many pathologic conditions of the integumentary system are classified based primarily on appearance and often include reference to the changes in skin color and texture. For example, the normal red appearance seen on the skin as the result of a burn is referred to as erythroderma, simply meaning red skin. The erythroderma shown here is the result of more serious skin disease.

Video Attribution

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UNIT 4 PREFIXES

As you can see from this table, we've covered most of the common prefixes in Medical English. So even though there are 33 prefixes used in Unit 4, only 11 are new. The previously used prefixes, as always, are shown in gray in this table.

Prefix	Definition
a-	not
ab-	off, from, away
allo-	other, different; different member of the same species
an-	not
anti-	against (an action)
auto-	self
bi-	two, double
cata-	down, downward
chemo-	using chemicals
cryo-	using cold
de-	from, down, out
dys-	bad
<u>e-</u>	
<u>ec-</u>	out, off, from, beyond
<u>ex-</u>	
epi-	on top of
eu-	good; well-
hetero-	other, different
homo-	same
hyper-	too much
hypo-	too little
in-	in, into (if derived from im-, can also mean "not")
intra-	inside
neo-	new
par-	
para-	beside; near
per-	through; by means of
pro-	in front of; in favor of
sub-	an underlayer; support
tel-	
telo-	an end; complete

Prefix	Definition
trans-	through
tri-	three
xeno-	stranger; a different species
xero-	dry

UNIT 4 ROOTS

Dermatologists use a lot of color and appearance terms to describe skin. Many of these roots, then, describe the color or appearance of skin.

Root Word	Definition
abrasion	scraping off skin
acne	skin eruption
actin	ray, beam
aden	gland
adip/o	fat
albin/o	the color white
alopec	bald
amine	chemical group containing nitrogen
andr/o	male
angi/o	vessel
bili	bile
carcin	cancer
carot	carrot; the color orange
cauter[ium]	device used to burn
cer/at	horn
cicatrix	scar
circ	a ring; circle
cision	cut
comedo	zit
cori/a	leather; skin
corn	horn
crine	secretion; release from cells or from the body
crust	crust; rind
crypt	hidden; concealed
cub	lie down
cutan/e/o	skin
cyan	the color blue
cyt cyte cyto	cell
dendr/o	tree

Root Word Definition

derm/at/o	skin
desicc	dry
ecchymosis	bleeding under the skin
electr/o	related to electricity
erosion	"gnawing away"
erythr/o	the color red
esthe	sensation; perception
fibr/o	fiber
fissure	deep crack
gangrene	bacterial disease causing necrosis of tissue
gen geno genet	birth (dealing with inheritance)
glabr	smooth
graft	transfer of tissue from one place to another
gran	grain
hem hemo hemat hemato	blood
hidr hidra hidro	sweat
hist/o	tissue
ichthy	fish
impetigo	skin eruption
incision	cut
jaun	the color yellow
kel	claw
kerat/o	horn
leuk/o	the color white
lip/o	fat
lith	stone

Root Word	Definition
lun	moon
macer	soften
melan/o	the color black
myc/o	fungus
myring/o	eardrum
necr	dead
nevus	mole
nidus	nest
nod[e]	knot
onych/o	nail; "little claw"
ops	seeing; appearance
pachy	thick
pap papilla	nipple
permeable	able to be penetrated
petechia	small, red spots on the skin
pheo	gray or dark in color
pigment	coloring agent
pil/o	hair
plaque	sticky deposit
plas[tic]	something molded
prur[is]	itch
pruritic	something which causes an itch
psōra	itch
py/o	pus
ras	scrape
regulat[e]	control or maintain
rhino	nose
rhytid/o	wrinkle
rub	red

Root Word	Definition
scale	scale
scess	going
scler/o	hard
seb seba sebo sebum	a waxy secretion from a skin gland
sens	sensation; perception
septic	filled with microorganisms
sicc	dry
some	body
squam	a shape wider than it is high (literally, "fish-scale")
steat/o	fatty
suction	using negative pressure
surgery	using tools to alter the structure of the body
sweat	perspiration
tect	roof
therapy	treatment to make something better
thermo	temperature
tinea	fungal infection
top	place
trich/o	hair
tumor	swelling
unguent	ointment; grease
urtica	nettle, stinger
vapora[te]	steam
vascul	little vessel
verruca	wart
vesic	bladder
vitaligo	skin disease
vulgaris	common

Root Word	Definition
xanth/o	yellow
xer/o	dry
zein	boil

UNIT 4 SUFFIXES

There are always a lot of suffixes to learn for each Unit. But unlike the root words, there's a lot of repetition in the use of suffixes. So while this table has 51 entries, only 18 are new. Of those 18, four are word endings from other languages, and don't really add much meaning to the terms. So you really only have a handful of new suffixes to learn in this Unit and in future Units. The ones you've learned already are in gray.

Suffix	Definition
-al	pertaining to
-algia	pain
-ar	pertaining to
-ary	pertaining to
-ate	word ending that turns a noun into an adjective that means "showing"
-ation	word ending that turns a verb into a noun
-cele	hernia
-ceous	pertaining to
-dice	French ending which puts "to be" before the word
-dynia	pain
-ectasia	dilated condition
-ectomy	cutting out of
-ema	Greek ending which puts "to be" before the word
-ene	chemical name ending
-er	word ending that turns a verb into a noun
-gen	arising from
-ia	disease
-iasis	presence of an abnormality
-ic	pertaining to
-icle	diminutive suffix
-in	chemical name ending for a protein
-ing	English verb ending forming a continuous tense (present participle)
-ion	word ending that turns a verb into a noun
-ism	belief; process word ending
-itic	pertaining to
-itis	inflammation
-itus	Latin suffix meaning the word takes the male gender
-ity	turns a noun into a state of being
-ium	anatomical structure
-ization	the process of

Suffix	Definition
-logy	study of
-lysis	breaking down; destroying
-malacia	softening due to disease
-megaly	enlargement due to disease
-o	Spanish word ending
-oid	resembling
-oma	tumor
-ops	face
-ory	pertaining to
-osis	process
-osum	full of, prone to
-osus	
-ous	word ending that turns a noun into an adjective
-pathy	disease process
-plasm	growth; development
-plasty	surgical revision
-poiesis	creation process
-rrhea	abnormal flow
-scopy	the process of looking inside
-tion	word ending that turns a verb into a noun
-tomy	cutting
-trophy	nourishment, growth
-ula	diminutive suffix
-ule	

UNIT 4 ABBREVIATIONS

Abbreviation	Meaning
ABCDE	mnemonic for assessing the cancer risk of skin lesions: asymmetry, border, color, diameter, evolving
BCC	basal cell carcinoma
Bx	biopsy
decub	decubitus ulcer
Derm	dermatology
I&D	incision & drainage
ID	intra dermal
IM	intramuscular
RF	radiofrequency (a type of electromagnetic energy)
SC	subcutaneous
SQ	
subQ	
SCC	squamous cell carcinoma
Tx	therapy
ung	ointment (unguent)

UNIT 4 WORD LIST

Word	Definition
abrasion	scrape
abscess	pus-filled wound
acne	skin disease characteristic of puberty
actinic	light-caused
adipocyte	fat cell
albinism	condition of white skin
albino	person who has a mutation in the melanin gene, causing changes in skin appearance
allograft	a skin tissue donation from an unrelated individual of the same species (same as homograft)
alopecia	baldness
anagen	stage of hair growth
androgen	steroid causing the development of male sexual characteristics
anesthetic	drug that removes feeling
angioma	blood vessel tumor
anhidrosis	lack of sweat
antihistamine	drug that fights inflammation (the chemical histamine is released in the inflammatory response)
antipruritic	drug that fights itching
antiseptic	chemical that fights microorganisms
atopic	an allergic reaction that occurs away from the site of exposure
atrachosis	absence of hair
autograft	a skin tissue donation from oneself
bilirubin	blood chemical formed from the breakdown and recycling of red blood cells
biopsy	tissue sample used for examination by a pathologist
bulla	blister
carcinoma	cancerous tumor
carotene	orange-colored precursor to vitamin A, found in carrots
catagen	stage of hair growth
cauterize	use a cautery to burn tissue
cautery	device used to burn tissue (e.g. to stop bleeding)
chemosurgery	surgery using chemicals

Word	Definition
chemotherapy	therapy using chemicals
cicatrix	scar
circulation	the way blood moves around the body in a circle
comedo	zit
corneocyte	a keratinocyte that has lost its nucleus and become a bag of keratin
crust	crust
cryosurgery	surgery using cold
cutaneous	pertaining to the skin
decubitus	from lying down
dendritic	pertaining to a tree (shape)
depigmentation	the process of removing pigment
dermabrasion	scraping the skin
dermatalgia	pain in the skin
dermatitis	inflammation of the skin
dermatodynia	pain in the skin
dermatofibroma	fibrous tumor of the skin
dermatology	study of the skin
dermatolysis	breakdown of the skin
dermatomycosis	fungal infection of the skin
dermatoscopy	visualizing the skin with an instrument
dermatosis	disease of the skin
dermis	the skin, specifically the internal connective tissue layer
dermopathy	skin disease
dermoscopy	visualizing the skin with an instrument
dysplastic	malformed
ecchymosis	bruise
eccrine	type of sweat gland used for thermoregulation
eczema	skin disease
electrocauterization	using electrical current to burn tissue
electrodesiccation	using electrical current to dry tissue

Word	Definition
epidermal	pertaining to the epidermis
epidermis	the skin, specifically the external layer
eponychium	cuticle
erosion	slowly wearing away
erythema	redness
erythrocyanosis	reddish-blue color of skin when the body is cold
erythroderma	red skin
eumelanin	pigment protein melanin which has a dark brown or black color
evaporation	the process of forming vapor (i.e. water entering the atmosphere)
excisional	pertaining to cutting out
excoriation	skin-picking disorder
fissure	deep crack
glabrous	hairless
granule	small grain
hematohidrosis	sweating blood (as when studying for Medical Terminology exams)
heterograft	a skin tissue donation from another species (same as xenograft)
hidradenoma	tumor in a sweat gland
hidradentitis	inflammation of a sweat gland
hidropoiesis	the process of forming sweat
homograft	a skin tissue donation from an unrelated individual of the same species (same as allograft)
hyperhidrosis	sweating too much
hyperkeratosis	abnormal skin thickness due to disease
hypermelanosis	abnormally high skin pigmentation due to disease
hyperpigmentation	abnormally high skin pigmentation due to disease
hypertrichosis	abnormal hair production due to disease
hypodermic	pertaining to the layer internal to the skin (below the dermis); hypodermic needle
hypodermis	layer of connective tissue below the dermis
hypohidrosis	sweating too little
hypomelanosis	abnormally low skin pigmentation due to disease

Word	Definition
hyponychium	skin under the free edge of the nail
hypopigmentation	abnormally low skin pigmentation due to disease
ichthyosis	fish-scale-like skin
impetigo	bacterial skin infection
incision	surgical cut
intra-dermal	inside the dermis
jaundice	yellowish color of the skin, usually due to liver disease
keloid	abnormal skin healing process
keratinocyte	the keratin-filled cells that form the layers of the epidermis; when they are shed they are called squames
keratoderma	abnormal thickening of the skin due to disease
keratogenic	promoting the formation of epidermis
keratosis	skin lesion
leukoderma	white skin
lipectomy	surgical removal of fat
lipoma	fatty tumor
liposuction	surgical removal of fat using negative pressure
lunula	"little moon": light-colored area at the base of each nail
macerate	to develop soft, wet, or soggy skin; usually because of long-term water exposure or abnormal healing
macule	discolored patch of flat skin
melanin	protein that causes skin pigmentation
melanocyte	cell that produces and stores melanin
melanoma	extremely dangerous cancer that arises from melanocytes
melanosome	melanin-filled pigment granules inside melanocytes
mycodermatitis	inflammation of skin due to fungal infection
mycosis	fungal infection
myringodermatitis	inflammation of the skin of the eardrum
necrosis	death
neoplasm	new growth; synonym for cancerous lesion
nevus	freckle

Word	Definition
nodule	abnormal bump in skin
onychalgia	pain in nails
onychectomy	surgical removal of a nail
onychchia	infection of the soft tissues around or under the nail
onychocryptosis	ingrown nail
onychodystrophy	malformed nail
onycholysis	breakdown of a nail
onychomalacia	softening of a nail
onychomycosis	fungal infection of a nail
onychopathy	disease of a nail
onychophagia	nail biting
onychotomy	cutting into a nail
pachyderma	thick skin
papilla	"nipples" or ripples in the dermis which help it attach to the epidermis
papillary	pertaining to papillae
papule	small raised area of skin
paronychia	infection of the nail folds or cuticle
percutaneous	through the skin
permeability	property of being able to penetrate
petechia	red pinpoint on the skin (usually used as the plural petichiae)
pheomelanin	pigment protein melanin which has a yellowish color
pigment	coloring agent
pilonidal	cyst which contains hair and pus, usually near the top of the gluteal fissure
pilosebaceous	the hair follicle and associated glands taken as a unit
plaque	dry, scaly area of skin
protection	defense against skin damage
pruritus	itchiness
psoriasis	skin disease
pus	white fluid, made up of white blood cells and dead bacteria
pustule	small, inflamed, pus-filled blisters on the skin (as in acne)

Word	Definition
pyoderma gangrenosum	pus under the surface of dying skin, caused by a bacterium
rhinoceros	a large mammal with a horn on its nose
rhytidermia	wrinkled skin
rhytidoplasty	surgical modification of wrinkles
scale	peeling or flaking of outer skin layers
sclerodermatitis	hard, thickened area of inflammation in the skin
scleronychia	hard, thickened nails
sebaceous	pertaining to the glands that make sebum
sebolith	stone in a sebaceous gland
sebopoiesis	the process of forming sebum
seborrhea	skin lesions (usually on scalp) that cause scaly patches, inflamed skin, and dandruff
seborrheic	pertaining to seborrhea
sebum	oily secretion
sensory	receiving information from the environment (e.g. the sense of touch)
squamous	cells that are wider than they are tall ("fish scales")
steatitis	inflammation of the fat tissues
steatoma	fatty tumor
subcutaneous	pertaining to the layer internal to the skin (below the dermis); same as hypodermis
subungual	beneath the nail
telangiectasia	small, widened blood vessels in the skin; "spider veins"
telogen	stage of hair growth
thermoregulation	keeping the body temperature within normal limits
tinea	a type of fungal infection
topical	applied to the skin
transdermal	across the skin
triceratops	a dinosaur with three horns on its face
trichomegaly	abnormally long eyelashes
trichomycosis	infection of the follicles
tumor	swelling

Word	Definition
ulcer	open sore
unguent	old name for ointment; butter
urticaria	large raised red patches of skin; hives
vascular	pertaining to vessels
verruca	wart
vesicle	small, fluid-filled, raised skin bump
vitiligo	autoimmune disease causing patchy skin depigmentation
vulgaris	common (e.g. acne vulgaris); something that everyone has
xanthoderma	yellow skin
xanthoma	yellow raised bumps due to fat accumulation under the skin
xanthosis	yellow discoloration of degenerating tissues
xenograft	a skin tissue donation from another species (same as heterograft)
xeroderma	dry skin
xerosis	dry skin

HOW THE UNIT 4 WORD LIST WAS BUILT

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin ab-, "off" + radere, "to scrape"	ab		ras			ion		abrasion
Latin ab-, "off" + cedere, "to withdraw", based on the idea that a bodily humor is escaping into the abscess	ab		scess					abscess
Latin aknas, 6th century misreading of Greek akmas, "point" (cf. acme)			acne					acne
			actin			ic		actinic
		adip/o	cyte					adipocyte
			albin			ism		albinism
Despite their depictions in fiction, albinos are not evil.			albin			o		albino
Graft from a different member of the same species	allo		graft					allograft
Greek alōpēx, "fox", because fox-mange (alōpēkia) is supposedly transferred to humans			alopec			ia		alopecia
	ana					gen		anagen
			andro			gen		androgen
	an		esthe			tic		anesthetic
			angi[o]			oma		angioma
	an		hidr			osis		anhidrosis
	anti		hist		amine			antihistamine
	anti		pruritic					antipruritic
	anti		septic					antiseptic
	a		top			ic		atopic
	a		trich			osis		atrachosis

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
a graft from one's own tissue (or an identical twin's)	auto		graft					autograft
			bili		rub	in		bilirubin
		bi[o]	ops			y		biopsy
								bulla
Greek karkinos, "crab" + -oma . The Greek physicians named cancers after a crab because their tendency to spread their "claws" into tissues and hold on tenaciously, like a crab. The plural of carcinoma is carcinomata.			carcin			oma		carcinoma
			carot[a]			ene		carotene
	cata					gen		catagen
The Latin word cauterium means "branding iron", from the Greek kauter, "burn".			cauter[ium]					cauterize
			cauter[ium]			y		cautery
	chemo		surgery					chemosurgery
	chemo		therapy					chemotherapy
			cicatrix					cicatrix
			circ			ula	tion	circulation
Latin comedere, "eat up"; originally used to describe parasitic worms, now used to describe the sebaceous secretions that resemble small worms that come out of zits (which, somehow, is even more disgusting than popping a zit.)			comedo					comedo

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin cubare, "to lie down" (cubicle comes from the same root)		corneo	cyte					corneocyte
			crust					crust
	cryo		surgery					cryosurgery
			cutane			ous		cutaneous
	de		cub			itus		decubitus
			dendr			itic		dendritic
	de		pigment			ation		depigmentation
		derm	abrasion					dermabrasion
			dermat			algia		dermatalgia
			dermat			itis		dermatitis
			dermato			dynia		dermatodynia
			dermato	fibr		oma		dermatofibroma
			dermato			logy		dermatology
			dermato			lysis		dermatolysis
			dermato	myc		osis		dermatomycosis
			dermato			scop	y	dermatoscopy
			dermat[o]			osis		dermatosis
			dermis					dermis
			dermo			pathy		dermopathy
			dermo			scopy		dermoscopy
Latin ecchymōsis, "pouring out"	dys		plas[ie]			tic		dysplastic
			ecchymosis					ecchymosis
	ec		crine					eccrine
Greek ek (ex) + krīnō, "separate"	ec		zema					eczema
Greek ek (ex) + zein "to boil"		electro	cauter[ium]			iz[e]	ation	electrocauterization
	de	electro	sicc			ation		electrodesiccation

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
	epi		derm			al		epidermal
	epi		dermis					epidermis
Greek ep[i]- + onúkhion, "little claw"	ep[i]		onych			ium		eponychium
Latin e[x]- + rodere "to gnaw"	e		rosion					erosion
			eryth			ema		erythema
		erythr/o	cyan			osis		erythrocytosis
		erythr/o	derm			a		erythroderma
	eu		melan[o]			in		eumelanin
Latin e[x]- + vapor "steam"	e		vapora[te]			tion		evaporation
	ex		cision			al		excisional
	ex		cori			ation		excoriation
			fissure					fissure
Latin glaber, "hairless, smooth, bald"			glabr			ous		glabrous
			gran			ule		granule
		hemat[o]	hidr[o]			osis		hematidrosis
"Graft" is taken from the Greek graphein, "to write", because of the resemblance of a plant graft to a writing pen	hetero		graft					heterograft
Latin & Greek hidrō, "sweat"		hidr	aden			oma		hidradenoma
Hidradenitis suppurativa is a chronic inflammatory skin condition		hidr	aden			itis		hidradenitis
			hidro			poiesis		hidropoiesis
same as allograft	homo		graft					homograft
	hyper		hidr			osis		hyperhidrosis
	hyper		kerato					hyperkeratosis

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
	hyper		melan			osis		hypermelanosis
	hyper		pigment			ation		hyperpigmentation
	hyper		trich			osis		hypertrichosis
	hypo		derm			icle		hypodermic
	hypo		dermis					hypodermis
	hypo		hidr			osis		hypohidrosis
	hypo		melan			osis		hypomelanosis
	hyp[o]		onych			ium		hyponychium
	hypo		pigment			ation		hypopigmentation
			ichthy			osis		ichthyosis
Latin impetere, "to attack" (cf. impetus)			impetigo					impetigo
Latin in- + cidere, "cut"			incision					incision
	intra		derm			al		intradermal
French jaunice, "yellowness" (jaune is French for "yellow")			jaun			dice		jaundice
			kel			oid		keloid
		kerat	cyte	cyte		in + o		keratinocyte
		kerat/o	derma					keratoderma
			kerat/o			gen	ic	keratogenic
			kerat			osis		keratosis
		leuk/o	derm			a		leukoderma
			lip			ectomy		lipectomy
			lip			oma		lipoma
			lip/o	suction				liposuction
			lun			ula		lunula
			macer			ate		macerate
								macule
			melan			in		melanin

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin nevus, "a mark"; now means a mole	neo	melan/o	cyte					melanocyte
			melan			oma		melanoma
			melan/o	some				melanosome
		myc/o	dermat			itis		mycodermatitis
			myc			osis		mycosis
		myring/o	derm/at			itis		myringodermatitis
			necr			osis		necrosis
						plasm		neoplasm
			nevus					nevus
			nod[e]			ule		nodule
			onych			algia		onychalgia
			onych			ectomy		onychectomy
			onych			ia		onychchia
			onych/o	crypt		osis		onychocryptosis
			onych/o + dys			trophy		onychodystrophy
			onych/o			lysis		onycholysis
			onych/o			malacia		onychomalacia
			onych/o	myc		osis		onychomycosis
			onycho			pathy		onychopathy
			onycho	phag		ia		onychophagia
			onycho			tomy		onychotomy
	pachy		derma					pachyderma
			papilla					papilla
			papill[a]			ary		papillary
			pap[illa]			ule		papule
	par		onych[o]			ia		paronychia
	per		cutane			ous		percutaneous
			permeable			ity		permeability

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
				petechia				petechia
		pheo		melan[o]		in		pheomelanin
				pigment				pigment
Latin pilus, "hair" + nidus, "nest"				pilo	nid	al		pilonidal
Latin pilus, "hair" + sebaceous gland				pil/o	seba	ceous		pilosebaceous
				plaque				plaque
	pro			tect		ion		protection
				prur[is]		itus		pruritus
				psor		iasis		psoriasis
Pus is taken directly from the Latin, and is related through PIE pu- "to rot, decay" to Sanskrit puyati, Greek puon, and Lithuanian pūvu. It's thought to be an imitation of the sound you make when you encounter pus.								pustule
Literally, "skin with pus, filled with gangrene"; gangrene comes from the Latin gangraena, from Medical Greek gangraina, "an sore that eats away [tissue]."		py/o		derma	gangren[e]	osum		pyoderma gangrenosum
				rhino	cer	osis		rhinoceros
				rhyti[d]	derm	ia		rhytidermia
				rhytid/o		plasty		rhytidoplasty
				scale				scale
		scler/o		dermat		itis		sclerodermatitis
		scler		onych		iasis		scleronychia

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin unguis, "nail, claw, hoof"			seba			ous		sebaceous
			sebo		lith			sebolith
			sebo			poiesis		sebopoiesis
			sebo			rrhea		seborrhea
			sebo			rrhe[a]	ic	seborrheic
			sebum					sebum
			sens			ory		sensory
			squam			ous		squamous
			steat[o]			itis		steatitis
			steat[o]			oma		steatoma
	sub		cutan/e			ous		subcutaneous
	sub		ingu			al		subungual
	tel		angi[o]			ectasia		telangiectasia
	tel/o		gen					telogen
		thermo	regulat[e]			ion		thermoregulation
Greek trikeratos "three-horned"			tinea					tinea
			topic			al		topical
	trans		derm			al		transdermal
	tri		cerat			ops		triceratops
			trich/o			megaly		trichomegaly
			trich/o		myc	osis		trichomycosis
			tumor					tumor
			ulcer					ulcer
			unguent					unguent
			urtica			r + ia		urticaria
Latin unguentum, "ointment"			vascul[um]			ar		vascular
			verruca					verruca

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Graft from another species ("stranger-graft")	xeno	xanth/o	ves			icle		vesicle
			viti					vitiligo
			vulgar					vulgaris
			derma					xanthoderma
			xanth[o]			oma		xanthoma
			xanth			osis		xanthosis
		xer/o	graft					xenograft
			derma					xeroderma
			xer[o]			osis		xerosis

PART V

UNIT 5: THE MUSCULOSKELETAL SYSTEM

UNIT 5 INTRODUCTION

The Musculoskeletal System is made up of both the muscular and skeletal systems, which work together to generate mobility, provide protection, and maintain stability. The musculoskeletal system includes bones, connective tissue (cartilage, ligaments, and tendons), and muscle tissue. Beyond serving as the framework for the body, bones also play important roles in mineral homeostasis, hematopoiesis, and energy storage.

Objectives

By the time you complete this unit, you will be able to:

- Identify word parts associated with the muscles, connective tissue, and bones (the *musculoskeletal system*).
- Assemble these word parts into words describing structures and diseases of the musculoskeletal system.
- Recognize abbreviations used to describe procedures and diseases related to the musculoskeletal system.

THE MUSCULOSKELETAL SYSTEM

Only [Med Terms] in the Building Episode 5



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://uen.pressbooks.pub/medicalenglish/?p=290>

The Muscular System provides locomotion (movement), stabilization (posture), and thermoregulation. Various muscles and muscle groups give us either gross (large) or fine (small, precise) motor movement. There are three types of muscle tissue in the body:

- skeletal
- cardiac
- smooth

Just given their names, you probably already have an idea about where a couple of them are located.

Skeletal muscle is named such because of its connection to the skeleton. This relationship allows contractions of the muscles to move the body or maintain posture. Skeletal muscle is under voluntary control, meaning that contraction is controlled by conscious thought and intentional stimuli from the nervous system. Skeletal muscle also helps warm the body by burning energy to release heat (commonly called shivering). Skeletal muscle is striated (striped due to sarcomeres).

Cardiac muscle is found in just one place – the heart. It is not under voluntary control and is probably the hardest-working muscle in the body, contracting about 60-100 times per minute, every minute, every hour, every day of our lives. Cardiac muscle is striated and branches (intercalated discs). The purpose of cardiac muscle contraction is to squeeze the heart, pushing blood out of the chambers and into the arteries of the body. The branching arrangement of cells allows for this type of movement. Like skeletal muscle, the coordinated shortening of thousands of sarcomeres results in the contraction of the heart muscle. Cardiac muscle is unique in that it responds to signals that arise from specialized cells within the heart itself. These cells send electrical signals about every 0.8 seconds causing the heart to contract about 72 times per minute (called heart rate or pulse). When under stress, hormones, nervous system signals, or both can cause heart rate to increase.

Smooth muscle gets its name because it lacks the sarcomeres that give skeletal and cardiac muscle their striations. Smooth muscle contractions are typically slower to initiate, but last longer than other muscle types. Smooth muscle is located in the walls of hollow internal structures, such as blood vessels, airways, and the digestive system. It is also seen in other areas like the eyes and reproductive organs. Like cardiac muscle, it is not under voluntary control. Smooth muscle is responsible for vasodilation and vasoconstriction (increasing and decreasing blood vessel diameter), bronchodilation and bronchoconstriction (increasing and decreasing



Unit 5 figure 2 Muscular System

airway diameter), and peristalsis (wave movements propelling food through the gut, sperm through the ductus deferens and urethra, and ova through the uterine tubes).

Video Attribution

- Justin Burr, Jim Hutchins & Maddison Johnston. (2024, December 20). *Only [Med Terms] in the Building Episode 5*. [Video]. Kaltura.

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UNIT 5 PREFIXES

Prefix	Definition
a-	not
ab-	going away (note added meaning)
ad-	going into
amphi-	around, on both sides
an-	not
anti-	against (an action)
apo-	from, off, away
brady-	slow
di-	separate, apart, double, two
dia-	between
dys-	bad
ec- (same as ex-)	out, off, from, beyond
endo-	inside
epi-	on top of
ex-	out, off, from, beyond
hyper- (before process)	too much
hypo- (before process)	too little
infra-	beneath
inter-	between, among
intra-	inside
iso-	equal
meta-	between, with, after; change
peri-	around
poly-	many; much
sub-	an underlayer; support
syn-	with, together

UNIT 5 ROOTS

Roots used in word building for Unit 5 (musculoskeletal system) are listed here.

Root	Definition
acet[um]	vinegar
agon	main actor; driver
alges[ia]	pain
ankylos	stiff
appendix	part attached to the end
arthr arthro	joint
articul[us]	joint
asthenia	weakness
axi	axis
blast	a rapidly dividing cell
burs bursa	purse; bag
carcin	cancer
carp	wrist
center	middle of a round object
cephal	pertaining to the head
cervic cervico	neck
chondr chondro	cartilage
circum	around
clast	breakdown
clonus	violent shaking
condyle	knob
cost costo	rib
cranio	head; skull
crepit[us]	crackling noise
cyt	cell
dactyl	finger
duct	to lead

Root	Definition
effusion	pouring out of
electro	related to electricity
erg[on]	energy
exten[d]	moving away from
facet	flat spot, like on a jewel
fasci fascio	bundle of fibers
femor femur	thigh
fissure	deep crack
fixate	remain in place
flex	reducing the angle of a joint
foramen	hole
fossa	shallow groove
fracture	broken bone
genu	knee
glen[e]	socket
gomph	tooth
graph	written record
hallux	big toe
hemo hemato	blood
hydr	water
inflammation	inflammation
kines kinet	movement
kyphosis	hunchback curve of spine
lacrim	crying; tears
lamella	thin plate
lemma	thin layer surrounding a cell
lordosis	swayback curve of spine

Root	Definition
lumb lumbo	lower back
luxation	dislocation
meatus	tube; passage
medull[a]	marrow; center
membran[e]	thin sheet
mere	a part
metric	measurement
musculo	muscle
my myo myos	muscle
myel	marrow
necro	death
neuro	nervous system
occipit[o]	back of the head
olisthes	slippage
ortho	straight; at right angles; true
os osse ost osteo	bone
pector[is]	chest
pelvis	hips; lower abdomen
phalanges	fingers and toes
physis	part of a bone
plasm	something molded; plastic or dough
polariz[e]	to go to one extreme or another
pollex	thumb
por[e]	hole
prone	face down
pyre	fever

Root	Definition
sarc sarco	muscle; flesh
scler	hard
scoliosis	lateral curvature of the spine
scop	instrument for looking inside
skelet	skeleton; bones collectively
spasm	uncontrollable contraction
sphincter	muscle surrounding an opening
spin spina spine	spine; vertebrae
spondyl spondylo	vertebrae
sprain	injury to ligaments
sten	abnormal narrowing
sterno sternum	sternum; breastbone
strain	injury to muscles or tendons
sulcus	groove
supine	face up
tars tarso tarsus	ankle
tax	moving
tempor	side of the head
ten teno tendo tendin tendon	tendon
thorac thorax	chest
tibia	bone of the lower extremity below the knee
tomo	cutting
ton tono	tone, strength

Root	Definition
trochanter	point on femur where large leg muscles attach
trochlea	piece of bone that acts as a pulley
troph	nourishment, growth
tuber	potato-like raised area on bone
valgum	knees closer than normal
varum	knees farther apart than normal
vertebra	bone of spinal column
zygoma	yoke; joins bones of face to those of skull

UNIT 5 SUFFIXES

Suffix	Definition
-al	pertaining to
-algia	pain
-ar	pertaining to
-ary	pertaining to
-atic	pertaining to
-ation	word ending that turns a verb into a noun
-bulum	suffix forming the name of a vessel or instrument
-centesis	puncture
-clasia	breakdown process leading to disease
-cle	diminutive suffix
-desis	binding together
-dynia	pain
-eal	pertaining to
-ectomy	cutting out of
-ella	diminutive suffix
-emia	present in blood, indicating disease
-genesis	the process of creating
-gram	a record of a test or procedure
-graph	instrument for recording a procedure
-graphy	a test or procedure
-ia	disease
-ic	pertaining to
-ification	word ending that turns a noun into a verb
-ing	verb ending that makes the continuous tense (ongoing process)
-ion	word ending that turns a verb into a noun
-is	anatomical structure
-ist	someone who studies
-itis	inflammation
-ium	anatomical structure
-logy	study of

Suffix	Definition
-lysis	breaking down; destroying
-malacia	softening due to disease
-meter	instrument for measuring
-oid	looks like; resembles
-oma	tumor
-on	unit of anatomical structure
-ory	pertaining to
-osis	process
-osity	word ending that turns a noun into an adjective and back into a noun again
-ous	word ending that turns a noun into an adjective
-pathy	disease process
-penia	low levels of
-physis	growth; form
-plasia	the process of molding or growing
-plasty	surgical revision
-poiesis	creation process
-ptosis	drooping
-rrhaphy	suturing
-scope	instrument for looking inside
-sion	the state of
-sium	anatomical word ending
-spasm	involuntary contraction
-tomy	cutting
-um -us	anatomical structure word ending
-y	disease

UNIT 5 ABBREVIATIONS

Abbreviation	Meaning
ACL	anterior cruciate ligament
C1-C7	cervical vertebrae (numbered superior to inferior)
CT	computed tomography
EMG	electromyogram
Fx	fracture
IM	intramuscular
L1-L5	lumbar vertebrae
LCL	lateral collateral ligament
MCL	medial collateral ligament
MRI	magnetic resonance imaging
NSAID	non-steroidal anti-inflammatory drug
OA	osteoarthritis
ORIF	open reduction / internal fixation
PCL	posterior cruciate ligament
PT	physical therapy
RA	rheumatoid arthritis
RICE	rest, ice, compression, elevation
ROM	range of motion
S1-S5	sacral vertebrae
T1-T12	thoracic vertebrae
WB	weight-bearing
WBAT	weight-bearing as tolerated

UNIT 5 WORD LIST

Word	Definition
abduction	moving a limb or body part away from the midline
acetabulum	hip socket
achondroplasia	failure in the development of cartilage (a type of dwarfism)
adactyly	absence of the fingers
adduction	moving a limb or body part toward the midline
agonist	the "prime mover": the muscle primarily responsible for the movement being studied
amphiarthrotic	slightly moveable joint
analgesic	drug that reduces pain
ankylosing	causing to be stiff
ankylosis	a disease that makes bones or joints stiff
antagonist	a muscle that opposes the action of the agonist
antiarthritic	drug that reduces the symptoms of arthritis
anti-inflammatory	drug that reduces inflammation
antipyretic	drug that reduces fever
apophysis	outgrowth or protuberance on a bone
appendicular	part of the skeleton that is attached to the axial skeleton (i.e. the upper and lower extremities)
arthralgia	pain in the joints
arthrectomy	surgical removal of a joint
arthrocentesis	puncture of a joint to remove fluid
arthroscope	device for examining the inside of a joint
arthrodesis	disease process binding a joint together
arthrodynia	pain in the joints
arthrodysplasia	bad formation of a joint
arthrogram	picture of a joint
arthrography	process for visualizing a joint
arthropathy	disease in a joint
arthroplasty	surgical modification of a joint
arthroscopic	pertaining to examination of a joint using an arthroscope
arthrotomy	cutting into a joint
articular	having to do with joints

Word	Definition
ataxia	inability to walk
atrophy	failure to develop, or loss of a structure, due to lack of nourishment
axial	pertaining to the center of the body (cf. appendicular)
bradykinesia	slow movement
bursa	"purse"; a bag of synovial fluid that cushions a ligament or tendon that would otherwise rub against a bone
bursectomy	surgical removal of a bursa
bursitis	inflammation of a bursa
bursotomy	cutting into a bursa
carpal	pertaining to the wrist bones
carpectomy	surgical removal of a wrist bone
carpitis	inflammation of a wrist bone
cervical	pertaining to the neck
cervicitis	inflammation of the neck
cervicodynia	pain in the neck
chondritis	inflammation of the cartilage
chondrodynia	pain in the cartilage
chondroma	tumor in the cartilage
chondromalacia	softening of the cartilage
chondroplasty	surgical modification of cartilage
circumduction	movement in a circle around a joint
condyle	knob on a bone
costal	pertaining to a rib
costalgia	pain in a rib
costectomy	surgical removal of a rib
costochondritis	inflammation of the rib and associated cartilage
craniectomy	surgical removal of a piece of the skull
craniomalacia	softening of the skull
craniometer	device for measuring the skull
cranosynostosis	disease where the skull bones of a child fuse too soon after birth

Word	Definition
craniotomy	cutting into the skull
crepitation	crackling noise in the joints
dactylalgia	pain in the fingers
dactylitis	inflammation of the fingers
diaphysis	the shaft of a long bone
diarthrotic	freely moveable joint
dyskinesia	trouble with movement
dysplasia	malformation of a structure
dystaxia	trouble with walking
dystonia	trouble with muscle tone (strength)
eccentric	force applied to a muscle as it is getting longer
effusion	the process of fluid leaking out of a structure
electromyogram	record of the electrical activity of a muscle
electromyography	obtaining a record of the electrical activity of a muscle
endochondral	occurring inside of cartilage (refers to a type of bone development)
endomysium	the innermost connective tissue sheath of a muscle
epimysium	the outermost connective tissue sheath of a muscle
epiphysis	the ends of a long bone (distal epiphysis or proximal epiphysis)
epicondyle	a bump next to a knob on a bone
exostosis	abnormal bone growth in the small ear bones (ossicles)
extension	increasing the angle of a joint
facet	a flattish surface on a bone where a joint articulates (it is supposedly similar to the flat surface of a cut jewel)
fascia	band of connective tissue
fasciectomy	removal of a fascia
fasciitis	inflammation of a fascia
fascioplasty	surgical modification of fascia
fasciorrhaphy	surgical repair of a fascia
fasciotomy	cutting into a fascia
femoral	pertaining to the thigh

Word	Definition
fissure	a deep crack
fixation	locking into place
flexion	reducing the angle of a joint
foramen	hole in a bone
fossa	shallow groove in a bone
fracture	breaking a bone
genu	knee
glenoid	the socket of the shoulder joint
gomphosis	the joint between a tooth and the jaw
graphospasm	writer's cramp
hallux	the big toe
hemarthrosis	blood in a joint
hematopoiesis	the process of blood cell formation in the bone marrow
hydrarthrosis	water (excess fluid) in a joint
hypercalcemia	abnormally high calcium in the blood
hyperextension	abnormal or excessive increase in the angle of a joint
hyperkinesia	abnormally high movement
hypertonia	abnormally high muscle tone (strength)
hypertrophy	abnormal growth of tissue
hypocalcemia	abnormally low calcium in the blood
hypokinesia	abnormally low movement
hypophyseal	pertaining to the tissue underneath the brain (i.e. the pituitary gland)
hypophysis	the pituitary gland
hypotonia	abnormally low muscle tone (strength)
infraspinatus	a muscle or space below the spine of the scapula
intercostal	between the ribs
interosseous	band of connective tissue between the radius and ulna or between the tibia and fibula
intramembranous	occurring within a membrane (refers to a type of bone development)
isometric	force applied to a muscle without a change in muscle length
kinesiology	the study of movement

Word	Definition
kinetic	pertaining to motion
kyphosis	"hunchback": a bend in the spine that results in a visible bump in the upper back
lacrimal	pertaining to crying or tears
lamella	thin plate
ligament	a band of connective tissue connecting bone to bone
lordosis	"swayback": a bend in the spine that results in a pronounced hollow space above the buttocks
lumbar	lower back
lumbodynia	pain in the lower back
meatus	tube or passage
medullary	pertaining to the center of a multilayered organ (in this case, the bone marrow)
metacarpal	the bones between the wrist and fingers
metacarpectomy	surgical removal of a metacarpal
metaphysis	transition between the end and the shaft of a long bone
metatarsalgia	pain in the bones between the ankle and toes
metatarsals	the bones between the ankle and toes
musculoskeletal	pertaining to the muscles and bones
myalgia	pain in the muscles
myasthenia	disease of muscle weakness
myectomy	surgical removal of a muscle
myoclonus	involuntary muscle twitch
myodynia	pain in a muscle
myofasciitis	inflammation of a muscle and its associated connective tissue
myography	obtaining a record of the contraction of a muscle
myolysis	breakdown of muscle
myoma	tumor derived from muscle
myomalacia	softening of muscle
myomectomy	removal of a tumor derived from muscle (in the uterine wall)
myoplasty	surgical modification of a muscle
myorrhaphy	surgical repair of a muscle
myosarcoma	cancer derived from muscle tissue

Word	Definition
myosclerosis	hardening of muscle
myositis	inflammation of muscle
myospasm	involuntary contraction of muscle
myotomy	cutting into a muscle
neuromuscular	pertaining to the nervous system and muscles
occipital	skull bone at the back of the head, next to (and attaches to) the cervical spine
orthotic	device to straighten the foot or leg
ossification	the process of making bone
ostalgia	pain in the bones
ostectomy	surgical removal of a bone
osteitis	inflammation of a bone
osteoarthritis	inflammation of bones and the joint formed by those bones
osteoblast	rapidly dividing cell of bone (responsible for bone formation)
osteocarcinoma	cancer derived from bone tissue
osteochondritis	inflammation of bone and its associated cartilage
osteochondroma	a tumor derived from bone and cartilage
osteoclasia	abnormal breakdown of bone
osteoclast	cell which breaks down bone
osteodynia	pain in a bone
osteogenesis	the process of forming bone
osteolysis	the process of breaking down bone
osteomalacia	abnormal softening of bone
osteomyelitis	inflammation of the bone marrow
osteon	microscopic anatomical unit of bone
osteopathy	disease of the bone
osteopenia	abnormal reduction in bone tissue (sign of osteoporosis)
osteoplasty	surgical modification of a bone
osteoporosis	disease resulting in an abnormal number of holes in bone
osteonecrosis	death of bone tissue
osteosarcoma	cancerous tumor derived from bone tissue

Word	Definition
osteosclerosis	hardening of bone
pectoral	pertaining to the chest
pelvis	bone of the lower abdomen (hip bone) which results from the fusion of six bones (three on each side)
perimysium	a connective tissue sheath of muscle that surrounds a bundle of muscle fibers (fascia)
periosteum	membrane surrounding the bone
phalanges	fingers and toes
pollex	the thumb
polydactyly	too many fingers (i.e. more than five on a hand)
polymyositis	inflammation of many muscles
pronation	turning the back of the hand facing forward; lying face down
rhabdomyolysis	abnormal breakdown of skeletal muscle
sarcolemma	the cell membrane surrounding a muscle cell
sarcomere	repeating microscopic anatomical unit of striated muscle
sarcoplasm	cytoplasm of a muscle cell
scoliosis	abnormal curvature of the vertebrae (side-to-side curvature)
sphincter	band of muscle surrounding an opening
spine	bones of the back; vertebrae
spondylitis	inflammation of the vertebrae
spondyloarthropathy	disease of the joints between vertebrae
spondylodynia	pain in the vertebrae
spondylolisthesis	abnormal slippage of the vertebrae past one another
spondylolysis	abnormal breakdown of the vertebrae
spondylomalacia	abnormal softening of the vertebrae
spondylosis	disease of the vertebrae
sprain	injury to ligaments
stenosis	abnormal narrowing of an opening
sternotomy	cutting into the sternum (breastbone)
sternum	breastbone; bone that ribs attach to
strain	injury to a muscle or its tendon

Word	Definition
subluxation	dislocation of a joint
sulcus	groove in a muscle
supination	turning the palm of the hand facing forward; lying face up
synarthrotic	immovable joint
syncytium	cells which join together to make a large, multinucleated cell (as in skeletal muscle)
syndactyly	abnormal joining of fingers
synergist	muscle which aids an agonist muscle in its normal function
talipes equinovarus	clubfoot
tarsal	pertaining to the ankle
tarsalgia	pain in the ankle
tarsectomy	surgical removal of a tarsal bone
tarsitis	inflammation of the ankle
tarsoclasia	breakdown of an ankle bone
tarsoptosis	condition of tarsal bones sagging; "flat feet" or "fallen arches"
tenalgia	pain in a tendon
tendectomy	surgical removal of a tendon
tendinitis	inflammation of a tendon
tendodesis	tendons bound together by a disease process
tendolysis	breakdown of a tendon
tendon	a band of connective tissue connecting muscle to bone
tendonitis	inflammation of a tendon
tendoplasty	surgical modification of a tendon
tenodynia	pain in a tendon
tenolysis	breakdown of a tendon
tenoplasty	surgical modification of a tendon
tenorrhaphy	surgical repair of a tendon
tenotomy	cutting into a tendon
thoracic	pertaining to the chest
tibia	bone of the lower extremity; larger of the two leg bones
tibialgia	pain in the tibia

Word	Definition
tomography	technique using low-dose X-rays from many angles to visualize a body part
tonograph	record of muscle contraction
torticollis	abnormal contraction of the neck on one side
trochanter	prominent knob on the femur (bone of the thigh) which forms from the attachment of muscles
trochlea	"pulley": bony structure which acts or looks like a pulley
tubercle	small bump on bone ("little potato")
tuberosity	bump on bone
valgum	abnormally close knees (genu valgum = "knock-knees")
varum	abnormally distant knees (genu varum = "bow-legged")
vertebra	bone of the spinal column
zygomatic	bone of the face

HOW THE UNIT 5 WORD LIST WAS BUILT

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin abducere, ab-, "off, away from" + ducere, "to lead"	ab		duct			ion		abduction
In Greek and Roman dining, an acetabulum was a shallow cup filled with vinegar. Vinegar (Latin <i>vinum acētum</i> ; French <i>vinaigre</i>) literally means "sour wine" and the vinegar you buy in the supermarket is 5% acetic acid (<i>acētum</i>). The suffix –bulum is one we'll see again; in Unit 7, we'll meet the <i>infundibulum</i> , a funnel-shaped organ.			acēt[um]/a			bulum		acetabulum
	a		chondro			plasia		achondroplasia
	a		dactyl			y		adactylous
	ad		duct			ion		adduction
			agon			ist		agonist
	amphi		arthr/o			t + ic		amphiarthrosis
	an		alges[ia]			ic		analgesic
"stiff lizard"			ankylo[s]	saurus				ankylosaurus
			ankylos			ing		ankylosing
Greek ankylos, "stiff" or "crooked"			ankyl[os]			osis		ankylosis
	ant[i]		agon			ist		antagonist
	anti		arthri			t + ic		antiarthritic
	anti–		inflammat[ion]			ory		anti-inflammatory
	anti		pyre			t + ic		antipyretic



Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Greek apo-, "from, off, away" + phúsis (physis), "growth"	apo		physis					apophysis
			append[ix]			icul[e]	ar	appendicu
			arthr			algia		arthralgia
			arthr[o]			ectomy		arthrectom
			arthro			centesis		arthrocent
			arthro			scope		arthroscop
			arthro			desis		arthrodesis
	dys		arthro			dynia		arthrodyn
			arthro			plasia		arthrodysp
			arthro			gram		arthrogram
			arthro			graphy		arthrograp
			arthro			pathy		arthropath
			arthro			plasty		arthroplasti
			arthro		scop	ic		arthroscop
			arthro			tomy		arthrotom
			articul[us]			ar		articular
	a		tax			ia		ataxia
	a		troph			y		atrophy
Latin axis, "axle; line that the earth rotates around"	brady		axi			al		axial
			kines			ia		bradykines
			bursa					bursa
			burs			ectomy		bursectom
			burs			itis		bursitis
			burso			tomy		bursotomy
			calci		ton[ic]	in		calcitonin
			calci		tri	ol		calcitriol
			carp[us]			al		carpal

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word		
Latin crepitus, "rattle"				carp		ectomy		carpectomy		
				carp		itis		carpitis		
				cervi[x]		c + al		cervical		
				cervic		itis		cervicitis		
				cervico		dynia		cervicodynia		
				chondr		itis		chondritis		
				chondro		dynia		chondrody		
				chondr[o]		oma		chondroma		
				chondro		malacia		chondromal		
				chondro		plasty		chondropl		
		circum		duct		ion		circumduc		
		condyle						condyle		
		cost			al			costal		
		cost			algia			costalgia		
		cost			ectomy			costectomy		
		costo			chondr	itis		costochon		
		crani[o]				ectomy		craniectom		
		cranio				malacia		craniomal		
		cranio				meter		craniomet		
	syn	cranio		ost[eo]		osis		cranosync		
				cranio		tomy		craniotom		
				crepit[us]		ation		crepitation		
	dia				dactyl		algia		dactylalgia	
					dactyl		itis		dactylitis	
					physis				diaphysis	
					di	arthro		t + ic		diarthrotic
					dys	kinesia				dyskinesia

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin, dys-, "abnormal, bad" + -plas-, "formation, development" + -ia	dys		plas			ia		dysplasia
	dys		taxia					dystaxia
	dys		ton			ia		dystonia
	ec		cent[e]r			ic		eccentric
Latin ef- (corrupted ex-), "out of" + fundere, "to pour"	ex → ef		fusion					effusion
		electro	my/o			gram		electromyography
		electro	myo			graphy		electromyography
	endo		chondr[o]			al		endochondral
	endo		my[o]			sium		endomysium
	epi		my[o]			sium		epimysium
Greek epi-, "on top of" + phúsis (physis), "growth"	epi		physis					epiphysis
	epi		condyle					epicondyle
	ex		osteo			osis		exostosis
Latin extensio, "extension"			exten[d]			sion		extension
			facet					facet
Ultimately, the political term "fascist" and the anatomical term "fascia" come from the same Latin word for a bundle of sticks with an axe attached			fascia					fascia
			fasci[a]			ectomy		fasciectomy
			fasci			itis		fasciitis
			fascio			rrhaphy		fasciorrhaphy
			fascio			tomy		fasciotomy
			fascio			plasty		fascioplasty

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin femur, "thigh"			femor			al		femoral
			fissure					fissure
			fix[ate]			ation		fixation
			flex			ion		flexion
Latin foramen, "hole"			foramen					foramen
Latin fosse, "ditch"			fossa					fossa
Latin fractura, "break"			fracture					fracture
Proto-Indo-European (PIE) genu, "knee". PIE is the primordial language from which all Western languages originated			genu					genu
Greek gléne, "socket"			glen[e]			oid		glenoid
			gomph			osis		gomphosis
			grapho			spasm		graphospasm
Latin allex, "big toe", with an h added along the way			hallux					hallux
		hem[o]	arthr[o]			osis		hemarthrosis
			hemato			poiesis		hematopoiesis
		hydr	arthr			osis		hydrarthrosis
	hyper		calc			emia		hypercalcemia
	hyper		exten[d]			sion		hyperextension
	hyper		kines			ia		hyperkinesia
	hyper		ton			ia		hypertonia
	hyper		troph			y		hypertrophy
	hypo		calc			emia		hypocalcemia
	hypo		kines			ia		hypokinesia
	hypo		phys[is]			eal		hypophyseal

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
pituitary gland, named because it lies underneath the brain ("physis cerebri")			physis					hypophysis
	hypo		ton			ia		hypotonia
	infra		spina			t + us		infraspina
	inter		cost			al		intercostal
	inter		osse			ous		interosseous
	intra		membran[e]			ous		intramembranous
	iso		metric					isometric
			kinesio			logy		kinesiology
Greek kyphos, "crooked"			kinet			ic		kinetic
			kyphosis					kyphosis
			lacrim[a]			al		lacrimal
Latin lamina, "thin metal plate" + -ella (dim. suffix)			lam[ina]			ella		lamella
			ligament					ligament
Greek lordos, "bent backward"; also indicates a sexually receptive posture in four-legged females			lordosis					lordosis
			lumb			ar		lumbar
			lumbo			dynia		lumbodynia
Latin meātus, "passage"			meatus					meatus
			medull[a]			ary		medullary
	meta		carp			al		metacarpal
	meta		carp[o]			ectomy		metacarpectomy
	meta		physis					metaphysis
	meta		tars			algia		metatarsalgia
	meta		tars[us]			al		metatarsal

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Greek asthenes, "weakness"			musculo		skelet	al		musculoskelet
			my			algia		myalgia
			my		asthenia			myasthenia
			my			ectomy		myectomy
			myo		clonus			myoclonus
			myo			dynia		myodynia
			myo		fasci[a]	itis		myofasciitis
			myo			graphy		myography
			myo			lysis		myolysis
			my[o]			oma		myoma
			myo			malacia		myomalacia
			my			om[a]	ectomy	myomectomy
			myo			plasty		myoplasty
			myo			rrhaphy		myorrhaphy
			myo		sarc[o]	oma		myosarcoma
			myo		scler[o]	osis		myosclerosis
			myos			itis		myositis
			myo		spasm			myospasm
			myo			tomy		myotomy
			neuro		muscul	ar		neuromuscular
			occipit[o]			al		occipital
			ortho			t + ic		orthotic
			os + s			ification		ossification
			ost			algia		ostalgia
			ost[e]			ectomy		oste(e)ctomy
			oste[o]			itis		osteitis
			osteo		arthr	itis		osteoarthritis
			osteo		blast			osteoblast

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
The Greek word phalanx (plural phalanges) describes a military formation with men closely packed together, like fingers and toes				osteo	carcin	oma		osteocarcin
				osteo	chondr[o]	itis		osteochond
				osteo	chondr[o]	oma		osteochond
				osteo	clas[t]	ia		osteoclasia
				osteo	clast			osteoclast
				osteo		dynia		osteodynia
				osteo		genesis		osteogenesis
				osteo		lysis		osteolysis
				osteo		malacia		osteomalacia
				osteo	myel	itis		osteomyelitis
				oste		on		osteon
				osteo		pathy		osteopathy
				osteo		penia		osteopenia
				osteo		plasty		osteoplasty
				osteo	por[e]	osis		osteoporosis
				osteo	necr[o]	osis		osteonecrosis
				osteo	sarc[o]	oma		osteosarcoma
				osteo	scler[o]	osis		osteosclerosis
				pector[is]		al		pectoral
				pelvis				pelvis
	peri			my[o]		sium		perimysium
	peri			oste[o]		um		periosteum
				phalanges				phalanges
				pollex				pollex
	poly			dactyl		y		polydactyl

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Greek rhabdo-, "rod" + myo + lysis. "Rod" in this context means striated muscle (i.e. skeletal muscle or voluntary muscle).	poly	rhabdo	myos			itis		polymyosi
			pron[e]		ation		pronation	
			myo			lysis		rhabdomy
			sarco	lemma			sarcolemm	
			sarco	mere			sarcomere	
			sarco	plasm			sarcoplas	
			scoliosis					scoliosis
			sphincter					sphincter
			spine					spine
			spondyl			itis		spondyliti
Greek spondylos, "vertebrae" + olisthánein, "slippage"			spondylo	arthro	pathy		spondyloa	
			spondylo		dynia		spondylo	
			spondyl	olisthes	is		spondyloli	
			spondylo		lysis		spondyloly	
			spondylo		malacia		spondylom	
			spondyl		osis		spondylos	
			sprain				sprain	
			sten		osis		stenosis	
			sterno		tomy		sternotom	
			sternum				sternum	
Latin luxationem, "a dislocation"	sub		strain				strain	
			luxation				subluxatio	
			sulcus				sulcus	

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
				supinat		ion		supination
	syn			arthro		t + ic		synarthrot
	syn			cyt		ium		syncytium
	syn			dactyl		y		syndactyl
	syn			erg[on]		ist		synergist
Talipes is formed from Latin tal[us], "ankle" + pes, "foot", in other words, "walking on the ankle". Equin[e], "horse" (walking on the toes like a horse) + varus, the direction the foot is turned in		talipes		equino	varus			talipes equ
				tars[us]		al		tarsal
				tars[us]		algia		tarsalgia
				tars[us]		ectomy		tarsectomy
				tars[us]		itis		tarsitis
				tarso		clasia		tarsoclasia
				tarso		ptosis		tarsoptosis
				ten		algia		tenalgia
				tend		ectomy		tendectomy
				tendin		itis		tendinitis
				teno		desis		tendodesis
				tendo		lysis		tendolysis
				tendon				tendon
				tendon		itis		tendonitis
				tendo		plasty		tendoplasty
				teno		dynia		tenodynia
				teno		lysis		tenolysis
				teno		plasty		tenoplasty
				teno		rrhaphy		tenorrhaphy

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
			teno			tomy		tenotomy
			thorac			ic		thoracic
			tibia					tibia
			tibi[a]			algia		tibialgia
			tomo			graphy		tomograph
			tono			graph		tonograph
literally, "twisted neck", from the same Latin roots as torque + collar			torti		collis			torticollis
			trochanter					trochanter
Latin trochlea, "pulley"			trochlea					trochlea
			tuber			cle		tubercle
			tuber			osity		tuberosity
knees closer together than normal			valgum valgus					valgum valgus
knees farther apart than normal			varum varus					varum varus
			vertebra					vertebra
Pertaining to the zygoma, from Greek zygon, "yoke". Named because it joins, or yokes, the bones of the face to the bones of the skull.			zygom[a]				atic	zygomatic

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PART VI

UNIT 6: THE NERVOUS SYSTEM

UNIT 6 INTRODUCTION

The nervous system consists of an estimated 80 billion neurons with an average of 10,000 connections each. It can receive information from the external environment or the internal environment. It processes and sends information to effectors, which can cause muscles to constrict or glands to secrete hormones. The receiving, processing, and sending of information is a neuron's specialty. Obviously, with so many neurons engaged in this activity, the brain's capacity for higher-order functions, such as learning, memory, or language, is really remarkable.

Most medical terminology textbooks split out the “special senses” (hearing, balance, and vision). That's a tradition in anatomy and physiology courses. We're not going to do it that way. We've included special senses in this chapter because they're still part of the nervous system and there's a lot of overlap in the terms used.

Objectives

After completing this unit, you will be able to:

- Recall terms used to describe the nervous system and its sensory organs (e.g. eye and ear).
- Analyze the parts of words used to describe the nervous system and nervous system diseases.

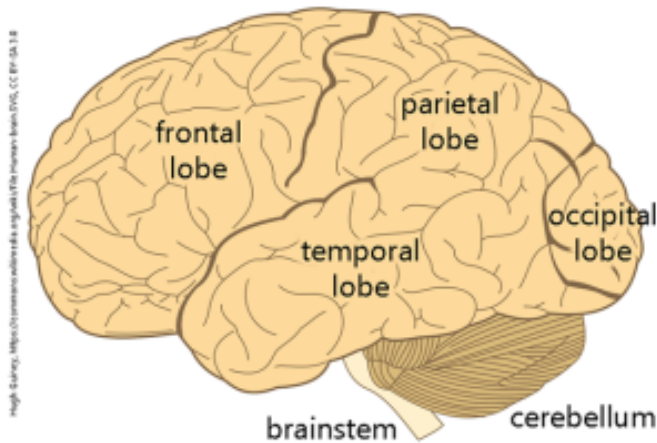
THE NERVOUS SYSTEM

Only [Med Terms] in the Building Episode 6



One or more interactive elements has been excluded from this version of the text. You can view them online here:

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Unit 6 figure 1 Human Brain

The nervous system consists of the brain and spinal cord. The job of the nervous system is to control body processes; it does this by

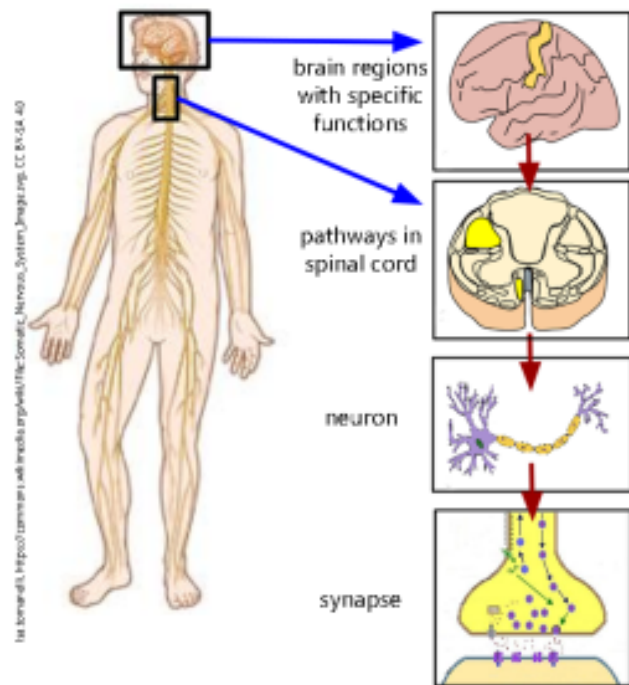
- receiving information from the internal and external environment;
- processing that information in billions of neurons;
- sending that information to the brainstem and spinal cord, and from there to the muscles of the body.

It's important to notice that the output of the nervous system is to move a muscle or change the

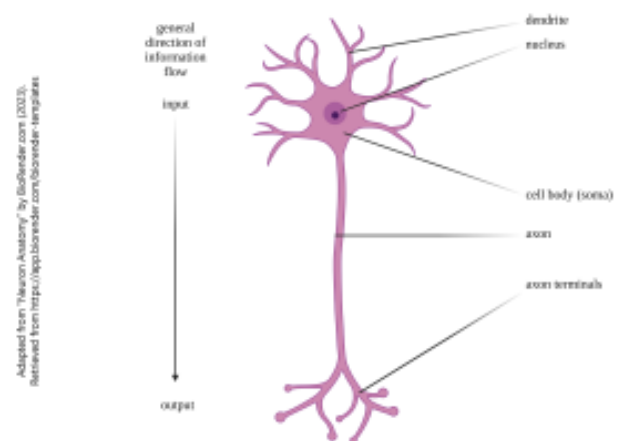
secretion of a gland. That's all it can do.

The nervous system is a triumph of miniturization as well as processing speed and efficiency. Imagine you're driving and you see the brake lights of the car ahead of you. It's time for a panic stop! Your visual system processes the lights and interprets what they mean. Visual association areas determine the correct action is to stomp on the brake as hard as possible. They send the command to an area called the premotor area, and the premotor area then writes up a work requisition and passes it to the **primary motor cortex** (the strip of brain shown in yellow in the top right of this diagram, also called the **prefrontal gyrus**). Nerve cables, called **axons**, carry signals through the spinal cord (second diagram from top right). The pathway (whose name is not important right now) is shown in yellow.

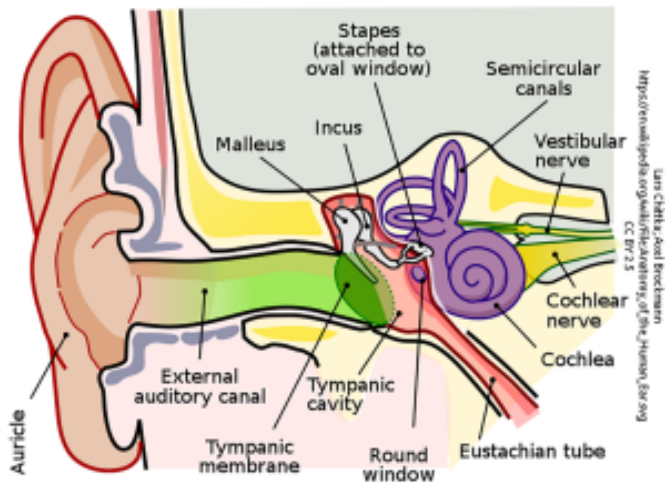
Eventually, the neurons that started off in the motor cortex end up in the **anterior (ventral) horn of the spinal cord**, where they send their signal to a motor neuron. This motor neuron picks up information at its **dendrites**. It has its **cell body (soma)** in the spinal cord, its **axon** in the nerves running to muscles, and ends in a synapse where it releases a chemical signal onto a muscle cell from its **axon terminals**. The chemical (acetylcholine) causes your thigh muscles to contract, and the brake is pressed, saving your life. That all happens in 100 milliseconds (0.1 second). Unfortunately, at 60 mph (27 m/sec) you have traveled about 1 car length while your brain has been processing the information. For this reason, neuroscientists don't like it when you tailgate.



Unit 6 figure 2 Nervous System Overview

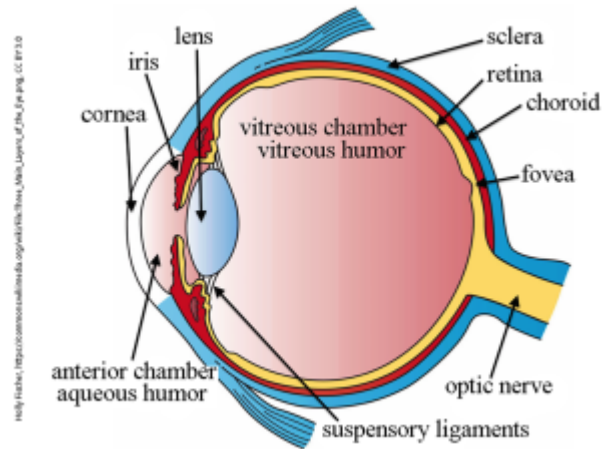


Unit 6 figure 3 Canonical Neuron



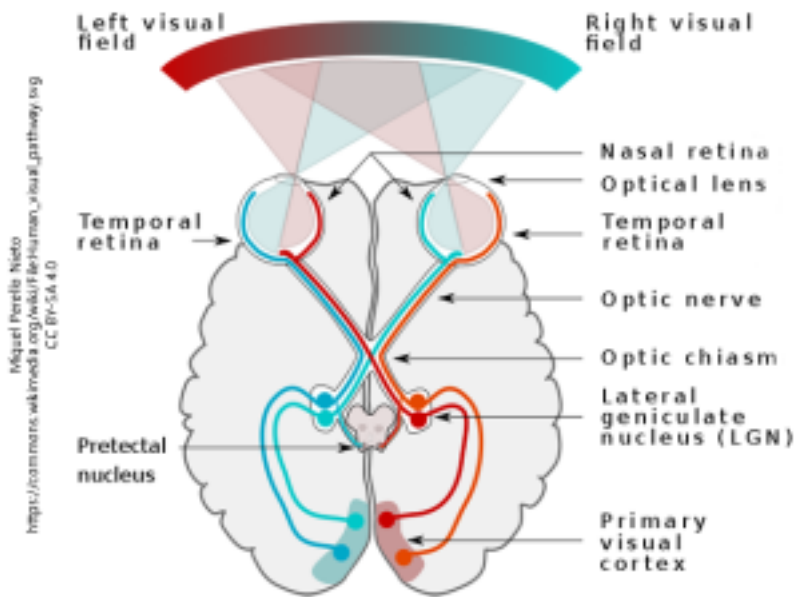
Unit 6 figure 4 Ear

The auditory (hearing) system transforms sound waves which travel through the air into an electrical signal via the hair cells of the cochlea. Sound waves from your favorite music are transmitted for your enjoyment via the structures of the outer, middle, and inner ear to the temporal lobe of the brain.



Unit 6 figure 5 Eye Cross Section

The eye is likely the most complicated of the sensory end-organs. It does an amazing job! It focuses a beam of photons onto the retina (a sheet of central nervous system cells at the back of the eye) and the retinal cells extract information (wavelength/color; edges; motion) from this beam of photons and send a report to the brain. The brain constructs a map of the visual world based on location and compares input from each eye. This helps us to more clearly perceive what we are seeing. You don't need to know this, but to give you a feeling for how wonderful the eye is, here are two factoids. The human eye can resolve things as small as $70\text{ }\mu\text{m}$ (10 red blood cells lined up side-to-side; 0.07 mm) at a distance of 25 cm . The human eye also operates over 14 log units of intensity and can detect anything from 10 photons (!) to 1,000,000,000,000,000 photons.



Unit 6 figure 6
Visual Pathways

An estimated 40% of the axons in the human brain are devoted to visual function in some way. This complexity and capability is why the visual system is the only sensory system that has a medical specialty (*ophthalmology*) devoted to it, and nothing else.

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UNIT 6 PREFIXES

Prefix	Definition
a-	not
af-	going into (was originally ad-)
an-	not
ana-	up, throughout, again
anti-	against (an action)
apo-	from, off, away
auto-	self
bi-	two, double
bin-	
cata-	down, downward
con-	together, with
de-	from, down, out
di-	separate, apart, double, two
dipl-	
dys-	bad
e-	out, off, from, beyond
ec-	
ex-	
ef-	going out of
en-	inside
end-	inside
endo-	
epi-	on top of
eso-	within, inward
eu-	good, well-
ex-	out, outside, without
exo-	
extra-	outside, more, beyond
hemi-	half
hyper-	too much
hypo-	too little
(before process)	

Prefix	Definition
hypo- (before anatomical structure)	below
im- in-	not
inter-	between
intra-	inside
macro-	large
micr- micro-	small
mid-	middle
mono-	one
multi-	several; many
para-	beside; near
peri-	around
poly-	many; too much
post-	behind (anatomically or in time)
pre-	in front of (anatomically or in time)
pro-	in front of; in favor of
quadri-	four
re-	again
sub-	an underlayer; support
sym- syn-	with, together
trans-	through
tri-	three
uni-	one
xero-	dry

UNIT 6 ROOTS

Root Word	Definition
acous acusis	hearing
aden	gland
aer	air
ambly	blunted
ambul	moving around
ampulla	flask with long, narrow neck
aneurysm[os]	dilation of an artery
angio	vessel
anxio	anxiety
arachn[id]	spider
arter arterio	artery
asthen	weakness
athero	encysted tumor
audio	sound
audit	hearing
aur	sound
aut[os]	self
axo[n] axon	nerve cell structure that carries a signal over a long distance
basi[s]	base (of the midbrain)
blephar/o	eyelid
brain	brain
cataract	cloudiness ("waterfall")
caud[a]	tail
central [sulcus]	in the center
cephal/o	pertaining to the head
cereb cerebr/o	brain
chemo	chemical

Root Word	Definition
chor[ion]	membrane (resembling the one surrounding the fetus)
clonus	violent shaking
cochlea	snail's shell
conjunctiva	joining (the eyelids and globe of the eye)
convuls[e]	jerking movement
corn[u]	horn
cortex cortico	outside of a multilayered structure ("bark of a tree")
cran/i/o	head; skull
cycl/o	circle; in this case, the ciliary body which forms a circle around the inside of the eye
cysticercus	larval stage of some tapeworms
cyst/o	fluid-filled space
dacry/o	tear (drop)
decussate	make a shape like the letter X
dendr/o	tree
depress	push down
derm/a derm/at	skin
duct	pipe
dura	tough
ectasis	dilation
edema	swelling
electro	related to electricity
embol	arterial blockage ("peg, stopper")
emulsion	mixture with large particle sizes
encephal/o	brain (literally, inside the head)
ependyma	lining of the ventricles
esthes	feeling
exter[i]o[r]	outside
externa	outside
fascia	sheet of fibers; bundle of stick-like structures

Root Word	Definition
fer[re]	carry; move
front	front
gangl gangl/i	cluster of nerve cell bodies
geminus	twins
gen	birth (dealing with inheritance)
glaucoma	increased pressure in eye
glia gli/o	glue
glosso	tongue
glyc/o	sugar
gnos	knowing
gustat[e]	taste
gyrus	elevation or bump on the brain (from the Greek word for "ring, circle")
[h]em[e] hem heme hemo hemat	blood
homunculus	"little man"; map of the body on the brain surface
hydro	water
hypno	sleep
ictus	seizure; sudden, unexpected illness
idios	self
insula	island
ir[is] irid[es] irido	iris (plural irides)
isch	keep back (shortening of Greek iskhein)
kerat/o	horn
kine/t	movement
labyrinth	maze
lacrim	crying; tears

Root Word	Definition
laryng/o	voice box
leps[is]	seizure; grasp
lex[is]	reading
ligand	signaling molecule or drug that binds a receptor
lir[a]	staying on track (literally, "furrow")
lith	stone
lob lobo	large region of the brain; roundish structure
lymph	here, the fluid inside the inner ear chambers
mechano	mechanical
medulla	inside of a multilayered structure ("marrow")
mening meningi meningo	coverings of the brain
mens ment	mind
metr	measurement
motor	movement
my[ein]	pupil constriction
myc	fungus
mydriasis	pupil dilation
myel/o	spinal cord
myo	muscle
myring/o	eardrum
narco	sleep
nas/o	nose
neur/o	nervous system
noci	harmful; noxious
nom	name
nucle	the center of something
nystagmus	nodding or jerking back-and-forth
occipit	back of the head

Root Word	Definition
ocul/o	eye
op ops opt/o	eye; vision
ophthalm/o	eye; vision
orexis	desire
osmē	smell
oste/o	bone
ot/o	ear
propagate	reproduce; breed
papilla	nipple
paresis	weakness
pariet	side of the head
path[os]	feeling
phac/o phak	lens
pharmaco	pertaining to drugs
phas	speaking
phon	sound
phor	carrier
phot/o	light
phren	skull; mind
physi	function
plas	something molded
pleg	stroke; weakness
plex	woven
pol[e] polar	one end of a round object, as in North Pole & South Pole
polio	degeneration of myelin
presby	elder; old person
propri[us]	understanding of one's own body position
prosop	face

Root Word	Definition
pseudo	false
psych/o	mind
py/o	pus
[re]ceptor receptor	a protein on the surface of a cell that turns a chemical on the outside to a signal on the inside of the cell
retin/a retin/o	net; the receptive surface of the eye
rhin/o	nose
saccul	small sac
salping/o	tube (e.g. Eustachian tube; Fallopian tube)
sciatic	nerve of the buttocks and thigh
scler/a scler/o	hard (white of the eye)
scot	darkness; blindness
soma	body
somn/o	sleep; drowsiness
sono	sound
spin[e]	spinal column
stigma	wound; point
stimulus	something which initiates an action
sulcus	groove in the brain
synapse	contact between a nerve cell and its target
syncope	fainting
tempor	side of the head around the ear
therm/o	temperature
thromb/o	clot
tinnitus	jingling or ringing noise
tox	poison
transmit	carry along
trop	turning toward
tympan/o	eardrum

Root Word	Definition
utricle	bag
varicos[e]	dilation; enlargement
vascul	little vessel
vertigo	spinning sensation
vestibul/o	inner ear structures controlling balance
vittr/e	glass

UNIT 6 SUFFIXES

Suffix or Ending	Definition
–al	pertaining to
–algia	pain
–ant	state of being
–ar	pertaining to
–asis	state of being
–ate	chemical name ending
–ation	word ending that turns a verb into a noun
–cele	hernia
–centesis	puncture
–dynia	pain
–ea	anatomical word ending
–eal	pertaining to
–ectomy	cutting out of
–ellum	diminutive suffix (cerebellum = "little brain")
–emia	present in blood, indicating disease
–ent	word ending that turns a verb into a noun
–etic	a drug or condition causing a process
–gram	a record of a test or procedure
–graphy	a test or procedure
–ia	disease
–ial	pertaining to
–iasis	presence of an abnormality
–iatrist	clinician who studies
–iatry	the act of studying
–ic	pertaining to
–ication	pertaining to + word ending that turns a verb into a noun
–icle	diminutive suffix
–ient	word ending which indicates an ongoing condition
–ing	at the end of a verb, means "ongoing"
–ion	word ending that turns a verb into a noun

Suffix or Ending	Definition
-ism	belief; process word ending
-ist	someone who studies
-itis	inflammation
-ity	turns a noun into a state of being
-ium	anatomical structure
-ive	to go
-ization	turns a noun into a verb and then back into a noun (which is weird)
-logist	one who studies
-logy	study of
-lysis	breaking down; destroying
-lytic	substance that causes a breakdown or destruction
-malacia	softening due to disease
-meter	instrument for measuring
-metry	the act of measuring
-oid	looks like; resembles
-ology	study of; o added to make it sound better
-oma	tumor
-on	unit of anatomical structure
-ory	pertaining to
-osis	process
-ous	word ending that turns a noun into an adjective
-pathy	disease process
-penia	low levels of
-pexy	surgical fixation
-plasty	surgical revision
-plegia	inability to move
-ptosis	drooping
-rrhag[e]	abnormal flow
-rrhaphy	suturing
-rrhea	abnormal flow

Suffix or Ending Definition

–scope	instrument for looking inside
–spasm	involuntary contraction
–stenosis	abnormal narrowing of an opening
–stomy	creating an opening (or "mouth") with surgery
–tatic	pertaining to stretching
–ter	noun ending
–tic	pertaining to; t added to make it sound better
–tome	instrument for cutting
–tomy	cutting
–trophic	pertaining to nourishment or growth
–trophy	nourishment, growth
–us	anatomical structure word ending
–y	disease

UNIT 6 ABBREVIATIONS

Abbreviation	Meaning
ALS	amyotrophic lateral sclerosis
AU	both ears (Latin auris uterque)
AD	right ear (Latin auris dextra)
AS	left ear (Latin auris sinistra)
CNS	central nervous system
CSF	cerebrospinal fluid
CVA	cerebrovascular accident; stroke
EEG	electroencephalogram
EENT	eye, ear, nose, and throat
ENT	ear, nose, and throat
EMG	electromyogram
HEENT	head, eyes, ears, nose, and throat
ICP	intracranial pressure (not Insane Clown Posse)
IOL	intraocular lens
IOP	intraocular pressure
LASIK	laser-assisted in situ keratomileusis
LOC	level of consciousness
LP	lumbar puncture
MRA	magnetic resonance angiography
MRI	magnetic resonance imaging
MS	multiple sclerosis
OU	both eyes (Latin oculus uterque)
OD	right eye (Latin oculus dexter)
OS	left eye (Latin oculus sinistra)
PET	positron emission tomography
PERRLA	pupils are equal, round, reactive to light, and accommodate
PNS	peripheral nervous system
TIA	transient ischemic attack

UNIT 6 WORD LIST

Word	Definition
acoustic	pertaining to hearing
aerotitis	air in the middle ear
afferent	traveling towards a structure
agnosia	not knowing
akinetopsia	motion-blindness
amblyopia	disease in which one eye's input is ignored by the brain
ampulla	round flask; structure in the vestibular system which resembles a flask
amyotrophic	pertaining to muscle tissue wasting
anastomosis	"backwards mouth"; two openings coming together to form one
anesthesia	lack of sensation (usually, pain)
anesthesiologist	physician who administers anesthesia
aneurysm	ballooning of a blood vessel
aniridia	disease marked by the absence of an iris
anorexia	disease marked by a lack of desire (usually, for food)
anosmia	disease marked by an absence of the ability to smell
anticonvulsant	drug which stops seizure activity
antidepressant	drug which regulates mood
antipsychotic	drug which reverses symptoms of psychosis (a loss of contact with reality)
anxiolytic	drug which "breaks down" or reduces anxiety
apathy	lack of feeling
aphakia	absence of the lens
aphasia	absence of the power of speech (also, written language)
aponeurosis	sheath of connective tissue joining muscle and bones
arachnoid	one of the meninges; layer covering the brain that resembles a wet spider web
arteriosclerosis	hardening of the arteries
asthenopia	eye strain or eye fatigue
astigmatism	defect in the optical properties of the eye
atherosclerosis	buildup of plaque in the arteries
audiogram	graphical representation of a patient's hearing
audiologist	clinician who studies hearing

Word	Definition
audiometry	measurement of a patient's hearing
auditory	pertaining to hearing
aural	pertaining to hearing
autism	alternative wiring of the brain resulting in characteristic behaviors and symptoms
autonomic	pertaining to body systems not under control of the conscious mind
axodendritic	a contact between an axon and a neuronal dendrite
axon	the part of a nerve cell responsible for its output
axosecretory	release of chemicals from an axon to the blood
axosomatic	a contact between an axon and neuronal cell body
basilar	running along the base of the brain (the name of an artery which does this)
binocular	using both eyes to perceive the visual world
bipolar	a neuron with two major extensions from the cell body
blepharedema	swelling of the eyelid
blepharitis	inflammation of the eyelid
blepharoconjunctivitis	inflammation of the eyelid and mucous membranes of the eye
blepharoplasty	surgical revision of the eyelid
blepharoplegia	paralysis of the eyelid
blepharoptosis	drooping of the eyelid
blepharopyorrhea	flow of pus from the eyelid
blepharospasm	involuntary contraction of the eyelid
blepharotomy	cutting into the eyelid
brain	the largest part of the central nervous system
bulimia	eating disorder characterized by the binge-purge-starve syndrome
cataract	cloudiness in the lens of the eye
catatonia	disease characterized by a lack of awareness
caudate	a structure in the brain which resembles a tail (Latin cauda)
cephalalgia	pain in the head
cephalodynia	pain in the head
cerebellar	pertaining to the part of the brain under the cerebrum
cerebellitis	inflammation of the cerebellum

Word	Definition
cerebellum	part of the brain under the cerebrum and attached to the brainstem
cerebral	pertaining to the cerebrum
cerebromeningitis	inflammation of the brain and the membranes that cover it
cerebroopathy	disease of the brain
cerebrospinal	pertaining to the brain and spinal cord
cerebrotomy	cutting into the brain
cerebrovascular	pertaining to the blood vessels of the brain
cerebrum	the large, wrinkled part of the brain; right and left hemispheres
cerumen	ear wax
ceruminolysis	process of breaking down ear wax
ceruminolytic	substance which breaks down ear wax
ceruminoma	tumor in the glands which produce ear wax
ceruminosis	condition of excessive ear wax
chemoreceptor	nerve cell which detects the presence of specific chemicals
choroid	a tissue resembling the chorionic membrane of the embryo; there is a choroid of the eye and a choroid plexus lining the ventricles of the brain
clonus	involuntary, rhythmic muscle contractions
cochlear	pertaining to the hearing apparatus of the inner ear (Latin "snail", because of its anatomical resemblance to a snail)
cochleitis	inflammation of the cochlea
conductive	arising from a defect in the structures which bring sound to the inner ear
conjunctiva	mucous membrane of the eye
conjunctivitis	inflammation of the mucous membrane of the eye
convulsion	sudden, violent, involuntary contraction of muscle
cornea	the clear window covering the iris and pupil
corticospinal	nerve cell pathway connecting the cerebral cortex and the spinal cord
craniosclerosis	hardening of the skull
craniosynostosis	premature fusion and hardening of the bones of a child's skull
cyclokeratitis	inflammation of the cornea and ciliary body of the eye
cycloplegia	paralysis of the ciliary body of the eye
cyclotomy	cutting into the ciliary body of the eye

Word	Definition
dacryoadenalgia	pain in the lacrimal gland (the gland that makes tears to lubricate the eye)
dacryoadenectomy	removal of a lacrimal gland
dacryoadenitis	inflammation of a lacrimal gland
dacryocystalgia	pain in the nasolacrimal sac (which stores tears to lubricate the eye)
dacryocystectomy	surgical removal of the nasolacrimal sac
dacryocystitis	inflammation of the nasolacrimal sac
dacryocystorhinostomy	surgery to enlarge the opening of the nasolacrimal sac
dacryocystotomy	cutting into the nasolacrimal sac
dacryohemorrhea	blood mixed with tears
dacryolith	stone in the lacrimal glands
dacryolithiasis	condition of stones in the lacrimal glands
dacryopyorrhea	flow of pus from the lacrimal glands
dacryorrhea	excessive flow of tears from the lacrimal glands
dacryostenosis	blockage of tears from the lacrimal glands
decrement	slow, steady decrease in size
decussation	crossing
delirium	confusion and disorientation
dementia	lack of orientation to person, place, or time, often with memory loss
depolarization	an accumulation of positive charges inside a nerve cell
dermatome	region on the surface of the body which sends information along a single spinal nerve
diencephalon	region of the brain between the cerebrum and the brainstem; thalamus and associated structures
diplopia	double vision
dopamine	chemical signaling molecule of the brain
dura	one of the meninges; layer covering the brain made up of tough connective tissue
duritis	inflammation of the dura, one of the coverings of the brain
dysesthesia	abnormal sensation
dyslexia	disruption in the ability to read
dysphasia	disruption in the ability to speak
dysphoria	abnormal sensation of dread or unease

Word	Definition
echoencephalography	the use of ultrasound to image the brain
ectropion	turning of the eyelid away from the eyeball
efferent	traveling away from a structure
electroencephalography	the use of electrical activity to image the brain
electrophysiology	the study of electrical activity in the brain or in brain cells
embolism	obstruction of an artery by a clot or air bubble
encephalitis	inflammation of the brain
encephalocele	herniation of the brain
encephalogram	image or trace of the electrical activity of the brain
encephalomyelitis	inflammation of the brain and spinal cord
encephalomyeloneuropathy	a disease of the brain, spinal cord, and nerves
encephalopathy	a disease of the brain
encephalopyosis	condition of pus in the brain
endarterectomy	removal of the inner lining of an artery along with a clot or plaque
endolymph	fluid of the inner ear
endoneurium	innermost connective tissue layer of a nerve
endophthalmitis	inflammation of the inside of the eye
endovascular	pertaining to the inner lining of a blood vessel
entropion	turning of the eyelid toward the eyeball
enucleation	removal of the eyeball
ependymal	pertaining to cells lining the ventricles (fluid-filled chambers of the brain)
epidural	pertaining to the space outside the dura, one of the coverings of the brain
epilepsy	disease characterized by the sudden onset of seizures
epineurium	outermost connective tissue of a nerve
esotropia	eye misalignment where both eyes turn inward
euphoria	sensation of well-being or happiness
exophthalmus	protrusion of the eyes out of the head
exotropia	eye misalignment where both eyes turn outward
externa	outermost
exteroceptor	nerve cell which detects the presence of stimuli outside the body

Word	Definition
extraocular	outside of the eye, especially referring to the muscles which move the eye
fascicle	bundle of nerve axons
frontal	bone at the front of the skull; lobe of the brain in the front
ganglioma	tumor of a ganglion
ganglion	a collection of nerve cells outside the central nervous system
ganglionitis	inflammation of a ganglion
glaucoma	abnormal increase in pressure within the eye
glossopharyngeal	pertaining to the tongue and oral cavity
gustation	the sensation of taste
gyrus	elevated bump on the surface of the cerebrum
hematoma	mass of blood
hemianopsia	abnormal visual perception, "half not seeing"
hemiparesis	muscle weakness affecting half of the body (e.g. right upper and lower extremity)
hemiplegia	paralysis affecting half of the body (e.g. right upper and lower extremity)
hemorrhagic	abnormal blood flow
homunculus	map of the body surface as represented in the brain (Latin "little man")
hydrocephalus	abnormal accumulation of cerebrospinal fluid
hyperacusis	abnormally increased ability to hear
hyperesthesia	abnormally increased sensation in skin
hyperopia	ability to see at a distance but not close up
hyperpolarization	an accumulation of negative charges inside a nerve cell
hypnotic	drug causing sleepiness
hypoacusis	abnormally reduced ability to hear
hypoglossal	underneath the tongue
ictus	seizure
idiopathic	a condition with no obvious cause
insomnia	inability to sleep
insula	a part of the cerebral cortex not visible from the outside of the brain (Latin, "island")
interictal	the time between seizures
interneuron	a nerve cell that is neither sensory nor motor

Word	Definition
interoceptor	nerve cell which detects the presence of stimuli inside the body
intracerebral	within the brain
intraocular	within the eyes
intra vitreal	within the clear, jelly-like substance which gives the eyeball its shape
iridalgia	pain in the iris
iridectomy	surgical removal of the iris
iridemia	blood surrounding the iris
iridocyclectomy	surgical removal of the iris and ciliary body
iridocyclitis	inflammation of the iris and ciliary body
iridokeratitis	inflammation of the iris and cornea
iridopathy	disease of the iris
iridotomy	cutting into the iris
iris	the colored part of the eye which defines the pupil
iritis	inflammation of the iris
ischemia	a lack of blood flow
ischemic	pertaining to a lack of blood flow
keratalgia	pain in the cornea
keratitis	inflammation of the cornea
keratomalacia	softening of the cornea
keratopathy	disease of the cornea
keratoplasty	surgical revision of the cornea
keratotomy	cutting into the cornea
labyrinthectomy	surgical removal of the vestibular (balance) apparatus of the inner ear
labyrinthitis	inflammation of the labyrinth
labyrinthotomy	cutting into the labyrinth
lacrimation	the process of forming tears and crying
ligand	chemical which binds to a cell surface receptor
lobe	major division of the cerebrum
lobectomy	surgical removal of a lobe of the brain
lobotomy	cutting into a lobe of the brain

Word	Definition
macrocephaly	abnormally large head
mechanoreceptor	nerve cell which detects the presence of movement, pressure, or other mechanical stimuli
medulla	the core of a multilayered organ (e.g. the brain)
meninges	coverings of the brain
meningioma	tumor of the coverings of the brain
meningitis	inflammation of the coverings of the brain
meningocele	hernation of the coverings of the brain
meningoencephalitis	inflammation of the brain and its coverings
meningopathy	disease of the coverings of the brain
microcephaly	abnormally small head
microtia	abnormally small ears
midbrain	part of the brainstem between the thalamus and pons
miosis	condition of a constricted (small) pupil
miotic	drug which causes miosis
monoparesis	weakness in a single extremity
monoplegia	paralysis of a single extremity
monosynaptic	characterized by a single contact between nerve cells
multipolar	a neuron with many major extensions from the cell body
mydriasis	condition of a dilated (large) pupil
mydriatic	drug which causes mydriasis
myelitis	inflammation of the spinal cord
myelocele	hernation of the spinal cord
myelogram	image of the spinal cord
myelomalacia	softening of the spinal cord
myelomeningocele	herniation of the spinal cord and its coverings
myelopathy	disease of the spinal cord
myopia	ability to see close up but not at a distance
myotatic	pertaining to muscle stretching
myringectomy	surgical removal of the eardrum
myringitis	inflammation of the eardrum

Word	Definition
myringodermatitis	inflammation of the eardrum and the adjacent skin of the ear canal
myringomycosis	fungal infection of the eardrum
myringoplasty	surgical revision of the eardrum
myringotomy	cutting into the eardrum
narcolepsy	condition of uncontrollable, sudden sleepiness
nasolacrimal	pertaining to the lacrimal (tear-producing) structures and the nose
neuralgia	pain in the nerves
neurasthenia	weakness or lack of energy in the nervous system
neurectomy	surgical removal of a nerve
neuritis	inflammation of a nerve
neuroarthropathy	loss of nerve sensation that leads to destruction of a joint
neurocysticercosis	"brain worm"; the larval stage of a parasitic worm found in the brain
neurodynia	pain in the nerves
neuroencephalomyelopathy	disease of the nerves, brain, and spinal cord
neurogenic	arising from the nervous system
neuroglycopenia	condition of low sugar (glucose) levels in the brain
neurolysis	breakdown of nervous system tissue
neuroma	tumor derived from the nervous system
neuron	anatomical unit of the nervous system
neuropathy	disease of the nervous system
neuropharmacology	study of the effect of drugs on the nervous system
neurophysiology	study of the (usually electrical) function of the nervous system
neuroplasty	surgical alteration of the nerves
neurorrhaphy	surgical repair of a nerve
neurosclerosis	abnormal hardening of the nerves
neurosis	abnormal change in the nervous system
neurosurgery	using surgical techniques to revise or repair the nervous system
neurotomy	cutting into a nerve
neurotransmitter	chemical which carries a signal from one neuron to another neuron, or to a muscle cell
nociceptor	nerve cell which detects the presence of damage

Word	Definition
nystagmus	eye movements that alternate between a rapid track to one side and a slow track to the other
occipital	bone at the back of the skull; lobe of the brain at the back
oculomycosis	fungal infection of the eye
oculopathy	disease of the eye
oculoplasty	surgical revision of the eye
ophthalmalgia	pain in the eye
ophthalmic	pertaining to the eye
ophthalmitis	inflammation of the eye
ophthalmologist	clinician who studies the eye
ophthalmomyitis	inflammation of the extraocular muscles
ophthalmopathy	disease of the eye
ophthalmoplegia	paralysis of the eye
ophthalmoscope	instrument for observing the eye
optic	pertaining to vision
optokinetic	pertaining to movement of the visual field
optometrist	clinician who measures the optical properties of the eye
optomyometer	instrument for measuring the strength of eye muscles
osmoreceptor	nerve cell which detects the strength of chemicals in the blood
osteoacsis	sound conduction via bone
otalgia	pain in the ear
otitis	inflammation of the ear
otodynia	pain in the ear
otolaryngologist	physician who studies the ear and throat
otoliths	microscopic stones in the ear that help detect head tilt
otomycosis	fungal infection of the ear
otoneurologist	physician who studies the ear and brain
otoplasty	surgical modification of the ear
otopyorrhea	flow of pus from the ear
otorhinolaryngologist	physician who studies the ear, nose, and throat

Word	Definition
otorrhea	flow of liquid from the ear
otosclerosis	hardening of the bones of the middle ear
otoscope	instrument for observing the ear
otosteal	pertaining to the bones of the middle ear
ototoxic	drug that destroys the nerve cells of the ear
papilledema	swelling of the optic nerve, visible through an ophthalmoscope
paraesthesia	pins-and-needles sensation in the skin caused by nerve damage
paralysis	inability to move
parasympathetic	pertaining to the division of the autonomic nervous system that maintains body functions at rest
paresis	weakness
parietal	bone on the side of the skull; lobe of the brain on the side
perilymph	fluid of the inner ear
perineurium	connective tissue layer of a nerve
phacoemulsification	surgical procedure that involves breaking apart the lens of the eye
phacomalacia	softening of the lens
phacosclerosis	hardening of the lens
phacoscope	instrument for observing the lens
phakitis	inflammation of the lens
phonasthenia	weakness or lack of energy in the voice
photon	particle-wave of light energy
photophobia	inability to tolerate bright light (literally, "fear of light")
photopic	pertaining to bright light conditions
photoreceptor	nerve cell which detects photons (particle-waves of light energy)
plexus	nerves which come together, mix, and separate
poliomyelitis	viral disease of the nervous system
polyneuritis	inflammation of several or many nerves
polyneuropathy	disease of several or many nerves
postcentral	behind the central sulcus
postganglionic	referring to nerve axons that are efferent to the autonomic ganglia

Word	Definition
postictal	after a seizure
postsynaptic	after a contact between nerve cells (i.e. the neuron receiving the information)
precentral	in front of the central sulcus
prefrontal	the rostral part of the frontal cortex ("the front of the front")
preganglionic	referring to nerve axons that are afferent to the autonomic ganglia
preictal	before a seizure
premotor	area of cortex that calculates, but does not execute, a motor program
presbycusis	elder hearing
presbyopia	elder vision
presynaptic	before a contact between nerve cells (i.e. the neuron sending the information)
propagate	spreading of a nerve signal
proprioceptor	nerve cell which detects the position of the body in space
prosopagnosia	inability to recognize faces
pseudesthesia	experiencing "false" feelings; feelings not caused by a stimulus
psychiatrist	physician who studies the mind
psychiatry	branch of medicine involved in the study of the mind
psychogenic	arising from the mind
psychologist	clinician who studies the mind
psychology	science of the mind
psychopathy	disease of the mind
psychopharmacology	study of drugs which affect the mind
psychosis	disease of the mind, usually one which involves a disconnection from reality
psychosomatic	pertaining to the connection between mind and body
psychotropic	a drug which acts on the mind
quadriplegia	paralysis of all four extremities
refractory	resistant to change
repolarization	restoration of the normal distribution of negative charges inside a nerve cell
retina	photosensitive nerve tissue lining the inside back of the eye
retinal	pertaining to the retina
retinitis	inflammation of the retina

Word	Definition
retinopathy	disease of the retina
retinopexy	surgical fixation of the retina
retinoscope	instrument for observing the retina
retinosis	condition of the retina
sacculæ	small sac; structure in the vestibular system which resembles a sack
sciatica	inflammation of the major nerve supplying the lower extremity
sclera	white of the eye (i.e. "hard part" of the eye)
sclerectasia	bulging and thinning of the sclera
scleritis	inflammation of the sclera
scleromalacia	softening of the sclera
sclerotomy	cutting into the sclera
scotoma	a "hole" in the visual field (literally, "dark tumor")
scotopic	pertaining to dim light conditions
segmental	broken into segments; a way of describing the pattern of nerve innervation on the body surface
sensorineural	pertaining to sensation and how it is processed by the brain
somnambulism	sleep-walking
somnography	measurement of a patient's sleep
stimulus	environmental energy that causes a response in the nervous system
strabismus	condition where the eyes don't look in the same direction
subarachnoid	between the brain surface and the arachnoid membrane
subdural	between the arachnoid membrane and the dura mater
sulcus	groove on the surface of the cerebrum
sympathetic	pertaining to the division of the autonomic nervous system that maintains body functions needed for escape or fighting
syncope	fainting
synesthesia	mixing up two or more senses (e.g. perceiving colors and sounds at the same time)
tastant	chemical which produces a taste sensation
temporal	bone on the side of the skull; lobe of the brain on the side
thermoreceptor	nerve cell which detects temperature

Word	Definition
thromboembolism	loss of blood supply due to a clot which dislodges from its source and lodges in a distant location
thromboembolus	clot which becomes dislodged, blocking the blood supply at a distant location
thrombolytic	drug which breaks apart a blood clot
thrombosis	condition of clotting
tinnitus	ringing in the ears
toxic	poisonous
transcranial	across the skull
transduction	the process of turning one kind of signal into another
transient	temporary; briefly passing
trigeminal	fifth cranial nerve, mostly carrying information from the face to the brain
tympanic	pertaining to the eardrum
tympanocentesis	puncture of the eardrum
tympanometry	process of measuring the eardrum
tympanoplasty	surgical alteration of the eardrum
tympanosclerosis	hardening of the eardrum
tympanostomy	creating a hole ("mouth") in the eardrum
unipolar	a neuron with one major extension from the cell body
utricle	bag-like structure in the vestibular system
varicosity	dilation or enlargement of a part (e.g. a nerve axon)
vertigo	sensation of spinning
vestibular	the organ of balance; part of the inner ear
vestibulitis	inflammation of the vestibular apparatus
vestibulocochlear	pertaining to the inner ear (balance & hearing apparatuses)
vestibulotomy	cutting into the vestibular apparatus
vitrectomy	surgical removal of the vitreous
vitreous	the clear, jelly-like material that gives the eyeball its shape
xerophthalmia	dry eye

HOW THE UNIT 6 WORD LIST WAS BUILT

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Greek akoustikos "pertaining to hearing"				acous			tic	acoustic
			aer	ot			itis	aerotitis
	ad → af			fer[re]			ent	afferent
	a			gnos			ia	agnosia
	a			kinet			opsia	akinetopsia
Greek amblys, "blunt" + ops, "sight"			ambly	op			ia	amblyopia
				ampulla				ampulla
	a			myo			trophic	amyotrophic
	ana			stom			osis	anastomosis
Greek aisthēsis, "feeling"	an			esthesi[o]			a	anesthesia
	an			esthesio			logist	anesthesiologist
Greek aneurysmos, "dilation"				aneurysm[os]				aneurysm
	an			irid			ia	aniridia
Greek orexis, "appetite, "desire"	an			orex			ia	anorexia
	an			osm[e]			ia	anosmia
	anti			convuls[e]			ant	anticonvulsant
	anti			depress			ant	antidepressant
	anti			psycho			tic	antipsychotic
				anxio			lytic	anxiolytic
	a			path			y	apathy
	a			phak			ia	aphakia
	a			phas			ia	aphasia
	apo			neur			osis	aponeurosis

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Greek autos, "self" + -ism	a			arachn			oid	arachnoid
				arterio		scler	osis	arteriosclerosis
				asthen			opia	asthenopia
				stigmat			ism	astigmatism
				athero		scler	osis	atherosclerosis
				audio			gram	audiogram
				audio			logist	audiologist
				audio			metry	audiometry
				audit			ory	auditory
	auto			aur			al	aural
				aut[os]			ism	autism
				nom[y]			ic	autonomic
			axo	dendrit			ic	axodendritic
				axon				axon
			axo	secret			ory	axosecretory
			axo	soma			tic	axosomatic
				basi[s]			l + ar	basilar
		bin		ocul			ar	binocular
		bi		pol[e]			ar	bipolar
				blepharo		edema		blepharedema
				blephar			itis	blepharitis
				blephro		conjunctiv	itis	blepharoconjunctivitis
				blepharo			plasty	blepharoplasty
				blepharo			plegia	blepharoplegia
				blepharo			ptosis	blepharoptosis
				blepharo		pyo	rrhea	blepharopyorrhea
				blepharo			spasm	blepharospasm
				blepharo			tomy	blepharotomy

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
				brain				brain
Greek bous , "ox" + limos , "hunger" (literally, "ox-hunger" , i.e. ravenous hunger)				bulimia			ia	bulimia
Latin "waterfall"				cataract				cataract
	cata			ton[o]			ia	catatonia
				caud[a]			ate	caudate
				ceph			algia	cephalgia
				cephalo			dynia	cephalodynia
				cerebell[o]			ar	cerebellar
				cerebell[o]			itis	cerebellitis
				cereb			ellum	cerebellum
				cerebr			al	cerebral
				cerebro		mening	itis	cerebromeningitis
				cerebro			pathy	cerebropathy
				cerebro		spin	al	cerebrospinal
				cerebro			tomy	cerebrotomy
			cerebro	vascul				cerebrovascular
				cerebr				cerebrum
Coined 1741 from Latin cera , "wax" + [bit]umen				cerumen				cerumen
				cerumino			lysis	ceruminolysis
				cerumino			lytic	ceruminolytic
				cerumin			oma	ceruminoma
				cerumin			osis	ceruminosis
			chemo	receptor				chemoreceptor

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Greek khorion, "membrane enclosing the fetus"				chor[ion]			oid	choroid
Greek klonos, "violent"				clonus				clonus
Latin cochlea, "snail shell"				cochlear				cochlear
				cochle			itis	cochleitis
	con			duct			ive	conductive
Latin, short for membrana conjunctiva, because it forms a conjunction between the lids and the globe of the eye				conjunctiva				conjunctiva
				conjunctiv			itis	conjunctivitis
Latin convulsus, "to pull this way and that"				convuls[e]			ion	convulsion
				corn[u]			ea	cornea
			cortico	spinal				corticospinal
				cranio	scler		osis	craniosclerosis
	syn			cranio	ost		osis	craniosynostosis
The ciliary body forms a ring, therefore cyclo				cyclo	kerat		itis	cyclokeratitis
				cyclo	pleg		ia	cycloplegia
				cyclo			tomy	cyclotomy
Greek dákrū, "tear[drop]"				dacry/o	aden		algia	dacryoadenalgia
				dacry/o	aden		ectomy	dacryoadenectomy

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
				dacry/o		aden	itis	dacryoadenitis
Greek dákrū, "tear[drop]" + Greek kústis, "cyst", "anatomical sac"				dacry/o		cyst	algia	dacryocystalgia
				dacry/o		cyst	ectomy	dacryocystectomy
				dacry/o		cyst	itis	dacryocystitis
			dacryo	cysto		rhino	stomy	dacryocystorhinostomy
				dacry/o		cysto	tomy	dacryocystotomy
				dacry/o		hemo	rrhea	dacryohemorrhea
				dacry/o		lith		dacryolith
				dacry/o		lith	iasis	dacryolithiasis
				dacry/o		pyo	rrhea	dacryopyorrhea
				dacry/o			rrhea	dacryorrhea
				dacry/o			stenosis	dacryostenosis
Latin de + crescere, "to grow less"					decrement[um]			decrement
Latin decussare, "form the letter X" (related to "decade" through Roman numeral X=10)					decussat[e]		ion	decussation
Latin for "to plow outside the furrow" i.e. "to go off tracks"	de				lir[a]		ium	delirium
Latin de- + ment, "mind", i.e. out of one's mind	de				ment		ia	dementia
					derma		tome	dermatome

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
	di			encephal			on	diencephalon
	dipl			opia			opia	diplopia
				dopa			[a]mine	dopamine
				dura				dura
				dur[a]			itis	duritis
	dys			esthes			ia	dysesthesia
	dys			lex			ia	dyslexia
	dys			phas			ia	dysphasia
	dys			phor			is	dysphoria
			echo	encephalo			graphy	echoencephalography
	ec			trop			ion	ectropion
	ef			fer[re]			ent	efferent
			electro	encephalo			graphy	electroencephalography
			electro	physi			ology	electrophysiology
				embol			ism	embolism
				encephal[o]			itis	encephalitis
				encephalo			cele	encephalocele
				encephalo			gram	encephalogram
				encephalo	myel		itis	encephalomyelitis
			encephal/o	myel/o	neur/o		pathy	encephalomyeloneuropathy
				encephalo			pathy	encephalopathy
				encephalo	py		osis	encephalopyosis
	end[o]			arter			ectomy	endarterectomy
	endo			lymph				endolymph
	endo			neur[o]			ium	endoneurium
	end			ophthalm			itis	endophthalmitis
	endo			vascul			ar	endovascular
	en			trop			ion	entropion
	e			nucle			ation	enucleation

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Greek epénduma, "upper garment"				ependym			al	ependymal
	epi			dur[a]			al	epidural
Latin lepsis, "seizure"	epi			leps			y	epilepsy
	epi			neur[o]			ium	epineurium
Greek eso= "within" + -tropos, "turn"	eso			trop			ia	esotropia
	eu			phor			ia	euphoria
	ex			ophthalmus				exophthalmus
	exo			trop				exotropia
Latin, "outside, outward"				externus				externa
			extero	ceptor				exteroceptor
				fasc			icle	fascicle
				front			al	frontal
				gangli			oma	ganglioma
				ganglion				ganglion
				ganglion			itis	ganglionitis
Greek glaukommatos, "gray-eyed"; glaukoma, "cataract"				glaucoma				glaucoma
Greek glossa or glotta, "tongue" + Greek phar anx, "cleft, chasm, gully, trench"			glosso	pharyng			eal	glossopharyngeal
				gustat[e]			ion	gustation
Greek gyros, "ring, circle"				gyrus				gyrus

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
				hemat[o]			oma	hematoma
	hemi	an		ops			ia	hemianopsia
	hemi			pares			is	hemiparesis
	hemi			pleg			ia	hemiplegia
				hemo			rrhag[e]	hemorrhagic
Latin, "tiny human being produced artificially"				homunculus				homunculus
			hydro	cephal[o]			us	hydrocephalus
Greek ákousis, "hearing"	hyper			acusis				hyperacusis
	hyper			esthes			ia	hyperesthesia
	hyper			op			ia	hyperopia
	hyper			polar			ization	hyperpolarization
				hypno			tic	hypnotic
	hypo			acusis				hypoacusis
	hypo			gloss			al	hypoglossal
Latin ictus, "strike, hit"				ictus				ictus
Greek idios, "one's own"			idio	path			ic	idiopathic
	in			somn[o]			ia	insomnia
				insula				insula
	inter			ict[us]			al	interictal
	inter			neuron				interneuron
			inter/o	ceptor				interoceptor
	intra			cerebr[o]			al	intracerebral
	intra			ocul			ar	intraocular
	intra			vitre			al	intra vitreal
				irid			algia	iridalgia

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Greek iris, "rainbow"				irid			ectomy	iridectomy
				irid			emia	iridemia
				irido		cycl	ectomy	iridocyclectomy
				irido		cycl	itis	iridocyclitis
				irido		kerat	itis	iridokeratitis
				irido			pathy	iridopathy
				irido			tomy	iridotomy
				iris				iris
				ir			itis	iritis
Greek iskhein, "to hold" + haima, "blood" (also spelled ischæmia)			isch	[h]em[e]			ia	ischemia
			isch	[h]em[e]			ic	ischemic
				kerat			algia	keratalgia
				kerat			itis	keratitis
				kerato			malacia	keratomalacia
				kerato			pathy	keratopathy
				kerato			plasty	keratoplasty
				kerato			tomy	keratotomy
				labyrinth			ectomy	labyrinthectomy
				labyrinth			itis	labyrinthitis
				labyrinth			tomy	labyrinthotomy
				lacrim			ation	lacrimation
				ligand				ligand
				lobe				lobe
				lob[o]			ectomy	lobectomy
				lobo			tomy	lobotomy
Latin lobus, "lobe"								

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Greek myein "to shut the eyes" + -osis	macro			cephal[o]			y	macrocephaly
				receptor				mechanoreceptor
				medulla				medulla
				mening			es	meninges
				meningi			oma	meningioma
				mening[o]			itis	meningitis
				meningo			cele	meningocele
			meningo	encephal			itis	meningoencephalitis
				meningo			pathy	meningopathy
				cephal[o]			y	microcephaly
				ot			ia	microtia
				brain				midbrain
				mi			osis	miosis
				miot			ic	miotic
				pares			is	monoparesis
				pleg			ia	monoplegia
				synap[se]			tic	monosynaptic
				pol[e]			ar	multipolar
From Greek múdro, "mass of red hot material in a forge", maybe because of the eye glowing red when the pupil is dilated				mydriasis			asis	mydriasis
				mydriat			ic	mydriatic
				myel[o]			itis	myelitis
				myelo			cele	myeloccele
				myelo			gram	myelogram

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
				myelo			malacia	myelomalacia
				myelo		meningo	cele	myelomeningocele
				myelo			pathy	myelopathy
				my		op	ia	myopia
		myo					tatic	myotatic
				myring			ectomy	myringectomy
				myring			itis	myringitis
				myringo		dermat	itis	myringodermatitis
		myringo		myc			osis	myringomycosis
				myringo			plasty	myringoplasty
				myringo			tomy	myringotomy
	narco			leps			y	narcolepsy
	naso			lacrim			al	nasolacrimal
				neur			algia	neuralgia
				neur[o]		asthen	ia	neurasthenia
				neur			ectomy	neurectomy
				neur[o]			itis	neuritis
		neuro		arthro			pathy	neuroarthropathy

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
This term describes a "brain worm", a horrifying result of infection with tapeworms (commonly, the pork tapeworm <i>Taenia solium</i>). The tapeworm typically resides in a chamber made by the brain called a cysticercus, from the Greek words cyst/i-, "fluid-filled space" + -kérkos, "tall". Greek neura, "nerve" + -on Neuro + Latin pharmacia, Greek pharmakeia, "herb, medicine, potion"			neuro	cysticerc	[us]		osis	neurocysticercosis
				neuro			dynia	neurodynia
			neuro	encephalo		myelo	pathy	neuroencephalomyelopathy
				neuro		gen	ic	neurogenic
				neuro		glyco	penia	neuroglycopenia
				neuro			lysis	neurolysis
				neur[o]			oma	neuroma
				neuron				neuron
				neuro			pathy	neuropathy
			neuro	pharmaco			logy	neuropharmacology
				neur/o		phys/i/o	logy	neurophysiology
				neuro			plasty	neuroplasty

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Latin nocere, "to hurt" Greek "to nod away in sleep"; the drooping of the chin then snapping back into place while you are trying to listen to a boring lecture				neuro			rrhaphy	neurorrhaphy
				neuro		scler	osis	neurosclerosis
				neur[o]			osis	neurosis
				neur/o		surgery		neurosurgery
				neuro			tomy	neurotomy
			neuro	transmit			ter	neurotransmitter
			noci	[re]ceptor				nociceptor
					nystagmus			nystagmus
				occiput			al	occipital
		oculo		myc			osis	oculomycosis
				oculo			pathy	oculopathy
				oculo			plasty	oculoplasty
				ophthalm			algia	ophthalmalgia
				ophthalm			ic	ophthalmic
				ophthalm			itis	ophthalmitis
				ophthalmo			logist	ophthalmologist
		ophthalm/o		my			itis	ophthalmomyitis
				ophthalmo			pathy	ophthalmopathy
				ophthalmo			plegia	ophthalmoplegia
				ophthalmo			scope	ophthalmoscope
				opt			ic	optic
		opto		kinet			ic	optokinetic
				opto		metr	ist	optometrist

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Greek <i>osmē</i>, "smell, scent, odor"			opto	my/o			meter	optomyometer
			osmo	receptor				osmoreceptor
				osteo		acusis		osteoacusis
				ot			algia	otalgia
				ot			itis	otitis
				oto			dynia	otodynia
				ot/o		laryng/o	logist	otolaryngologist
				oto		liths		otoliths
				ot/o		myc	osis	otomycosis
			oto	neuro			logist	otoneurologist
				oto			plasty	otoplasty
				ot/o		py/o	rrhea	otopyorrhea
			oto	rhino		laryngo	logist	otorhinolaryngologist
				oto			rrhea	otorrhea
				oto		scler	osis	otosclerosis
				oto			scope	otoscope
			ot	oste			al	otosteal
				oto		tox	ic	ototoxic
				papill		edema		papilledema
Greek & Latin paralysis, Greek <i>lyein</i>, "loosen"	para			esthes			ia	paraesthesia
				para			lysis	paralysis
	para	sym		path[os]			etic	parasympathetic
				paresis				paresis
				pariet			al	parietal
	peri			lymph				perilymph

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Latin emulsionem, ex- + mulgere "to milk" (milk is an emulsion)	peri			neur[o]			ium	perineurium
				phaco		emulsif	ication	phacoemulsification
				phaco			malacia	phacomalacia
				phaco		scler	osis	phacosclerosis
				phaco			scope	phacoscope
				phak			itis	phakitis
				phon		asthen	ia	phonasthenia
				phot			on	photon
				photo		phob	ia	photophobia
				photo + p			ic	photopic
			photo	receptor				photoreceptor
				plex			us	plexus
			polio	myel[o]			itis	poliomyelitis
Greek polios, "gray"; named because of the inflammation of the spinal cord	poly			neur[o]			itis	polyneuritis
	poly			neuro			pathy	polyneuropathy
	post			central				postcentral
	post			ganglion			ic	postganglionic
	post			ict[us]			al	postictal
	post			synap[se]			tic	postsynaptic
	pre			central				precentral
	pre			front			al	prefrontal
	pre			ganglion			ic	preganglionic
	pre			ict[us]			al	preictal
	pre			motor				premotor

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
	pre			presby			acusis	presbycusis
				presby		op	ia	presbyopia
				synap[se]			tic	presynaptic
Latin pro- + pag, "to fasten"; reproduce one's kind	pro			pag			ate	propagate
				propri[us]	[re]ceptor			proprioceptor
Greek prosopon, "face" + a, "not" + gnosis, "knowing"	a		prosop	gnos			ia	prosopagnosia
				pseud	esthes		ia	pseudesthesia
				psych			iatrist	psychiatrist
				psych			iatry	psychiatry
				psycho		gen	ic	psychogenic
				psycho			logist	psychologist
the mind is not separate from the brain, but this terminology dates from a time in history when psychology and neurology were thought to be separate sciences				psycho			logy	psychology
				psycho			pathy	psychopathy
				psycho	pharmaco		logy	psychopharmacology
				psych[o]			osis	psychosis
				psycho	somat		ic	psychosomatic
				psycho		trop	ic	psychotropic
	quadri			pleg			ia	quadriplegia

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Latin refractarius, "obstinate, stubborn"				refract			ory	refractory
	re			polar			ization	repolarization
the eye is divided into three concentric layers, or tunics; the retina is the innermost, net-like (Latin rete) tunic				retina				retina
				retin			al	retinal
				retin			itis	retinitis
				retino			pathy	retinopathy
				retino			pexy	retinopexy
				retino			scope	retinoscope
				retin			osis	retinosis
				saccul				saccul
				sciatica				sciatica
				sclera				sclera
				scler		ectas[is]	ia	sclerectasia
				scler			itis	scleritis
Greek éktasis, "stretched out", "dilated"				sclero			malacia	scleromalacia
				sclero			tomy	sclerotomy
				scot			oma	scotoma
				scot		op	ic	scotopic
				segment			al	segmental
			sensori	neur			al	sensorineural
			somn	ambul			ism	somnambulism

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Greek toxicon, "poison to use on arrows", from Scythian toxon, "bow"; this may be the only Scythian word found in Medical English				tox			ic	toxic
	trans			cran			ial	transcranial
Latin trans- + ducere, "to lead"	trans			duct			ion	transduction
				trans			ient	transient
Latin geminus, "twins" (as in the constellation Gemini)	tri			gemin			al	trigeminal
				tympan[um]			ic	tympanic
				tympano			centesis	tympanocentesis
				tympano			metry	tympanometry
				tympano			plasty	tympanoplasty
				tympan/o	scler		osis	tympanosclerosis
				tympano			stomy	tympanostomy
	uni			pol[e]			ar	unipolar
Latin ūter, "leather bag" + diminutive suffix -icle				utricle				utricle
				varicos[e]			ity	varicosity
Latin, "to whirl around"				vertigo				vertigo
				vestibul			ar	vestibular
				vestibul			itis	vestibulitis
				vestibulo	cochlear			vestibulocochlear

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix Word End	Word
Latin, "glass-like"				vestibulo			tomy	vestibulotomy
				vitr			ectomy	vitrectomy
				vitre			ous	vitreous
	xer[o]			ophthalm			ia	xerophthalmia

PART VII

UNIT 7: THE ENDOCRINE SYSTEM

UNIT 7 INTRODUCTION

The functions of organs, tissues, and the cells of which they are made are coordinated by chemicals. One category of these regulatory chemicals is hormones. Hormones are produced by the cells of endocrine glands. The medical term, endo|crine, is made from the word parts, endo, which means “inside,” and crin which means “to secrete.” So, endocrine glands secrete hormones inside the body into the fluid that surrounds the cells. From this extracellular fluid, hormones travel into and through the bloodstream to affect cells, tissues, and organs throughout the body. The endocrine system is comprised of the organs (glands) that secrete hormones.

Objectives

After completing this unit, you should be able to:

- State word parts associated with the endocrine (hormonal) system.
- Recognize word parts used to describe diseases of the endocrine system.
- Name abbreviations used in association with the endocrine system.

THE ENDOCRINE SYSTEM

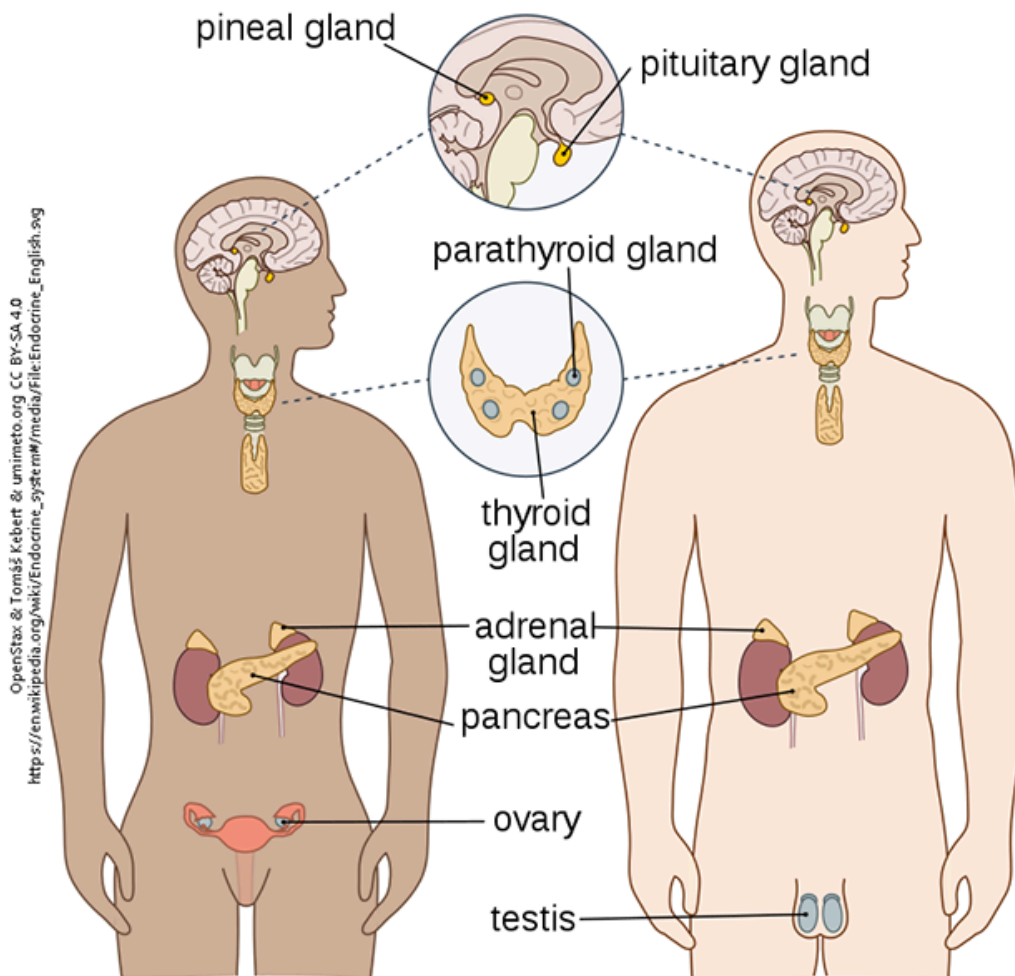
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Unit 7 figure 1
Endocrine System

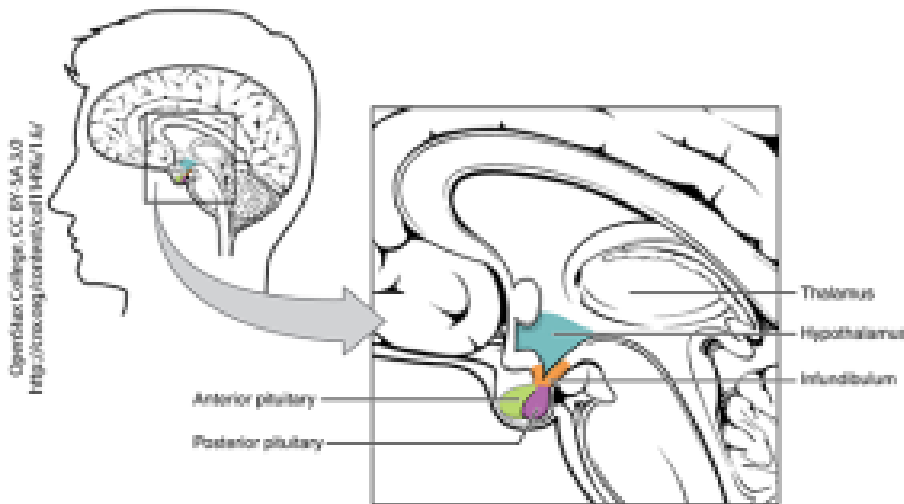


The Endocrine System, together with the nervous system, controls everything that occurs in the body's internal environment. Some endocrine organs act independently, without nervous system control. Other times, the nervous system controls the endocrine gland. When the nervous system causes a gland to secrete, that gland is part of the *endocrine system*, and the secreted substance is a *hormone*.

The endocrine system is comprised of primary endocrine glands (hormone secretion is their only job) and secondary endocrine glands/tissues (secreting hormones is their second job).

Each hormone has specific cell types that it targets to produce a particular action. The table below provides a limited number of hormones to learn.

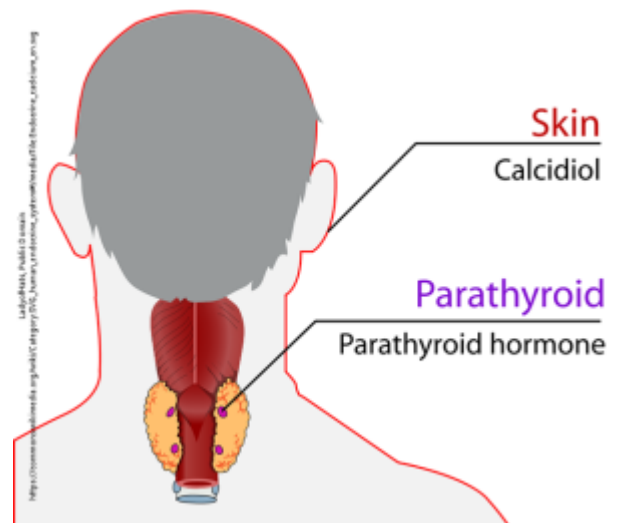
Hormone	Origin	Target	Action
adrenaline epinephrine	adrenal medulla	cells in many organs	stimulates sympathetic nervous system (i.e. “fight-or-flight response” when the body is under stress)
adrenocorticotropic hormone (ACTH)	anterior pituitary	adrenal cortex	stimulates adrenal cortex
antidiuretic hormone (ADH)	hypothalamus & posterior pituitary	kidneys	stimulates the kidneys to conserve water so that less urine is produced
calcitonin	a subset of thyroid cells	bones & kidneys	helps to regulate calcium levels in the blood
cortisol	adrenal cortex	cells in many organs	stimulates metabolism; inhibits inflammation; part of a class of hormones called <i>glucocorticoids</i>
estrogens	ovaries	uterine lining; many other cells	stimulate development of female sexual characteristics & helps the female body prepare for pregnancy; a class of hormones, the most common of which is 17- β -estradiol
glucagon	pancreas	primarily liver cells	stimulates the creation of glucose (gluconeogenesis) and breakdown of the storage form of glucose (glycogenolysis)
growth hormone	anterior pituitary	primarily bones and muscles	promotes growth
insulin	pancreas	most cells in the body	stimulates cells to take in glucose, which decreases blood sugar levels
oxytocin	hypothalamus & posterior pituitary	uterus, mammary glands, & brain cells	stimulates the uterus to contract during childbirth; stimulates the mammary glands to release milk; promotes bonding to another person
parathyroid hormone (PTH)	parathyroid glands	bones & kidneys	helps to regulate calcium levels in the blood
releasing hormones (many different kinds)	hypothalamus	anterior pituitary	stimulate the cells of the anterior pituitary to make and release hormones
testosterones	testes	many cells in the body	stimulate development of male sexual characteristics; stimulate development of sperm; a class of hormones, the most common of which is dihydrotestosterone
thyroid stimulating hormone (TSH)	anterior pituitary	thyroid gland	stimulates the thyroid gland to produce T4 and T3
thyroxine (T4) & triiodothyronine (T3)	thyroid gland	cells throughout the body	stimulate metabolism

Unit 7 figure 2
Hypothalamus and
Pituitary

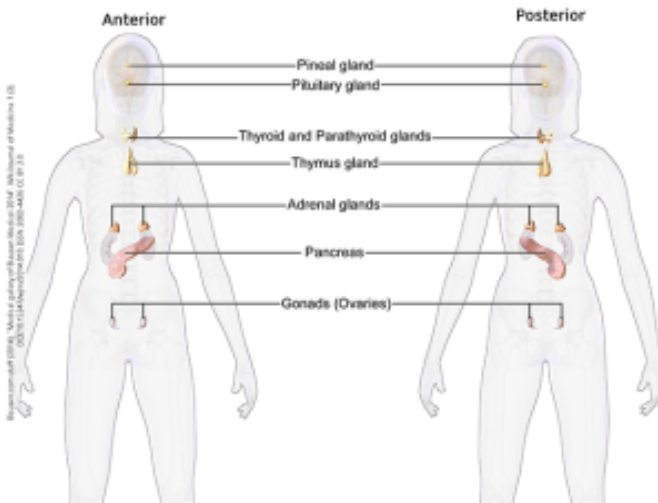
The hypothalamus is a part of the brain that produces hormones which either target or are stored in the pituitary gland. The pituitary gland is located inferior to the hypothalamus and has two parts, the anterior (*adenohypophysis*) and posterior (*neurohypophysis*). The hypothalamus controls the anterior pituitary (adenohypophysis) through a “private” blood system called the *hypophyseal portal system*. Releasing hormones are dropped into this private bloodstream and carried to the anterior pituitary, where they encounter cells that release hormones such as adrenocorticotropic hormone, growth hormone, and thyroid stimulating hormone.

The posterior pituitary (neurohypophysis) works differently. Neurons in the hypothalamus make the hormones antidiuretic hormone and oxytocin. They ship the hormone down their nerve cell axons which end in the posterior pituitary, where the release of hormones into the bloodstream occurs.

Another job of the endocrine system is regulating the amount of calcium carried in the blood. A small subset of thyroid gland cells releases *calcitonin*, and the nearby *parathyroid glands* release *parathyroid hormone* (PTH). The kidneys release *calcitriol* (vitamin D), a closely related compound, calcidiol, is made by the skin. Calcitonin moves calcium from the blood to the bones, lowering blood calcium levels; parathyroid hormone moves calcium from the bones to the blood, increasing blood calcium levels. Calcitriol (vitamin D) helps in the absorption of calcium in the intestines, which also increases blood calcium levels.



Unit 7 figure 3 Calcium Homeostasis



Unit 7 figure 4 Female Endocrine System

is *thelarche*.

Hormones are also used to regulate sexual function. Genes on the X or Y chromosome will direct brain cells (through a complex, multi-step process) to begin producing hormones, which in turn promote the development of primary sexual characteristics (the gonads and external genitalia). Primary sexual characteristics are those a person is born with. After a pause of several years, another burst of hormones at puberty promotes the development of secondary sexual characteristics (for example, menstruation and breast development in the female). The beginning of menstruation is called *menarche* and the beginning of breast development

Video Attribution

- Justin Burr, Jim Hutchins & Maddison Johnston. (2024, December 20). *Only [Med Terms] in the Building Episode 7*. [Video]. Kaltura.

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UNIT 7 PREFIXES

Prefix	Definition
auto–	self
con–	together, with
dys–	bad
endo–	inside
epi–	on top of
eu–	good, well–
ex–	out, off, from, beyond
exo–	out, outside, without
hyper–	too much
hypo–	too little
in–	in, into (compare Unit 6 definition; depends on context)
pan–	all; everything or everyone
para–	beside, near
per–	through; by means of
poly–	many; much
pro–	in front of; in favor of
pseudo–	false, lying
retro–	backwards; behind
syn–	with, together

UNIT 7 ROOTS

Root	Definition
acid	acid; presence of hydrogen ions
acro	high
aden adeno	gland
adren adreno	adrenal gland (on top of the kidney)
agon	driver of an action (Greek "contestant")
ald[ehyde] + o	chemical group
alkal[i]	alkali; absence of hydrogen ions
angio	vessel
arche	beginning
bol	throw (as in anabolism, catabolism, metabolism)
calc calci	calcium
carcin	cancer
catechol	chemical group
chlor	chlorine
chole	bile
cortic cortico	adrenal cortex (the outside of the adrenal gland)
crine crino	secretion
cyst	fluid-filled space
diabet	polydipsia & polyuria; Greek "go through"
dips	drinking
drome	to run
fus	pouring into
galacto	milk
genit	reproduction; birth
gluc gluco glucos	sugar

Root	Definition
glyc glyco	sugar
glycogen	sugar molecules assembled into larger structure
gonad gonado	sex organs
grade	movement (in time or space)
gyneco	female
[h]em[o]	blood
hirsut[e]	hairy
hormon + o	hormone
hypophys hypophysis	pituitary gland
infundi	stalk of the pituitary gland
insul insulin	island; the hormone insulin, which is made in the pancreatic islets
kal[ium]	potassium
ket keto keton	chemical group
lact	milk
laparo	flank, loin
lipid lipo	fat
lith	stone
mast	breast
men	monthly
mineralo	pertaining to sodium or potassium
natr	sodium
neo	new
nephr	kidney
neuro	brain
ophthalmos	the eye
ox	oxygen

Root	Definition
pancreat pancreato	pancreas (digestive gland)
parathyroid	parathyroid gland (behind the thyroid)
permiss	allowing to pass through
phag	eat
phosphat	phosphorus-containing chemical group
pituitar	pituitary gland
receptor	a protein on the surface of a cell that turns a chemical on the outside to a signal on the inside of the cell
scop	device for looking inside
somato	body
stat[us]	standing
sterol	chemical group
sufficiens	Latin "adequate"
thel	nipple; breast
thym	thymus gland (in the upper chest)
thyr thyro thyroid	thyroid gland
tox	poison
trop	turning towards
troph	nourishing
ur	urine
viril	maleness; machismo

UNIT 7 SUFFIXES

Suffix	Definition
-al	pertaining to
-algia	pain
-amine	chemical group
-bulum	suffix forming the name of a vessel or instrument
-cele	hernia
-ectomy	cutting out of
-em	present in blood (shortening of [h]em[e])
-emia	present in blood, indicating disease
-genesis	the process of creating
-genic	giving rise to
-graphy	a test or procedure
-ia	disease
-iasis	presence of an abnormality
-ic	pertaining to
-in	chemical name ending for a protein
-ine	chemical name ending
-ion	word ending that turns a verb into a noun
-is	anatomical structure
-ism	belief; process word ending
-itis	inflammation
-lith	stone
-logist	one who studies
-lysis	breaking down; destroying
-megaly	enlargement due to disease
-oid	looks like; resembles
-oma	tumor
-osis	process
-pathy	disease process
-penia	low levels of
-plasia	the process of molding or growing

Suffix	Definition
-poiesis	creation process
-ptosis	drooping
-rrhea	abnormal flow
-sive	capable of
-sterone	chemical name ending
-tion	word ending that turns a verb into a noun
-tomy	cutting
-toxin	poisonous to
-tropic	influencing the activity of a gland
-tropin	chemical which influences the activity of a gland
-y	disease

UNIT 7 ABBREVIATIONS

Abbreviation	Meaning
ACTH	adrenocorticotrophic hormone
ADH	antidiuretic hormone
BG	blood glucose
BS	blood sugar
CSII	continuous subcutaneous insulin infusion
CGM	continuous glucose monitor
DI	diabetes insipidus
DM	diabetes mellitus
ERCP	endoscopic retrograde cholangiopancreatography
FBS	fasting blood sugar
GDM	gestational diabetes mellitus
GH	growth hormone
GTT	glucose tolerance test
HgA1C	hemoglobin A1C test (used by diabetics to monitor blood sugar levels)
HRT	hormone replacement therapy
IDDM, T1	insulin dependent diabetes mellitus (type 1)
NIDDM, T2	non-insulin dependent diabetes mellitus (type 2)
PTH	parathyroid hormone
RAIU	radioactive iodine uptake test
T3	triiodothyronine
T4	thyroxine (tetraiodothyronine)
TFT	thyroid function test
TSH	thyroid stimulating hormone (thyrotropin)

UNIT 7 WORD LIST

Word	Definition
acidemia	condition of acid in the blood
acromegaly	disease involving abnormally high levels of human growth hormone after puberty
adenectomy	surgical removal of a gland
adenitis	inflammation of a gland
adenocarcinoma	cancerous tumor of a gland
adenohypophysis	anterior pituitary gland
adenoma	tumor of a gland
adenomegaly	abnormal enlargement of a gland
adenopathy	disease of a gland
adenosis	abnormal change in a gland
adrenalectomy	surgical removal of the adrenal gland
adrenalgia	pain in the adrenal gland
adrenaline	chemical released from the adrenal gland (same as epinephrine)
adrenatitis	inflammation of the adrenal gland
adrenarche	the onset of adrenal function
adrenocortical	pertaining to the adrenal cortex
adrenocorticohyperplasia	abnormal growth of the adrenal cortex
adrenocorticotropic	acting on the adrenal cortex
adrenomegaly	enlargement of the adrenal gland
aldosterone	steroid hormone released from the adrenal gland that affects sodium (salt) balance
alkalemia	condition of alkali in the blood
autoreceptor	a protein on the surface of cells that responds to chemicals released by that cell
calcemia	condition of calcium in the blood
calciuria	condition of calcium in the urine
catecholamine	category of chemicals that include dopamine, adrenaline/epinephrine, and noradrenaline/norepinephrine
chloremia	condition of chlorine in the blood
cholangiopancreatography	imaging the gallbladder, pancreas, and associated vessels
congenital	condition occurring at birth
corticotropic	acting on the adrenal cortex

Word	Definition
corticotropin	chemical acting on the adrenal cortex
diabetic	Greek "the funnel", pertaining to a condition where patients drink too much and urinate too much
dysmetabolic	abnormal function of the metabolism
endocrine	relating to the release of hormones
endocrinologist	physician who studies the release of hormones
endoscopic	pertaining to an instrument that visualizes the digestive tract
epinephrine	chemical released from the adrenal gland (same as adrenaline)
euglycemia	ideal level of sugar in the blood
euthyroid	ideal level of thyroid hormones
exocrine	relating to the release of substances from the body
exophthalmos	protrusion of the eyes
galactorrhea	abnormal flow of milk from the nipple
glucagon	hormone that regulates blood sugar levels
glucocorticoid	steroid hormone released from the adrenal gland that affects sugar levels
gluconeogenesis	the creation of new sugar from other dietary compounds
glucosuria	abnormal condition of sugar in the urine
glycemia	level of sugar in the blood
glycemic	pertaining to the level of sugar in the blood
glycogenolysis	the breakdown of glycogen (a storage form of glucose, consisting of multiple connected glucose molecules)
glycolysis	the breakdown of sugar by cells
glycopenia	low levels of glucose
goiter	enlargement of the thyroid gland
gonadocorticoid	steroid hormone released from the adrenal cortex that affects the reproductive organs
gonadogenesis	the creation of reproductive organs
gonadopathy	disease of the reproductive organs
gonadotropic	pertaining to a hormone acting on the gonads (directly or indirectly)
gonadotropin	a chemical acting on the reproductive organs
gynecomastia	enlargement of the male breasts due to hormone imbalance or hormone therapy
hirsutism	condition of increased hair

Word	Definition
hormonopoiesis	the beginning of hormone secretion
hypercalcemia	increased level of calcium in the blood
hypercholesterolemia	increased level of cholesterol in the blood
hyperglycemia	increased level of sugar in the blood
hypergonadism	condition of overactive reproductive organs
hyperkalemia	increased level of potassium in the blood
hyperlipidemia	increased level of fats (lipids) in the blood
hyponatremia	increased level of sodium in the blood
hyperparathyroidism	condition of enlarged or overactive parathyroid glands
hyperphosphatemia	increased level of phosphate in the blood
hyperpituitarism	condition of enlarged or overactive pituitary glands
hyperprolactinemia	increased level of prolactin in the blood
hyperthyroidism	condition of enlarged or overactive thyroid glands
hypocalcemia	decreased level of calcium in the blood
hypoglycemia	decreased level of sugar in the blood
hypoglycemic	condition of decreased sugar levels
hypogonadism	condition of decreased function of the reproductive organs
hypogonadotropic	pertaining to a condition of decreased hormone acting on the reproductive organs
hypokalemia	decreased level of potassium in the blood
hyponatremia	decreased level of sodium in the blood
hypophosphatemia	decreased level of phosphate in the blood
hypophysectomy	surgical removal of the pituitary (hypophysis)
hypophysitis	inflammation of the pituitary (hypophysis)
hypopituitarism	condition of low pituitary function
hypothyroidism	condition of low thyroid function
infundibulum	the stalk that connects the pituitary to the hypothalamus
infusion	method for putting fluids (such as drugs or antibiotics) into the bloodstream through an IV
insufficiency	condition of not enough of a hormone or organ secretion
insulin	hormone that regulates blood sugar levels

Word	Definition
insulinoma	tumor which releases insulin
ketoacidosis	condition of ketones (organic compounds) and acid in the blood
ketogenesis	the process of making ketones
ketogenic	giving rise to ketones
ketonuria	ketones in the urine
ketosis	condition of ketones (organic compounds) in the blood
laparoscopic	surgical approach that uses a small tube to introduce surgical instruments
lipolysis	breakdown of fats (lipids)
menarche	the onset of menstruation
metabolism	the breakdown of food compounds, and the building up of body components from food
mineralocorticoid	steroid hormone, released from the adrenal cortex, which affects salt (sodium, potassium, magnesium, or calcium) levels
neurohypophysis	posterior pituitary
pancreatalgia	pain in the pancreas
pancreatectomy	surgical removal of the pancreas
pancreatic	pertaining to the pancreas
pancreatitis	inflammation of the pancreas
pancreatolith	stone in the pancreas
pancreatolithectomy	surgical removal of a pancreatic stone
pancreatolithiasis	condition of stones in the pancreas
panhypopituitarism	a low level of all hormones secreted from the pituitary gland
parathyroid	a gland next to (behind) the thyroid gland
parathyroidectomy	surgical removal of the parathyroid gland
parathyroidoma	tumor of the parathyroid gland
permissive	allowing something to happen
polyadenopathy	a disease affecting several glands
polydipsia	increased fluid intake
polyphagia	increased food intake
polyuria	increased urine production
prolactin	hormone that promotes the development of milk glands in the breast

Word	Definition
pseudocyst	a fluid-filled cavity that resembles a cyst but lacks a wall or lining
retrograde	operating or moving in reverse
somatostatin	hormone that stops the secretion of growth hormone
somatotroph	cell in the pituitary that makes growth hormone
syndrome	signs and symptoms that occur together without an obvious relationship
thelarche	the onset of breast development
thymectomy	surgical removal of the thymus
thymoma	tumor of the thymus
thyrocele	fluid-filled space in the thyroid gland
thyroidectomy	surgical removal of the thyroid gland
thyroiditis	inflammation of the thyroid gland
thyroidotomy	incision into the thyroid gland
thyroidotoxin	a substance which damages the thyroid gland
thyromegaly	enlargement of the thyroid gland
thyroparathyroidectomy	surgical removal of the thyroid and parathyroid glands
thyroptosis	downward displacement of an enlarged thyroid
thyrotoxicosis	dangerous condition of increased thyroid hormone levels
thyrotropin	a hormone acting on the thyroid gland
thyroxine	thyroid hormone (specifically, tetraiodothyronine or T ₄)
uremia	increased levels of nitrogen wastes (including urea) in the blood
virilism	a condition involving the development of male characteristics

HOW THE UNIT 7 WORD LIST WAS BUILT

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
ad- , "near" + renal , "pertaining to the kidney"			adeno	acid			[h]em[e]	ia	acidemia
				acro			megaly		acromegaly
				aden			ectomy		adenectomy
				aden			itis		adenitis
				adeno	carcin		oma		adenocarcinoma
				hypophysis					adenohypophysis
				aden			oma		adenoma
				adeno			megaly		adenomegaly
				adeno			pathy		adenopathy
				aden			osis		adenosis
				adrenal			ectomy		adrenalectomy
				adren			algia		adrenalgia
				adrenal			ine		adrenaline
				adrenal			itis		adrenalitis
				adren	arche				adrenarche
				adreno	cortic		al		adrenocortical
				hyper	adreno	cortico	plasia		adrenocorticohyperplasia
				adreno	cortico	trop	ic		adrenocorticotropic
				adreno			megaly		adenomegaly
Root adren + (see word root table) + Greek arkhē , "beginning"				ald[ehyde] + o			sterone		aldosterone
				alkal			emia		alkalemia
				receptor					autoreceptor
				calc			emia		calcemia
				calci			uria		calciuria
				catechol			amine		catecholamine

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
				chlor			[h]em[e]	ia	chloremia
			chol	angio		pancreato	graphy		cholangiopancreatography
	con			genit			al		congenital
				cortico		trop	ic		corticotropic
				cortico		trop	in		corticotropin
				diabet			ic		diabetic
	dys		meta	bol			ic		dysmetabolic
	endo			crine					endocrine
	endo			crino			logist		endocrinologist
	endo			scop			ic		endoscopic
	epi			nephr			ine		epinephrine
	eu			glyc			[h]em[e]	ia	euglycemia
	eu			thyroid					euthyroid
	exo			crine					exocrine
	ex			ophthalmos					exophthalmos
				galacto			rrhea		galactorrhea
				gluc		agon			glucagon
			gluco	cortic			oid		glucocorticoid
				gluco		neogenes	is		gluconeogenesis
				glucos			uria		glucosuria
				glyc			[h]em[e]	ia	glycemia
				glyc		[h]em[o]	ic		glycemic
				glycogeno			lysis		glycogenolysis
				glyco			lysis		glycolysis
				glyco			penia		glycopenia
Latin guttur, "throat"									goiter
			gonado	cortic			oid		gonadocorticoid
				gonad/o			genesis		gonadogenesis

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin hirsutus, "rough, shaggy, bristly"				gonad/o			pathy		gonadopathy
				gonad/o			trop	ic	gonadotropic
				gonad/o			trop	in	gonadotropin
				gyneco	mast		ia		gynecomastia
				hirsut[e]			ism		hirsutism
				hormono			poiesis		hormonopoiesis
	hyper			calc			[h]em[e]	ia	hypercalcemia
	hyper			chole	sterol		[h]em[e]	ia	hypercholesterolemia
	hyper			glyc			[h]em[e]	ia	hyperglycemia
	hyper			gonad			ism		hypergonadism
	hyper			kal			[h]em[e]	ia	hyperkalemia
	hyper			lipid			[h]em[e]	ia	hyperlipidemia
	hyper			natr			[h]em[e]	ia	hypernatremia
	hyper			parathyroid			ism		hyperparathyroidism
	hyper			phosphat			emia		hyperphosphatemia
	hyper			pituitar			ism		hyperpituitarism
	hyper	pro		lact			in	emia	hyperprolactinemia
	hyper			thyroid			ism		hyperthyroidism
	hypo			calc			[h]em[e]	ia	hypocalcemia
	hypo			glyc			[h]em[e]	ia	hypoglycemia
	hypo			glyc			[h]em[e]	ic	hypoglycemic
	hypo			gonad			ism		hypogonadism
	hypo			gonad/o			trop	ic	hypogonadotropic
	hypo			kal			[h]em[e]	ia	hypokalemia
	hypo			natr			[h]em[e]	ia	hyponatremia
	hypo			phosphat			[h]em[e]	ia	hypophosphatemia
				hypophys			ectomy		hypophysectomy
				hypophys			itis		hypophysitis

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
<u>Latin infundere,</u> <u>"to pour into" +</u> <u>ending -bulum,</u> <u>suffix indicating an</u> <u>instrument;</u> <u>"funnel-shaped</u> <u>organ"</u>	hypo					pituitar	ism		hypopituitarism
	hypo					thyroid	ism		hypothyroidism
						infundi	bulum		infundibulum
	in					fus	ion		infusion
	in					sufficienc	y		insufficiency
						insul	in		insulin
						insulin	oma		insulinoma
						keto	acid	osis	ketoacidosis
						keto		genesis	ketogenesis
						keto		genic	ketogenic
<u>Greek men,</u> <u>"month" + arkhē,</u> <u>"beginning"</u>						keton	ur	ia	ketonuria
						ket		osis	ketosis
						laparo	scop	ic	laparoscopic
						lipo		lysis	lipolysis
						men	arche		menarche
			meta			bol		ism	metabolism
			mineralo			cortic		oid	mineralocorticoid
			neuro			hypophys		is	neurohypophysis
						pancreat		algia	pancreatalgia
						pancreat		ectomy	pancreatectomy
						pancreat		ic	pancreatic
						pancreat		itis	pancreatitis
						pancreato		lith	pancreatolith
						pancreato	lith	ectomy	pancreatolithectomy

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
<u>Greek thēlē,</u> <u>"nipple" + arkhē,</u> <u>"beginning"</u>				pancreato		lith	iasis		pancreatolithiasis
	pan	hypo		pituitar			ism		panhypopituitarism
	para			thyroid					parathyroid
				parathyroid			ectomy		parathyroidectomy
				parathyroid			oma		parathyroidoma
				permiss			ive		permissive
	poly			adeno			pathy		polyadenopathy
	poly			dips			ia		polydipsia
	poly			phag			ia		polyphagia
	poly			ur			ia		polyuria
	pro			lact			in		prolactin
	pseudo			cyst					pseudocyst
	retro			grade					retrograde
				somato		stat[us]	in		somatostatin
				somato		troph			somatotroph
	syn			drome					syndrome
				thel		arche			thelarche
				thym			ectomy		thymectomy
				thym			oma		thymoma
				thyro			cele		thyrocele
				thyroid			ectomy		thyroidectomy
				thyroid			itis		thyroiditis
				thyroido			tomy		thyroidotomy
				thyroid/o		toxin			thyroidotoxin
				thyro			megaly		thyromegaly
				thyr/o		parathyroid	ectomy		thyroparathyroidectomy
				thyr/o			ptosis		thyroptosis
				thyro		tox	ic	osis	thyrotoxicosis

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
				thyr/o		trop	in		thyrotropin
				thyr		ox	ine		thyroxine
				ur			[h]em[e]	ia	uremia
				viril			ism		virilism

PART VIII

UNIT 8: THE CARDIOVASCULAR, BLOOD, AND LYMPHATIC SYSTEMS

UNIT 8 INTRODUCTION

The Cardiovascular System is a collection of blood vessels (the pipes – arteries, arterioles, capillaries, venules, veins) and cardiac muscle (the pump – the heart) that pushes blood throughout the body. Blood flowing through these blood vessels (vasculature) carries oxygen, nutrients, and waste to/from tissues throughout the body. Additionally, since the Lymphatic System is complementary to the circulatory system (returning excess fluid from tissues to the blood) and is intimately involved with our immune system (much of which also circulates in the blood), it will be discussed in this unit as well.

Objectives

After you complete this unit, you should be able to:

- List word parts associated with the cardiovascular, blood, and lymphatic systems (including the immune system).
- Recognize words which describe diseases of the cardiovascular, blood, and lymphatic systems.
- Identify abbreviations used to describe diseases and treatments for the cardiovascular, blood, and lymphatic systems.

THE CARDIOVASCULAR, BLOOD, AND LYMPHATIC SYSTEMS

Only [Med Terms] in the Building Episode 8

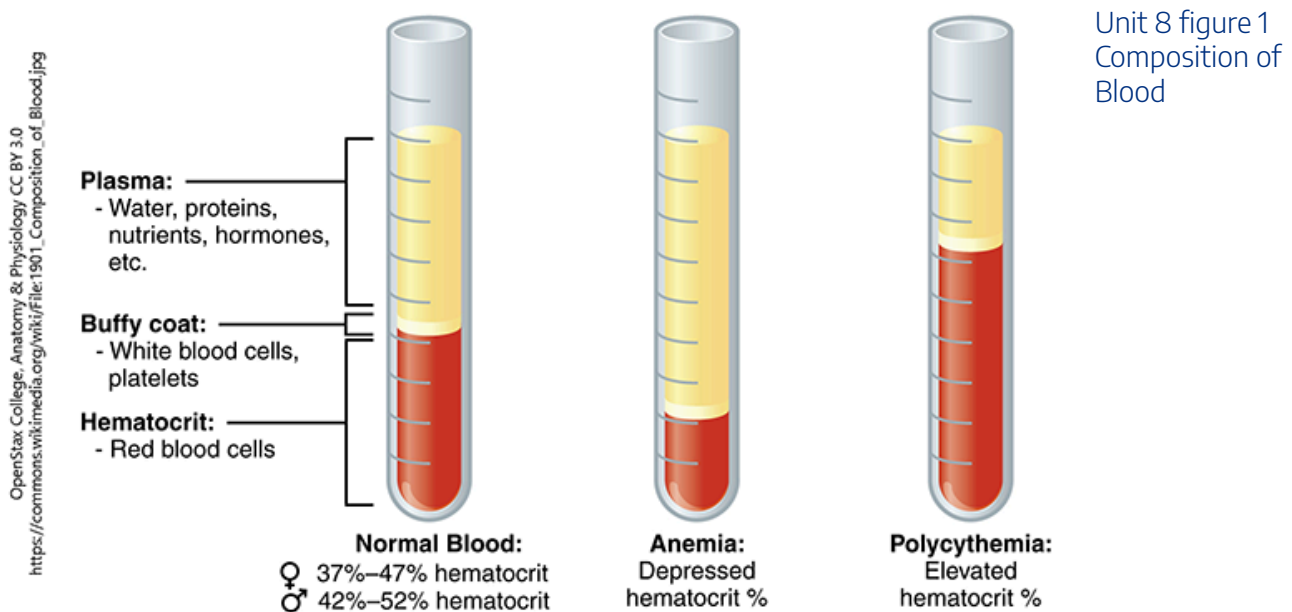


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Blood

Blood is one component of the cardiovascular system, the others being the heart and blood vessels. As the body's transport medium, the blood provides a means to transport oxygen, carbon dioxide, hormones, nutrients, and waste. The blood also contains most of the cells, proteins, and chemicals required to initiate and support resistance to diseases. Body temperature and pH are also regulated by the blood.



Blood is composed of *plasma* (55%) and *formed elements* (45%). *Plasma* is the straw colored fluid that is made

up of water, salts, and proteins. The formed elements consist of the red blood cells (*erythrocytes*), the white blood cells (*leukocytes*), and the platelets (*thrombocytes*). Red blood cells contain a protein called *hemoglobin* that helps transport oxygen in blood and is responsible for blood's bright red color. White blood cells help fight infection and consist of:

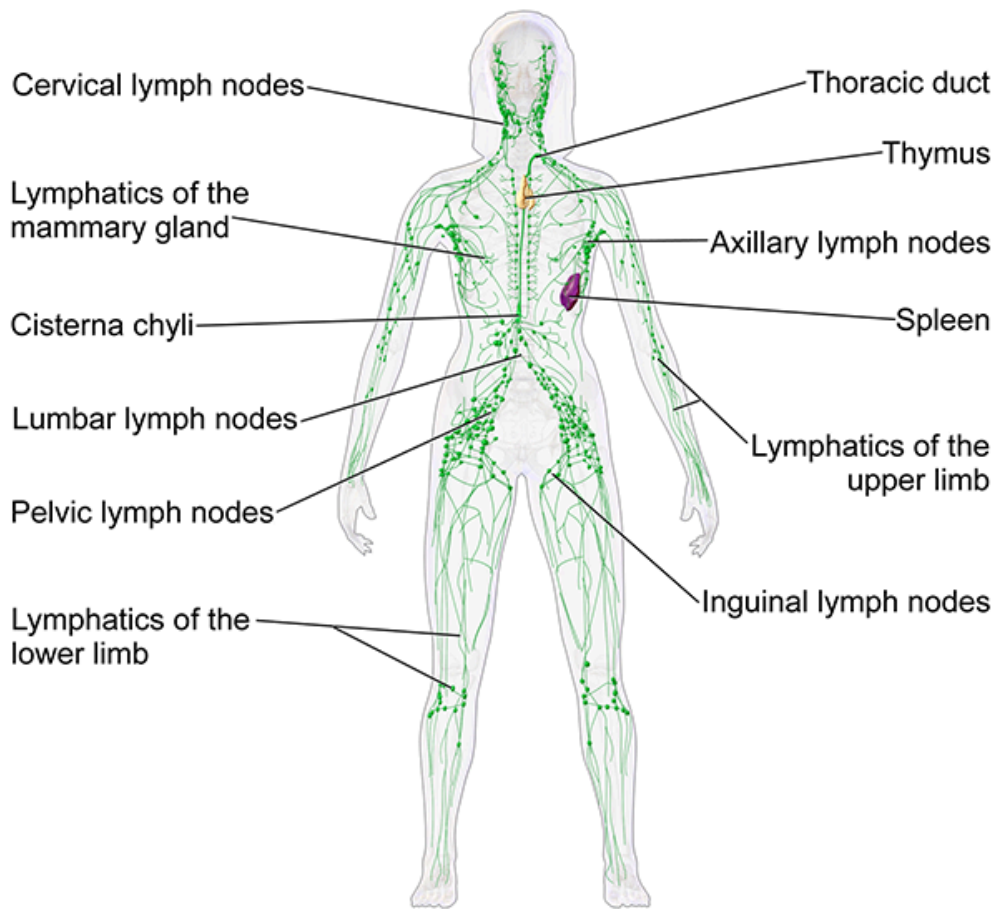
- *neutrophils*,
- *eosinophils*
- *basophils*
- *monocytes*
- *lymphocytes*

Platelets are responsible for clotting blood.

Lymphatic System

The lymphatic system is a system that runs throughout the body and is made up of lymphatic vessels and supporting organs that drain excess interstitial fluid, transport dietary lipids, and facilitate immune responses. The lymphatic system includes:

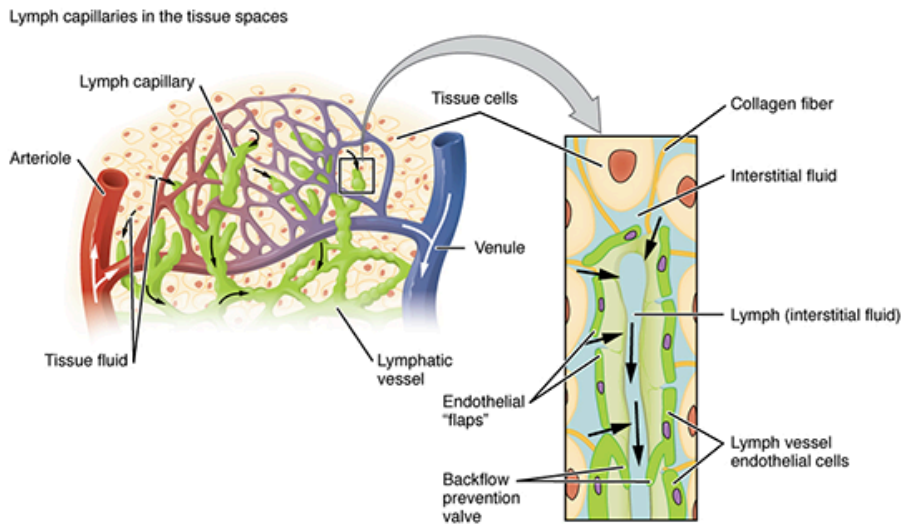
- *lymphatic fluid (lymph)*
- *lymphatic capillaries*
- *lymph vessels*
- *trunks*
- *ducts*
- *lymph nodes*
- *red bone marrow*
- *thymus gland*
- *spleen*



Lymph fluid comes from the blood and contains white blood cells. As oxygen and nutrients are diffused at the capillary level to all cells, lymph capillaries are also present. As fluid goes between cells the blood becomes interstitial fluid and then either enters back into the bloodstream or into the lymph capillaries. Some fluid diffuses into the lymph capillaries and then into the lymph vessels. The lymph vessels transport lymph to the lymph nodes to be filtered. Lymph nodes are clusters of lymph tissue in the neck (*cervical lymph nodes*), armpit (*axillary lymph nodes*), and groin (*inguinal lymph nodes*). There are also clusters in the intestines called Peyer patches. After the lymph is filtered it is returned to the bloodstream through a duct in the chest.

Unit 8 figure 3
Lymphatic
Capillaries

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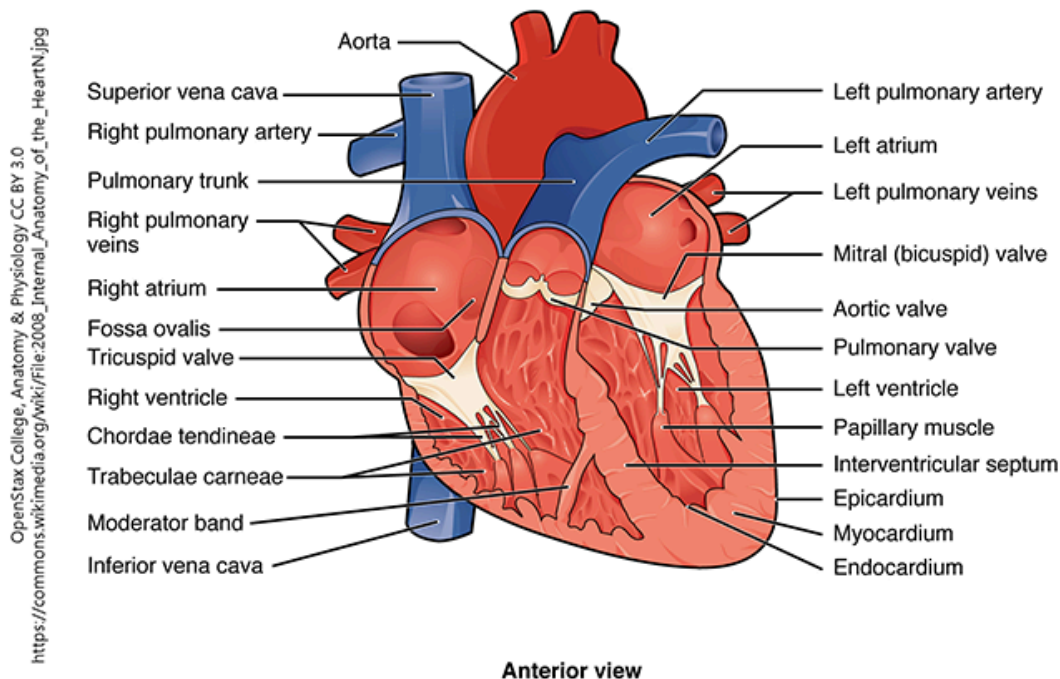
Lymphocytes are produced in the bone marrow and matured in different parts of the body. *B lymphocytes* mature in the bone marrow and *T lymphocytes* mature in the thymus. Both B cells and T cells are part of the immune response and coordinate different ways the body attacks foreign invaders such as viruses and bacteria.

The *tonsils* are located at the back of the throat and contain white blood cells which help trap and kill germs. The *spleen* is located in the left upper quadrant and is also made of lymphatic tissue. It filters blood and removes old and damaged red blood cells.

Heart & Blood Vessels

The heart is a strong, muscular pump just a little over the size of a fist. The heart pumps about 3600 gallons of blood through the body in one day, beating ~100,000 times a day, about 39 million times per year. By the time you reach 65 years old, the heart has beaten two billion times. Each heartbeat ejects approximately 70 ml of blood into the systemic circulation.

The human heart is located within the thoracic cavity, medially between the lungs and the pleural membranes that cover them in the space known as the *mediastinum*. The heart is covered by a tough membrane known as the *pericardium* (pericardial sac) and sits in its own space called the *pericardial cavity*.

Unit 8 figure 4
Internal Anatomy
of the Heart

The human heart consists of four chambers: The left side and the right side are divided by a *septum* and each have one *atrium* and one *ventricle*. Each of the upper chambers, the right atrium (plural = atria) and the left atrium, acts as a receiving chamber and contracts to push blood into the lower chambers, the right ventricle and the left ventricle. The right atrium receives deoxygenated blood from the body; the left atrium receives oxygenated blood from the lungs. The ventricles serve as the primary pumping chambers of the heart, propelling blood to the lungs or to the rest of the body.

The right ventricle receives blood from the right atrium. It then pumps deoxygenated blood into the pulmonary trunk, which leads toward the lungs and bifurcates into the left and right pulmonary arteries. These vessels in turn branch many times before reaching the pulmonary capillaries, where gas exchange occurs: carbon dioxide exits the blood and oxygen enters. The pulmonary trunk arteries are the only arteries in the body that carry relatively deoxygenated blood. Highly oxygenated blood returning from the pulmonary capillaries in the lungs passes through a series of vessels that join together to form the pulmonary veins—the only veins in the body that carry highly oxygenated blood. Understanding the flow of blood through the pulmonary (heart to lungs back to heart) and systemic (heart to body back to heart) circuits is critical to all health professions.

The heart has four valves. The *atrioventricular valves* control the flow of blood between the atria and ventricles. There are two atrioventricular (AV) valves in the heart: the *tricuspid valve* that is located between right atrium and right ventricle, and the *bicuspid valve* (*mitral valve*) that is located between left atrium and left ventricle. The *semilunar valves* control the blood leaving the heart from the ventricles. There are also two of these valves: the *pulmonary valve* is located between the right ventricle and pulmonary artery, and the *aortic*

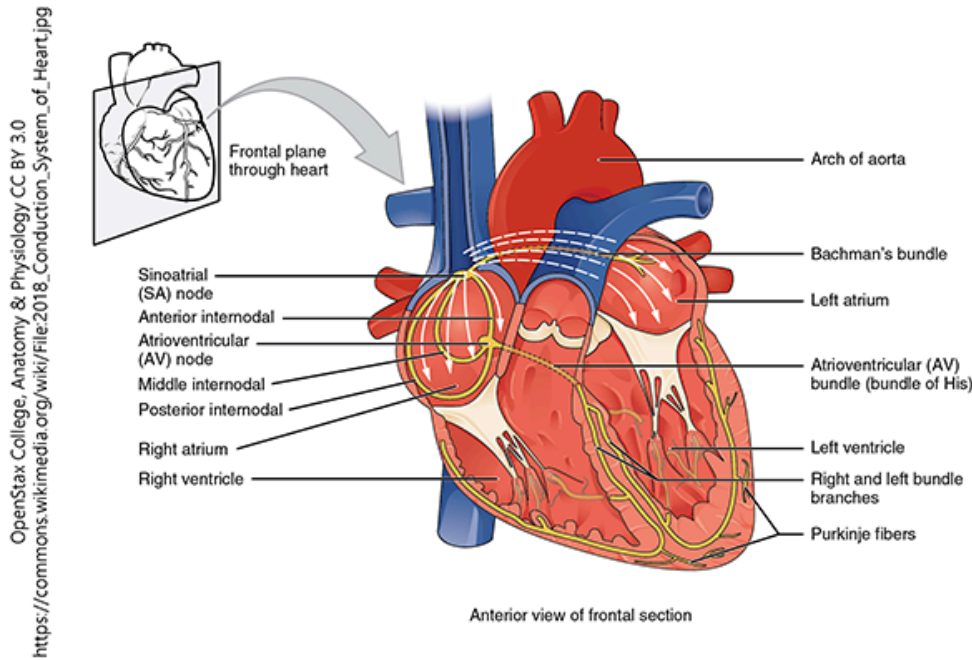
valve is located between the left ventricle and aorta.

Circulation of the blood through the body is achieved through blood vessels. There are three main types of blood vessels: arteries, veins, and capillaries. Arteries take blood away from the heart and carry oxygenated blood. The largest artery in the body is the aorta and the smallest arteries are called arterioles. Veins return blood back to the heart from the body. The largest vein is the vena cava. The superior vena cava returns blood from the arms and head, and the inferior vena cava returns blood from the trunk and legs. The smallest veins are called venules. Where arterioles and venules meet up in the body are called capillaries and this is where gas exchange occurs.

Circulation through the Heart – Summary

1. Blood from the superior vena cava and the inferior vena cava enters the right atrium
2. Blood flows from the right atrium to the right ventricle via the *tricuspid valve*
3. Blood is pumped from right ventricle to pulmonary artery (trunk) via the *pulmonary valve*
4. Blood is oxygenated in the pulmonary capillaries of the lungs
5. Blood returns to heart via pulmonary veins
6. Oxygenated blood from lungs returns to the left atrium
7. Blood flows through the *mitral valve* (also called the *bicuspid valve*) into the left ventricle
8. Blood is pumped through the *aortic valve* into the aorta
9. Blood distributed to body through branches of the aorta and other vessels into capillaries
10. Blood returns to heart via systemic veins and the venae cavae

The heart wall consists of three layers. The deepest layer is the *endocardium* which provides a smooth surface for blood as it flows through the chambers and the valves of the heart. The thick middle layer is the *myocardium*, the muscular wall of the heart. The muscular layer makes up 95% of the heart and is responsible for the pumping action. The thin, outermost layer of the heart is called the *epicardium*.

Unit 8 figure 5
Conduction
System of Heart

The heart has its own electrical conduction system that causes it to pump. The *sinoatrial (SA) node* controls the rhythm of the heartbeat and is known as the pacemaker of the heart. The SA sends an impulse to the *atrioventricular (AV) node*, and then travels to the *bundle of His*, then to the *bundle branches*, and finally to the *Purkinje fibers* causing the ventricles to contract. The heart muscle contraction is called *systole* (giving rise to the *systolic blood pressure*) and the relaxation is *diastole* (giving rise to the *diastolic blood pressure*). An *electrocardiogram (ECG or EKG)* is a recording of the electrical changes of the heart. Abnormal ECGs show problems within the conduction pathways of the heart. An ECG can show if the heart is enlarged, if certain regions of the heart are damaged, and the cause of chest pain.

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UNIT 8 PREFIXES

Prefix	Definition
a-	not
af-	going into
an-	not
ana-	up, throughout, again
anti-	against (an action)
auto-	self
baro-	weight; pressure
bi-	two, double
brady-	slow
con-	together, with
de-	from, down, out
dia-	between
dys-	bad
endo-	inside
epi-	on top of
hemi-	half
hyper-	too much
hypo- (before process)	too little
inter-	between
isch-	keep back
iso-	equal
mega-	large
micro-	small
mono-	one
oligo-	several; a handful
pan-	all; everything or everyone
peri-	around
pluri-	more than one; several
poly-	many; too much

Prefix	Definition
pre–	in front of
pro–	in front of; in favor of
re–	again
semi–	half
sub–	an underlayer; support
tachy–	fast
tel–	an end; complete
trans–	through
tri–	three

UNIT 8 ROOTS

Root Word	Definition
abdomin	belly
ability	capacity to do something
adapt	change
aden aden/o	gland
albumin	protein found in egg white; major blood protein
amine	chemical group containing nitrogen
aneurysm	dilation of an artery
ang ang/i ang/i/o	vessel
angina	chest pain
aort aort/o	the large vessel leading from the heart
apex	the highest point
apheresis	separation of blood components
append	hanging upon, used in the name of the vermiform ("wormlike") appendix, usually shortened to "appendix"
arrest	to stop
arter arter/i arter/i/o	artery
ather ather/o	plaque in an artery
atri atrio atrium	the chamber of the heart where blood first enters (by analogy to the atrium of a Roman villa)
axilla	armpit
baso	a chemical which accepts a hydrogen ion
bili	bile
body	protein molecule
brach brach/i brach/io	arm (thanks, Jay!)
capill[us]	small blood vessel

Root Word	Definition
card card/i card/i/o	heart
catheter	tube used to gain access to the urethra or a blood vessel
cava	cave
cephal	head
chol	bile
chordæ tendineæ	cords (strings) + tendon
circula	going around in a circle
circum	going around
clav[icula]	clavicle
clon[e]	exact (genetically identical) copy of a cell or organism
coagul coagul/a coagul/o	clot
compatible	getting along with
competent	capable of doing something
compromise	weaken (in this context)
concave	bent inward
conduct	carry electricity
congest	accumulation of material
constrict constrict	make an opening smaller
continu	ongoing
contract	make a muscle shorter
coron	like a crown; blood vessels which encircle the heart
crit	judge; measure to analyze
cuspid	pointed end (as of a tooth or a heart valve)
cyan	blue
cyt cyte cyt/o	cell

Root Word	Definition
diastole	relaxation phase of the heart cycle
dilate	to make an opening larger
diuresis diuret	excessive urine production
duct	tube
echo	return of sound waves
edema	excessive fluid production
effusion	pouring out of
electr/o	pertaining to the flow of electrical charges
ellipto	shaped like an oval
embol embolus	arterial blockage
enzyme	protein that speeds up chemical reactions in the cell
eosin/o	red stain used to visualize structures in cells; also referred to as acid
erythro	red
esophag	esophagus (tube from the mouth to the stomach)
externa	outermost
extrinsic	outside of
farc	stuffing
femor femur	thigh
fenestra	window
fibr fibre	fiber
fibrill	fluttering; rapid beating
flex	bend
fusion	joining together
gest	carrying (a baby)
glob	glob
glycer	sugar
granule granul/o	particle; microscopic cell structures that resemble small grains of sand

Root Word	Definition
gurgita[re]	flood
hapten	molecule that fastens other molecules together
hem hem/at hem/at/o hem/o	blood
hepat/o	liver
hist hist/o	tissue
hydr/o	water
ilia	flank
immun immun/o	defensive system of the body which protects against foreign invaders
inflamma	literally, "on fire"; the body's non-specific response to real or perceived invasion
inguin	groin
innate	something that one knows already; not learned
intercalate	inserted between
interstit interstit/i	the space between
intima	innermost
intrinsic	internal; built-in
jugul	referring to the collarbone & neck
kary/o	nucleus of the cell
kine	movement or action
lapar/o	belly; abdomen
laps[e]	fall forward
leuk leuk/o	white
lith	stone
lunar	shaped like the moon
lymph lymph/o	vessels that drain fluid from around the cells
macro	large

Root Word	Definition
mandibul	jawbone
mano	pressure
margin	edge; border
media	middle
mediastinum	cavity that encloses the heart, the major vessels of the heart, and the thymus
mitr[a]	a type of hat, with a crease in the center, worn by Roman Catholic bishops; the mitral valve is named for its resemblance to this hat
murmur	rumbling sound
myel myel/o	marrow (in this context, bone marrow)
myo	muscle
nephr/o	kidney
neutr/o	neither acid nor base; referring to Wright's stain for blood cells which uses hematoxylin (base) and eosin (acid)
nitr/o	chemical group containing nitrogen
norm/o	normal
nucl/e	the cell nucleus
opson[in]	a substance that makes bacteria more susceptible to immune attack
oxygen	a chemical element necessary for cell metabolism
palpita	trembling or quivering
pector	chest
perfor[ate]	poke holes in
perfu	flood throughout
petechia	small, red spots on the skin
phage phago	eat
phil	love
phleb phleb/o	vein
phoresis	movement of particles
phylaxis	protection; safeguard
placenta	shared organ between a mother and a fetus

Root Word	Definition
plasm plasm/a	liquid part of the blood, including clotting factors
plast	mold or shape (compare the suffix –plasty, meaning surgical alteration)
plat	flat
platelet	the cell fragment responsible for blood clotting; "little flat thing"; same as thrombocyte
poikilo	taking on many different shapes
poplite[us]	inside of the knee
potent	the power to do something; in this context, the ability to make a particular cell type
pressure	the force exerted by a liquid or gas on the walls of a vessel
pulmon	lung
receptor	a protein on the surface of a cell that turns a chemical on the outside to a signal on the inside of the cell
reticulo	like a little net
rhythm rrhythm	having a regular beat
rub	red
scler	hard
sept sept/o	wall separating two cavities
septic	filled with microorganisms
serum	fluid portion of blood with clotting components removed (e.g., the yellow fluid that exudes from a scraped knee after clotting has occurred)
sin sin/o	open space
spasm	involuntary contraction
sphero	sphere
sphygmo	pulse
spleen splen splen/o	spleen
stasis static	standing still
sten	blockage
steroid	lipid signaling molecule based on cholesterol

Root Word	Definition
stom	mouth
striat[us]	having stripes
restrict	restrict, bind, draw together, hold back
suscita	give life to
systole	contraction phase of the heart cycle
tamponade	plug up; smash into
tens	stretched
thorac	chest
thromb thromb/o	clot
thym thym/o	thymus
titer	amount of an antibody needed to have a desired effect
ton	strength; tone
tonsil tonsil/l	organs that trap and kill germs in the throat
tope	place
tox	poison
troph	nourishment, growth
tunica	tunic, coat
umbilic[us]	cord connecting the placenta to the baby
vagus	the nerve that innervates many thoracic and abdominal organs
valv valv/o valvul valvul/o	valve
varic varic/o	dilated vein
vas/o	vessel, tube, duct
vascul	little vessel
vegeta	growth

Root Word	Definition
vein ven vena ven/o	vein
ventr ventric/ul ventric/ulo	diminutive of ventrum, "belly"; so "little belly"
ver	turn
vol	volume
zygon	yoked, paired

UNIT 8 SUFFIXES

Suffix	Definition
–al	pertaining to
–algia	pain
–amine	chemical group
–ant	word ending that indicates a substance having that property
–ar	pertaining to
–ary	pertaining to
–ase	name for an enzyme, a protein or RNA molecule which promotes the forming or breaking of chemical bonds
–ated	having that quality
–centesis	puncture
–dynia	pain
–eal	pertaining to
–eau	ending used in some words of French origin
–ectasia	dilated condition
–ectasis	dilation
–ectomy	cutting out of
–em	present in blood (shortening of [h]em[e])
–emia	present in blood, indicating disease
–esis	process
–gen	arising from
–genesis	the process of creating
–genic	giving rise to
–gram	a record of a test or procedure
–graphy	a test or procedure
–ia	disease
–iac	pertaining to
–ial	pertaining to
–ic	pertaining to
–icle	diminutive suffix
–ile	word ending meaning "able to"

Suffix	Definition
-in	chemical name ending for a protein
-ion	word ending that turns a verb into a noun
-is	anatomical structure
-ism	belief; process word ending
-ist	someone who studies
-itis	inflammation
-ity	turns a noun into a state of being
-ium	word ending for an anatomical structure
-ive	to go
-ize	to make; cause to become
-ization	the process of making or causing to become
-logist	one who studies
-logy	study of
-lysis	breaking down; destroying
-malacia	softening due to disease
-megaly	enlargement due to disease
-meter	instrument for measuring
-oid	looks like; resembles
-ole	diminutive suffix
-oma	tumor
-one	chemical name ending
-or	substance or muscle which has that action
-ose	having that quality
-osis	process
-osus	Latin word ending
-ous	word ending that turns a noun into an adjective
-pathy	disease process
-penia	low levels of
-pexy	surgical fixation
-plasia	the process of molding or growing

Suffix	Definition
-plasty	surgical revision
-poiesis	creation process
-ptosis	drooping
-rrhage	bursting forth
-rrhaphy	suturing
-rrhexis	rupture
-scope	instrument for looking inside
-scopy	the process of looking inside
-sion	word ending that turns a verb into a noun
-stenosis	abnormal narrowing of an opening
stomy	creating an opening (or "mouth") with surgery
-tion	word ending that turns a verb into a noun
-tomy	cutting
-us	anatomical structure word ending
-y	disease

UNIT 8 ABBREVIATIONS

Abbreviation	Meaning
AAA	abdominal aortic aneurysm
A-Fib	atrial fibrillation
AV	atrioventricular
AIDS	acquired immunodeficiency syndrome
AED	automated external defibrillator
ALL	acute lymphoblastic leukemia
AML	acute myeloid leukemia
ASD	atrial septal defect
BMT	bone marrow transplant
BP	blood pressure
CA	cardiac arrest
CABG	coronary artery bypass graft
CAD	coronary artery disease
CBC	complete blood count
CC	cardiac catheterization
CHF	congestive heart failure
CLL	chronic lymphoblastic leukemia
CML	chronic myeloid leukemia
CO	cardiac output
DVT	deep vein thrombosis
ECHO	echocardiogram
ECG EKG	electrocardiogram
Hct	hematocrit
Hgb	hemoglobin
HIV	human immunodeficiency virus
HR	heart rate
IVC	inferior vena cava
IVIg	intravenous immunoglobulin
IVS	interventricular septum

Abbreviation	Meaning
LA	left atrium
LV	left ventricle
MI	myocardial infarction
MVP	mitral valve prolapse
NSR	normal sinus rhythm
PCI	percutaneous coronary intervention
PLT	platelet count
RA	right atrium
RBC	red blood cell
RV	right ventricle
SA	sinoatrial
SV	stroke volume
SVC	superior vena cava
TEE	transesophageal echocardiogram
VSD	ventricular septal defect
WBC	white blood cell

UNIT 8 WORD LIST

Word	Definition
abdominal	pertaining to the abdomen
adaptive	able to respond to changing circumstances; describes the immune response which is specific to a particular organism or cell surface marker
adenoid	a collection of immune tissue at the top back of the oral cavity
adenoidectomy	surgical removal of the adenoids
adenoiditis	inflammation of the adenoids
affinity	attraction for another person or substance
agranulocyte	an immune cell without microscopic granules
albumin	the most abundant protein in the liquid component of blood (plasma); also found in egg whites
anaphylaxis	a condition in which the innate immune system goes out of control and threatens life
anastomosis	the joining of two or more vessels
anemia	a condition of low oxygen carrying capacity in the blood (literally, "without hemoglobin in the blood")
aneurysmectomy	surgical removal of an aneurysm
angina pectoris	pain in the chest, possibly indicating a myocardial infarction ("heart attack")
angiocarditis	inflammation of the heart and the large blood vessels
angioedema	swelling due to leakage from blood vessels (often from an abnormal immune response)
angiogenesis	the process of creating blood vessels
angiogram	a record of the anatomy of blood vessels
angiography	a procedure for visualizing the anatomy of blood vessels
angiolith	a calcified clot in the blood vessels
angioplasty	surgical revision of the blood vessels
angiopoiesis	the process of creating blood vessels
angiorrhaphy	suturing blood vessels
angiosclerosis	hardening of blood vessel walls due to disease
angioscope	instrument for looking inside blood vessels
anisocytosis	condition where red blood cells are not all the same size or shape
antianginal	drug which relieves angina pectoris
antiarrhythmic	drug which helps re-establish normal heart rhythm
antibody	protein which is an essential part of the adaptive immune defense; made by activated B lymphocytes

Word	Definition
anticoagulant	drug which interferes with the clotting response
antidiuretic	drug which prevents the excessive production of urine
antigen	a biological substance which provokes an immune response, especially one involving the formation of antibodies
antihypertensive	drug which reduces blood pressure
aortalgia	pain in the aorta
aortectasia	abnormal dilation of the aorta
aortic	pertaining to the aorta
aortitis	inflammation of the aorta
aortogram	a record of the anatomy of the aorta
aortolith	a calcified clot in the aorta
aortorrhaphy	suturing the aorta
aortotomy	cutting into the aorta
apex	the highest point
apheresis	removal of the blood, separation of one or more blood components, followed by the return of the remainder
aplastic	unable to form (e.g. blood in the bone marrow)
appendix	an immune organ which is attached to the large intestine near the junction with the small intestine; short for vermiform ("wormlike") appendix
arrest	a stopping (e.g. cardiac arrest, stopping the heart cycle)
arrhythmia	abnormal rhythm of the heart
arterectomy	surgical removal of an artery
arteriogram	a record of the anatomy of arteries
arteriole	small artery
arteriolith	a calcified clot in the arteries
arteriopathy	disease of the arteries
arterioplasty	surgical revision of the arteries
arteriorrhaphy	suturing of an artery
arteriorrhexis	rupture of an artery
arteriosclerosis	hardening of arteries due to disease
arteriosus	containing many arteries

Word	Definition
arteritis	inflammation of one or more arteries
artery	vessel which leads away from the heart
asplenia	condition of no spleen
atherectomy	surgical removal of an atherosclerotic plaque
atherogenesis	the process of creating atherosclerotic plaques
atherosclerosis	the formation of hardened plaques (collections of fat, cholesterol, cellular waste, calcium, and fibrin) in arteries
atrial	pertaining to the atrium
atrioventricular	pertaining to the atrium and ventricle
atrioventricular bundle	electrical conduction path between the atrioventricular node and the heart muscle
atrioventricular node	secondary pacemaker for the heart; a way for the heart to maintain rhythm if the sinoatrial node fails
atrium	two of the four chambers of the heart; where the blood first enters the heart (by analogy with the atrium of a Roman villa)
autoimmune	immune attack on one's own tissues
autoregulation	maintenance of one's blood pressure or flow regardless of changing conditions (e.g. when standing)
autorhythmicity	pacemaker activity of the heart
axillary	pertaining to the armpit
azygos	unpaired vein in the trunk
baroreceptor	receptor for detecting blood pressure
basophil	white blood cell whose granules stain intensely for basic dyes (e.g. hematoxylin, a dark blue or purple dye)
biconcave	having indentations on two sides (such as the shape of a normal red blood cell)
bicuspid	having two sharp leaflets (such as the mitral valve of the heart); the bicuspid and mitral valves are two names for the same thing
bilirubinemia	condition of excess bilirubin in the blood (bilirubin is a product of the breakdown of red blood cells)
brachiocephalic	supplying the arm and head
bradycardia	condition of slow heartbeat
bundle branches	the part of the electrical conduction pathway of the heart that branches off the atrioventricular bundle (bundle of His) and leads to the Purkinje fibers
bundle of His	atrioventricular bundle

Word	Definition
capillary	blood vessel where oxygen and glucose are exchanged for waste products; joins an arteriole and a venule
cardiac	pertaining to the heart
cardiomegaly	enlargement of the heart due to disease
cardiomyocyte	a heart muscle cell
cardiomyopathy	disease of the heart muscle
cardiomyotomy	cutting into the heart muscle
cardiothoracic	pertaining to the heart and chest
cardiotonic	a drug that strengthens the heart
cardiotoxic	poisonous to the heart
cardiovascular	pertaining to the heart and blood vessels
cardioversion	restoring a normal heart rhythm
carditis	inflammation of the heart
catheterization	introducing a tube into a vessel in order to add drugs, remove blood, or insert surgical instruments
cephalic	pertaining to the head
chordæ tendineæ	tiny strands of connective tissue which anchor the heart valves in place
circulation	the movement of blood around in a continuous circuit
circumflex	bent like a circle
coagulation	the process of blood clotting
coagulopathy	disease of the clotting pathway
conduction	movement of electrically charged particles in the heart
congestive	heart disease characterized by the buildup of fluid
contractile	able to shorten in response to electrical stimulation
coronary	the arteries of the heart, named because of their resemblance to a crown
cyanosis	abnormal blue color of the skin or other organs
cytapheresis	removal of the blood with separation of blood cells, followed by the return of the plasma (fluid portion of blood)
cytokine	extracellular chemical signals which promote movement or differentiation of immune cells
cytopenia	abnormally low levels of cells in the blood
cytotoxic	poisonous to cells

Word	Definition
deoxygenated	blood that has delivered its oxygen cargo to cells in capillaries
diaphoresis	excessive sweating as a symptom of another illness
diastole	relaxation of the heart
diastolic	pertaining to relaxation of the heart, e.g. the lowest value in blood pressure
differential	the process of taking things apart to analyze them, for example a list of diagnoses, or blood cells
differentiation	the process by which a cell becomes different from its parent (stem) cell
dilate	enlargement of a round structure
ductus	Latin word for tube or channel
dysrhythmia	disease characterized by an abnormal heart rhythm pattern
echocardiogram	a record of the anatomy of the heart, obtained using sound waves
echocardiography	a procedure for visualizing the anatomy of the heart, obtained using sound waves
effusion	fluid leaking from a vessel or organ
electrocardiogram	a record of the electrical activity of the heart
electrocardiography	a procedure for observing the electrical activity of the heart
elliptocytosis	condition where red blood cells have an abnormal oval shape
embolectomy	surgical removal of an embolus
embolus	a plug in a vessel, often made up of air, fat, or a clot
endocarditis	inflammation of the inner lining of the heart
endocardium	innermost layer of the heart
endoscopy	using a tube-like camera to look inside an organ
eosinophil	white blood cell whose granules stain intensely for acidic dyes (e.g. eosin, a bright red dye)
epicardium	outermost layer of the heart
epitope	place on the outer surface of an antigenic molecule which attracts binding of an antibody
erythrocyte	a red blood cell
erythrocytosis	abnormally high number of red blood cells
erythropoiesis	the process of creating red blood cells
erythropoietin	protein hormone which promotes the formation of red blood cells
externa	outside
extrinsic	coming from outside

Word	Definition
femoral	pertaining to the thigh (femur)
fenestrated	containing many microscopic holes ("windows")
fibrillation	abnormally rapid heart rhythm which interferes with the effective pumping action of the heart
fibrin	protein which forms clots
fibrinogen	protein which is a precursor to fibrin
fibrinolysis	breakdown of fibrin clots
fibrous	filled with fibers
granulocyte	an immune cell with microscopic granules that can be seen with staining protocols
granzyme	an enzyme found in the granules of granulocytes which is used to disable invaders
hapten	a molecule that attaches to a non-antigenic molecule, making it into an effective antigen
hematocrit	a test to determine the number of red blood cells in a blood draw
hematology	the study of blood and blood diseases
hematopoiesis	the process of creating blood (i.e. blood cells)
hemiazygos	vein in the trunk which half-resembles the azygos vein
hemoglobin	the protein responsible for carrying oxygen (and to a lesser extent, carbon dioxide) in red blood cells
hemoglobinopathy	disease of hemoglobin production
hemolysis	breakdown of blood, particularly red blood cells
hemophilia	disease characterized by an abnormally poor clotting response (literally, "love of blood" because of the amount of blood produced by wounds)
hemorrhage	abnormally large flow of blood (literally, "blood bursting forth")
hemostasis	blood stoppage due to clotting or the use of a surgical procedure (e.g. cautery) or instrument (e.g. a hemostat)
hemostatic	tending to promote the stoppage of blood
hepatosplenitis	inflammation of the liver and spleen
histamine	extracellular chemical signal released in the inflammatory response
histocompatibility	the ability of transplanted tissue to evade the body's immune defenses (literally, "the ability of tissue to get along with others")
hydrostatic	pressure due to the pumping action of the heart
hyperbilirubinemia	condition of too much bilirubin in the blood
hypercoagulability	condition of too much blood clotting

Word	Definition
hypersplenism	condition of too much spleen action (e.g. removal of normal red blood cells)
hypertension	condition where blood pressure is too high
hypertrophic	pertaining to a condition of too much growth
hypertrophy	condition of too much growth
hypervolemia	condition of too much blood volume
hypoperfusion	condition of too little blood delivery to tissues
hypotension	condition where blood pressure is too low
hypovolemia	condition of too little blood volume
iliac	pertaining to the lumbar region
immunity	the process by which the body defends itself against microbial invaders
immunocompetent	able to mount an effective immune response
immunocompromised	unable to mount an effective immune response
immunodeficient	able to mount only a weak or ineffective immune response
immunoglobulin	a protein involved in the immune response; antibody
immunology	the study of the immune system and its dysfunction
immunosuppression	the process of lowering or incapacitating the immune response
infarction	interruption of blood supply, for example due to a clot
inflammation	the innate immune response; the four cardinal signs are redness, heat, swelling, and pain
inguinal	pertaining to the groin
innate	"built-in"; describes the immune response which is not specific to a particular organism or cell surface marker
intercalated	inserted in between layers
interleukin	extracellular chemical signal which affects white blood cells primarily
interstitial	the space between cells (including the fluid which resides there)
interventricular	between the ventricles of the heart
intima	inside
intrinsic	coming from factors within a tissue or organ
ischemia	lack of blood supply
jugular	vein of the neck which carries blood from the brain and head back to the heart
laparosplenectomy	surgical removal of the spleen through a small incision in the abdomen

Word	Definition
leukemia	a cancer characterized by a large number of abnormally formed white blood cells
leukocyte	a white blood cell
leukocytosis	abnormally high levels of white blood cells
leukopenia	abnormally low levels of white blood cells
lymphadenectomy	surgical removal of lymph nodes (lymph glands)
lymphadenitis	inflammation of lymph nodes
lymphadenopathy	disease of the lymph nodes
lymphadenotomy	cutting into the lymph nodes
lymphangiectasia	abnormal dilation of the lymphatic vessels
lymphangiogram	a record of the anatomy of the lymphatic vessels
lymphangiography	a procedure for visualizing the anatomy of the lymphatic vessels
lymphangitis	inflammation of one or more lymphatic vessels
lymphedema	swelling of the lymphatic vessels
lymphocyte	a subtype of white blood cell that either controls an adaptive immune attack or makes antibodies
lymphoid	resembling the lymph organs
lymphoma	a cancer characterized by abnormalities in the lymphocytes, a subset of white blood cells
lymphopenia	low levels of lymphocytes (a subset of white blood cells)
macrocytosis	condition where red blood cells are abnormally large
macrophage	defensive cell in the connective tissues that engulfs and destroys invaders
marginal	pertaining to the edges
mast cell	defensive cell in the connective tissues that releases histamine and other chemicals
media	in the middle
mediastinum	the cavity which contains the heart, great vessels, part of the esophagus, and the thymus; essentially everything in the chest except lungs
megakaryocyte	a cell which gives rise to platelets (thrombocytes)
microcytosis	condition where red blood cells are abnormally small
mitral	a heart valve that is shaped like a Roman Catholic bishop's mitre (peaked cap with a crease in the middle); same as bicuspid
monoclonal	a group of cells that are all genetically identical (usually pertains to B cells which all make the same type of antibody molecule)
monocyte	circulating white blood cell which transforms into a macrophage in the tissues

Word	Definition
mononucleosis	disease characterized by a large number of abnormal lymphocytes; caused by Epstein-Barr virus
murmur	heart sound that resembles a rumble or groan or machinery noise
myeloid	a family of blood cells which includes red blood cells and granulocytes
myeloma	a cancer arising from the bone marrow
myelopoiesis	the process of creating bone marrow
myocardial	pertaining to the heart muscle
myocarditis	inflammation of the heart muscle
myocardium	the heart muscle itself (i.e. not including the inner lining or outer lining or membranes)
nephrosplenopexy	surgical fixation of the kidney and spleen
neutropenia	abnormally low levels of neutrophils
neutrophil	white blood cell whose granules stain equally for acidic and basic dyes (e.g. both hematoxylin and eosin in the Wright stain)
nitroglycerin	drug which dilates blood vessels to relieve angina pectoris
normocyte	a red blood cell that is normal in size and hemoglobin content
normotension	normal blood pressure
occlusion	blockage
oligocythemia	disease characterized by too few ("a handful of") red blood cells
opsonization	process by which a bacterial invader becomes "tagged" and set up for destruction by the immune system
oxygenated	blood that has picked up its oxygen cargo in the lungs
palpitation	the physical feeling from a fluttery, irregular heartbeat
pancytopenia	abnormally low levels of all blood cells
pectoralgia	pain in the chest, possibly indicating a myocardial infarction ("heart attack")
pectoris	the chest
perforin	protein that attacks invading cells and pokes holes in their cell membrane
perfusion	blood moving through a tissue to supply oxygen and glucose and remove wastes
pericardial	pertaining to the membrane surrounding the heart
pericardiocentesis	removing fluid from the pericardial cavity
pericardiotomy	cutting into the pericardium
pericarditis	inflammation of the pericardium

Word	Definition
pericardium	the bag-like membrane surrounding the heart
petechia	red pinpoint on the skin (usually seen as the plural petichae)
phagocytosis	process by which a macrophage or other immune cell surrounds, swallows, and destroys an invading cell
phlebalgia	pain in the veins
phlebarteriectasia	abnormal dilation of the veins and arteries
phlebectomy	surgical removal of a vein
phlebitis	inflammation of a vein
phlebologist	a person who studies vein disease
phlebology	the study of vein disease
phlebophlebostomy	making a connection ("mouth") between two veins
phlebosclerosis	hardening of veins due to disease
phlebostenosis	narrowing of a vein
phlebotomist	a person who cuts into a vein (i.e. punctures veins to remove blood for analysis)
phlebotomy	cutting into a vein (i.e. venipuncture to remove blood for analysis)
placenta	the tissue which supplies oxygen and nutrients to the fetus; it results from a collaboration between mother and fetus
plasma	the liquid component of blood (including soluble clotting factors)
plasmapheresis	removal of the blood with separation of plasma, followed by the return of the cells and platelets
plasmin	protein component of the blood which breaks down fibrin when activated
plasminogen	precursor protein which gives rise to plasmin when activated
plateau	a flat area of a graph, such as a cardiac voltage trace
platelet	cell fragment that is necessary for blood clotting (literally, "a little plate"); same as thrombocyte
plateletpheresis	removal of the blood with separation of platelets, followed by the return of the other cellular components and plasma
pluripotent	a stem cell which is capable of making several different types of daughter cells
poikilocytosis	condition where red blood cells are of many different sizes and shapes
polycythemia	disease characterized by too many red blood cells
popliteal	pertaining to the dorsal surface of the knee
precapillary	the segment of an arteriole before its entry into a capillary

Word	Definition
pressure	physical property that results from the collision of molecules in a liquid or gas with the walls of the container; an increase in the number of molecules results in an increase in pressure
progesterone	hormone which acts in favor of gestation (carrying a child to term)
prolapse	disease characterized by a floppy valve which does not close completely, permitting blood to regurgitate
prothrombin	precursor protein which gives rise to thrombin when activated
prothrombinase	enzyme which acts on prothrombin
pulmonary	pertaining to the lungs
Purkinje fibers	the part of the electrical conduction system that spreads over the entire surface of the ventricle
regurgitation	movement in a direction opposite to the normal flow
reperfusion	restoring blood to a tissue or organ that has lost blood supply temporarily
restrictive	type of heart muscle disease characterized by stiffness of the heart muscle
resuscitation	bringing someone back to life
reticulocyte	the stage of red blood cell development just before maturation (named for its net-like appearance)
semilunar	having a half-moon shape
septal	pertaining to the wall within an organ such as the heart
septicemia	abnormally large number of microorganisms in the blood
septoplasty	surgical revision of the chamber walls of the heart
septum	the wall between the chambers of the heart (or other organs)
serum	the liquid component of blood after soluble clotting factors are removed
sinoatrial	pertaining to the atrium of the heart, and the nearby coronary sinuses (which receive venous blood from the heart)
sinusoid	resembling a sinus (large, hollow space)
spherocyte	an abnormally round red blood cell
spherocytosis	condition where red blood cells are abnormally rounded
sphygmomanometer	device for measuring blood pressure (blood pressure cuff)
spleen	organ in the upper left abdomen, important in the immune response and in removing aged red blood cells
splenalgia	pain in the spleen
splenectomy	surgical removal of the spleen
splenitis	inflammation of the spleen

Word	Definition
splenodynia	pain in the spleen
splenolysis	breakdown of the tissue of the spleen
splenomalacia	softening of the spleen due to disease
splenomegaly	enlargement of the spleen due to disease
splenopathy	disease of the spleen
splenoptosis	drooping of the spleen (movement of the spleen out of its normal position)
splenorrhexis	rupture of the spleen
stasis	holding back
stenosis	narrowing of a tube or vessel
striated	showing obvious stripes
subclavian	pertaining to an area below the clavicle (collarbone)
submandibular	pertaining to an area below the mandible (lower jaw)
systemic	affecting the entire system
systole	contraction of the heart
systolic	pertaining to contraction of the heart, e.g. the peak value in blood pressure
tachycardia	condition of fast heartbeat
tamponade	pressure on the heart due to the buildup of fluid in the pericardial sac
telangiectasia	abnormal dilation of the "end" (smallest) blood vessels
thrombin	protein component of the blood which makes up a key part of the clotting cascade
thrombocyte	platelet
thrombocytopenia	abnormally low levels of platelets (thrombocytes)
thrombocytosis	abnormally high levels of platelets (thrombocytes)
thromboembolism	a plug in a vessel from a clot
thrombogenic	drug or substance which promotes blood clotting
thrombophlebitis	inflammation of a vein due to formation of a clot
thromboplastin	blood protein which is part of the clotting pathway
thrombus	clot
thymic	pertaining to the thymus
thymopathy	disease of the thymus
thymus	immune organ in the mediastinum; responsible for the maturation of T lymphocytes

Word	Definition
titer	the amount of antibody needed to cause a particular response, such as inactivation of a foreign protein
tonsil	a collection of immune tissue on either side of the throat
tonsillectomy	surgical removal of the tonsils
tonsillitis	inflammation of the tonsils
transesophageal	across the esophagus (especially where it is close to the heart in the mediastinum)
transfusion	moving blood from one person to another
tricuspid	having three sharp leaflets (such as the tricuspid valve of the heart)
tunica	tunic or coat; refers to layers of a blood vessel
umbilical	pertaining to the connection between placenta and fetus
vagus	nerve which innervates most of the thoracic and abdominal organs
valvectomy	surgical removal of a heart valve
valvotomy	surgical revision of a heart valve (same as valvuloplasty or valvulotomy); the surgical revision involves making cuts into it
valvular	pertaining to a valve
valvulitis	inflammation of a heart valve
valvuloplasty	surgical revision of a heart valve (same as valvotomy or valvulotomy)
valvulotomy	surgical revision of a heart valve (same as valvotomy or valvuloplasty); the surgical revision involves making cuts into it
varicose	abnormally dilated
varicotomy	cutting into a varicose vein
vascular	pertaining to the blood vessels
vasculitis	inflammation of the blood vessels
vasoconstriction	the process of reducing the diameter of a blood vessel
vasoconstrictor	a drug which reduces the diameter of a blood vessel
vasodilation	the process of increasing the diameter of a blood vessel
vasodilator	a drug which increases the diameter of a blood vessel
vasopressin	a hormone which increases blood pressure (same as antidiuretic hormone)
vasopressor	a drug which increases blood pressure
vasospasm	an abnormal contraction in the muscles of the arterial wall, potentially causing a loss of blood supply and/or increase in blood pressure
vegetation	a growth on a heart valve

Word	Definition
vein	blood vessel leading to the heart
vena cava	the largest blood vessels leading to the right atrium of the heart (superior vena cava; inferior vena cava)
venectomy	surgical removal of a vein
venogram	a record of the anatomy of the veins
venosclerosis	hardening of veins due to disease
venospasm	an abnormal contraction in the muscles of the walls of veins
venostasis	blood which pools and does not move normally within veins
venosus	containing many veins
venous	pertaining to the veins
ventricle	two of the four chambers of the heart; where the blood is pumped out of the heart to the lungs or body
ventriculotomy	cutting into a heart ventricle

HOW THE UNIT 8 WORD LIST WAS BUILT

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
				abdomin			al		abdominal
				adapt			ive		adaptive
				aden			oid		adenoid
				adenoid			ectomy		adenoidectomy
				adenoid			itis		adenoiditis
Latin, ad- or af- + finis, "on the border"	af			fin[is]			ity		affinity
	a			granul/o		cyte			agranulocyte
Latin albumen ovi, "white of an egg" (an almost-pure source of this protein)				albumin					albumin
Phylaxis is a Greek word meaning "protection", as in the phylacterion, an amulet or safeguard against disease (ana- meaning "against" or "backward" here)	ana			phylaxis					anaphylaxis
Greek anastomosis, "to furnish with a mouth"	ana			stom[a]			osis		anastomosis
	an			[h]em			ia		anemia
				aneurysm			ectomy		aneurysmectomy
Latin angina, "strangling" + pectoris, "chest"			angina	pectoris					angina pectoris
				ang/i/o		card	itis		angiocarditis
				ang/i/o		edema			angioedema
				ang/i/o			genesis		angiogenesis
				ang/i/o			gram		angiogram
				ang/i/o			graphy		angiography
				ang/i/o		lith			angiolith
				ang/i/o			plasty		angioplasty
				ang/i/o			poiesis		angiopoiesis

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
				ang/i/o			rrhaphy		angiorrhaphy
				ang/i/o		scler	osis		angiosclerosis
				ang/i/o			scope		angioscope
	an	iso		cyt			osis		anisocytosis
	anti			angin			al		antianginal
	anti	a		rrhythm			ic		antiarrhythmic
	anti			body					antibody
	anti			coagul			ant		anticoagulant
	anti			diuret			ic		antidiuretic
	anti						gen		antigen
	anti	hyper		tens			ive		antihypertensive
				aort			algia		aortalgia
				aort			ectasia		aortectasia
				aort			ic		aortic
				aort			itis		aortitis
				aort/o			gram		aortogram
				aort/o		lith			aortolith
				aort/o			rrhaphy		aortorrhaphy
				aort/o			tomy		aortotomy
				apex					apex
Greek aphairein, "to take away"				apher			esis		apheresis
	a			plast			ic		aplastic
Latin appendere, ap- + pendere, "hang upon"				append			ix		appendix
Latin arrestare, "to stop"				arrest					arrest
	a/r			rhythm			ia		arrhythmia
				arter			ectomy		arterectomy
				arter/i/o			gram		arteriogram

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word				
-osus is added to the end of a noun to make an adjective meaning "full of"	a			arter	i		ole		arteriole				
				arter	i/o	lith			arteriolith				
				arter	i/o		itis		arteriopathy				
				arter	i/o		plasty		arterioplasty				
				arter	i/o		rrhexis		arteriorrhexis				
				arter	i/o		rrhaphy		arteriorrhaphy				
				arter	i/o	scler	osis		arteriosclerosis				
				arteri	[a]		osus		arteriosus				
				arter			itis		arteritis				
				arter			y		artery				
Greek athērē, "porridge, groats"; from the appearance of plaques in blood vessels				splen			ia		asplenia				
				ather			ectomy		atherectomy				
				ather	/o		genesis		atherogenesis				
				ather	/o	scler	osis		atherosclerosis				
				atri			al		atrial				
				atrio	ventricul			ar	atrioventricular				
					atrium				atrium				
				Latin atrium, splendid entrance hall in Roman architecture	auto			immune					autoimmune
								regulat[e]		ion		autoregulation	
								rhythm[us]		ic	ity	autorhythmicity	
axill[a]		ary						axillary					

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
A-, "not" + Greek "zygos", yoke or pair. The azygos vein is named this because it only appears on one side, usually the right.	a			zyg[on]				os	azygos
	baro			receptor					baroreceptor
				baso		phil			basophil
Latin concavus, con- (here, used for emphasis) + cavus, "hollow"	bi			concave					biconcave
	bi			cuspid					bicuspid
Bile + Latin rubor, "red" + in, chemical word ending + -emia, blood disease				bili		rub	in	emia	bilirubinemia
pertaining to the arm and head; "brachi/o-" is a variant of "brachy" which means "short" because the Greeks correctly identified the arm as shorter than the leg (thanks, Lily!). The older name for this artery is the <i>innominate</i> which means "no name"; how can it have no name when the name is "innominate"?				brachi/o		cephal	ic		brachiocephalic
	brady			card			ia		bradycardia
Latin, "pertaining to the hair"				capill[us]			ary		capillary
Latin cardiacus				cardiac					cardiac
				card/i/o			megaly		cardiomegaly
			card/i/o	myo		cyte			cardiomyocyte
			card/i/o	myo			pathy		cardiomyopathy

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word									
Latin "to go in a circle" Latin coagulatus, "to cause curdling"; "congeal" Latin congestus, "to pile up" Greek phorēsis, "movement of small particles by some unseen force"			card/i/o	myo			tomy		cardiomyotomy									
									card/i/o	thorac	ic	cardiothoracic						
									card/i/o	ton	ic	cardiotonic						
									card/i/o	tox	ic	cardiotoxic						
									card/i/o	vascul	ar	cardiovascular						
									card/i/o	ver	sion	cardioversion						
									card		itis	carditis						
									catheter		iz[e]/a	tion	catheterization					
									cephal		ic	cephalic						
			chordæ	tendineæ					chordæ tendineæ									
									circula		tion	circulation						
									circum	flex		circumflex						
									coagul/a		tion	coagulation						
									coagul/o		pathy	coagulopathy						
									conduc[t]		tion	conduction						
									congest		ive	congestive						
									contract		ile	contractile						
									coron		ary	coronary						
			de									oxygen						
ated	deoxygenated																	
dia																	phoresis	
																	diaphoresis	

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin & Greek, "dilation"					diastole				diastole
					diastol[e]		ic		diastolic
					different		ial		differential
					different			tion	differentiation
Latin dilatare "make wider, enlarge"					dilate				dilate
					duct		us		ductus
	dys				rrhythm		ia		dysrhythmia
Latin echo, from Greek ēkhē, sound			echo		card/i/o		gram		echocardiogram
			echo		card/i/o		graphy		echocardiography
					effusion				effusion
			electr/o		card/i/o		gram		electrocardiogram
			electr/o		card/i/o		graphy		electrocardiography
			ellipto		cyt		osis		elliptocytosis
Latin embolus, "piston of a pump"					embol[us]		ectomy		embolectomy
					embolus				embolus
	endo				card		itis		endocarditis
	endo				card		ium		endocardium
	endo						scopy		endoscopy
					eosin/o	phil			eosinophil
	epi				card		ium		epicardium
	epi				tope				epitope
					erythro	cyte			erythrocyte
					erythro	cyt	osis		erythrocytosis
					erythro		poiesis		erythropoiesis
					erythro		poiet[ic]	in	erythropoietin
					externa				externa
					extrinsic				extrinsic

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin fenestra, "window"				femor			al		femoral
				fenestra			[a]ted		fenestrated
				fibrill/a			tion		fibrillation
				fibr			in		fibrin
				fibr			in/o	gen	fibrinogen
				fibr			in/o	lysis	fibrinolysis
				fibr[e]			ous		fibrous
granule + enzyme				granul/o		cyte			granulocyte
				gran[u]le		[en]zyme			granzyme
Greek háptein, "to fasten"				haptēn					haptēn
hemato + Greek kritēs, "judge"				hem/at/o		crit			hematocrit
				hem/at/o			logy		hematology
				hem/at/o			poiesis		hematopoiesis
Named after the azygos vein; the bottom half (hemi-) resembles the azygos vein.	hemi			zyg[on]				os	hemiazzygos
				hem/o		glob	in		hemoglobin
				hem/o		glob	in/o	pathy	hemoglobinopathy
				hem/o			lysis		hemolysis
				hem/o		phil	ia		hemophilia
hemo + Greek stásis, "to stand"; ancient Greek αἱμόστασις (haimóstasis) means "styptic drug"				hem/o			rrhage		hemorrhage
				hem/o		stasis			hemostasis
				hem/o		static			hemostatic
				hepat/o		splen	itis		hepatosplenitis
		hist		amine					histamine

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Greek statikos, "causing to stand"; "weigh"			hist/o		compatibil[e]		ity		histocompatibility
			hydro		static				hydrostatic
	hyper			bili		rub	in	emia	hyperbilirubinemia
Proto-Indo-European ten-, "to stretch"	hyper			coagul		ability			hypercoagulability
	hyper			splen			ism		hypersplenism
	hyper			ten[s]			sion		hypertension
	hyper			troph			ic		hypertrophic
	hyper			troph			y		hypertrophy
	hyper			vol		[h]em	ia		hypervolemia
	hypo			perfu			sion		hypoperfusion
	hypo			ten[s]			sion		hypotension
	hypo			vol		[h]em	ia		hypovolemia
				il[ia]			iac		iliac
Latin ilia, "groin, flank"				immun			ity		immunity
			immun/o	compet			ent		immunocompetent
			immun/o	compromis			ed		immunocompromis
The original meaning was "to make a promise together", but this meaning is "expose to risk or hazard" which came later			immun/o	deficient					immunodeficient
				immun/o		glob	in		immunoglobulin
				immun/o			logy		immunology
				immun/o		suppres	sion		immunosuppression

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
The Latin word <i>farcire</i> means "to stuff"; the English word "farce" comes from the same root. In 16th century France, a "farce" was a comedic interlude "stuffed" into a more serious play.	in			farc			tion		infarction
Latin <i>inflammare</i>, "to set on fire"; compare English <i>inflammable</i>, "capable of being set on fire"				inflamma			tion		inflammation
				inguin			al		inguinal
				innate					innate
Latin <i>intercalatus</i>, a day inserted into the calendar, now used to indicate something inserted generally				intercalat			ed		intercalated
	inter			leuk			in		interleukin
Latin <i>interstitius</i>, "the space between"				interstit/i				al	interstitial
	inter			ventric[le]/ul				ar	interventricular
				intima					intima
Latin <i>intrinsic</i>, "internal"				intrinsic					intrinsic
Greek <i>iskhein</i>, "to hold" + <i>haima</i>, "blood" (also spelled <i>ischæmia</i>)	isch			[h]em[e]			ia		ischemia
Latin <i>jug[um]</i>, "yoke" + <i>ulum</i>, diminutive suffix				jugul			ar		jugular
Greek <i>lapara</i>, "flank, loins, soft part between the ribs and hips", i.e. the abdomen			lapar/o	splen			ectomy		laparosplenectomy

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
The great German histologist Paul Ehrlich called these Mastzellen, "well-fed cells" because he thought they had absorbed large amounts of nutrients into their distinctive granules				leuk			emia		leukemia
			leuk/o	cyte					leukocyte
			leuk/o	cyt			osis		leukocytosis
				leuk/o			penia		leukopenia
			lymph	aden			ectomy		lymphadenectomy
			lymph	aden			itis		lymphadenitis
			lymph	aden/o			pathy		lymphadenopathy
			lymph	aden/o			tomy		lymphadenotomy
			lymph	ang/i			ectasia		lymphangiectasia
			lymph	ang/i/o			gram		lymphangiogram
			lymph	ang/i/o			graphy		lymphangiography
			lymph	ang			itis		lymphangitis
			lymph	edema					lymphedema
				lymph/o	cyte				lymphocyte
				lymph			oid		lymphoid
				lymph			oma		lymphoma
				lymph/o			penia		lymphopenia
		macro		cyt			osis		macrocytosis
		macro		phage					macrophage
				margin			al		marginal
				mast					mast cell
				media					media

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin mediastinus, "a common servant"; probably a version of "standing in the middle"					mediastin				mediastinum
	mega				kary/o	cyte			megakaryocyte
	micro				cyt		osis		microcytosis
Latin mitra, "turban"; "hat" based on the valve's resemblance to a Roman Catholic bishop's mitre					mitr		al		mitral
Greek klōn, "twig; sprout"	mono				clon[e]		al		monoclonal
	mono				cyte				monocyte
	mono				nucl/e		osis		mononucleosis
Latin "to grumble" or "hum"					murmur				murmur
					myel[os]		oid		myeloid
					myel		oma		myeloma
					myel/o		poiesis		myelopoiesis
			myo		card/i		al		myocardial
			myo		card		itis		myocarditis
			myo		card		ium		myocardium
					nephro/o	splen/o	pexy		nephrosplenopexy
					neutr/o	[phil]	penia		neutropenia
Neutrophils, the most numerous white blood cells, are the body's normal response to bacterial infection. "Neutrophils dive into pus and die." – Linda Clayton					neutr/o	phil			neutrophil
			nitro		glycer		in		nitroglycerin
					norm/o	cyte			normocyte

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
			norm/o		ten[s]		sion		normotension
Latin occludere, "fact of being stopped up"									occlusion
	oligo				cyt	hem	ia		oligocythemia
The original term opsonins (substances that make bacteria more susceptible to immune attack) comes from the Greek opsōnéō, "to buy provisions, to cater"					opson[in]		iz[e]/a	tion	opsonization
					oxygen		ated		oxygenated
Latin palpitare, "to flutter"					palpita		tion		palpitation
	pan				cyto		penia		pancytopenia
					pector		algia		pectoralgia
Latin pectus, "chest"					pector		is		pectoris
					perfor[ate]		in		perforin
Latin perfundere, "to pour over"					perfu		sion		perfusion
	peri				card/i		al		pericardial
	peri				card/i/o		centesis		pericardiocentesis
	peri				card/i/o		tomy		pericardiotomy
	peri				card		itis		pericarditis
	peri				card		ium		pericardium
Latin "freckle" or "spot"					petechia				petechia
					phago	cyt	osis		phagocytosis
					phleb		algia		phlebalgia
					phleb	arteri	ectasia		phlebarteriectasia
					phleb		ectomy		phlebectomy
					phleb		itis		phlebitis

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
					phleb/o		logist		phlebologist
					phleb/o		logy		phlebology
					phleb/o	phleb/o	stomy		phlebophlebostomy
					phleb/o	scler	osis		phlebosclerosis
					phleb/o	sten	osis		phlebostenosis
					phleb/o		tom	ist	phlebotomist
					phleb/o		tomy		phlebotomy
					placenta				placenta
					plasma				plasma
					plasm	apheresis			plasmapheresis
					plasm[a]		in		plasmin
					plasm		in/o	gen	plasminogen
					plat		eau		plateau
					plate		let		platelet
					platelet	[a]pheresis			plateletpheresis
Latin pluris (stem of plus), "more than one, several, many" Greek poikílos, "spotted; embroidered"	pluri				potent				pluripotent
					poikilo	cyt	osis		poikilocytosis
	poly				cyt	hem	ia		polycythemia
					poplite[us]		al		popliteal
	pre				capill[us]		ary		precapillary
					pressure				pressure
	pro				gest	ster[oid]	one		progesterone
					laps[us]		e		prolapse
					thromb		in		prothrombin
					thromb		in	ase	prothrombinase
pro- + Latin gestare, "to carry about"					pulmon		ary		pulmonary

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin regurgitare, "to overflow"; from re- + gurgitare, "engulf, flood"	re			gurgita[re]			tion		regurgitation
	re			perfu			sion		reperfusion
Latin restrictivus, "restrict, bind, draw together"				restrict			ive		restrictive
Latin resuscitare, re- + "give life to, raise"	re			suscita			tion		resuscitation
				reticulo		cyte			reticulocyte
	semi			lunar					semilunar
				sept			al		septal
Greek sēptikos, "full of putrefaction (rotten)"				septic			emia		septicemia
				septo			plasty		septoplasty
Latin, "wall separating two cavities"				septum					septum
Latin, "watery animal fluid"				serum					serum
		sino		atrial					sinoatrial
				sinus			oid		sinusoid
				sphero		cyte			spherocyte
				sphero		cyt	osis		spherocytosis
Word coined from sphygmo-, "pulse" + manometer, instrument for measuring the pressure of a gas				sphygmo		mano	meter		sphygmomanometer
				spleen					spleen
				splen			algia		splenalgia
				splen			ectomy		splenectomy
				splen			itis		splenitis

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
					splen/o		dynia		splenodynia
					splen/o		lysis		splenolysis
					splen/o		malacia		splenomalacia
					splen/o		megaly		splenomegaly
					splen/o		pathy		splenopathy
					splen/o		ptosis		splenoptosis
					splen/o		rrhexis		splenorrhexis
					stasis				stasis
					sten		osis		stenosis
					striat[us]		ed		striated
	sub				clav[icula]		ian		subclavian
	sub				mandibul		ar		submandibular
					system		ic		systemic
					systole				systole
Greek, "contraction"					systol[e]		ic		systolic
	tachy				card		ia		tachycardia
					tamponade				tamponade
					ang/i		ectasia		telangiectasia
					thromb		in		thrombin
					thromb/o	cyte			thrombocyte
					thromb/o	cyto	penia		thrombocytopenia
					thromb/o	cyt	osis		thrombocytosis
					thromb/o	embol	ism		thromboembolism
					thromb/o	gen	ic		thrombogenic
Originally meant the application of tampons in surgery to control bleeding; now means "plug up"; "smash into"					thromb/o	phleb	itis		thrombophlebitis
					thromb/o	plast	in		thromboplastin

Etymology	Prefix1	Prefix2	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
				vas/o		dila[te]	tion		vasodilation
				vas/o		dilat[e]	or		vasodilator
				vas/o		press[ure]	in		vasopressin
				vas/o		press[ure]	or		vasopressor
				vas/o		spasm			vasospasm
Latin vegetabilis, "growing, flourishing"				vegeta			tion		vegetation
Latin vena, "vein"				vein					vein
Latin vena, "vein" + cavus, "hollow"			vena	cava					vena cava
				ven			ectomy		venectomy
				ven/o			gram		venogram
				ven/o		scler	osis		venosclerosis
				ven/o		spasm			venospasm
				ven/o		stasis			venostasis
				ven[a]			osus		venosus
				ven[a]			ous		venous
Latin, "little belly"				ventr			icle		ventricle
				ventric[le]/ulo				tomy	ventriculotomy

PART IX

UNIT 9: THE RESPIRATORY SYSTEM

UNIT 9 INTRODUCTION

The central function of the Respiratory System is moving oxygen into the body and carbon dioxide out. This is performed through ventilation (mechanical movement of air) and respiration (exchange of gas across a membrane). Oxygen is essential for the survival of cells throughout the body, with brain and heart muscle cells being particularly vulnerable to hypoxia (low oxygen), only surviving a few minutes. Oxygen is continually made into carbon dioxide by cell metabolic processes, and that carbon dioxide must be removed.

Objectives

After completing this unit, you will be able to:

- Name the word parts used to describe the structures and diseases of the respiratory system.
- Recognize abbreviations used to describe diseases and treatments associated with the respiratory system.

THE RESPIRATORY SYSTEM

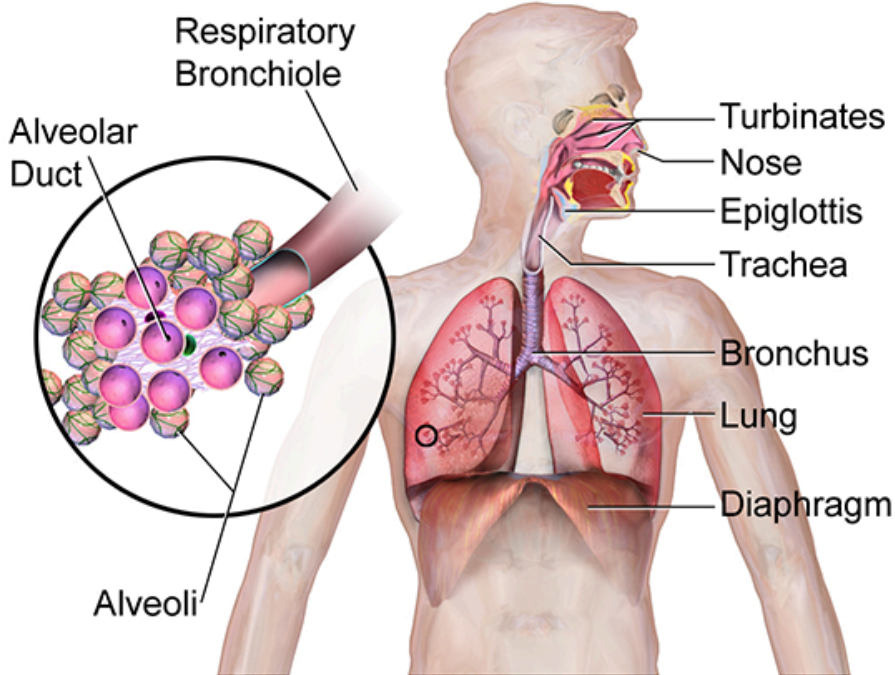
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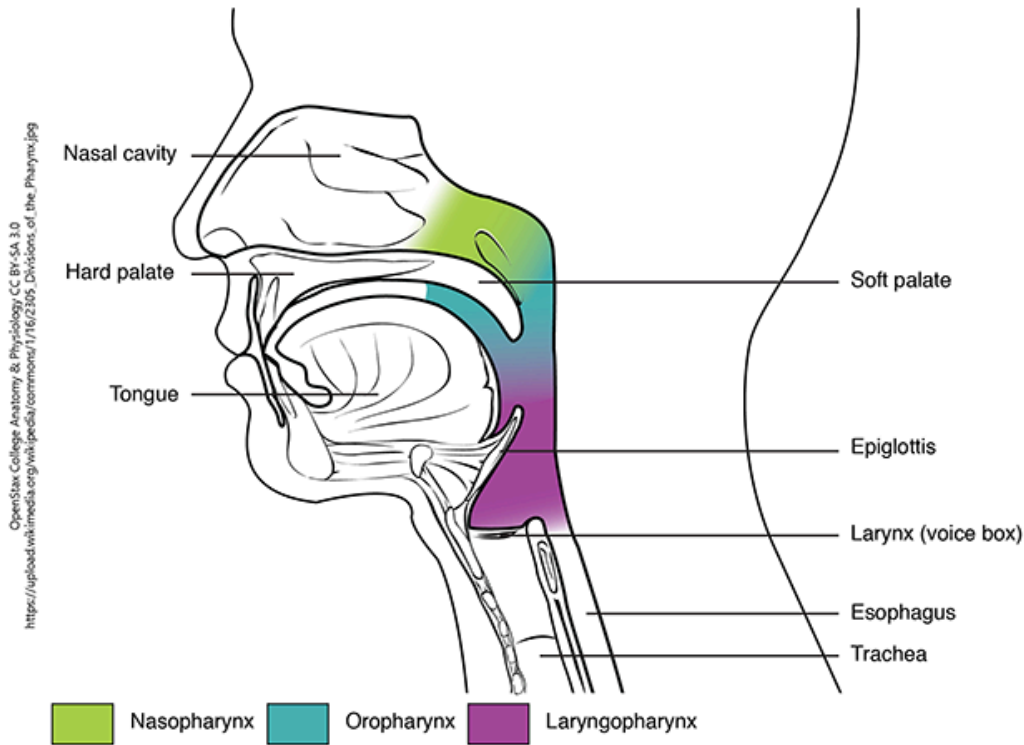
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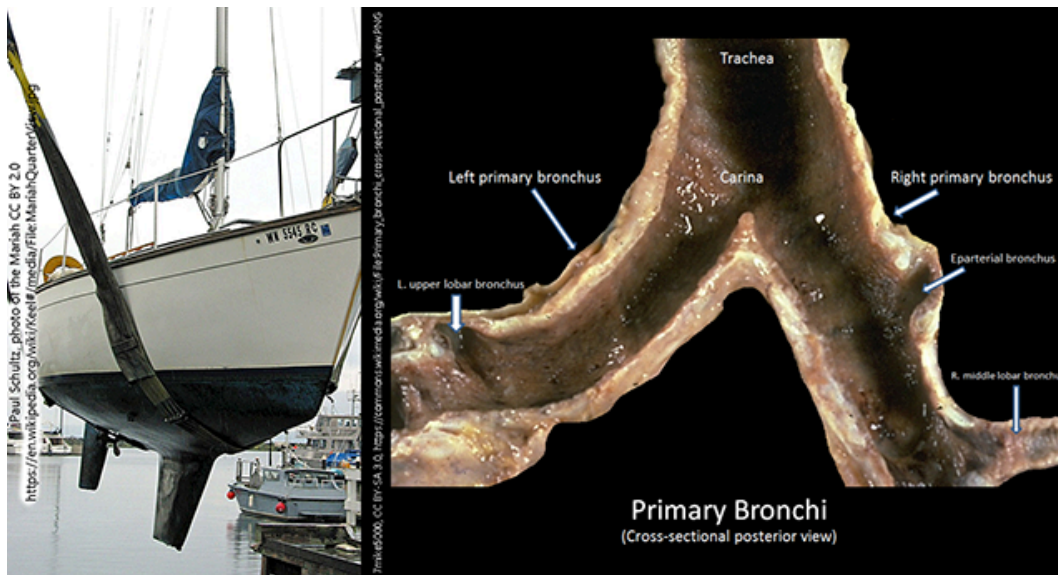


Unit 9 figure 1
Respiratory
System Overview

The respiratory system is a system that plays several roles in the physiology of our bodies. It helps us to move air in and out of our lungs (a process called *ventilation*) and enables our bodies to exchange gases between our blood and the air that we breathe (a process called *respiration*). Oxygen (O_2) is exchanged for carbon dioxide (CO_2) in the millions of tiny air sacs called *alveoli* that fill our lungs. Our respiratory system enables us to produce sound through our vocal chords and to smell along with receptors in our nasal passages. The respiratory system also heats and humidifies the air that we breathe. Cilia, which are part of the *mucociliary escalator* (a lining of cilia along the bronchi and trachea), help to trap tiny particles that we may inhale and move them upwards towards our mouth so that the air that reaches the alveoli is free of dust or dirt.

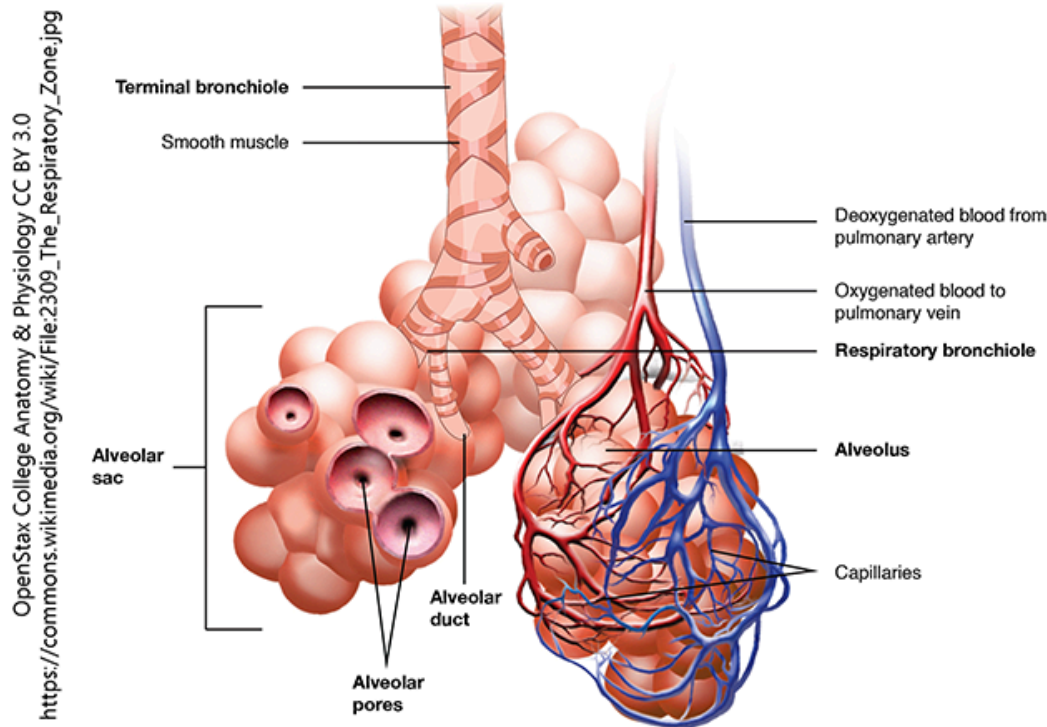
Unit 9 figure 2
Divisions of the
Pharynx

The structure of the respiratory system can be divided into the upper and lower respiratory tracts. The *nasal cavity*, *oral cavity*, and *pharynx* make up the *upper respiratory system*. The *larynx*, or voice box, separates the upper from lower respiratory system. The *trachea*, *bronchi*, and *lungs* make up the lower respiratory system. We often designate respiratory infections based on this classification. For example, when we talk about upper airway infection (URI), we refer to infection in the parts of the respiratory system from the mouth or nose to the larynx.

Unit 9 figure 3
Coat Keel and
Carina in Trachea

The trachea divides into two primary bronchi (one for each lung, right and left) at an anatomical feature called the carina, after the Latin word for boat keel.

Unit 9 figure 4
Respiratory Zone



Oxygen and carbon dioxide are exchanged in the *lower respiratory system*. The main bronchi branch into smaller air pathways called *bronchioles* and then *terminal bronchioles* and *respiratory bronchioles* which eventually terminate at the small air sacs called *alveoli*. We divide respiration (the exchange of gases such as O_2 and CO_2) into two categories: *external respiration* and *internal respiration*. External respiration is the exchange of gases between the *pulmonary capillaries* and the alveoli which contain atmospheric air. CO_2 diffuses from the pulmonary capillaries into the alveoli; O_2 diffuses from the alveoli into the pulmonary capillaries. Internal respiration is the exchange of gases between the tissues and the systemic capillaries. Oxygen diffuses from the systemic capillaries into the tissues; carbon dioxide diffuses from the tissues into the bloodstream. Carbon dioxide is carried back to the lungs, first by venules, then veins, then one of the two venae cavae, to be exchanged with oxygen. As we exhale, we release that carbon dioxide back into the air.

Video Attribution

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UNIT 9 PREFIXES

Prefix	Meaning
a-	not
anti-	against (an action)
brady-	slow
dys-	bad
em-	in
endo-	inside
eu-	good; well-
ex-	out, off, from, beyond
hyper-	too much
hypo-	too little
in-	in, into
inter-	between, among
ortho-	straight; at right angles; true
pan-	all; everything or everyone
poly-	many; too much
re-	again
tachy-	fast

UNIT 9 ROOTS

Root Word	Definition
adenoid	a collection of immune tissue at the top back of the oral cavity
alveol	a small hollow space
apneustic	abnormal breathing pattern: deep, gasping inspiration, followed by a pause, followed by a brief expiration
asthma	lung disease causing shortness of breath
ausculta	listening carefully
bronch bronch/i bronch/i/o bronch/o	tubes joining the trachea to the bronchioles
bronchiol	tubes joining the bronchi to the alveoli
capn capn/o	carbon dioxide
carb	carbon (as in carbon dioxide)
card/i/o	heart
carina	the keel of a ship
chylo	lymph fluid
concha	shell; the curved bones of the respiratory sinus walls
coni	dust
cost cost/o	rib
cyst	fluid-filled space
diaphragm	muscular sheet separating the thoracic and abdominal cavities
dilator	a muscle or drug that increases the diameter of an opening
fibr	fiber
gastr/o	stomach
glott	the vocal cords and the opening between them
hem/o	blood
hilum	inner curve of a bean-shaped organ like the lung
kinet	movement
laryng laryng/o larynx	the vocal apparatus; voice-box

Root Word	Definition
lob lobe lob/o	one of the two (left) or three (right) major divisions of the lung
loquy	speak
lung	lung
melan	black
mesotheli	flat cells of the pleural membrane between the lungs and chest wall
muco	mucus
nares	nostrils
naso	nose
nebul	cloud
nerve	nerve
obstruct	to stand in the way
oro	mouth
ortho	straight; at right angles; true
ox ox/i	oxygen
palato	palate; the roof of the mouth
pect pector pector/i	chest
percus	hit hard or beat
perfus	blood moving through a tissue
pharyng pharyng/o pharynx	the oral cavity (oropharynx) and nasal cavity (nasopharynx) and the place where they join (laryngopharynx)
phon phon/a	sound
phren phren/o	diaphragm
pleur pleura pleur/o	membrane between the lungs and chest wall
pnea	breath, respiration

Root Word	Definition
pneum pneum/at pneum/ato pneum/o pneum/on	respiration
ptysis	the act of spitting
pulm pulm/o pulm/o/n pulm/o/ n/o	lungs
py py/o	pus
rhin rhin/o	nose
rhythm	regular beat
sept	wall between parts of an organ
sinus sinus/o	open space
somn somn/o	sleep
spasm	involuntary contraction
spir spir/a spir/at spir/o	breath
sputum	that which is brought up from the lungs
stern stern/o	sternum; breastbone
syncope	fainting
tel	end
thorac thorac/o thorax	chest
trach trach/e trach/e/o	the tube between the pharynx and lungs; windpipe
tuber	root vegetable, like a potato

Root Word	Definition
turbin	spinning wheel with vanes
tuss	cough
ventil ventil/a	movement of air

UNIT 9 SUFFIXES

Suffix	Definition
-al	pertaining to
-algia	pain
-ant	word ending that indicates a substance having that property
-ar	pertaining to
-ary	pertaining to
-ate	shaped like
-ation	word ending that turns a verb into a noun
-cele	hernia
-centesis	puncture
-dynia	pain
-ectasis	dilation
-ectomy	cutting out of
-ema	Greek ending which puts "to be" before the word
-emia	present in blood, indicating disease
-genic	giving rise to
-gram	a record of a test or procedure
-graphy	a test or procedure
-ia	disease
-ic	pertaining to
-icity	condition or quality of being the thing indicated

Suffix	Definition
-ion	word ending that turns a verb into a noun
-is	anatomical structure
-itis	inflammation
-ive	to go
-izer	device that performs the action indicated
-logist	one who studies
-logy	study of
-lytic	substance which breaks down
-malacia	softening due to disease
-meter	instrument for measuring
-metry	the act of measuring
-ole	diminutive suffix
-oma	tumor
-ory	pertaining to
-osis	process
-pexy	surgical fixation
-plasty	surgical revision
-plegia	inability to move
-ptosis	drooping

Suffix	Definition
-rrhagia	bursting forth in disease
-rrhea	abnormal flow
-scope	instrument for looking inside
-scopy	the process of looking inside
-sion	word ending that turns a verb into a noun
-spasm	involuntary contraction
-stenosis	abnormal narrowing of an opening
-stomy	creating an opening with surgery
-tion	word ending that turns a verb into a noun
-tomy	cutting
-us	anatomical structure word ending

UNIT 9 ABBREVIATIONS

Abbreviation	Meaning
ABG	arterial blood gas
ARDS	acute respiratory distress syndrome
BiPAP	bi-level positive airway pressure (treatment for one form of sleep apnea)
BPD	bronchopulmonary dysplasia
BS	breath sounds
Bx	biopsy
CF	cystic fibrosis (a genetic disorder affecting the lungs and digestive system)
CO ₂	carbon dioxide
COPD	chronic obstructive pulmonary disease
CPAP	continuous positive airway pressure (treatment for sleep apnea)
CPR	cardiopulmonary resuscitation
CXR	chest X-ray
DPT	diphtheria pertussis tetanus
ET	endotracheal
ETT	endotracheal tube
IRDS	infant respiratory distress syndrome
LTB	laryngotracheobronchitis
NG	nasogastric
O ₂	oxygen
OSA	obstructive sleep apnea
PE	pulmonary embolism
PFT	pulmonary function test
PSG	polysomnography
SIDS	sudden infant death syndrome
SOB	shortness of breath
T&A	tonsillectomy and adenoidectomy
TB	tuberculosis
URI	upper respiratory infection
V/Q	ventilation-perfusion ratio

UNIT 9 WORD LIST

Word	Definition
acidosis	condition of high acid [in the blood]
adenoidectomy	surgical removal of an adenoid gland
adenoiditis	inflammation of an adenoid gland
alkalosis	condition of high alkali or low acid [in the blood]
alveolar	pertaining to the air sacs of the lungs
alveolitis	inflammation of the air sacs of the lungs
alveolus	an air sac of the lungs
antitussive	a drug that suppresses coughing
apnea	a condition where the patient repeatedly stops breathing at night
apneustic	pertaining to a part of the brain that triggers an apneustic breathing pattern
asthma	autoimmune disease causing inflammation of the air passages
atelectasis	alveoli fail to dilate, making breathing impossible; collapsed lung
auscultation	listening to breath sounds through a stethoscope
bradypnea	slow breathing
bronchiectasis	dilation of the bronchi
bronchiogenic	cancer that arises from the bronchi
bronchiole	air passage that connects bronchi to alveoli
bronchiolectasis	dilation of the bronchioles
bronchiolitis	inflammation of the bronchioles
bronchiostenosis	narrowing of the bronchi
bronchitis	inflammation of the bronchi
bronchodilator	drug that widens the bronchi
bronchoplasty	surgical revision of the bronchi
bronchorrhea	flow of liquid from the bronchi
bronchoscope	an instrument for looking into the bronchi
bronchoscopy	looking into the bronchi
bronchospasm	abnormal contraction of the bronchi
bronchus	one of the tubes that connect the trachea to the bronchioles
capnography	a test of the carbon dioxide concentration in the breath
capnometer	instrument for measuring carbon dioxide concentration

Word	Definition
cardiopulmonary	pertaining to the heart and lungs
carina	the anatomical feature associated with the division of the trachea into two primary bronchi (literally, "a boat keel")
caseous	cheese-like
chylothorax	accumulation of lymphatic fluid in the pleural space
conchae	the interior walls of the nasal sinuses that resemble the chambers of a conch shell
costophrenic	pertaining to the ribs and diaphragm
cystic fibrosis	a genetic condition that results from a mutation in a chloride channel protein; affects the lungs and digestive system with thick, sticky mucus
diaphragmatocele	herniation of the diaphragm
dysphonia	abnormal speech
dyspnea	abnormal breathing
empyema	pus in the pleural space
endoscope	an instrument for looking inside [the respiratory passages]
endotracheal	tube inserted into the trachea
epistaxis	nosebleed
eupnea	normal breathing
expectorant	drug which aids in thinning mucus secretions
expectoration	the process of spitting
expiration	breathing out
glottis	the vocal cords and the opening between them
hemoptysis	coughing blood
hemothorax	blood in the pleural space
hilum	the inner curve of a bean-shaped organ such as the lungs
hypercapnia	too much carbon dioxide in the blood
hypercarbia	too much carbon dioxide in the blood
hyperpnea	abnormally rapid breathing
hyperventilate	to breathe too rapidly
hyperventilation	the result of breathing too rapidly
hypocapnia	too little carbon dioxide in the blood
hypocarbia	too little carbon dioxide in the blood

Word	Definition
hypopnea	abnormally slow breathing
hypoventilate	to breathe too slowly
hypoventilation	the result of breathing too slowly
hypoxemia	low oxygen levels in the blood
hypoxia	low oxygen levels
innervation	the connection of nerves to an organ
inspiration	breathing in
intercostal	between the ribs
intubate	insert an endotracheal tube
kinetic	pertaining to movement
laryngectomy	surgical removal of the larynx
laryngitis	inflammation of the larynx
laryngopharynx	the portion of the pharynx that surrounds the larynx
laryngoplasty	surgical revision of the larynx
laryngospasm	abnormal contraction of the larynx
laryngotracheobronchitis	inflammation of the larynx, trachea, and bronchi
larynx	the organ of speech; voice-box
lobe	one of the major divisions of the lung (two on the left, three on the right)
lobectomy	surgical removal of a lung lobe
lobotomy	cutting into a lung lobe
lung	the organ of respiration
mesothelioma	cancer of the pleural lining that usually results from asbestos exposure
mucolytic	drug that breaks apart mucus (i.e. thins the mucus)
nares	nostrils
nasoendoscope	an instrument for looking inside the nose and nasal sinuses
nasogastric	a tube that enters the nose and nasopharynx and is inserted into the stomach
nasopharyngoscope	an instrument for looking inside the nose and nasopharynx
nasopharynx	the portion of the pharynx that is within the nasal sinuses
nebulizer	device for breaking liquid into microscopic droplets (1/1000 to 5/1000 of a millimeter in diameter)

Word	Definition
obstructive	characterized by blockage of air flow
oropharynx	the portion of the pharynx that is within the oral cavity
orthopnea	shortness of breath when lying down
oximetry	measurement of oxygen
palatoplasty	surgical revision of the palate
pansinusitis	inflammation of all nasal sinuses
pectoriloquy	abnormal resonance of the voice through lung structures, heard through a stethoscope (literally, "chest talking")
pectus carinatum	deformity of the chest with the sternum and ribs pushed outward, like a boat keel
pectus excavatum	deformity of the chest with the sternum and ribs pushed inward
percussion	tapping the chest
perfusion	passing liquid (such as blood) through an organ
pharyngitis	inflammation of the pharynx
pharyngostenosis	narrowing of the pharynx
pharynx	formed from the joining of the nasal and oral cavities; divides into the esophagus and trachea inferiorly
phonation	the act of making vocal sounds
phrenic	pertaining to the diaphragm
phrenoplegia	paralysis of the diaphragm
phrenoptosis	drooping of the diaphragm
phrenospasm	abnormal contraction of the diaphragm
pleura	the membrane lining the space between the lungs and chest wall
pleural	the space between the lungs and chest wall
pleuralgia	pain in the pleural membrane
pleurectomy	surgical removal of the pleura
pleurisy	inflammation of the pleura
pleuritis	inflammation of the pleura
pleurodynia	pain in the pleural membrane
pleuropexy	surgical fixation of the pleura
pneumatocele	herniation of the lungs
pneumoconiosis	condition of dust in the lungs

Word	Definition
pneumohemothorax	air and blood in the pleural space
pneumomelanosis	black spots in the lungs due to the accumulation of coal dust or carbon particles
pneumonectomy	surgical removal of the lung
pneumonia	inflammation of the lungs (usually infectious)
pneumonitis	inflammation of the lungs (usually non-infectious)
pneumotaxic	pertaining to movement of the lungs
pneumothorax	air in the pleural space (leads to a collapsed lung, or atelectasia)
polysomnography	test which uses several modalities (EEG, pulse ox, pulse rate, respiratory rate) during sleep
pulmonary	pertaining to the lungs
pulmonologist	a person who studies the lungs
pyothorax	pus in the pleural space
respiration	the process of exchanging gases (between room air and the blood in the lungs; between blood and the extracellular fluid in the tissues)
respiratory	pertaining to the process of exchanging gases
retraction	the process of pulling back
rhinitis	inflammation of the nasal passages
rhinoplasty	surgical revision of the nose ("a nose job")
rhinorrhagia	a profuse nosebleed
rhinorrhea	flow of liquid from the nose ("a runny nose")
rhythmicity	the quality of being rhythmic; having a regular beat
septectomy	surgical removal of the wall between the nasal sinuses
sinoatrial node	primary pacemaker, whose electrical activity sets a pace for the heartbeat
sinus	a large cavity within a bone or other tissue
sinusitis	inflammation of the nasal sinuses
sinusotomy	cutting into the nasal sinuses
spirogram	a record of breathing
spirometry	the process of measuring the volume and rate of the breath
sputum	thick fluid expelled from the lungs
sternocostal	pertaining to the sternum and ribs
syncope	fainting; loss of consciousness

Word	Definition
tachypnea	fast breathing
thoracalgia	pain in the thorax
thoracentesis	puncture of the thorax with removal of fluid
thoracic	pertaining to the thorax
thoracocentesis	puncture of the thorax with removal of fluid
thoracoplasty	surgical revision of the thorax
thoracostomy	creating an opening in the thorax
thoracotomy	cutting into the thorax
thoracoscopy	looking inside the thorax
trachea	tube connecting the laryngopharynx to the bronchi; windpipe
tracheitis	inflammation of the trachea
tracheomalacia	abnormal softening of the trachea
tracheostenosis	narrowing of the trachea
tracheostomy	creating an opening in the trachea (e.g. to restore breathing in a choking person)
tracheotomy	cutting into the trachea
tubercle	lesion characteristic of tuberculosis
tuberculosis	lung disease caused by <i>Mycobacterium tuberculosis</i>
turbinate	shaped like a turbine (round wheel with vanes)
ventilation	the process of moving air in and out of the body

HOW THE UNIT 9 WORD LIST WAS BUILT

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix	Word
			acid		osis	acidosis
			adenoid		ectomy	adenoidectomy
			adenoid		itis	adenoiditis
			alkal[i]		osis	alkalosis
Latin alveolus, "tray, trough, basin; bed of a small river; small hollow or cavity"			alveol		ar	alveolar
			alveol		itis	alveolitis
			alveol		us	alveolus
	anti		tuss		ive	antitussive
	a		pnea			apnea
a- + Greek pneustikós, a pattern of breathing with deep inspiration, a pause, then a brief expiration	a		pneustic			apneustic
Greek asthma, "shortness of breath, panting"			asthma			asthma
	a		tel		ectasis	atelectasis
Latin auscultationem, "listen attentively to"			auscult		ation	auscultation
	brady		pnea			bradypnea
			bronchi		ectasis	bronchiectasis
			bronchio		genic	bronchiogenic
			bronchi		ole	bronchiole
			bronchiol		ectasis	bronchiolectasis
			bronchiol		itis	bronchiolitis
			bronchio		stenosis	bronchiostenosis
			bronch		itis	bronchitis
			broncho	dilator		bronchodilator
			broncho		plasty	bronchoplasty
			broncho		rrhea	bronchorrhea
			broncho		scope	bronchoscope

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix	Word
			broncho		scopy	bronchoscopy
			broncho	spasm		bronchospasm
			bronchus			bronchus
			capno		graphy	capnography
			capno		meter	capnometer
			card/i/o	pulm/o/n	ary	cardiopulmonary
			carina			carina
structure that protrudes like the keel of a ship						carinatum
cheese-like; note association with Spanish queso						caseous
			chylo	thorax		chylothorax
			concha			conchae
			costo	phren	ic	costophrenic
			cyst		ic	cystic
"ato" here just for the sound of it			diaphragm	ato	cele	diaphragmatocele
	dys		phon		ia	dysphonia
	dys		pnea			dyspnea
From Greek emphysema "to inflate"			emphysema			emphysema
	em		py		ema	empyema
	endo				scope	endoscope
	endo		trache		al	endotracheal
From Greek epistazo meaning "to drip or upon", a nose bleed			epistaxis			epistaxis
	eu		pnea			eupnea
Latin, from which we get English ex- + -cav[e]- + -ate						excavatum
	ex		pector		ant	expectorant

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix	Word
	ex		pector		ation	expectoration
	ex		[s]pir[e]		ation	expiration
			fibr		osis	fibrosis
			glott[a]		is	glottis
			hem/o	ptysis		hemoptysis
			hem/o	thorax		hemothorax
			hilum			hilum
	hyper		capn		ia	hypercapnia
	hyper		carb		ia	hypercarbia
	hyper		pnea			hyperpnea
	hyper		ventil		ate	hyperventilate
	hyper		ventil		ation	hyperventilation
	hypo		capn		ia	hypocapnia
	hypo		carb		ia	hypocarbia
	hypo		pnea			hypopnea
	hypo		ventil		ate	hypoventilate
	hypo		ventil		ation	hypoventilation
	hyp[o]		ox		emia	hypoxemia
	hyp[o]		ox		ia	hypoxia
	in		nerv[e]		ation	innervation
	in		spira[re]		tion	inspiration
	inter		cost		al	intercostal
	in		tub		ate	intubate
			kinet[os]		ic	kinetic
			laryng		ectomy	laryngectomy
			laryng		itis	laryngitis
		laryngo	pharynx			laryngopharynx
			laryngo		plasty	laryngoplasty
			laryngo	spasm		laryngospasm

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix	Word
		laryngo	tracheo	bronch	itis	laryngotracheobronchitis
			larynx			larynx
			lobe			lobe
			lob		ectomy	lobectomy
			lobo		tomy	lobotomy
			lung			lung
			meso	theli	oma	mesothelioma
			muco		lytic	mucolytic
			nares			nares
	naso		endo		scope	nasoendoscope
	naso		gastr/o		ic	nasogastric
	naso		pharyngo		scope	nasopharyngoscope
	naso		pharynx			nasopharynx
Latin nebula, "cloud"			nebul		izer	nebulizer
			obstruct		ive	obstructive
	oro		pharynx			oropharynx
Greek orthópnoia, "breathing only in an upright posture"		ortho	pnea			orthopnea
			oxi		metry	oximetry
			palato		plasty	palatoplasty
	pan		sinus		itis	pansinusitis
Latin pectus, "chest" + loquor, "speak"			pectori	loquy		pectoriloquy
Latin pectus, "chest"			pectus			pectus
Latin "to strike"			percus		sion	percussion
			perfus[us]		ion	perfusion
			pharyng/o		itis	pharyngitis
			pharyngo		stenosis	pharyngostenosis
			pharynx			pharynx
			phon/a		tion	phonation

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix	Word
The proper term in Latin is pleuritis, but medicine adopted this word directly from the French pleurésie			phren		ic	phrenic
			phreno		plegia	phrenoplegia
			phreno		ptosis	phrenoptosis
			phreno		spasm	phrenospasm
			pleura			pleura
			pleur		al	pleural
			pleur		algia	pleuralgia
			pleur		ectomy	pleurectomy
			pleur		isy	pleurisy
			pleur		itis	pleuritis
			pleuro		dynia	pleurodynia
			pleuro		pexy	pleuropexy
			pneumato		cele	pneumatocele
			pneumo	coni	osis	pneumoconiosis
		pneum/o	hem/o	thorax		pneumohemothorax
Greek pneumo- + táxis, "arrangement"			pneumo	melan	osis	pneumomelanos
			pneumon		ectomy	pneumonectomy
			pneumon		ia	pneumonia
			pneumon		itis	pneumonitis
			pneumo		tax[is]	pneumotaxic
			pneumo	thorax		pneumothorax
		poly	somno		graphy	polysomnography
			pulmon		ary	pulmonary
			pulmono		logist	pulmonologist
			pyo	thorax		pyothorax
Latin respiratio, "breathing"	re		spira		tion	respiration
			spirat		ory	respiratory

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix	Word
Latin sputum, "spittle"	re		trac		tion	retraction
			rhin		itis	rhinitis
			rhino		plasty	rhinoplasty
			rhino		rrhagia	rhinorrhagia
			rhino		rrhea	rhinorrhea
			rhythm[us]		icity	rhythmicity
			sept		ectomy	septectomy
			sinus			sinus
			sinus		itis	sinusitis
			sinuso		tomy	sinusotomy
			spiro		gram	spirogram
			spiro		metry	spirometry
			sputum			sputum
			sterno	cost	al	sternocostal
			syncope			syncope
	tachy		pnea		tachypnea	
			thorac		algia	thoracalgia
			thora[c]		centesis	thoracentesis
			thorac		ic	thoracic
			thorac/o		centesis	thoracocentesis
			thorac/o		plasty	thoracoplasty
			thorac/o		stomy	thoracostomy
			thorac/o		tomy	thoracotomy
			thorac/o		scopy	thoracoscopy
			trachea			trachea
			trache		itis	tracheitis
			trache/o		malacia	tracheomalacia
			trache/o		stenosis	tracheostenosis
			trache/o		stomy	tracheostomy

Etymology	Prefix	"Pre-Root"	Root Root	"Post-Root"	Suffix	Word
Literally, "little potato". Disease characterized by the presence of inflammatory nodules that resembled little potatoes. Note the addition of a letter "u" just to make the word sound better.			trache/o		tomy	tracheotomy
			tuber		[i]cle	tubercle
			tubercul[e]		osis	tuberculosis
			turbin		ate	turbinate
			ventila		tion	ventilation

PART X

UNIT 10: THE DIGESTIVE SYSTEM

UNIT 10 INTRODUCTION

The Digestive System is a series of organs within the body that are responsible for the transportation, digestion, absorption, and elimination of food. Absorbed nutrients are passed on to the vascular system for distribution throughout the body. Waste products and unusable substances are eliminated from the body in the form of feces.

Objectives

After you complete this unit, you will be able to:

- Identify prefixes, word roots, and suffixes related to the diagnosis and treatment of gastrointestinal diseases.
- Recall abbreviations related to the diagnosis and treatment of gastrointestinal diseases.

THE DIGESTIVE SYSTEM

Only [Med Terms] in the Building Episode 10

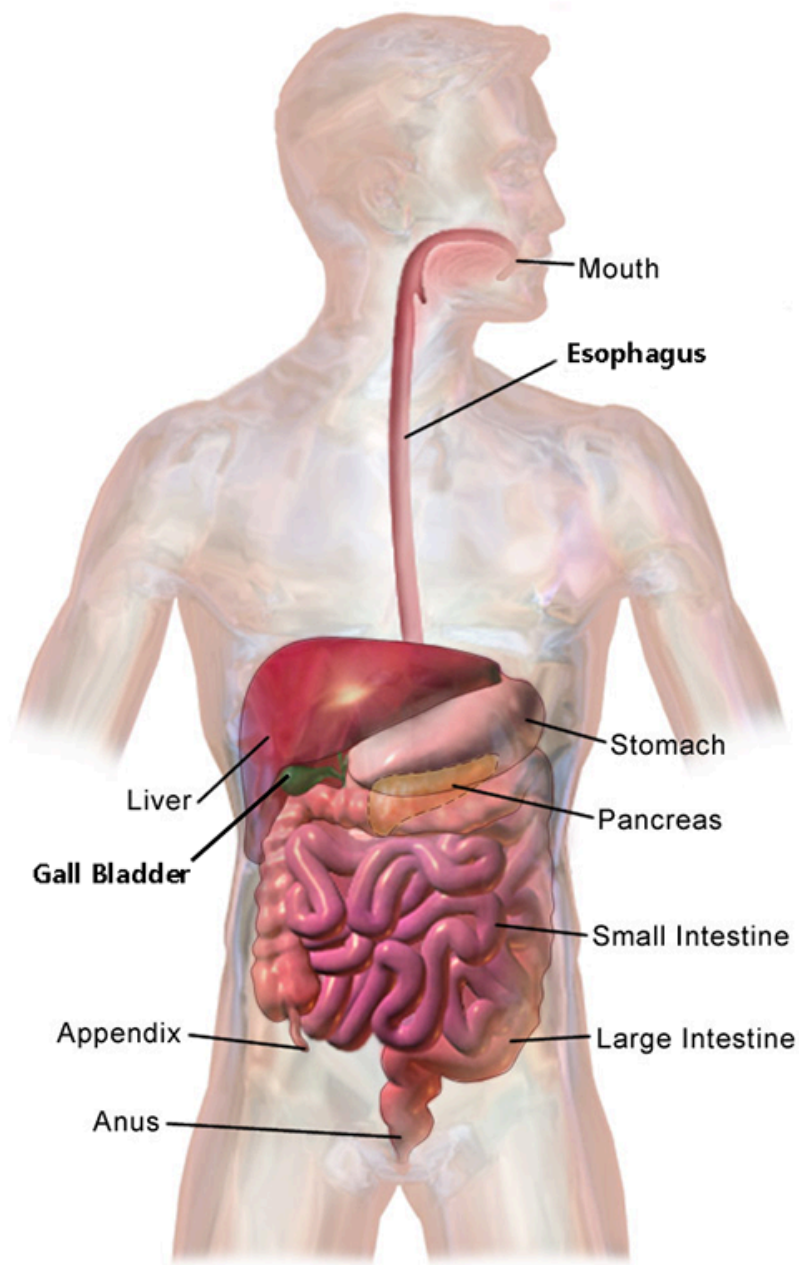


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The digestive system has but a single overarching function: to take in food, break it down into nutrients that are absorbed by the stomach and intestines, and then excrete any leftover waste as feces.

In order to accomplish this complex function, the digestive system has two basic subdivisions: the gastrointestinal tract and the accessory organs.



Digestive System

Gastrointestinal tract organs include:

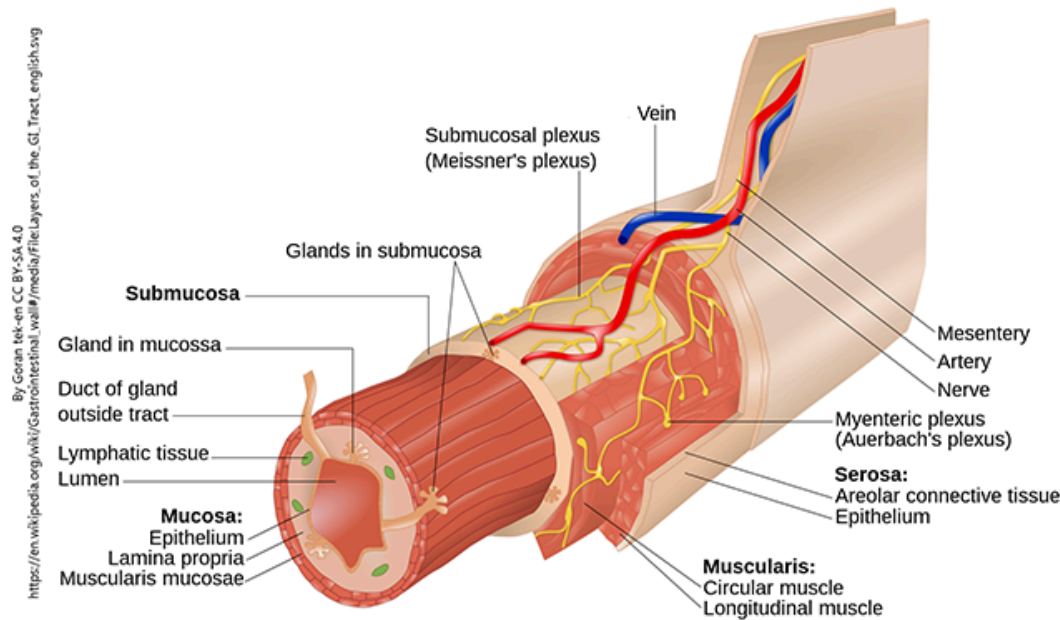
- mouth
- pharynx
 - oropharynx
 - laryngopharynx

- esophagus
- stomach
- small intestine
 - duodenum
 - jejunum
 - ileum
- large intestine
 - [vermiform] appendix
 - cecum
 - ascending colon
 - transverse colon
 - descending colon
 - sigmoid colon
 - rectum
- anus

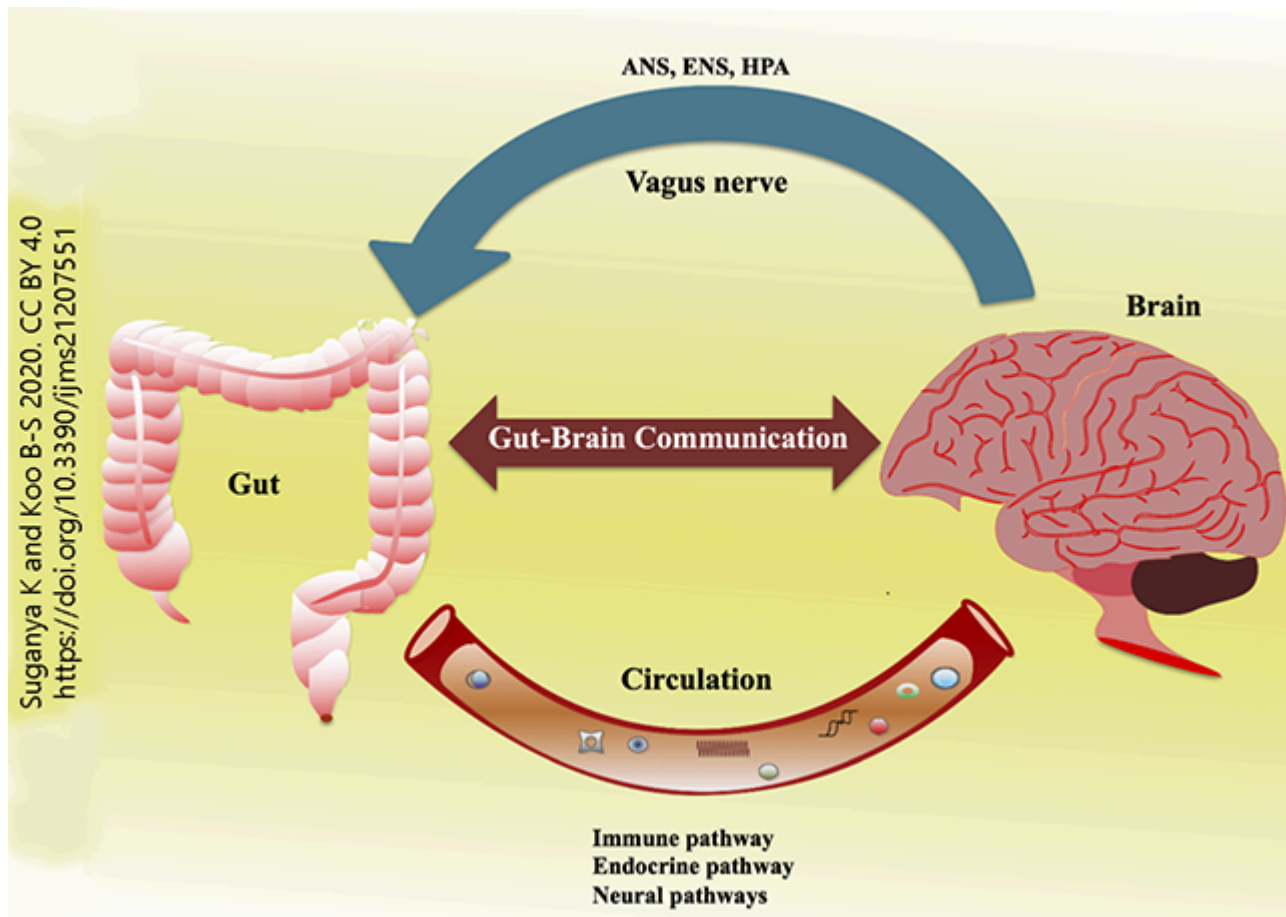
Accessory organs include:

- teeth
- tongue
- salivary glands
- liver
- gall bladder
- pancreas

In keeping with its straightforward function, the microscopic structure of the stomach and intestines are almost identical throughout the gut tube. There are variations, for example, the addition of acid secreting cells in the stomach, and the specializations involved in extracting water in the large intestine, but the basic structure is the same, as shown in the diagram below.

Unit 10 figure 2
Layers of the
Alimentary canal

In this diagram, note the presence of two plexi: the *submucosal plexus* (Meissner's plexus) and the *myenteric plexus* (Auerbach's plexus). The structures are part of an important control system: the *enteric nervous system*. There is semi-independent control of gut functions (such as *peristalsis*, the milking action of the gut wall that moves substances from *esophagus* to *anus*). The enteric nervous system (ENS) both receives signals from, and sends signals to, the central nervous system and the autonomic nervous system (ANS: the sympathetic and parasympathetic divisions we studied in Unit 6). The hypothalamic-pituitary-adrenal (HPA) axis sends information to the gut as well. These nerve signals primarily travel along the tenth cranial nerve, the vagus nerve. The immune system also contributes to, and is affected by, the function of the gut. Finally, there is a gut microbiome, consisting of all the bacteria which exist in a cooperative relationship with the human cells. For example, bacteria make vitamin K, a clotting factor that cannot be made by human cells. All these relationships are encompassed in the gut-brain-immune pathways diagrammed here.



Unit 10 figure 3 Gut Brain Axis

Video Attribution

- Justin Burr, Jim Hutchins & Maddison Johnston. (2024, December 20). *Only [Med Terms] in the Building Episode 10*. [Video]. Kaltura.

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UNIT 10 PREFIXES

Prefix	Meaning
a-	not
an-	
ad-	going into
ant-	against (an action)
anti-	
con-	together, with
de-	from, down, out
dia-	through
dys-	bad
en-	inside; put in or on (note additional new meaning)
epi-	on top of
eu-	good, well-
hemi-	half
hyper-	too much
hypo-	too little
in-	in, into
mal-	badly
ortho-	straight; at right angles; true
para-	beside; near
peri-	around
re-	again
sub-	an underlayer; support
un-	not

UNIT 10 ROOTS

Root Word	Definition
abdomin/o	abdomen
absorption	taking up nutrients
acid	a chemical that gives off hydrogen ions
aden aden/o	gland
aero	air
agog	causing secretion
agon	main actor; driver
aliment	food
amel	covering
amyl	chemical name for starch
ang ang/i ang/i/o	vessel
an/o anus	anus
ascites	abdominal swelling
bar	weight, pressure
bile bil bil/i	bile
bladder	bladder
bucca	cheek
canal	canal
carboxy[l]	chemical group name
carcin	cancer
cardia	heart
cathar/t	purge
cauda	tail
cec[um]	dead-end pouch between ileocecal valve and appendix
celi celi/o	bowels
cement	cement

Root Word	Definition
chol chol/e	bile
chondr	cartilage
chyme chym/o	liquid formed in the intestine from partially digested food
cirrh	inflammation of connective tissue
clast	breakdown
col col/o colon colon/o	colon
constipa	crowded together, as feces in the intestines
cyst cyst/o	bladder; fluid-filled space
cyte	cell
decidu	destined to fall
deglutition	swallowing
dent dent/i	tooth
digest	digest
diverticul	pocket
doch/o	common bile duct
dont	tooth
duoden duoden/o	the first segment (of three) in the small intestines
elast	elastic
emesis	vomiting
endo	inside
enter enter/o	intestine
esophag esophag/o	tube between the mouth and stomach
falc	sickle (shape)
feces	feces

Root Word	Definition
filum	thread
fist	abnormal opening between two spaces
flatus	fart
flux	flow
folium	leaf
frice	substance used for scrubbing
fundus	the bottom of a bottle-shaped object
fungi	fungi
gall	bile
gastr gastr/o	stomach
ghrel	acronym for "growth hormone releasing"
gingiv gingiv/o	gums
gloss gloss/o	tongue
gluc glyc	sugar
haustum	small pouch (pl. haustra) which make the large intestine look segmented
hem/o	blood
hepat hepat/o	liver
hepatic/o	hepatic duct
hernia herni/o	abnormal outpocketing of an organ
hydr	water
icterus	jaundice
il il/e/o il/i	the third and last segment in the small intestines
inguin	groin
insula	island; the hormone insulin, which is made in the pancreatic islets
intestin	intestine

Root Word	Definition
islet	small island of tissue in the pancreas
jaundice	yellow color of skin
jejun jejun/o	the second segment (of three) in the small intestines
jug[are]	join
kin	move
labi[a]	lips
lact	milk
lamina propria	"one's own layer", one of the histological layers of the digestive tract
lapar/o	flank, loin
ling/u	tongue
lip	fat
lith lith/o	stone
liver	liver
lob[e]	lobe
lumb	lower back
lyso	breakdown
macro	large
mes[os]	in the middle
mucosa	layer of tissue with mucus-secreting cells
muscular	muscular
my my/o my/o/s	muscle
myc	fungus
nas/o	nose
occlus[us]	position of teeth with mouth closed
odont odont/o	tooth
or	mouth
oti[s]	ear

Root Word	Definition
oxy	oxygen
pancr pancr/e pancr/e/at pancr/e/at/o	pancreas
pariet	the outer wall of the two-walled membranes around body organs
peps peps/in peps/in/o	digestion
peptid	digestion
periton periton/eo	membrane lining the abdominal cavity
pernici[es]	gradual harm
phag	eat
plicae circulares	literally "circular pleats", folds in the intestinal membrane (sing. plica circularis)
proct proct/o	anus
propuls	movement driven by muscular action
pylor pylorus	region of stomach where it meets the duodenum
quadrate	four-sided
rect rect/o	final segment of large intestine
rub[er]	red
saliva	saliva
scend	climb (up or down)
scleros	hard
secret[ion]	release from a cell or the body
segment	segment
serosa	membrane lining the contents of the abdominal cavity
sial sial/o	saliva
sigm[a]	Greek letter which resembles English "s"
sigmoid/o	shaped like the letter s

Root Word	Definition
splen splen/o	spleen
stalsis	contraction
steato	fatty matter
sten	blockage
sterco	feces
stomat stomat/o	mouth
sucr[e]	sugar
Taenia	genus of tapeworms
tongue	tongue
trich	hair
trypsin	digestive enzyme; one of the earliest enzymes discovered
ulcer	open sore
umbilic	cord connecting the placenta to the baby
vagin	infolded body part
vallate	like a moat
verd	green
vermi[s]	worm
villus	small, hair-like feature of the intestinal surface
vit[a]	related to life
volunt[us]	of one's free will
zyme	ending of the word "enzyme"

UNIT 10 SUFFIXES

Suffix	Definition
-ac	pertaining to
-al	pertaining to
-algia	pain
-amine	chemical group containing nitrogen
-ar	pertaining to
-ary	pertaining to
-ase	name for an enzyme
-ate	shaped like
-ated	having that quality
-ation	word ending that turns a verb into a noun
-ative	word ending that turns a verb into an adjective
-cele	hernia
-centesis	puncture
-crine	secretion
-dynia	pain
-eal	pertaining to
-ectasis	dilation
-ectomy	cutting out of
-emia	present in blood, indicating disease
-form	shaped like
-gen	arising from
-genesis	the process of creating
-gram	a record of a test or procedure
-graphy	a test or procedure
-ia	disease
-iac	pertaining to
-ial	pertaining to
-iasis	presence of an abnormality
-ic	pertaining to
-ics	

Suffix	Definition
-in -in/o	chemical name ending for a protein
-ing	verb ending that makes the continuous tense (ongoing process)
-ion	word ending that turns a verb into a noun
-is	anatomical structure
-ist	someone who studies
-istry	the practice of studying
-itis	inflammation
-ive	to go
-lith	stone
-logist	one who studies
-logy	study of
-malacia	softening due to disease
-megaly	enlargement due to disease
-oid	looks like; resembles
-oma	tumor
-orus	anatomical structure
-osis	process
-ous	word ending that turns a noun into an adjective
-paresis	weakness
-pathy	disease process
-pexy	surgical fixation
-phony	sound-making
-plasty	surgical revision
-plegia	weakness
-ptosis	drooping
-rrhaphy	suturing
-rrhea	abnormal flow
-scope	instrument for looking inside
-scopy	the process of looking inside

Suffix	Definition
-stomy	creating an opening with surgery
-tion	word ending that turns a verb into a noun
-tomy	cutting
-tripsy	rubbing, friction, pounding
-ula -ule	diminutive suffix
-um	anatomical structure
-uronic	chemical name ending
-y	disease

UNIT 10 ABBREVIATIONS

Abbreviation	Meaning
ANS	autonomic nervous system
Ba	barium
BE	barium enema
BM	bowel movement
CCK	cholecystokinin
EGD	esophagogastroduodenoscopy
ENS	enteric nervous system
ERCP	endoscopic retrograde cholangiopancreatography
EUS	endoscopic ultrasound
FOBT	fecal occult blood test
FBS	fasting blood sugar
GERD	gastroesophageal reflux disease
GI	gastrointestinal
GTT	glucose tolerance test
HPA	hormonal connections between the hypothalamus, pituitary, and adrenal gland (HPA axis)
IBS	irritable bowel syndrome
LFT	liver function test
LLQ	left lower quadrant
LUQ	left upper quadrant
N&V	nausea and vomiting
NGT	nasogastric tube
NPO	nothing by mouth (Latin: nihil per os)
PUD	peptic ulcer disease
RLQ	right lower quadrant
RUQ	right upper quadrant

UNIT 10 WORD LIST

Word	Definition
abdominocentesis	puncture of the abdominal wall
abdominoplasty	surgical revision of the abdomen
adenocarcinoma	glandular cancer
aerodontalgia	pain in the teeth when air is blasted at them
alimentary	pertaining to the process of nourishment
amylase	enzyme which breaks down starch
anal	pertaining to the anus
anhydrase	enzyme which helps form stomach acid
anophony	the sound made by the expulsion of flatus (i.e. a fart)
anoplasty	surgical revision of the anus
anosigmoidoscopy	the process of looking into the anus and sigmoid colon
antacid	drug that reduces the acidity of the stomach
antiemetic	drug that reduces vomiting
anus	the posterior end of the digestive tract
aphagia	not eating
ascending	the segment of the large intestine on the right side between the appendix and the transverse colon
ascites	swelling of the abdomen, usually because of an underlying disease
bariatrics	medical specialty that focuses on weight control
bile	fluid made by the cells of the liver and stored in the gall bladder; when released into the duodenum, it helps digest fats
biligenesis	the process of creating bile
bilirubin	breakdown product of red blood cells; named for its reddish color
biliverdin	breakdown product of red blood cells; named for its greenish color
buccal	pertaining to the cheek
canaliculus	small canals between liver cells where bile is excreted
carboxypeptidase	enzyme which breaks down proteins
cardia	the portion of the stomach where the esophagus enters
cathartic	drug used to purge the gastrointestinal tract; laxative
caudate	liver lobule shaped like a tail
celiac	pertaining to the abdomen

Word	Definition
celiomyositis	inflammation of the abdominal muscles
celiopathy	abdominal disease
celiotomy	cutting into the abdomen
cementum	mineralized tissue layer covering the root of a tooth
choangiogastrostomy	surgical creation of a communication between the bile duct and the stomach
cholangiogram	x-ray image of the bile ducts after ingestion of a x-ray dense dye
cholangiography	process of obtaining a cholangiogram
cholangioma	tumor of the bile ducts
cholangiopancreatography	process of obtaining an image of the bile ducts and pancreas
cholangitis	inflammation of the bile duct
cholecystalgia	pain in the gallbladder
cholecystectomy	surgical removal of the gallbladder
cholecystitis	inflammation of the gallbladder
cholecystogram	x-ray image of the gallbladder after ingestion of a x-ray dense dye
cholecystokinin	hormone that causes contraction of the gallbladder; also used as a neurotransmitter in the brain
choledochoceles	herniation of the common bile duct
choledochointerostomy	surgical creation of a communication between the common bile duct and the jejunum
choledocholithectomy	surgical removal of a stone in the common bile duct
choledocholithiasis	the presence of a stone in the common bile duct
choledochotomy	cutting into the common bile duct
cholelith	gallstone
cholelithiasis	the presence of a stone in the gall bladder
cholelithotomy	cutting into the gall bladder to remove a gallstone
cholelithotripsy	using sound waves to pound a gallstone into small fragments that can be passed normally
chyme	fluid found in the intestines that results from the partial digestion of food
chymotrypsin	enzyme that breaks down proteins
cirrhosis	inflammation of the connective tissue of the liver
colectomy	surgical removal of the colon
colitis	inflammation of the colon

Word	Definition
colon	large intestine
colonectomy	surgical removal of the colon
colonoscopy	the process of looking into the colon
colorectal	pertaining to the colon and rectum
colostomy	surgical creation of a communication between the colon and the outside of the body
colovaginal	pertaining to the colon and vagina; these two organs share a common wall that can be breached, resulting in a colovaginal fistula
conjugated	bilirubin that is made water-soluble so it may be carried in the bloodstream
constipation	inability to defecate
cystic	a disease that causes cysts (e.g. cystic fibrosis, which affects both the lungs and digestive tract)
deciduous	teeth that are normally lost (i.e. "baby teeth")
defecation	the process of removing feces from the body
deglutition	swallowing
dentalgia	pain in the teeth
dentifrice	toothpaste
dentin	hard bony tissue beneath the enamel that forms the bulk of the tooth
dentist	person who treats diseases of the oral cavity
dentistry	practice of treating diseases of the oral cavity
descending	the segment of the large intestine on the left side between the transverse colon and the sigmoid colon
diarrhea	abnormal flow of feces
digestive	about the act of digestion
diverticulitis	outpocketing of the intestines, with inflammation
diverticulosis	outpocketing of the intestines, without inflammation
duodenectomy	surgical removal of the duodenum
duodenitis	inflammation of the duodenum
duodenum	initial segment of the small intestine, between the stomach and jejunum
dysentery	disease involving problems with the intestines
dyspepsia	disease involving problems with the stomach
elastase	enzyme that breaks down elastin, a protein in connective tissue

Word	Definition
emesis	vomiting
enamel	outer coating of teeth
enterectomy	surgical removal of a segment of the intestines
enteric	pertaining to the intestines
enterocele	herniation of the intestines
enterocyte	intestinal cell
enterodynia	pain in the intestines
enteroendocrine	relating to the hormones which regulate digestion
enteropathy	disease of the intestines
enterorrhaphy	suturing the intestines
enterotomy	cutting into the intestines
epigastric	one of the nine abdominal regions, lying along the midline at the top of the abdomen
esophagalgia	pain in the esophagus
esophageal	pertaining to the esophagus
esophagectomy	surgical removal of a segment of the esophagus
esophagitis	inflammation of the esophagus
esophagogastroduodenoscopy	the process of looking into the esophagus, stomach, and initial segment of the intestines
esophagogastroplasty	surgical revision of the esophagus and stomach
esophagoscopy	the process of looking into the esophagus
euglycemia	condition of perfect blood sugar
eupepsia	condition of perfect digestion
falciform	liver ligament that is shaped like a sickle
fecal	pertaining to feces
feces	the waste expelled from the anus as the last step of digestion
filiform	the most common type of papillae on the tongue; shaped like hairs and do not contain taste buds
fistula	abnormal connection between two structures, such as the rectum and vagina
flatus	expulsion of intestinal gases from the anus
foliate	papillae on the tongue, with taste buds, that are leaf-shaped
fundus	the part of a bottle-shaped structure like the stomach that is opposite the opening

Word	Definition
fungiform	papillae on the tongue, with taste buds, that are mushroom-shaped
gallbladder	the organ where bile made by the liver is stored, until it is released into the duodenum via the common bile duct
gastralgia	pain in the stomach
gastrectomy	surgical removal of all or part of the stomach
gastric	pertaining to the stomach
gastritis	inflammation of the stomach
gastroduodenoscope	instrument for looking inside the stomach and duodenum
gastroduodenostomy	surgical creation of a communication between the stomach and duodenum
gastrodynia	pain in the stomach
gastroenteritis	inflammation of the stomach and intestines
gastroenterocolitis	inflammation of the stomach, intestines, and colon
gastroenterologist	medical specialist who studies the stomach and intestines
gastroenterology	medical specialty that focuses on the stomach and intestines
gastroenterostomy	surgical creation of a communication between the stomach and intestines
gastroesophageal	pertaining to the stomach and esophagus
gastrointestinal	pertaining to the stomach and intestines
gastrojejunostomy	surgical creation of a communication between the stomach and jejunum
gastromalacia	abnormal softening of the stomach due to disease
gastroparesis	weakness in the muscular contractions of the stomach
gastropexy	surgical fixation of the stomach
gastroplasty	surgical revision of the stomach
gastroscope	instrument for looking inside the stomach
ghrelin	hormone that affects appetite
gingivalgia	pain in the gums
gingivectomy	surgical removal of part of the gums
gingivitis	inflammation of the gums
gingivoglossitis	inflammation of the gums and tongue
gingivoplasty	surgical revision of the gums
gingivostomatitis	inflammation of the gums and mouth

Word	Definition
glossopathy	disease of the tongue
glossoplasty	surgical revision of the tongue
glossoplegia	weakness of the tongue muscles
glossorrhaphy	suturing the tongue
glossotomy	cutting into the tongue
glossotrichia	disease characterized by hairy appearance of the tongue
glucagon	hormone involved in glucose metabolism; causes glycogen to be broken down and glucose to be made from other sources, raising blood sugar
glucuronic acid	chemical group used to modify water insoluble compounds so that they are now water soluble and may be passed in urine
haustrum	periodic outpocketings of the colon that give it a segmented appearance
hematemesis	vomiting blood
heme	oxygen-holding chemical group; there are four heme groups per hemoglobin molecule
hemicolectomy	surgical removal of a segment of the colon
hemorrhoid	painful varicose veins in the anus
hemorrhoidectomy	surgical removal of a hemorrhoid
hepatectomy	surgical removal of the liver
hepatic	pertaining to the liver
hepaticogastrostomy	surgical creation of a communication between the hepatic duct and the stomach
hepaticotomy	surgical incision of the hepatic duct
hepatitis	inflammation of the liver
hepatocarcinoma	liver cancer
hepatocytes	cells of the liver
hepatoma	tumor of the liver
hepatomalacia	abnormal softening of the liver due to disease
hepatomegaly	abnormal enlargement of the liver
hepatopancreatic	pertaining to the liver and pancreas
hepatopexy	surgical fixation of the liver
hepatoptosis	displacement of the liver out of its normal position
hepatosclerosis	abnormal hardening of the liver due to disease
hepatosplenitis	inflammation of the liver and spleen

Word	Definition
hepatosplenomegaly	abnormal enlargement of the liver and spleen
hernia	displacement of an organ from its normal position
herniorrhaphy	suturing the site of a herniation (e.g. in the abdominal wall) to keep it from recurring
hyperemesis	abnormally frequent vomiting
hypochondriac	one of the nine abdominal regions, lying laterally at the top of the abdomen
hypogastric	one of the nine abdominal regions, lying along the midline at the bottom of the abdomen
hypoglossal	underneath the tongue
icterus	yellow color of the skin and eyes due to liver malfunction
ileitis	inflammation of the ileum
ileocecal	junction of the ileum and the cecum of the large intestine
ileocolitis	inflammation of the ileum and colon
ileocolostomy	surgical creation of a communication between the ileum and the colon
ileorrhaphy	suturing the ileum
ileostomy	surgical creation of a communication between the ileum and the body surface
ileotomy	cutting into the ileum
ileum	final segment of the small intestine, between the jejunum and large intestine
ingestion	taking food into the body
inguinal	pertaining to the groin
insulin	hormone involved in glucose metabolism; causes glucose to be moved from bloodstream into cells, lowering blood sugar
insulinotropic	drug which affects insulin levels
intestinal	pertaining to the intestines
islet	small collections of endocrine cells in the pancreas that manufacture insulin, glucagon, and other hormones
jaundice	yellow color of the skin and eyes due to liver malfunction
jejunitis	inflammation of the jejunum
jejunoileitis	inflammation of the jejunum and ileum
jejunorrhaphy	suturing the jejunum
jejunostomy	surgical creation of a communication between the jejunum and the body surface
jejunotomy	cutting into the jejunum

Word	Definition
jejunum	middle segment of the small intestine, between the duodenum and ileum
labial	pertaining to the lips
lacteal	lymph vessel which runs in the center of each intestinal villus and carries lipids into the bloodstream for processing by the liver and other organs
lamina propria	a layer specific to the digestive system; lies between the epithelium lining the intestinal lumen and the muscular layer underneath
laparocele	herniation of the abdominal wall
laparoenterostomy	surgical creation of a communication between the intestines and the body surface through the abdominal wall
laparoscope	instrument for looking inside the abdomen
laparoscopy	the process of looking into the abdomen through a small incision
laparotomy	cutting into the abdomen
lingual	pertaining to the tongue
lipase	enzyme which breaks down fats
liver	organ which has primary responsibility for making blood proteins; processing non-glucose sugars and toxins; managing lipids; converting blood breakdown products to bile
lobule	divisions of the liver or other organs with visible creases
lumbar	pertaining to the lower back
lumen	anatomical passageway; the open part of a tube-like organ
lysozyme	enzyme used within cells to break down toxins and other unwanted substances
macrocyte	abnormally enlarged red blood cell (see Unit 8); found in pernicious anemia
malabsorption	poor or altered absorption of nutrients, for example due to a genetic or metabolic condition
mesenteric	pertaining to the middle part of the intestines
mesial	used in place of medial to refer to teeth; towards the midline
mucosa	the epithelium, lamina propria, and muscular layer of the gut tube; collectively, the three mucus-secreting layers surrounding the lumen
muscularis	layer of the gut tube with prominent smooth muscle cells
myenteric	one of the plexi of the enteric nervous system
nasogastric	a tube introduced through the nose, pharynx, and esophagus to access the stomach
occlusal	tooth surface that contacts other teeth
odontalgia	pain in the teeth

Word	Definition
odontectomy	surgical removal of a tooth
odontoclasia	breakdown of teeth
odontodynia	pain in the teeth
oral	pertaining to the mouth cavity
orthodontic	pertaining to the straightening and spacing of teeth
orthodontist	dentist who specializes in straightening and spacing teeth
oxygenase	enzyme that adds an oxygen to help in processing of heme
pancreatoduodenectomy	surgical removal of the head of the pancreas and the duodenum to treat pancreatic cancer
pancreatolith	a stone in the pancreas
pancreatolithectomy	surgical removal of a stone from the pancreas
pancreatolithiasis	the presence of a stone in the pancreas
parietal	the layer of membranes lining the abdominal organs that is closest to the body wall
parotid	the largest of the salivary glands
pancreatography	process of imaging the pancreas
pepsin	the first enzyme discovered; found in the stomach, where it breaks down proteins
pepsinogen	protein precursor to pepsin; when it reacts with stomach acid, pepsin is produced
periodontal	pertaining to the connective tissue holding the teeth in place
periodontitis	inflammation of the connective tissue holding the teeth in place
peristalsis	autonomic "milking" movements of the muscular layer of the gut tube resulting in movement of substances from mouth to anus
peritoneoscopy	laparoscopy
peritoneotomy	laparotomy
peritoneum	membrane lining the abdominal cavity
peritonitis	inflammation of the peritoneum
pernicious	slow but destructive process
plicae circulares	"circular pleats": folds in the intestines that help in the mixing of chyme with enzymes and microorganisms
proctitis	inflammation of the rectum
proctologist	medical specialist who studies the rectum
proctology	medical specialty of the rectum

Word	Definition
proctopexy	surgical fixation of the rectum
proctoptosis	displacement of the rectum out of its normal position
proctoscope	instrument for looking inside the rectum
propria	"proper"; "one's own"; a layer unique to the gut tube, between the epithelium lining the lumen and the muscularis layer beneath
propulsion	movement of substances through the gut tube
purgative	drug used to purge the gastrointestinal tract; laxative
pyloric	pertaining to a region of the stomach near its exit to the duodenum (e.g. the pyloric valve which regulates passage of materials from stomach to duodenum)
pylorus	a region of the stomach near its exit to the duodenum
quadrate	liver lobule shaped like a square
rectalgia	pain in the a**
rectitis	inflammation of the rectum
rectopexy	surgical fixation of the rectum
rectoplasty	surgical revision of the rectum
rectum	the final segment of the colon; between the sigmoid colon and the anus
reflux	abnormal backward movement of material in the gut tube; usually refers to stomach acid in the esophagus
saliva	watery mixture produced by salivary glands that aids in digestion
sclerosing	causing the abnormal hardening
secretin	hormone released by the duodenum (to indicate increased acidity) and which acts on the liver and pancreas
segmentation	mixing or churning movement in the intestines
serosa	membrane lining the organs of the abdominal cavity
sialadenectomy	surgical removal of a salivary gland
sialadenitis	inflammation of a salivary gland
sialadenosis	swelling of a salivary gland not caused by inflammation or cancer
sialagogic	drug that causes increased production of saliva
sialoadenectomy	surgical removal of a salivary gland
sialoadenitis	inflammation of a salivary gland
sialoadenosis	swelling of a salivary gland not caused by inflammation or cancer
sialoangiectasis	dilation of the blood vessels of a salivary gland

Word	Definition
sialolith	stone in a salivary gland
sialolithiasis	the presence of a stone in a salivary gland
sialolithotomy	cutting into a salivary gland to remove a stone
sialorrhea	abnormal flow from a salivary gland
sialostenosis	a narrowed or blocked duct in a salivary gland
sigmoid	shaped like the Greek letter sigma; S-shaped
sigmoidoscope	instrument for looking inside the sigmoid colon
sigmoidoscopy	the process of looking into the sigmoid colon
steatorrhea	fatty stools
stenosis	narrowing or blockage of a passage, such as the intestines
stercobilin	excreted blood breakdown products in the feces; gives the feces a brown color
stercobilinogen	chemical that gives rise to stercobilin
stomatitis	inflammation of the mouth
stomatodynia	pain in the mouth
stomatogastric	pertaining to the mouth and stomach
stomatomycosis	fungal infection of the mouth
stomatoplasty	surgical revision of the mouth
stomatosis	disease of the oral cavity
sublingual	underneath the tongue
submucosal	one of the plexi of the enteric nervous system
sucrase	enzyme which breaks down sucrose into glucose and fructose
Taenia	genus name for tapeworms; Taenia saginata is the beef tapeworm while Taenia solium and Taenia asiatica are pork tapeworms found in undercooked meat
tenesmus	straining, as when a patient tries to empty an already empty bowel
trypsin	the second enzyme discovered; digestive enzyme that breaks down proteins
ulcerate	cause the formation of an open wound
ulcerative	causing the formation of an open wound
umbilical	one of the nine abdominal regions, lying in the center of the grid and centered on the umbilicus ("belly button")
unconjugated	bilirubin that is not water-soluble (its normal state)
vallate	papillae on the tongue, with taste buds, that are shaped like a castle and moat (valley)

Word	Definition
vermiform	shaped like a worm
villus	tiny, threadlike projection of the intestinal wall which increases the surface area for absorption and digestion
vitamin	chemical needed to help in the body's biochemical reactions; generally aids enzymes to work more efficiently
voluntary	under a person's conscious control

HOW THE UNIT 10 WORD LIST WAS BUILT

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Cancer of the glands in the breast, stomach, prostate, lung, pancreas, colon, & rectum					abdomin/o			centesis		abdominocentesis
					abdomin/o			plasty		abdominoplasty
					aden/o	carcin		oma		adenocarcinoma
					aero		dont	algia		aerodontalgia
					aliment			ary		alimentary
					amyl			ase		amylase
					an[us]			al		anal
	an				hydr[o]			ase		anhydrase
					ano			phony		anophony
					ano			plasty		anoplasty
				ano		sigmoid/o	scopy		anosigmoidoscopy	
ant				acid					antacid	
anti				eme[sis]/t			ic		antiemetic	
				anus					anus	
a				phag			ia		aphagia	
a[d]				scend[ere]			ing		ascending	
From Greek askos referring to a bag made from goat skin used to hold wine										ascites
					bar		iatr	ics		bariatrics
					bile					bile
					bil/i			genesis		biligenesis
					bil/i		rub[er]	in		bilirubin
					bil/i		verd	in		biliverdin
Latin bucca, "cheek"					bucc[a]			al		buccal

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
					canal			icul	us	canaliculus
German peptid, from German Pepton & Greek pepton, "digested"				carboxy[l]	peptid			ase		carboxypeptidase
					cardia					cardia
Greek katharsis, "purging, cleaning"					cathar/t			ic		cathartic
					caud[a]			ate		caudate
Latin coeliacus, Greek koiliakos, "bowels"					celi			ac		celiac
					celio		my/o/s	itis		celiomyositis
					celio			pathy		celiopathy
					celio			tomy		celiotomy
					cement			um		cementum
				chol	ang/i/o		gastr/o	stomy		choleangiogastronomy
				chol	ang/i/o			gram		choleangiogram
				chol	ang/i/o			graphy		choleangiography
				chol	ang/i/o			oma		choleangioma
a procedure for visualizing the anatomy of the bile ducts and pancreatic ducts				chol	ang/i/o		pancreat/o	graphy		choleangiography
				chol	ang			itis		cholangitis
				chol	cyst			algia		cholecystalgia
				chol/e	cyst			ectomy		cholecystectomy
				chol/e	cyst			itis		cholecystitis
				chol/e	cyst/o		cysto	gram		cholecystogram
				chol/e	cyst/o		kin	in		cholecystokinase
				chol/e	doch/o			cele		cholelethectomy
				chol/e	doch/o		enter/o	stomy		choledochostomy

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
common bile duct Greek choledochos "bile duct"				chol/e	doch/o		lith	ectomy		choledocholithiasis
				chol/e	doch/o		lith	iasis		choledocholithiasis
				chol/e	doch/o			tomy		choledochotomy
				chol/e		lith				cholelith
				chol/e		lith		iasis		cholelithiasis
				chol/e		lith/o		tomy		cholelithotomy
				chol/e		lith/o		tripsy		cholelithotripsy
Greek kirros, "red-yellow, yellow-brown". The "h" is a mistake.						chyme				chyme
						chym/o	trypsin			chymotrypsin
						cirr		osis		cirrhosis
						col		ectomy		colectomy
						col		itis		colitis
						colon				colon
						colon		ectomy		colonectomy
						colon/o		scopy		colonoscopy
						col/o	rect	al		colorectal
						col/o		stomy		colostomy
Latin com + iugare, "to yoke together"										colovaginal
	con					jug[are]		ated		conjugated
	con					stipa[re]		tion		constipation
						cyst[o]		ic		cystic
Latin deciduus, "that which falls down", referring to leaves, petals, teeth, etc.						decidu		ous		deciduous

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
	de				fec[es]			ation		defecation
Latin deglutitionem, "to swallow down"	de				gluti[re]			tion		deglutition
					dent			algia		dentalgia
Latin dens, "tooth" + fricare, "rub"					denti		frice			dentifrice
					dent			in		dentin
					dent			ist		dentist
					dent			istry		dentistry
	de				scend[ere]			ing		descending
	dia							rrhea		diarrhea
Latin digerere (dis + gerere), "separate, divide, arrange"					digest			ive		digestive
Latin diverticulum, "a bypath"					diverticul			itis		diverticulitis
					diverticul			osis		diverticulosis
					duoden			ectomy		duodenectomy
					duoden			itis		duodenitis
Shortening of Latin duodenum digitorum, "space of twelve digits", after Greek physician (c. 300 BC) Herophilus who called it dodkadaktylon					duoden			um		duodenum
	dys				enter			y		dysentery
	dys				peps			ia		dyspepsia
					elast			[in]	ase	elastase
Greek emesis, "vomiting"					emesis					emesis

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
French en- + amailler, shortened to émail (as in "covering in chain mail"); "mail" ultimately comes from Latin macula because chain mail armor looked spotted	en				amel					enamel
					enter			ectomy		enterectomy
					enter			ic		enteric
					enter/o			cele		enteroceles
					enter/o	cyte				enterocytes
					enter/o			dynia		enterodynia
					enter/o	endo		crine		enteroendocrine
					enter/o			pathy		enteropathy
					enter/o			rrhaphy		enterorrhaphy
					enter/o			tomy		enterotomy
	epi				gastr			ic		epigastric
					esophag			algia		esophagalgia
					esophag			eal		esophageal
					esophag			ectomy		esophagectomy
					esophag			itis		esophagitis
				esophag/o	gastr/o	duoden/o		scopy		esophagogastroduodenoscopy
				esophag/o	gastr/o			plasty		esophagogastropasty
					esophag/o			scopy		esophagoscopy
	eu				glyc			emia		euglycemia
	eu				peps			ia		eupepsia
Latin falx, "sickle"					falc			iform		falciform
					fec[es]			al		fecal

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin fæces, "sediment, dregs" which entered English in 17th century. The word for "feces" in Latin was "caca".					feces					feces
					fil[um]			iform		filiform
Latin fistula, "pipe, flute"					fist			ula		fistula
Latin flatus, "a blowing" (from the bowels)					flatus					flatus
					foli[um]			ate		foliate
					fundus					fundus
					fungi			form		fungiform
					gall		bladder			gallbladder
					gastr			algia		gastralgia
					gastr			ectomy		gastrectomy
					gastr			ic		gastric
					gastr			itis		gastritis
					gastr/o		duoden/o	scope		gastroduode
					gastr/o		duoden/o	stomy		gastroduode
					gastr/o			dynia		gastrodynia
					gastr/o		enter	itis		gastroenteri
				gastr/o	enter/o		col	itis		gastroentero
					gastr/o		enter/o	logist		gastroentero
					gastr/o		enter/o	logy		gastroentero
					gastr/o		enter/o	stomy		gastroentero
					gastr/o		esophag	eal		gastroesoph
				gastr/o	intestin[e]			al		gastrointesti
					gastr/o		jejun/o	stomy		gastrojejuno
					gastr/o			malacia		gastromalaci

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
PIE ghrel, "to grow", but doing double-duty as an acronym for growth hormone-releasing polypeptide					gastr/o			paresis		gastroparesis
					gastr/o			pexy		gastropexy
					gastr/o			plasty		gastroplasty
					gastr/o			scope		gastroscope
					ghrel			in		ghrelin
					gingiv			algia		gingivalgia
					gingiv			ectomy		gingivectomy
					gingiv			itis		gingivitis
					gingiv/o	gloss		itis		gingivogloss
					gingiv/o			plasty		gingivoplast
					gingiv/o	stomat		itis		gingivostom
					gloss/o			pathy		glossopathy
					gloss/o			plasty		glossoplasty
Latin haustum, "scoop for drawing water"					gloss/o			plegia		glossoplegia
					gloss/o			rrhaphy		glossorrhaph
					gloss/o			tomy		glossotomy
					gloss/o	trich		ia		glossotrichia
					gluc			agon		glucagon
					gluc			uronic		glucuronic a
					haustrum					haustrum
					gingiv/o	stomat		itis		hematemesis
					heme					heme
	hemi				col			ectomy		hemicolecto

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Greek haimorrhoides, "[veins] likely to discharge blood"						hemorrhoid				hemorrhoid
						hemorrhoid		ectomy		hemorrhoid
					hepat			ectomy		hepatectomy
					hepat			ic		hepatic
				hepatic/o	gastr/o			stomy		hepaticogast
					hepatic/o			tomy		hepaticotom
					hepat			itis		hepatitis
					hepat/o	carcin		oma		hepatocarcin
					hepat/o	cyte				hepatocytes
					hepat			oma		hepatoma
					hepat/o			malacia		hepatomalac
					hepat/o			megaly		hepatomega
				hepat/o	hepat/o	pancr/e/at		ic		hepatopancr
					hepat/o			pexy		hepatopexy
					hepat/o			ptosis		hepatoptosis
					hepat/o			sclerosis		hepatosclero
					hepat/o	splen		itis		hepatosplen
					hepat/o	splen/o		megaly		hepatosplen
Latin hernia, "rupture"						hernia				hernia
						herni/o		rrhaphy		herniorrhaph
	hyper					emesis				hyperemesis
	hypo					chondr		iac		hypochondr
	hypo					gastr		ic		hypogastric
	hypo					gloss		al		hypoglossal

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Greek ikteros, "jaundice", but also the name of a yellow bird that was supposed to cure jaundice by picking up the yellow color and then dying						icterus				icterus
						ile		itis		ileitis
Latin intestinum cæcum, "blind gut"				ileo		cec[um]		al		ileocecal
						ileo	col	itis		ileocolitis
						ileo	colo	stomy		ileocolostomy
						ileo		rrhaphy		ileorrhaphy
						ileo		stomy		ileostomy
						ileo		tomy		ileotomy
						ili[a]		um		ileum
	in					gest[us]		ion		ingestion
						inguin			al	inguinal
Latin insula, "island" (i.e. islets of Langerhans)						insul[a]		in		insulin
						insul[a]		in/o	tropic	insulinotropic
						intestin[e]		al		intestinal
						islet				islet
French jaunice, "yellowness", with "d" added for no particular reason						jaundice				jaundice
						jejun		itis		jejunitis
						jejun/o	ile	itis		jejunoileitis
						jejun/o		rrhaphy		jejunorrhaphy
						jejun/o		stomy		jejunostomy
						jejun/o		tomy		jejunotomy

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word		
Latin lumen, "light" (i.e., a passageway; the part of a tube-like structure you can see light through)										jejunum		
										labi[a]	al	labial
										lact	eal	lacteal
							propria			lamina		lamina propria
										laparo	cele	laparoccele
							enter/o			laparo	stomy	laparoentero
										laparo	scope	laparoscope
										laparo	scopy	laparoscopy
										laparo	tomy	laparotomy
										lingual		lingual
										lip	ase	lipase
										liver		liver
										lob[e]	ule	lobule
										lumb	ar	lumbar
Latin absorptionem, "a swallowing" Mes-, "middle" + -enter[on], "intestine"										lumen		lumen
										lyso	zyme	lysozyme
										macro	cyte	macrocyte
	mal						absorption					malabsorpti
	mes						enter			ic		mesenteric
							mes[os]			ial		mesial
							mucosa					mucosa
							muscular			is		muscularis
					my		enter			ic		myenteric
					nas/o		gastr			ic		nasogastric

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Both the Latin dont and Greek odont are based on PIE əd-ent "uhd-ent" sound (which means "biting"), and there is no real logic which is which in Medical English					occlus[us]			al		occlusal
					odont			algia		odontalgia
					odont			ectomy		odontectomy
					odont/o			clasis		odontoclasia
					odont/o			dynia		odontodynia
					or			al	al	oral
	ortho				dont			ic		orthodontic
	ortho				dont			ist		orthodontist
					oxy			gen		oxygenase
					pancreato	duoden			ectomy	
Greek peptikos, "able to digest". Pepsin and trypsin were the first two enzymes discovered and do not follow the normal naming convention of ending in –ase.					pancreato	lith				pancreatolithiasis
					pancreato	lith		ectomy		pancreatolithotomy
					pancreato	lith		iasis		pancreatolithiasis
					pariet			al		parietal
	par[a]				oti[s]				d	parotid
					pancreato			graphy		pancreatography
					peps			in		pepsin
					pepsin/o			gen		pepsinogen

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin peritonæum, "part stretched over" Latin perniciēs, "death, destruction, ruin", related to per- ("complete") + noxi ("harm")	peri				odont			al		periodontal
	peri				odont			itis		periodontiti
	peri				stalsis					peristalsis
					periton/eo			scopy		peritoneosco
					periton/eo			tomy		peritoneoto
					peritoneum					peritoneum
					periton			itis		peritonitis
					pernici[es]			ous		pernicious
					plica		circularis			plicae circul
					proct			itis		proctitis
Greek pylōros, "gatekeeper" (pylē "gate, pylon" + ouros "watcher, guardian")					proct/o			logist		proctologist
					proct/o			logy		proctology
					proct/o			pexy		proctopexy
					proct/o			ptosis		proctoptosis
					proct/o			scope		proctoscope
					propria					propria
					propuls			ion		propulsion
					purg			ative		purgative
					pylor[us]			ic		pyloric
					pylorus					pylorus
					quadrate					quadrate
					rect			algia		rectalgia
					rect			itis		rectitis

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin intestinum rectum, "straight part of bowel", because it is the straight segment of the bowel (this is a Latin translation of Galen's Greek name apeuthysmeon enteron, because he only dissected animals with truly straight rectums)										
Latin fluxus, "flowing"	re									
Sial/o is "saliva"; sialadeno- is the salivary gland										
Sialo- + Greek agōgós, "leading" (as in pedagogy)										

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
The sigmoid colon is shaped somewhat like the Greek letter sigma (ς when it appears at the end of a word), hence "sigma-oid"					sial/o			lith	iasis	sialolithiasis
					sial/o			lith/o	tomy	sialolithotomy
					sial/o			rrhea		sialorrhea
					sial/o		sten	osis		sialostenosis
					sigm[a]			oid		sigmoid
					sigmoid/o			scope		sigmoidoscopy
					sigmoid/o			scopy		sigmoidoscopy
					steato			rrhea		steatorrhea
					sten			osis		stenosis
					sterco		bil	in		stercobilin
					sterco		bil	in	ogen	stercobilinogen
					stomat			itis		stomatitis
					stomat/o			dynia		stomatodynia
					stomat/o		gastr	ic		stomatogastria
					stomat/o		myc	osis		stomatomycosis
					stomat/o			plasty		stomatoplasty
					stomat			osis		stomatosis
	sub				ling/u			al		sublingual
	sub				mucos[a]			al		submucosal
					sucr[e]			ase		sucrase
Latin tænia, "flat band"; Tænia or Taenia is the name of a genus of tapeworms					Taenia					Taenia
Greek teinesmos, "straining"					tenesmus					tenesmus

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Trypsin is one of the first digestive enzymes isolated, by Kühne (1876). The name derives from Greek tripsis ("rubbing") + -in, apparently because he isolated it by rubbing the pancreas with glycerin.						trypsin				trypsin
						ulcer		ate		ulcerate
						ulcer		ative		ulcerative
						umbilic		al		umbilical
	un	con				jug[are]		ated		unconjugate
						vallate				vallate
						vermi[s]		form		vermiform
Latin villus (pl. villi), "shaggy hair"						villus				villus
						vit[a]		amin[e]		vitamin
						volunt[us]		ary		voluntary

PART XI

UNIT 11: THE RENAL SYSTEM

UNIT 11 INTRODUCTION

The Urinary System filters the blood in order to eliminate waste from the body (in the form of urine) and regulate fluid volume and electrolytes levels. The structures of the urinary system include the kidneys, ureters, bladder, and urethra.

Objectives

After you complete this unit, you will be able to:

- Recognize prefixes, root words, and suffixes related to the diagnosis and treatment of kidney and bladder diseases.
- Recall abbreviations related to the diagnosis and treatment of kidney and bladder diseases.

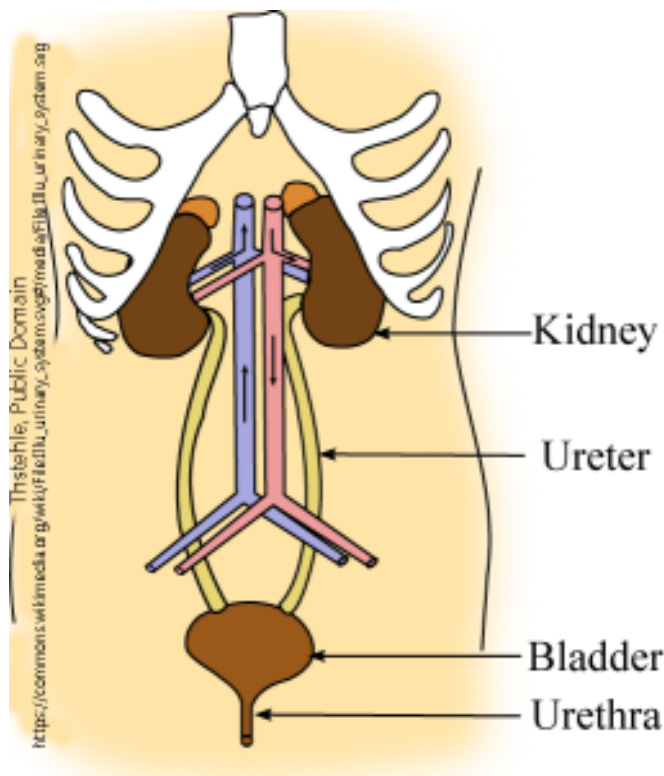
THE RENAL SYSTEM

Only [Med Terms] in the Building Episode 11



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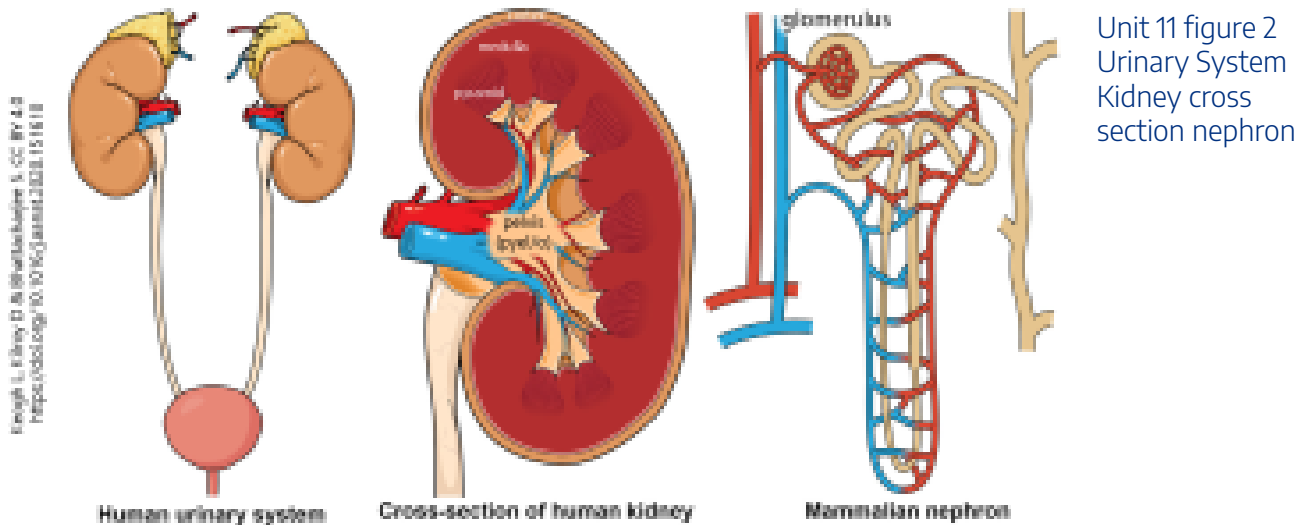


Unit 11 figure 1 Urinary System

The *kidneys* filter about 180 liters of our blood per day, producing *urine* as a waste by-product. Other urinary system structures, including the ureters, bladder, and urethra, provide a drainage and storage system for urine. The urinary system is not simply a waste removal system; it also plays a vital role in maintaining homeostasis by assisting in the control of the body's acid/base balance (pH), electrolytes, metabolites, fluid volume, and fluid concentration. Each kidney has about a million nephrons, which are the functional units of the kidney that filter, reabsorb, and secrete fluids. A dysfunctional urinary system will have effects on all other systems of the body.

The kidney is a kidney-bean-shaped organ, which probably comes as no surprise to anyone who has seen a kidney or a kidney bean. It has a *renal pelvis* (the inner curve of the “bean”, root word *pyel/o*) where the *renal artery* enters and the *renal vein* and

ureter leave. The slightly darker color in the middle figure shows the *renal pyramids*, each of which feeds urine into a *minor calyx*; the urine from the minor calyces drains into a larger *major calyx*. The major calyces, in turn, join to form the *ureter*. There is one ureter per kidney, and both the right and left ureter form a functional connection between the kidney and the *urinary bladder* (root word *cyst/o* or *vesic/o*) where urine is stored before it is expelled from the body.



Each nephron consists of a *glomerulus* (Latin: “little ball of thread”), shaped like a little ball of thread with ample blood supply. That blood is filtered by the glomerulus and the filtrate is collected in the *glomerular capsule* before it passes into a series of tubes, each of which prepares the filtrate in different ways:

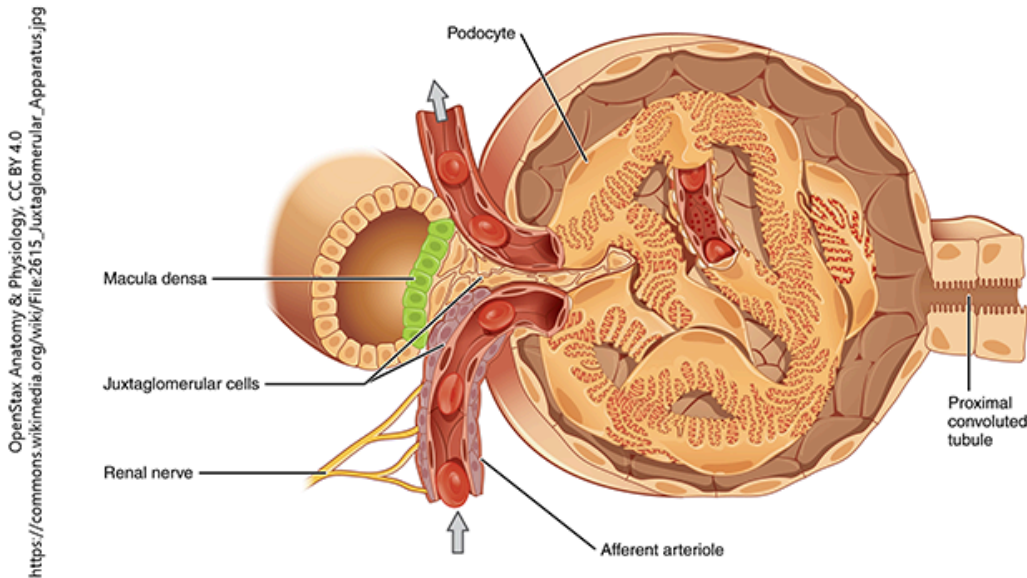
- the proximal convoluted tubule
- the nephron loop (loop of Henle)
 - descending limb of nephron loop
 - ascending limb of nephron loop
- distal convoluted tubule
- collecting duct

A network of blood vessels (*vasa recta*) runs parallel to the nephron loop to receive any materials coming out of the nephron loop. Each nephron ends in a single collecting duct, and it is those hundreds of thousands of collecting ducts that come together to form the minor calyx we saw earlier.

The kidney has a major role in the fluid, acid-base, and salt balance of the body. Under the influence of hormones (Unit 7) such as antidiuretic hormone, excess water is secreted by the body. If the blood is too high in solutes (i.e. too salty), then water is retained to reduce its osmolarity. Hydrogen, sodium and potassium ions are actively moved by the kidney either into the nephron loop or into the *vasa recta* (nearby bloodstream) to keep the blood concentration of these ions constant.

Glucose passes through the glomerular filtration system and into the filtrate within the nephron, and therefore must be replaced to avoid depletion of blood glucose.

Protein metabolism results in the production of *urea* and other nitrogen-containing (*azo*) compounds. The production of these wastes is tracked by examining the lab value called *blood urea nitrogen (BUN)*. The kidney is responsible for removing these compounds from the blood and placing them in the urine.



The kidney regulates blood pressure as well. Its own pressure is controlled by receptors in the *juxtaglomerular apparatus*, which controls the amount of filtration that occurs in the glomerulus. There, an area of the distal convoluted tubule called the “dense spot” (*macula densa*) joins with juxtaglomerular cells in the lining of the arteriole supplying the nephron with blood to be filtered. This allows detection and regulation of blood pressure, both going into the nephron but also for the body generally. The juxtaglomerular cells secrete the hormone *renin*, which participates in a complex pathway that ultimately controls blood pressure.

The kidney also plays a key role in blood levels of calcium; it responds to parathyroid hormone (PTH; Unit 7) and vitamin D by increasing retention of calcium to keep blood calcium levels within normal limits.

Video Attribution

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UNIT 11 PREFIXES

Prefix	Meaning
an-	not
anti-	against (an action)
dys-	bad
en-	inside
extra-	outside, more, beyond
hemi-	half
hyper-	too much
hypo-	too little
in-	not
intra-	inside
juxta-	beside; close to
oligo-	several; a handful
para-	beside; near
peri-	around
poly-	many; too much
re-	again
retro-	backwards; behind
trans-	through
ultra-	beyond; extremely

UNIT 11 ROOTS

Root Word	Definition
absorp	taking up
albumin	protein found in egg white; major blood protein
anastomosis	the joining of two or more vessels
azo/t azo/to	chemical group with two nitrogen atoms joined together
calyx	part of the kidney's urine collection system
capsule	a membranous sac
cele	hernia
contin	holding together; holding back (as urine)
convolute	twisted together
corpus	body
cortex	outside of a two-layered organ
cyan	blue
cyst cyst/o	bladder
densa	dense; tightly packed cells
detrusor	muscle that expels urine
dialysis	the process of separating (waste products out of blood)
dips dips/o	drinking
facult facult/a	ability, opportunity
fascia	connective tissue
fulgur/a	electrocautery
glomer glomer/ul glomer/ul/o	filtering system of the nephron
glucos	sugar
glycos	sugar
hem hem/at hem/at/o hem/o	blood
hydr/o	water

Root Word	Definition
ileo	flank
keto keton	chemical made in the liver when fats are broken down
lapar/o	abdomen
lith lith/o	stone
macula	spot
meatus meat/o	tube, such as the urethra
medulla	center of a two-layered organ
micturi	desire to urinate
natr	sodium
nephr nephr/o	kidney
noct	night
obligat	a duty; a binding agreement
periton	membrane surrounding the intestinal organs
py py/o	pus
pyel pyel/o	renal pelvis
pyramid	renal pyramid
rectum	end of the gastrointestinal tract
ren	kidney
resect	cut away
scler	hardening
sono	sound waves
spadon	fissure
spasm	involuntary contraction
stoma	anatomical mouth; opening
toxin	substance which is damaging
trigone	triangular region
tripsy	using sound waves to crush or break apart

Root Word	Definition
tube	tube
tubule tubul/o	small tube
ur ur/o	urine
ureter ureter/o	tube between the kidney and bladder
urethr urethr/o	tube between the bladder and the outside world
urine	waste fluid created by the body
vesic/o	bladder
void	empty
xanth	yellow

UNIT 11 SUFFIXES

Suffix	Definition
-al	pertaining to
-algia	pain
-ar	pertaining to
-ary	pertaining to
-ation	word ending that turns a verb into a noun
-cele	hernia
-cle	diminutive suffix
-dynia	pain
-eal	pertaining to
-ectasia	dilated condition
-ectasis	dilation
-ectomy	cutting out of
-emia	present in blood, indicating disease
-ence	word ending that turns an adjective into a noun
-esis	process
-etic	a drug causing a process
-genic	giving rise to
-gram	a record of a test or procedure
-graphy	a test or procedure
-ia	disease
-iasis	presence of an abnormality
-ic	pertaining to
-in	chemical name ending
-ing	at the end of a verb, means "ongoing"
-ion	word ending that turns a verb into a noun
-ism	belief; process word ending
-itis	inflammation
-lith	stone
-logist	one who studies
-logy	study of

Suffix	Definition
-lysis	breaking down; destroying
-malacia	softening due to disease
-megaly	enlargement due to disease
-oma	tumor
-on	unit of anatomical structure
-or	muscle that has that action
-osis	process
-pathy	disease process
-pexy	surgical fixation
-plasty	surgical revision
-plegia	inability to move
-poiesis	creation process
-ptosis	drooping
-rrhaphy	suturing
-rrhea	abnormal flow
-rrhexis	rupture
-scope	instrument for looking inside
-scopy	the process of looking inside
-spasm	involuntary contraction
-stenosis	abnormal narrowing of an opening
-stomy	creating an opening with surgery
-tion	word ending that turns a verb into a noun
-tive	to go
-tomy	cutting
-trophy	nourishment, growth
-ule	diminutive suffix
-us	anatomical structure ending

UNIT 11 ABBREVIATIONS

Abbreviation	Meaning
ARF	acute renal failure
BUN	blood urea nitrogen
cath	catheter
cysto	cystoscopy
ESWL	extracorporeal shock wave lithotripsy
HD	hemodialysis
I&O	intake and output
IVP	intravenous pyelography
OAB	overactive bladder
PKD	polycystic kidney disease
RP	retrograde pyelogram
SUI	stress urinary incontinence
UA	urinalysis
UTI	urinary tract infection
VCUG	voiding cystourethrogram
VUR	vesicoureteral reflux

UNIT 11 WORD LIST

Word	Definition
albuminuria	the presence of albumin (protein) in the urine, indicating kidney disease
antispasmodic	drug that suppresses muscle spasms causing urinary urgency
anuria	absence of urine
azotemia	increase in BUN (blood urea nitrogen), indicating kidney dysfunction
azotorrhea	excessive secretion of urea nitrogen in the urine
azoturia	excessive secretion of urea nitrogen in the urine
calyx	part of the urine collection system of the kidney, leading to the urethra
capsular	pertaining to the glomerular capsule
capsule	the membranous sac surrounding the filtration system of the glomerulus
convoluted	twisted in a complex way
corpuscle	"small body"; the glomerulus plus the glomerular (Bowman's) capsule
cortical	nephrons with a shorter nephron loop that penetrates only the superficial renal medulla
cystalgia	pain in the bladder
cystectomy	surgical removal of the bladder
cystitis	inflammation of the bladder
cystocele	herniation of the bladder (anterior vaginal prolapse)
cystodynia	pain in the bladder
cystogram	x-ray image of dye inserted into the bladder
cystography	process of obtaining a cystogram
cystolith	bladder stone
cystolithectomy	surgical removal of a bladder stone
cystoma	tumor of the bladder
cystoplegia	weakness of the bladder muscles
cystoptosis	drooping of the bladder
cystorrhexis	rupture of the bladder
cystoscopy	the process of looking into the bladder
cystospasm	sudden contraction of the bladder
cystostomy	surgical creation of a communication that allows urine to drain from the bladder
cystotomy	cutting into the bladder
cystoureteritis	inflammation of the bladder and ureter

Word	Definition
cystourethrocele	hernation of the bladder and urethra (anterior vaginal prolapse)
cystourethrogram	x-ray image of dye inserted into the bladder and urethra
detrusor	muscle which expels urine from the bladder
dialysis	process to artificially remove wastes from the blood, replicating this kidney function
dipsogenic	drug which increases thirst
diuresis	increased urination
diuretic	drug which increases the production of urine
dysuria	pain or difficulty with urination
enuresis	nighttime urination (literally, "urine in bed")
extracorporeal	from outside the body
facultative	having the ability to do something
fascia	connective tissue sheet
fulguration	using an electric current during surgery (e.g. to destroy stones or seal blood vessels); electrocauterization
glomerular	pertaining to the glomerulus
glomerulonephritis	inflammation of the kidney glomeruli
glomerulopathy	disease of the kidney glomerulus
glomerulosclerosis	hardening of the kidney glomeruli
glomerulus	filtration system of the kidney
glucosuria	sugar in the urine
glycosuria	sugar in the urine
hematuria	blood in the urine
heminephrectomy	surgical removal of part of a kidney
heminephroureterectomy	surgical removal of a duplex kidney (congenital condition where an "extra" kidney forms)
hemodialysis	process to artificially remove wastes from the blood, replicating this kidney function
hydronephrosis	swelling of kidneys due to urine buildup
hypernephroma	tumor of the renal tubules
hyponatremia	low blood sodium
hypospadias	congenital urethral opening on the underside of the penis
incontinence	inability to hold urine and expel it appropriately

Word	Definition
intracorporeal	from inside the body
juxtaglomerular apparatus	hormone-secreting organ near the glomerulus
juxtamedullary	nephrons with a longer nephron loop that penetrates deep into the renal medulla
ketolysis	breakdown of ketones
ketonuria	ketones in the urine
laparonephrectomy	surgical removal of a kidney through an incision in the abdominal wall
lithectomy	surgical removal of a stone
lithiasis	the presence of a stone
lithocystotomy	cutting into the bladder to remove a stone
lithonephritis	inflammation of the kidney due to a stone
lithonephrotomy	cutting into the kidney to remove a stone
lithotripsy	breaking apart stones with high-frequency sound waves
macula densa	"dense spot"; area of the distal convoluted tubule which forms part of the juxtaglomerular apparatus
meatal	pertaining to an anatomical tube, such as the urethra or ureter
meatoplasty	surgical revision of an anatomical tube
meatorrhaphy	suturing an anatomical tube
meatoscope	instrument for looking inside an anatomical tube
meatoscopy	the process of looking into the urinary meatus (urethra) or the ureteral orifice
meatotomy	cutting into the urethra or ureter (for example, to remove a stone)
micturition	urination
nephralgia	pain in the kidney
nephrectomy	surgical removal of a kidney
nephritis	inflammation of the kidney
nephrocele	herniation of the kidney
nephrocystanastomosis	surgical creation of a connection between the kidney and bladder
nephrogram	x-ray image of dye inserted into the kidney
nephrography	process of obtaining a nephrogram
nephrohypertrophy	abnormal growth of the kidney
nephrolithiasis	the presence of a stone in the kidney

Word	Definition
nephrolithotomy	cutting into the kidney to remove a stone
nephrologist	a person who studies kidney disease
nephrology	the study of kidneys
nephroma	tumor of the kidney cortex
nephromalacia	softening of the kidney due to disease
nephromegaly	enlargement of the kidney due to disease
nephron	functional unit of the kidney; a glomerulus, proximal convoluted tubule, nephron loop, distal convoluted tubule, and collecting duct
nephropathy	disease of the kidney
nephropexy	surgical fixation of the kidney
nephroptosis	drooping of the kidney
nephrorrhaphy	suturing the kidney
nephrosclerosis	hardening of the kidney
nephroscopy	the process of looking at the kidney
nephrosis	kidney disease (especially when accompanied by edema, abnormal swelling of the tissues); nephrotic syndrome
nephrosonography	process of imaging the kidney with sound waves
nephrostomy	surgical creation of a communication that allows urine to drain from the kidney
nephrotomy	cutting into the kidney
nephrotoxin	a chemical which is damaging to the kidney
nephroureterectomy	surgical removal of a kidney and its ureter
nocturia	nighttime urination
obligatory	required
oliguria	sparse urine production
paraphimosis	condition of having the foreskin trapped behind the head of the penis
pedicels	"little feet"; microscopic structures supporting the filtration system of the kidney
peritubular	near the tubules
phimosis	inability to retract the foreskin of the penis
podocytes	"foot cells"; cells of the glomerular apparatus that extend pedicels to support the filtration membrane
polycystic	disease process characterized by many fluid-filled structures

Word	Definition
polydipsia	frequent drinking
polyuria	frequent urination
pyelitis	inflammation of the pelvis region and calyces of the kidney
pyelocystitis	inflammation of the pelvis region of the kidney and the bladder
pyelocystostomosis	surgical creation of a communication between the kidney and bladder
pyelogram	x-ray image of dye inserted into the kidney calyces
pyelolithotomy	cutting into the kidney calyces to remove a stone
pyelonephritis	inflammation of the pelvis region, calyces, and kidney tissue
pyelopathy	disease of the kidney calyces
pyeloplasty	surgical revision of the kidney calyces
pyelostomy	surgical creation of a communication that allows urine to drain from the kidney calyces
pyelotomy	cutting into the kidney calyces
pyeloureterectasia	dilation of the kidney calyces and ureter
pyonephritis	inflammation of the kidney with pus formation
pyonephrolithiasis	the presence of pus and stones in the kidney
pyopyeloectasis	dilation of the kidney calyces with the formation of pus
pyramid	structure in the kidney medulla
pyuria	pus in the urine
reabsorption	moving substances from the filtrate in the nephron to the vasa recta (and thus back into the blood)
renal	pertaining to the kidney
renin	hormone released from the juxtaglomerular apparatus of the kidney
resection	surgical removal of all or part of an organ
resectoscope	instrument for looking inside the body to resect (remove) a structure
retroperitoneal	behind the peritoneal cavity
transrectal	through or across the rectum (e.g. ultrasound or surgery)
transurethral	through or across the urethra
trigone	anatomical triangle
tubule	small tube, as in the nephron
tubuloglomerular	pertaining to the nephron

Word	Definition
ultrasonography	process of imaging with sound waves (same as sonography)
urea	waste product formed from protein metabolism
uremia	buildup of waste products in the blood because of kidney dysfunction
ureteralgia	pain in one or both ureters
ureterectomy	surgical removal of a ureter
ureteritis	inflammation of the ureter
ureterocele	hernation of the ureter
ureteroileostomy	surgical creation of a communication that allows urine to drain from the ureters through the body wall
ureterolithiasis	the presence of a stone in the ureter
ureteronephrectomy	surgical removal of a kidney and its ureter
ureteroplasty	surgical revision of the ureter
ureteropyelitis	inflammation of the ureter and calyces of the kidney
ureteropyelonephritis	inflammation of the ureter, calyces, and kidney tissue
ureterorrhaphy	suturing the ureter
ureteroscopy	the process of looking into the ureter
ureterstenosis	narrowing or blockage of the ureter
urethrectomy	surgical removal of the urethra
urethritis	inflammation of the urethra
urethrocystitis	inflammation of the urethra and bladder
urethrodynia	pain in the urethra
urethrogram	x-ray image of dye inserted into the urethra
urethropexy	surgical fixation of the urethra
urethroplasty	surgical revision of the urethra
urethrorrhea	abnormal flow from the urethra
urethroscope	instrument for looking inside the urethra
urethroscopy	the process of looking into the urethra
urethrospasm	sudden contraction of the urethra
urethrostenosis	narrowing or blockage of the urethra
urethrotomy	cutting into the urethra

Word	Definition
urinalysis	shortening of urine + analysis
urinary	pertaining to the organs which produce urine
urocyanosis	blue color in the urine
urodynia	pain during urination
urologist	medical specialist who studies the urinary system
urology	the study of the urinary system
uropathy	disease in the urinary system
uropoiesis	the process of creating urine
urostomy	surgical creation of a communication that allows urine to drain outside the body
uroxanthin	pigment which gives the urine its yellow color
vesicocele	herniation of the bladder (into the vaginal canal)
vesicostomy	surgical creation of a communication that allows urine to drain from the bladder
vesicotomy	cutting into the bladder
vesicoureteral	pertaining to the bladder and ureter
voiding	emptying the bladder (i.e. urinating)

HOW THE UNIT 11 WORD LIST WAS BUILT

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
			albumin		ur	ia		albuminuria
	anti		spasm			od	ic	antispasmodic
	an		ur			ia		anuria
			azo/t			emia		azotemia
			azo/to			rrhea		azotorrhea
			azo/t		ur	ia		azoturia
Greek kalyx, "seed pod, husk, outer covering"			calyx					calyx
			capsul[e]			ar		capsular
French "a membranous sac" from Latin capsula, "box or chest" + -ula diminutive suffix			capsule					capsule
Latin convolutus, "rolled together", made from com- + volvere, "to roll"			convolute			d		convoluted
			corpus			cle		corpuscle
			cortic[is]			al		cortical
			cyst			algia		cystalgia
			cyst			ectomy		cystectomy
			cyst			itis		cystitis
			cyst/o			cele		cystocele
			cyst/o			dynia		cystodynia
			cyst/o			gram		cystogram
			cyst/o			graphy		cystography
			cyst/o			lith		cystolith
			cyst/o		lith	ectomy		cystolithectomy

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
			cyst/o			oma		cystoma
			cyst/o			plegia		cystoplegia
			cyst/o			ptosis		cystoptosis
			cyst/o			rrhexis		cystorrhesis
			cyst/o			scopy		cystoscopy
			cyst/o		spasm			cystospasm
			cyst/o			stomy		cystostomy
			cyst/o			tomy		cystotomy
			cyst/o		ureter	itis		cystoureteritis
			cyst/o		urethr/o	cele		cystourethrocele
			cyst/o		urethr/o	gram		cystourethrogram
Latin dētrūdō, "to drive, force, or thrust away"			detrusor					detrusor
Greek dialysis, "dissolution, separation"			dialysis					dialysis
			dips/o			genic		dipsogenic
	di		ur			esis		diuresis
	di		ur			etic		diuretic
	dys		ur			ia		dysuria
Greek enourein, "to urinate in" (en- + ourein); urinating in bed	en		ur			esis		enuresis
	extra		corp[us]/r			eal		extracorporeal

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin <i>facultatem</i> means "the power, ability, opportunity". Ultimately, it comes from facere, "to do". The same meaning in cell biology is carried by the word "regulated" (literally, "by rule". No one knows why different words are used to mean the same thing.								
			facult/a			tive		facultative
			fascia					fascia
Latin <i>fulgur</i>, "lightning"			fulgur			ation		fulguration
			glomerul			ar		glomerular
			glomerul/o	nephr		itis		glomerulonephritis
			glomerul/o			pathy		glomerulopathy
			glomerul/o	scler		osis		glomerulosclerosis
Latin <i>glomer</i>, "ball of thread" + -ul[e] diminutive suffix + -us anatomical structure word ending			glomer			ul[e]	us	glomerulus
			glucos	ur		ia		glucosuria
			glycos	ur		ia		glycosuria
		hem/at	ur			ia		hematuria
	hemi		nephr			ectomy		heminephrectomy
	hemi		nephr/o	ureter		ectomy		heminephroureterectomy
		hem/o	dia	lysis				hemodialysis

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
			hydr/o		nephr	osis		hydronephrosis
	hyper		nephr			oma		hypernephroma
	hypo		natr			emia		hyponatremia
Greek hupospadías (from Galen), hypo- + spadon , "fissure"	hypo		spad[on]			ias[is]		hypospadias
Latin contenere , "to hold back, check"	in		contin			ence		incontinence
	intra		corp[us]/r			eal		intracorporeal
	juxta		medull[a]			ary		juxtamedullary
			keto			lysis		ketolysis
		keton	ur			ia		ketonuria
			lapar/o		nephr	ectomy		laparonephrectomy
			lith			ectomy		lithectomy
			lith			iasis		lithiasis
			lith/o		cyst/o	tomy		lithocystotomy
			lith/o		nephr	itis		lithonephritis
			lith/o		nephr/o	tomy		lithonephrotomy
			lith/o		tripsy			lithotripsy
			macula		densa			macula densa
			meat[us]			al		meatal
			meat/o			plasty		meatoplasty
			meat/o			rrhaphy		meatorrhaphy
			meat/o			scope		meatoscope
			meat/o			scopy		meatoscopy
			meat/o			tomy		meatotomy

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Latin micturiturum, "desire to urinate". Original meaning was "strong urge to urinate", but medically, it's used just as a fancier way of saying "urination".			micturi			tion		micturition
			nephr			algia		nephralgia
			nephr			ectomy		nephrectomy
			nephr			itis		nephritis
			nephr/o			cele		nephrocele
	nephr/o		cyst		anastomosis			nephrocystanastomosis
			nephr/o			gram		nephrogram
			nephr/o			graphy		nephrography
			nephr/o	hyper		trophy		nephrohypertrophy
			nephr/o	lith		iasis		nephrolithiasis
			nephr/o	lith/o		tomy		nephrolithotomy
			nephr/o			logist		nephrologist
			nephr/o			logy		nephrology
			nephr			oma		nephroma
			nephr/o			malacia		nephromalacia
			nephr/o			megaly		nephromegaly
			nephr			on		nephron
			nephr/o			pathy		nephropathy
			nephr/o			pexy		nephropexy
			nephr/o			ptosis		nephroptosis
			nephr/o			rrhaphy		nephrorrhaphy
			nephr/o	scler		osis		nephrosclerosis

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
			nephr/o			scopy		nephroscopy
			nephr			osis		nephrosis
			nephr/o	sono		gram		nephrosonography
			nephr/o			stomy		nephrostomy
			nephr/o			tomy		nephrotomy
			nephr/o	toxin				nephrotoxin
			nephr/o	ureter		ectomy		nephroureterectomy
			noct	ur		ia		nocturia
The Latin <i>obligare</i> means "to bind", like a bandage on a wound. In cell biology, the same idea is expressed by the word "constitutive". No one knows why we use different words to mean the same thing.								
			obligator[ius]			y		obligatory
	olig[o]		ur			ia		oliguria
	para		phimosi					paraphimosis
	pedi					cels		pedicels
	peri		tubul[e]			ar		peritubular
From Greek for "muzzle"								
			phimosi					phimosi
	podo					cytes		podocytes
	poly		cyst			ic		polycystic
	poly		dips			ia		polydipsia
	poly		ur			ia		polyuria
			pyel			itis		pyelitis
		pyel/o	cyst			itis		pyelocystitis
		pyel/o	cyst/o	stom[a]		osis		pyelocystostomosis

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
			pyel/o			gram		pyelogram
			pyel/o		lith/o	tomy		pyelolithotomy
			pyel/o		nephr	itis		pyelonephritis
			pyel/o			pathy		pyelopathy
			pyel/o			plasty		pyeloplasty
			pyel/o			stomy		pyelostomy
			pyel/o			tomy		pyelotomy
		pyel/o	ureter			ectasia		pyeloureterectasia
		py/o	py/o		nephr	itis		pyonephritis
			nephr/o		lith	iasis		pyonephrolithiasis
			py/o		pyel/o	ectasis		pyopyeloectasis
			pyramid					pyramid
			py		ur	ia		pyuria
Latin absorbere, "to swallow up"	re		absorp			tion		reabsorption
			ren			al		renal
			ren			in		renin
Latin resectionem, "action of cutting off or away"			resect			ion		resection
			resect/o			scope		resectoscope
	retro		periton			eal		retroperitoneal
	trans		rect[um]			al		transrectal
	trans		urethr			al		transurethral
			trigone					trigone
Greek trigōnos, "triangular"			tub[e]			ule		tubule
		tubulo	glomer			ul[us]	ar	tubuloglomerular
	ultra		sono			graphy		ultrasonography

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
So much fun, they wrote a song about it.			urea					urea
			ur			emia		uremia
			ureter			algia		ureteralgia
			ureter			ectomy		ureterectomy
			ureter			itis		ureteritis
			ureter/o			cele		ureteroceles
			ureter/o	ileo		stomy		ureteroileostomy
			ureter/o	lith		iasis		ureterolithiasis
			ureter/o	nephr		ectomy		ureteronephrectomy
			ureter/o			plasty		ureteroplasty
			ureter/o	pyel		itis		ureteropyelitis
		ureter/o	pyel/o	nephr		itis		ureteropyelonephritis
			ureter/o			rrhaphy		ureterorrhaphy
			ureter/o			scopy		ureteroscopy
			ureter			stenosis		ureterstenosis
			urethr			ectomy		urethrectomy
			urethr			itis		urethritis
			urethr/o	cyst		itis		urethrocystitis
			urethr/o			dynia		urethrodynia
			urethr/o			gram		urethrogram
			urethr/o			pexy		urethropexy
			urethr/o			plasty		urethroplasty
			urethr/o			rrhea		urethrorrhea
			urethr/o			scope		urethroscope
			urethr/o			scopy		urethroscopy
			urethr/o			spasm		urethrospasm
			urethr/o			stenosis		urethrostenosis
			urethr/o			tomy		urethrotomy

Etymology	Prefix	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End	Word
Urin[e] + [an]alysis			urin[e]		[an]alysis			urinalysis
			urin			ary		urinary
			ur/o		cyan	osis		urocyanosis
			ur/o			dynia		urodynia
			ur/o			logist		urologist
			ur/o			logy		urology
			ur/o			pathy		uropathy
			ur/o			poeiesis		uropoiesis
			ur/o			stomy		urostomy
			ur/o		xanth	in		uroxanthin
			vesic/o			cele		vesicoccele
			vesic/o			stomy		vesicostomy
			vesic/o			tomy		vesicotomy
			vesic/o		ureter	al		vesicoureteral
			void			ing		voiding

PART XII

UNIT 12: THE REPRODUCTIVE SYSTEM

UNIT 12 INTRODUCTION

The Reproductive System refers to the tissues, glands, and organs involved in producing offspring. In females, it includes the ovaries, fallopian tubes, uterus, cervix, vagina, and vulva. In males, it includes the prostate, seminal vesicles, scrotum, testes, epididymis, vas deferens, urethra, and penis.

Objectives

After you complete this unit, you will be able to:

- Define terms that relate to the male and female reproductive systems.
- Recognize prefixes, root words, and suffixes related to the diagnosis and treatment of reproductive disorders.
- Recall abbreviations related to the diagnosis and treatment of reproductive disorders.

THE REPRODUCTIVE SYSTEM

Only [Med Terms] in the Building Episode 12

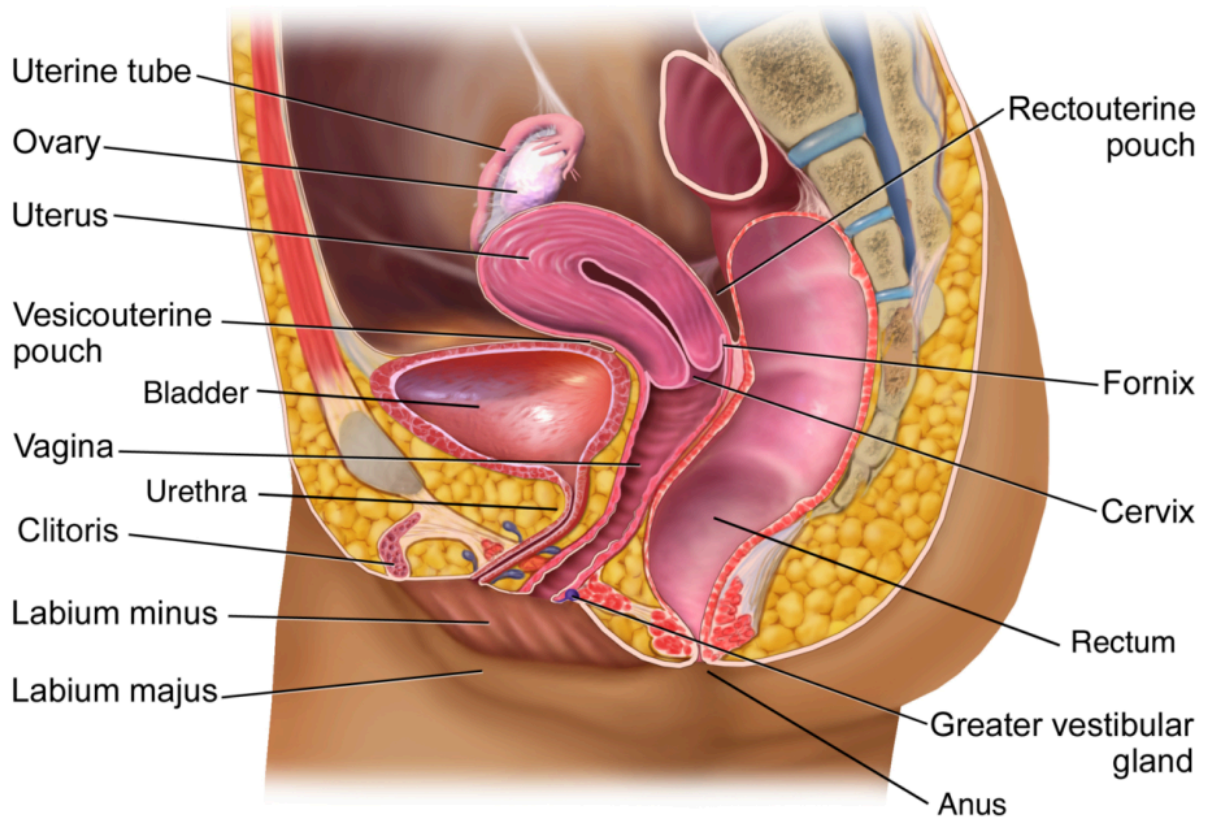


One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://uen.pressbooks.pub/medicalenglish/?p=349>

The Reproductive System is the only system in which you see major differences between male and female. All other body systems are virtually the same, anatomically and physiologically, between female and male (with the exception of a minor sex difference in the Urinary System). It is also the only system not necessary for the survival of an individual organism. Yes, it's necessary for the survival of the species, but neither you nor I depend on our Reproductive System to survive as individuals. Every other body system is absolutely necessary for life.

Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". Wikijournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436 CC BY 3.0
https://commons.wikimedia.org/wiki/File:Blausen_0400_FemaleReproSystem_02.png



Unit 12 figure 1 female reproductive anatomy

Female Reproductive Organs

Ovaries: produce and store the female sex cells (eggs), produce estrogen and progesterone

Fallopian Tubes: tunnels for egg cells to travel from the ovaries to the uterus, common site of fertilization

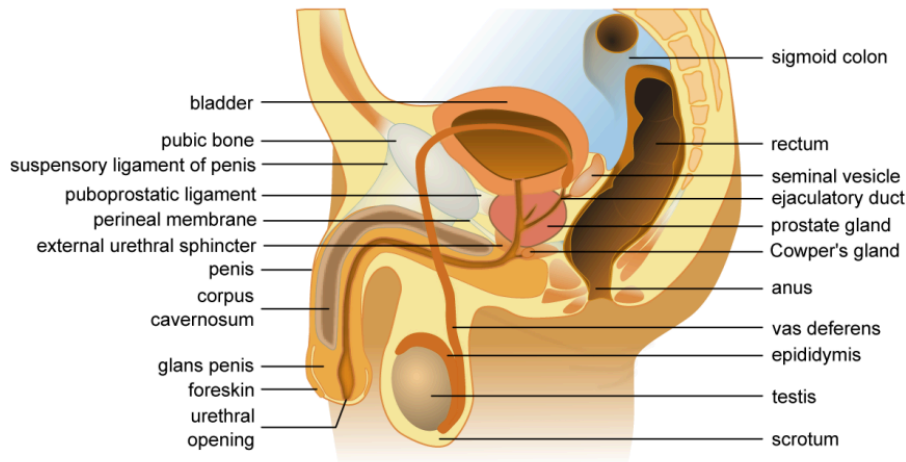
Uterus: also called the “womb”, site of implantation into the uterine lining, home to the developing fetus during pregnancy

Cervix: narrow canal at the lower part of the uterus that opens to the vagina

Vagina: also called the “birth canal”, joins the cervix to the outside of the body

Vulva: the external portion of the female genital organs

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https://en.wikipedia.org/wiki/File:Male_reproductive_system/Male_anatomy_en.svg



Male Reproductive Organs

Prostate: produces fluid to contribute to the ejaculate that nourishes sperm and helps sperm move through the female reproductive system (double check)

Seminal Vesicles: produces fluid to contribute to the ejaculate that provides a source of energy for sperm

Scrotum: sac-like organs containing the testes, protect and provide climate control for the testes

Testes: responsible for testosterone and sperm production, also called testicles

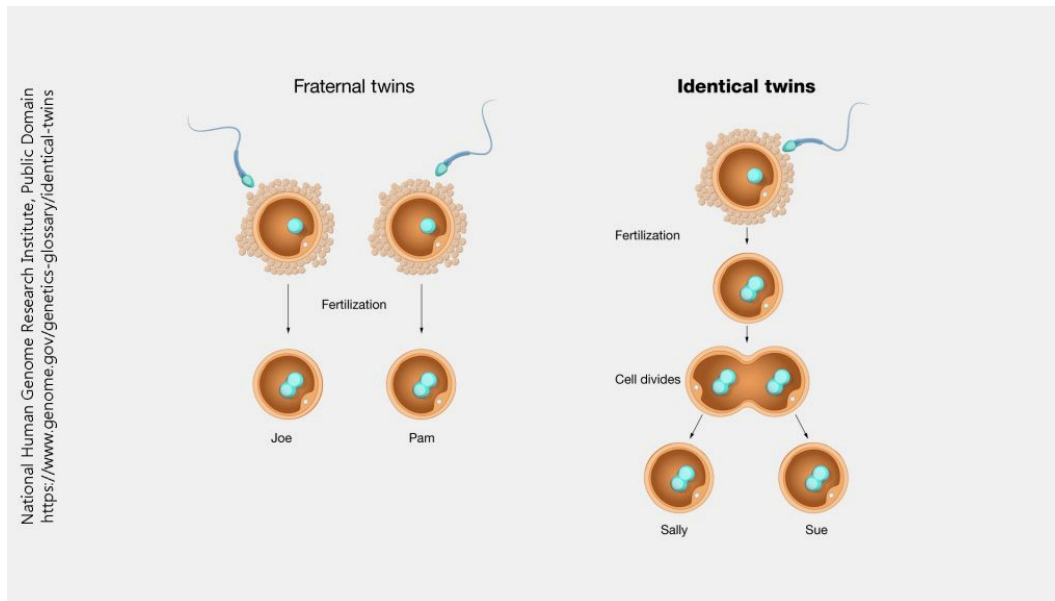
Epididymis: transports and stores sperm produced in the testes, brings sperm to maturity

Vas Deferens: muscular tube transporting mature sperm from the epididymis to the urethra

Urethra: tube responsible for expelling semen (as well as carrying urine from the bladder to outside the body)

Penis: composed of erectile tissues (corpora cavernosum and [corpus spongiosum](#)), sperm is expelled through the urethral opening at the tip of the penis.

Fertilization (Conception)



Unit 12 figure 3
twins

During sexual intercourse, when a male sperm penetrates the female egg, *fertilization* results, forming a *zygote*. This normally occurs at about 14 days after the last menstrual cycle, accounting for the difference between gestational age (used by most people and their doctors) and embryonic age (used in textbook descriptions of embryology). The image above shows the process as it occurs for *dizygotic* (fraternal) twins as well as *monozygotic* (identical) twins.

The zygote implants in the female's uterine wall after about a week, resulting in an embryo. At about 1/4 of the way during development (10 weeks gestational age), the embryo is called a fetus.

Video Attribution

- Justin Burr, Jim Hutchins & Maddison Johnston. (2024, December 20). *Only [Med Terms] in the Building Episode 12*. [Video]. Kaltura.

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UNIT 12 PREFIXES

Prefix	Meaning
a– an–	not
ante–	in front of; before
brady–	slow
con–	together, with
contra–	against, opposite
di–	separate, apart, double, two
dys–	bad
ec–	out, off, from, beyond
ecto–	outside
endo–	inside
epi–	on top of
eu–	good, well–
extra–	outside, more, beyond
hyper–	too much
hypo–	too little
in–	not
inter–	between
intra–	inside
macro–	large
meso–	in the middle
micro–	small
mono–	one
neo–	new
ob–	in front of; against
oligo–	several; a handful
par– para–	beside; near
peri–	around
poly–	many; too much
post–	behind (anatomically or in time)

Prefix	Meaning
pre-	in front of (anatomically or in time)
pro-	in front of; in favor of
pseudo-	false, lying
trans-	through

UNIT 12 ROOTS

Root Word	Definition
abort abort/i	early termination of a pregnancy
abruptio	Latin word for sudden breaking off
abstinence	choosing to not have sex
actin	cytoskeletal protein
albicans	Latin word for whitening
albumen albus	(egg) white
amnio amnion amnios	membrane surrounding the fetus
angi	vessel
anomaly	something out of the ordinary
antron	an anatomical cavity
arche	beginning
areola	pigmented circle around a nipple
balan balan/o	glans of the penis
blast blast/o	a rapidly dividing cell
bulb/o	the bulb (base) of the penis
Cæserian	from a Roman emperor (Julius Cæsar)
carcin	cancer
card/i/o	heart
cavernosum	Latin word for cavernous, open space
cele	hernia
cephal/o	the head
cervic cervic/o cervix	the muscular sphincter between the uterus and vagina
Chlamydia	genus of obligate intracellular parasites
chorio chorion	outermost membrane surrounding a fetus; contributes to the formation of the placenta
circumcision	removal of the prepuce (foreskin) of the penis with a circular incision

Root Word	Definition
clitoris	female genital erectile tissue that plays a major role in sexual arousal
coitus	the act of sexual intercourse
colp colp/o	vagina
conception	the process of sperm penetrating egg which begins human development
condom	a sheath over the penis used for disease prevention and/or contraception
corpora	body (plural)
corpus	body (singular)
cremaster	involuntary muscle causing retraction of the testicles
crypt	hidden chamber
cyesis	pregnancy
cyst	fluid-filled space
cyt/e cyt/o	cell
dartos	thin layer of smooth muscle under the male scrotum or female labia majora
decidua	the mucous membrane of the uterus; shed during menstruation
derm	skin (in this context, refers to one of the layers of the developing embryo)
didymis	testicle
dyna	force (of uterine contractions); cf. dynamic
eclampsia	convulsions of the mother during pregnancy caused by high blood pressure
ejacula	expulsion of semen from the male urethra
embryo/n	a developing infant from conception to week 10 (of 40)
epision	perineum, the area between the vaginal opening and the anus
fertil[e]	capable of bearing or creating a child
fetus	a developing infant from week 10 to birth (at about 40 weeks)
fil[um]	thread-like structure
fimbria	structures between the uterine (Fallopian) tubes and ovaries that help an ovum enter the tubes
fraternal	related like brothers
gastrula	embryonic stage characterized by an inpocketing and folding
genitus	the time of birth
gest	carrying an infant

Root Word	Definition
glans	enlarged, pigmented area at the end of the penis or clitoris
gon/ad/o	organs of reproduction (male testis and/or female ovaries)
granulosa	type of cell in the uterus
gravida	pregnancy
gravidarum	pregnant person
gynec/o	women
hem/at/o hem/o	blood
hermaphrodite	individual showing physical characteristics of both sexes
herpes	virus that causes a blistering, painful rash
hydr hydr/o	water
hymen	membrane covering the vaginal canal
hyster hyster/o	uterus
in vitro	"in glass", that is, performed in a laboratory
inhib[it]	to hold back
labia labi/o	lips, in this context, the lips around the vulva
lact/o	breast milk
lapse	to slip or fall
lei/o	smooth (muscle)
leuk/o	white or colorless (in this context, refers to a vaginal discharge)
ligat[e]	tied together
lith/o	stone
luteum	Latin word for yellow
majora	Latin word for major
mamm/o	breast
mast/o	breast
megal/o	enlargement
men/o	menstruation
menstru[s]	menstruation

Root Word	Definition
metr/i metr/o	womb
minora	Latin word for minor
my my/o	muscle
myc	fungus
nat nat/o	birth
nephr	kidney
nucleus	in this context, the center of a cell, where DNA is stored and RNA is made
null	zero; none
oo	egg
oophor oophor/o	ovary (oo + phor, "egg carrier")
orch orch/i orch/i/o orchid orchid/o	testicle
organ/o	internal organs
orifice	opening
ovar/i/o	ovary
ovum	egg; female gamete
oxy	oxygen
Papanicolaou	name of a Greek-American physician who is credited with the invention of the Pap smear
papilla	nipple
para -parous parus	bearing (as in, "bearing a child")
pareun	sexual intercourse (literally, "lying with")
partum	Latin, giving birth
pause	coming to a stop
pelv/i	the hip bone and structures contained in it (pelvis)
penis	the male sexual organ

Root Word	Definition
perine/o	the area between the genitals and anus
phleb	vein
placenta	placenta (singular)
placentae	placenta (plural)
prepuce	foreskin
previa	Latin, previous, in front of
Priapos	embodiment of male fertility in Greek mythology
prostat prostat/o	prostate gland
pubo	pubic
py/o	pus
recto	rectum
relax	the opposite of contraction of a muscle
rete	net
salping salping/o salpinx	uterine tubes; Fallopian tubes
scrot	scrotum
semen semin	sperm plus glandular secretions
sex	the physical characteristics of males or females
som som/ato	body
sono	sound
speculum	instrument for looking inside the vagina
sperm sperm/at/o sperm/i sperm/o	sperm; male gamete
spermatid	developmental stage of a spermatozoon
spongiosum	Latin, "like a sponge"
sten	abnormal narrowing
stet/r	standing; with ob-, standing in front of, as a midwife

Root Word	Definition
suckl[e]	draw milk from mother's breast
sustentacul[um]	a body part that suspends another
syphilis	sexually transmitted disease
terat/o	mutations or malformations (especially of a fetus)
test testic	testicle
thele	nipple
toc toc/o	labor; active childbirth
top/o	place
tox	poison
Toxoplasma	genus of single-celled parasites
Trichomonas	genus of single-celled parasites
tropho	nourishment, growth
tunica	tunic, coat
urethr/o	tube between the bladder and the outside world
uter	womb
vagin/o	vagina (the passage from the uterus to the vulva)
varico	dilated [vein]
vas deferens	vessel that conducts sperm from the testicle to the urethra
vas/o	vessel
vesico	urinary bladder
vesicul/o	seminal vesicle
virus	infectious packet of nucleic acid (DNA or RNA)
vulv/o	the outer genitalia of the female
zoo	life, living
zygote	fertilized egg before the first division (i.e. stage between fertilization and embryo)

UNIT 12 SUFFIXES

Suffix	Definition
–al	pertaining to
–algia	pain
–ar	pertaining to
–ation	word ending that turns a verb into a noun
–cele	hernia
–centesis	puncture
–cide	substance which kills
–cyte	cell
–dynia	pain
–ectomy	cutting out of
–facient	substance that promotes a process
–gen	arising from
–genesis	the process of creating
–genic	giving rise to
–gram	a record of a test or procedure
–graph	instrument for recording a procedure
–graphy	a test or procedure
–ia	disease
–ial	pertaining to
–ian	pertaining to
–iasis	presence of an abnormality
–ic	pertaining to
–iferous	having, bearing, or containing
–in	name for a protein
–ine	pertaining to (also a chemical name ending, depending on the context)
–ing	verb ending that makes the continuous tense
–ion	word ending that turns a verb into a noun
–ism	belief; process word ending
–itis	inflammation

Suffix	Definition
-ity	turns a noun into a state of being
-ium	anatomical structure
-ize	turns a noun into a verb
-logist	one who studies
-logy	study of
-lysis	breaking down; destroying
-lytic	substance that causes a breakdown or destruction
-megaly	enlargement due to disease
-metry	the act of measuring
-oma	tumor
-osis	process
-osum	full of, prone to
-osus	
-pathy	disease process
-pexy	surgical fixation
-plasia	the process of molding or growing
-plasty	surgical revision
-ptosis	drooping
-rrhage	bursting forth
-rrhagia	bursting forth in disease
-rrhaphy	suturing
-rrhea	abnormal flow
-rrhexis	rupture
-scope	instrument for looking inside
-scopy	the process of looking inside
-stenosis	abnormal narrowing of an opening
-stomy	creating an opening with surgery
-tion	word ending that turns a verb into a noun
-tomy	cutting
-trop	influencing the activity of

Suffix	Definition
-ula -ule	diminutive suffix
-um -us	anatomical structure word ending
-y	disease
-zoa -zoon	living thing

UNIT 12 ABBREVIATIONS

Abbreviation	Meaning
AIDS	acquired immunodeficiency syndrome
BPH	benign prostatic hyperplasia
BRCA	gene which, when mutated, causes one form of breast cancer; pronounced "brack-uh"
CPD	cephalopelvic disproportion
CS C-section	Cesarean section
DRE	digital rectal examination
ED	erectile dysfunction
G	gravida, gravidity (number of pregnancies, followed by a number)
GYN	gynecology
HIV	human immunodeficiency virus
HPV	human papilloma virus
HRT	hormone replacement therapy
HSV	herpes simplex virus
IUD	intra-uterine device
IVF	in vitro fertilization
LMP	last menstrual period
OB-GYN	obstetrics and gynecology
P	para, parity (number of pregnancies carried > 20 weeks, followed by a number)
PID	pelvic inflammatory disease
PIH	pregnancy-induced hypertension (eclampsia)
PMS	premenstrual syndrome
PSA	prostate-specific antigen
RDS	respiratory distress syndrome
STD STI	sexually transmitted disease (infection)
TSS	toxic shock syndrome
TURP	transurethral resection of the prostate

UNIT 12 WORD LIST

Word	Definition
abortifacient	substance that speeds or promotes abortion
abortion	early termination of a pregnancy
abruptio placentae	separation of the placenta from the uterine wall before birth
abstinence	refraining from something (in this case, sexual intercourse)
amastia	failure of breast tissue to develop
amenorrhea	failure of menstruation to occur
amniocentesis	sampling amniotic fluid during pregnancy
amnion	one of the membranes surrounding the developing fetus
amniorrhea	abnormal flow of amniotic fluid
amniorrhexis	rupture of the amniotic sac ("breaking water")
amnioscope	instrument for looking inside the amniotic sac
amniосcopy	process of looking inside the amniotic sac
amniotomy	cutting into the amniotic sac
anomaly	a person, structure, or process that is out of the ordinary
anorchidism	condition of no testis
anovulation	condition of no ovulation
antepartum	time period before birth
antral	ovarian follicles in a resting state and visible by ultrasound (i.e. 2-9 mm diameter)
areola	pigmented area around the nipple
aspermia	no sperm
azoospermia	no living (moving) sperm
balanitis	inflammation of the glans penis
balanoplasty	surgical revision of the glans penis
balanorrhea	abnormal flow (discharge) from the glans penis
Bartholin glands	glands in the female vagina that produce a lubricating fluid (analogous to male bulbourethral gland)
blastocele	fluid-filled cavity inside the embryo at the blastocyst stage
blastocyst	stage of embryonic development where the embryo forms a fluid-filled cavity (about 5-6 days after fertilization)
bradytocia	abnormally slow process of childbirth

Word	Definition
bulbourethral	glands found next to the urethra in the male; also called Cowper's gland; produce Cowper's fluid ("pre-cum")
cardiotocograph	instrument for recording the child's heartbeat during childbirth
cephalopelvic	pertaining to the head of the baby in relationship to the pelvis of the mother
cervicectomy	surgical removal of the cervix
cervicitis	inflammation of the cervix
cervicocolpitis	inflammation of the cervix and vagina
cervicodynia	pain in the cervix
cervicography	procedure for photographing the cervix after application of acetic acid (vinegar)
cervicovaginitis	inflammation of the cervix and vagina
cervix	the opening of the uterus into the vagina
cesarean section	childbirth performed with surgical methods
chancre	ulcer on the genitalia caused by syphilis
Chlamydia	bacterial genus causing a sexually transmitted disease or the disease caused by this organism
chorioamnionitis	inflammation of the fetal chorion and amniotic sac
chorioangioma	tumor of the blood vessels supplying the chorion
choriocarcinoma	cancerous tumor of the chorion
chorion	outermost membrane surrounding the embryo; contributes to the formation of the placenta
chorionitis	inflammation of the fetal chorion
circumcision	removal of the foreskin
clitoris	erectile, sexually responsive tissue in the female
coitus	the process of sexual intercourse
colpitis	inflammation of the vagina
colpocystitis	inflammation of the vagina and bladder
colpopexy	surgical fixation of the vagina
colpoplasty	surgical revision of the vagina
colpoptosis	drooping (displacement) of the vagina
colposcope	instrument for looking inside the vulva at the vagina and cervix
colposcopy	process of looking inside vulva at the vagina and cervix
colpostenosis	narrowing of the vagina

Word	Definition
conception	the process of fertilization; when sperm meets egg
condom	device for preventing the emission of sperm by covering the penis
congenital	occurring or apparent at birth
contraception	a drug or device that reduces the rate of conception
contraction	shortening of a muscle (in this context, especially uterine smooth muscle)
corpora cavernosum	erectile tissue in the male penis forming the bulk of the penis
corpora albicans	remnant of a shed egg seen on the surface of the ovary
corpus hemorrhagicum	remnant of a shed egg seen on the surface of the ovary with evident blood
corpus luteum	remnant of a shed egg seen on the surface of the ovary which secretes hormones for about two weeks after the egg is shed
corpus spongiosum	erectile tissue in the male penis surrounding the urethra
cremaster	muscle that covers the testes and spermatic cord in males
cryptorchidism	condition of hidden testis (i.e. undescended testis)
cytomegalovirus	virus related to herpes that can cause birth defects
dartos	muscle that underlies the scrotum or labia majora
decidua	inner lining of uterus that is shed during childbirth
dermoid cyst	common congenital cyst
dizygotic	type of twin that results from two zygotes; fraternal twins
dysmenorrhea	abnormally painful menstrual flow
dyspareunia	pain during sexual intercourse
dysplasia	malformation of a structure
dystocia	difficulty during the process of childbirth
eclampsia	seizures resulting from high blood pressure with pregnancy
ectoderm	one of the layers of the developing embryo which forms epidermis and nervous tissue
ectopic	development of an embryo anywhere except the uterine lining
ejaculation	emission of semen as the result of sexual stimulation
embryonic	pertaining to the first 10 weeks after conception
endocervicitis	inflammation of the lining of the cervical canal
endoderm	one of the layers of the developing embryo which forms the gastrointestinal tract, lungs, and urinary system
endometriosis	displacement of the inner lining of the uterus to places where it does not belong

Word	Definition
endometritis	inflammation of the inner lining of the uterus
endometrium	the inner lining of the uterus
epididymectomy	surgical removal of the epididymis
epididymis	the "waiting room" where sperm go to mature between being made in the testes and ejaculated
epididymitis	inflammation of the epididymis
epididymo-orchitis	inflammation of the epididymis and testes
epididymotomy	cutting into the epididymis
episiorrhaphy	suturing the opening of the vulva near the perineum, especially to repair episiotomy
episiostenosis	narrowing of the opening of the vulva
episiotomy	cutting into the opening of the vulva near the perineum (to decrease the chance that these structures will tear)
erectile	tissues that fill with blood during sexual arousal
eutocia	perfect process of childbirth
extraembryonic	membranes outside the embryo
fecund	fertile or productive
fecundability	ability to bear children
fertilization	the union of a spermatozoon with a female egg, producing a zygote
fertilized	an egg which has united with a spermatozoon
fetometry	process of measuring a fetus
fimbria	finger-like extensions of the uterine (Fallopian) tubes; help move the egg shed from the ovary into the tube
fraternal	type of twin that results from two zygotes; share the same amount of DNA as any two non-twin siblings
gastrulation	the process of infolding in the human embryo that forms three layers; occurs at about day 16 after fertilization or day 30 after the last menstrual period
gestation	the process of carrying a fetus
glans	the pigmented, enlarged end of the male penis or female clitoris
gonad	an organ of sexual reproduction
gonadarche	the beginning of gonadal development
gonadotropin	a hormone which acts on the gonads
gonorrhea	a sexually transmitted disease characterized by abnormal flow from the gonads

Word	Definition
granulosa	description of small, scattered cells in a tissue
gravida	number of pregnancies
gravidarum	associated with pregnancy or pregnant persons
gravidity	number of pregnancies (compare to parity)
gynecologist	medical specialist who studies the female reproductive system
gynecology	medical specialty of female reproduction
gynecomastia	enlarged breasts in the male
hematosalpinx	blood in the uterine (Fallopian) tubes
hemorrhagicum	pertaining to blood
hermaphrodite	a person who has congenital external characteristics that lie between male and female; intersex
herpes	virus that infects the nervous system, often lying dormant for years
hydrocele	a fluid-filled sac inside the scrotum
hydrocelectomy	surgical removal of a hydrocele
hydrosalpinx	a fluid-filled sac in the wall of the uterine (Fallopian) tube
hymen	thin membrane covering the female vulva at birth
hyperadrogenism	condition of high male hormone levels
hypermastia	condition of excessively large breasts
hypomastia	condition of excessively small breasts
hysteralgia	pain in the uterus
hysterectomy	surgical removal of the uterus
hysterocele	herniation of the uterus
hysterodynia	pain in the uterus
hysteroigraphy	procedure for imaging the uterus, especially as part of a fertility test
hysteropexy	surgical fixation of the uterus
hysteroptosis	drooping (displacement) of the uterus
hysterorrhexis	rupture of the uterus
hysterosalpingectomy	surgical removal of the uterus and uterine (Fallopian) tubes
hysterosalpingogram	x-ray image of the uterus and uterine (Fallopian) tubes
hysteroscope	instrument for looking inside the uterus

Word	Definition
hysteroscopy	process of looking inside the uterus
hysterotomy	cutting into the uterus
implantation	the process of an embryo burrowing into the uterine wall; occurs at about 6 days after fertilization or about 20 days after last period
in vitro	occurring in a laboratory
infertile	unable to bear children
infertility	inability to conceive offspring
inhibin	hormone that acts to reduce gonadotropin secretion
intersex	an individual who does not perfectly fit the anatomical definition of "male" or "female" which predominates in their society
intraepithelial	cells on the surface of the cervix; used to describe the location of abnormal cervical cells in a Pap smear
intrapartum	time period during birth
intrauterine	type of insemination that involves direct insertion of semen into the uterus
labia majora	external genitalia of the female
labia minora	external genitalia of the female
labioscrotal	pertaining to the tissues which develop into the labia in females and the scrotum in males
lactation	the process of producing breast milk
lactogenic	substance which promotes the formation of breast milk
lactorrhea	abnormal flow of breast milk
leiomyoma	tumor of the smooth muscle layer of the uterus
leukorrhea	abnormal flow of whitish fluid from the vagina
ligation	tying together with a suture, such as a sterilization procedure in females (tubal ligation)
macromastia	enlarged breasts
macrosomia	abnormally large body size in a fetus
mammogram	image of breast tissue
mammoplasty	surgical revision of the breast
mastalgia	pain in the breast
mastectomy	surgical removal of the breast
mastitis	inflammation of the breast
mastopexy	surgical fixation of the breast

Word	Definition
mastoptosis	drooping (displacement) of the breast
menarche	the beginning of menstrual cycles in a female
menopause	the end of menstrual cycles in a female
menorrhagia	menstrual flow lasting longer than 7 days
menorrhalgia	(abnormally intense) pain with menstruation
menorrhea	flow of menstrual blood
menstruation	the process of shedding blood and tissue from the uterine lining on a monthly cycle
mesoderm	one of the layers of the developing embryo which forms muscle, connective tissue, and kidneys
mesonephric duct	an embryonic structure which forms the epididymis, vas deferens, and seminal vesicles
metrocolpocele	herniation of the uterus and vagina
metromenorrhagia	excessive uterine bleeding
metrophlebitis	inflammation of the uterine veins
metrorrhagia	abnormal uterine bleeding
microcephalus	condition of a fetus with a smaller than normal head
micromastia	condition of small breasts
monozygotic	twins resulting from the splitting of a fertilized egg early in embryonic development; identical twins
myoepithelial	cells of the breast which share muscle and epithelial tissue characteristics
myometritis	inflammation of the muscular layer of the uterus
myometrium	smooth muscle layer of the uterus
neonatal	pertaining to the time immediately after childbirth
neonatologist	medical specialist who studies newborn infants
neonatology	medical specialty of newborn infants
neoplasia	process of cancer formation
nulliparous	has not ever given birth before
obstetric	pertaining to childbirth
obstetrician	medical specialist who studies childbirth
oligohydramnios	condition of an unusually sparse amount of amniotic fluid
oligomenorrhea	condition of unusually sparse flow of menstrual fluid
oligospermia	condition of unusually low number of sperm

Word	Definition
oocyte	female gamete; egg cell
oogonia	stage in development of the oocyte
oophorectomy	surgical removal of the ovary
oophoritis	inflammation of the ovary
oophorocystectomy	surgical removal of an ovarian cyst
oophorocystosis	condition of ovarian cysts
oophoroma	tumor of the ovary
oophorosalphingectomy	surgical removal of the uterine (Fallopian) tube and ovary
oophorotomy	cutting into the ovary
orchidalgia	pain in the testis
orchidectomy	surgical removal of the testis
orchiditis	inflammation of the testis
orchidopexy	surgical fixation of the testis
orchidoptosis	drooping (displacement) of the testis
orchidotomy	cutting into the testis
orchiectomy	surgical removal of the testis
orchiepididymitis	inflammation of the testis and epididymis
orchiodynia	pain in the testis
orchioopathy	disease of the testis
orchiopey	surgical fixation of the testis
orchiooplasty	surgical revision of the testis
orchitis	inflammation of the testis
organogenesis	the formation of organs during embryonic development
orifice	opening
ovaralgia	pain in the ovary
ovarialgia	pain in the ovary
ovarian	pertaining to the ovary
ovariocentesis	surgical puncture of an ovary or ovarian cyst
ovariocyesis	ectopic pregnancy involving implantation of an embryo on the ovary
ovariorrhesis	rupture of an ovarian follicle or cyst

Word	Definition
ovariostomy	creating an opening in an ovarian cyst
ovaritis	inflammation of the ovary
ovary	the gonads of the female; organ which creates, matures, and sheds female gametes (eggs) and secretes hormones
ovulation	process of developing and shedding mature oocytes (ova)
ovum	mature oocyte
oxytocin	polypeptide hormone that increases uterine contractions during childbirth; mediates milk let-down during suckling; promotes pair bonding between mates
Pap smear	clinical test for obtaining a biopsy of cervical tissue for microscopic examination; used to detect precancerous and cancerous conditions
papilloma	wart, especially one caused by human papilloma virus (HPV)
para	bearing a child; pregnancy that becomes obvious (cf. parity)
paramesonephric ducts	Müllerian ducts; embryonic structures that develop into the uterine (Fallopian) tubes, uterus, cervix, and part of the vagina
paraurethral glands	Skene's glands; become engorged with blood and secrete fluid during female sexual arousal
parity	the number of live births (compare to gravidity)
pelvic	pertaining to the bones and organs near the hips
pelvicephalometry	measurement of the mother's pelvis in relationship to the baby's head
pelvimetry	measurement of the pelvis
penis	male organ used for sexual intercourse
perimetritis	inflammation of the membranes surrounding the uterus
perimetrium	membrane surrounding the uterus (i.e. visceral peritoneum of the uterus)
perinatal	pertaining to the time period immediately before and after birth
perinatologist	medical specialist who studies the period immediately before and after birth
perinatology	medical specialty of the period immediately before and after birth
perineocele	herniation of the perineum
perineoplasty	surgical revision of the perineum
perineorrhaphy	suturing of the perineum, especially to repair an episiotomy
perineotomy	cutting into the perineum (to decrease the chance that these structures will tear)
perineum	the tissue between the genitals and anus
placenta previa	condition where the baby's exit is blocked by the placenta
placental abruption	condition in which the placenta tears free of the uterine lining before labor and delivery

Word	Definition
polyhydramnios	abnormally large amount of amniotic fluid
polymenorrhea	abnormally large flow of menstrual fluid
polyspermy	abnormal condition where multiple sperm penetrate a single egg (usually incompatible with fertilization)
postnatal	time period after birth (from the baby's perspective)
postpartum	time period after labor and delivery (from the mother's perspective)
preeclampsia	high blood pressure which has not yet, but may lead to, the complications of eclampsia
prenatal	time period before birth
prepuce	foreskin of the penis
priapism	condition of permanent erection
prolapse	condition where nearby organs (uterus, bowel, bladder, or upper vagina) bulge into the vagina
pronucleus	transient structure derived from the egg or sperm nucleus during fertilization
prostate	gland surrounding the male urethra near its exit from the bladder
prostatectomy	surgical removal of the prostate gland
prostatitis	inflammation of the prostate gland
prostatocystitis	inflammation of the prostate gland and bladder
prostatolith	stone in the prostate gland
prostatolithotomy	surgical removal of a stone in the prostate
prostatomegaly	enlargement of the prostate gland
prostatorrhea	abnormal flow from the prostate
prostatovesiculectomy	surgical removal of the prostate and seminal vesicle
prostatovesiculitis	inflammation of the prostate gland and seminal vesicle
pseudocyesis	false pregnancy
pseudohermaphrodite	inconsistency between the apparent sex of an individual and their genetic makeup
pubarche	the beginning of the development of the pubic region
pubic	pertaining to the anterior wall of the pelvis
pubis	the anterior wall of the pelvis
pubocervical ligament	ligament attaching the cervix to the pubic symphysis (in the midline, where the pelvic bones join)
pyosalpinx	pus in the uterine (Fallopian) tubes

Word	Definition
rectocele	herniation of the rectum
relaxin	hormone that aids childbirth by relaxing the mother's muscles
rete testis	network of tubes between the testicle and epididymis
salpingectomy	surgical removal of the uterine (Fallopian) tube
salpingitis	inflammation of the uterine (Fallopian) tube
salpingocele	herniation of the uterine (Fallopian) tube
salpingocyesis	ectopic pregnancy involving implantation of an embryo in the uterine (Fallopian) tubes
salpingo-oophorectomy	surgical removal of the uterine (Fallopian) tube and ovary
salpingo-oophoritis	inflammation of the uterine (Fallopian) tube and ovary
salpingopexy	surgical fixation of the uterine (Fallopian) tube
salpingoscope	instrument for looking inside the uterine (Fallopian) tubes
semen	mixture of sperm and secretions of the seminal vesicles, prostate, and bulbourethral glands
seminal vesicles	glands which provide about 2/3 of the semen by volume
seminiferous tubules	network of tubes in the testicle that collect sperm cells
seminoma	tumor of the seminal vesicles
somatomammotropin	hormone released from the placenta that acts on the mother's metabolism, protecting the fetus from malnutrition
sonohysterography	procedure for visualizing the uterus using ultrasound, especially as part of a fertility test
speculum	medical instrument used to enlarge the vagina for easier inspection by a clinician
spermatid	stage in the development of a spermatozoon
spermatocoele	fluid-filled sac near the top of the testis; spermatic cyst
spermatocyte	stage in the development of a spermatozoon
spermatogenesis	the process of creating spermatozoa
spermatogenic cells	cells in the testis that are responsible for sperm production
spermatogonia	stage in the development of a spermatozoon
spermatolysis	breakdown of spermatozoa
spermatozoa	mature sperm cells (plural)
spermatozoon	a mature sperm cell (singular)
spermicide	substance that kills or incapacitates sperm
spermolytic	substance that breaks down sperm

Word	Definition
stroma	fibrous connective tissue
suckling	the process used by an infant to draw milk from a mother's breast
sustentacular cells	Sertoli cells; cells that support spermatogenesis and respond to follicle-stimulating hormone
syphilis	sexually transmitted disease caused by the bacterium <i>Treponema pallidum</i>
teratogen	substance that causes birth defects
teratogenic	pertaining to birth defects
teratology	medical specialty of the causes of birth defects
teratoma	striking abnormality in the development of a fetus
testes	the gonads of the male; organ which creates, matures, and sheds male gametes (sperm) and secretes hormones
testicle	testis
testicular	pertaining to the testicles
testis	singular of testes; testicle
testitis	inflammation of the testis
testotoxicosis	condition of excessive testosterone production in an child leading to early puberty
tocodynamograph	instrument for recording the force of uterine contractions
tocograph	instrument for recording the force of uterine contractions
tocography	procedure for recording the force of uterine contractions
tocolytic	substance that slows or stops the process of childbirth
torsion	twisting something in a spiral fashion
toxoplasmosis	disease caused by the single-celled parasite <i>Toxoplasma gondii</i> , commonly found in cat feces; can cause birth defects
transvaginal	procedure that operates through the vaginal wall (e.g. ultrasound or surgery)
trichomoniasis	sexually transmitted disease caused by the single-celled parasite <i>Trichomonas vaginalis</i> ; as the name implies, it's usually asymptomatic in males
trophoectoderm	cells surrounding the blastocyst that contact the uterine lining at implantation (about days 5-6)
tunica albuginea	connective tissue layer around the corpus cavernosum of the penis; also the connective tissue layer around the scrotum or ovaries
urethra	tube that carries urine from the bladder to the outside world
urethrocele	herniation of the urethra (urethral prolapse)
uterine	pertaining to the uterus

Word	Definition
uterus	female organ that houses the embryo during gestation; womb
vagina	female organ used for sexual intercourse and as a passage during childbirth
vaginitis	inflammation of the vagina
vaginodynia	pain in the vagina
vaginomycosis	fungal infection of the vagina
vaginoperineoplasty	surgical revision of the vagina and perineum
vaginoperineorrhaphy	suturing the vagina and perineum
vaginoperineotomy	cutting into the vagina and perineum
vaginoplasty	surgical revision of the vagina
vaginoscope	instrument for looking inside the vagina
vaginosis	abnormal flora (bacteria) in the vagina
varicocele	enlargement of the veins draining the testis
vas deferens	tube which carries sperm from the epididymis to the urethra
vasectomy	surgical removal or ligation of the vas deferens
vasovasostomy	creating an opening between two cut ends of the vas deferens; reversing a vasectomy
vesicovaginal	pertaining to the bladder and vagina
vesiculectomy	surgical removal of the seminal vesicle
vesiculitis	inflammation of the seminal vesicle
vesiculotomy	cutting into the seminal vesicle
vulva	outer opening of the vagina
vulvitis	inflammation of the vulva
vulvodynia	pain in the vulva
vulvovaginitis	inflammation of the vulva and vagina
zygote	fertilized but not yet implanted embryo

HOW THE UNIT 12 WORD LIST WAS BUILT

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
						abort/i		facient	
From Latin abortionem, "miscarriage" or "abortion"						abort		ion	
Latin abruptionem, "sudden breaking off" + placentæ						abruptio	placentæ		
Latin abstinencia, literally ab- + tenere "to hold off"						abstinence			
	a					mast			
	a					men/o		rrhea	
						amnio		centesis	
Greek amnion, "membrane around a fetus"						amnion			
						amnio		rrhea	
						amnio		rrhexis	
						amnio		scope	
						amnio		scopy	
						amnio		tomy	
From Greek anōmalos meaning "uneven" or "irregular"						anomaly			
	an					orchid		ism	
	an					ov[a]		ula	tion
Latin ante partum, "before birth", with space removed	ante					partum			
Greek antron, "a cave"						antr[on]		al	
shortening of Latin areola papillaris, "small area around a nipple"						areola			
	a					sperm		ia	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
Greek zoian, "animal" (hence, alive)	a			zoo	sperm			ia	
Greek balanus, "glans of the penis"; from bálanos, "acorn"					balan			itis	
					balan/o			plasty	
					balan/o			rrhea	
first described 1677 by Caspar Bartholin the Younger					Bartholin		glands		
					blast/o			cele	
					blast/o		cyst		
	brady				toc			ia	
					bulb/o		urethr	al	
					card/i/o		toco	graph	
					cephal/o		pelv		
					cervic			ectomy	
					cervic			itis	
					cervic/o		colp	itis	
					cervic/o			dynia	
					cervic/o			graphy	
					cervic/o		vagin	itis	
					cervix				
Likely refers to Roman law that required a child to be cut from the mother's womb if the mother died in childbirth					Cæserean		section		
">French chancre, "cancer"; same etymology as "canker sore"					chancre				

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
Greek "military cloak", because of its ability to cloak itself inside of cells as a vacuole						Chlamydia			
						chorio	amnion	itis	
						chorio	angi	oma	
						chorio	carcin	oma	
						chorion			
						chorion		itis	
Latin circumcisionem, made from circum–, "around" + cædere, "cut"						circumcision			
						clitoris			
						coitus			
Greek kolpos "bay, gulf of the sea" (i.e. vagina)						colp		itis	
						colp/o	cyst	itis	
						colp/o		pexy	
						colp/o		plasty	
						colp/o		ptosis	
						colp/o		scope	
						colp/o		scopy	
						colp/o	sten	osis	
Latin conceptionem, "to take in and hold; become pregnant" (con– + capere, "to take")						conception			
						condom			
	con					genit[us]			al
	contra					[con]ception			
Latin contractionem, "to draw together" or "shorten"	con					contrac[t]		tion	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
					corpora		cavern	osum	
Latin albicans, "becoming white"					corpus		albicans		
					corpus		hemorrhagicum		
Latin luteus, "yellow" (lutum is a weed used by the Romans to dye fabric yellow)					corpus		luteum		
Latin for "spongy body"					corpus		spong[e]/i	osum	
					cremaster				
				crypt	orchid			ism	
Congenital CMV can cause long-term health problems				cyt/o	megal/o		virus		
From Greek dartós, "flayed, skinned"					dartos				
Latin dēciduus, "falling down or off"					decidua				
					derm			oid	
	di				zygot[e]			ic	
	dys				men/o			rrhea	
dys- + Greek páreunos, "lying beside, lying with" (a medical term for intercourse)	dys				pareun			ia	
	dys							plasia	
	dys				toc				
Greek eklampsis, "shining forth" or "to burst forth violently", apparently because of sudden onset					eclampsia				
	ecto				derm				
Greek, ektopos, literally "out of place", from ek- + -topos	ec				top			ic	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
Latin eiaculatus, "to shoot out"						ejacula		tion	
Greek embryon, "a young one" (en- + bryein, "to swell")						embryo/n		ic	
	endo					cervic		itis	
	endo					derm			
	endo					metri		osis	
	endo					metr		itis	
	endo					metr		ium	
						epididym		ectomy	
Greek epi + didymos, "testicle" (literally, "double"; "two-fold"); probably coined by the Greek anatomist Herophilus	epi					didymis			
						epididym		itis	
						epididym/o	orch	itis	
						epididym/o		tomy	
Greek epision, "the pubic region"						episio[n]		rrhaphy	
						episio[n]		stenosis	
						episio[n]		tomy	
French érectile, from Latin erect "to raise or set up"	eu					toc		ia	
	extra					embryo/n		ic	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
<u>">Latin fēcundus, "fertile, productive". Among the words that come from the same Latin root are: felare, "to suck"; femina, "woman, she who suckles"; felix, "happy, auspicious, fruitful"; fetus, "pregnancy"; filia/filius, "daughter/ son"</u>						fecund			
						fecund	ability		
<u>Latin fertilis, "fruitful, productive" from ferre, "to bear"</u>						fertil[e]		iz[e]	ation
						fertil[e]		iz[e]	ed
						fet[us]/o		metry	
<u>Latin, "fringe, border, threads"</u>						fimbria			
						fraternal			
						gastr		ula	tion
<u>Latin gestationem, "carrying" (it used to mean "riding around on horseback")</u>						gest		ation	
<u>Latin glans, "acorn"</u>						glans			
<u>Coined from Greek gonos, "child, offspring, seed"</u>						gonad			
				gonad		arche			
						gonad/o		trop	in
						gono		rrhea	
						granulosa			
<u>Latin gravidus, "loaded, full, swollen"</u>						gravida			
<u>Latin gravidarum, "pregnant [woman]"</u>						gravidarum			

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
						hyster/o		rrhexis	
						hyster/o	salping	ectomy	
						hyster/o	salping/o	gram	
						hyster/o		scope	
						hyster/o		scopy	
						hyster/o		tomy	
French implantation, "the action of inserting or engrafting"						implantation			
Latin, "in glass" (i.e. in a glass dish or tube)				in		vitro			
	in					fertile			
	in					fertil[e]		ity	
Latin inhibere, "to keep back"						inhib[it]		in	
	inter					sex			
	intra	epi				thel[e]		ial	
	intra					partum			
	intra					uterine			
						labia	majora		
						labia	minora		
				labio		scrot		al	
						lact		ation	
						lact/o		genic	
						lact/o		rrhea	
"smooth muscle tumor"				lei/o		my		oma	
						leuko		rrhea	
Latin ligatus, "bound"						ligat[e]		ion	
	macro					mast			
	macro					som			

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
Latin mamma, "breast" (and, not coincidentally, what babies in many cultures call mothers, "ma + ma"), probably because "ma" is one of the first sounds babies make					mammo			gram	
					mammo			plasty	
					mast			algia	
					mast			ectomy	
					mast			itis	
					masto			pexy	
					mast/o			ptosis	
German menarche from Greek men-, "month" + Greek arkhē, "beginning"				men	arche				
				men/o	pause				
					men/o			rrhag[e]	ia
					men/o			rrh[age]	algia
					men/o			rrhea	
					menstru[s]			ation	
	meso				derm				
	meso				nephr[os]			ic	
					metr/o		colp/o	cele	
					metr/o		men/o	rrhagia	
					metr/o		phleb	itis	
					metr/o			rrhagia	
	micro				cephal			us	
	micro				mast				
	mono				zygot[e]			ic	
				my/o	epithel[e]			ium	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
				my/o	metr			itis	
				my/o	metr			ium	
	neo				nat			al	
	neo				nat/o			logist	
	neo				nat/o			logy	
	neo							plasia	
Latin nulli- + -parous				null/i	par[us]			ous	
Latin obstetricus, "pertaining to a midwife", literally "one who stands opposite [the woman giving birth]: ob- "opposite" + stare, "stands"	ob				stet/r			ic	
	ob				stet/r			ic	ian
	oligo				hydr		amnios		
	oligo				men/o			rrhea	
	oligo				sperm			ia	
					oo		cyte		
					oo		gon	ia	
					oophor			ectomy	
					oophor			itis	
					oophor/o		cyst	ectomy	
					oophor/o		cyst	osis	
					oophor			oma	
					oophor/o		salping	ectomy	
					oophor/o			tomy	
					orchid			algia	
					orchid			ectomy	
					orchid			itis	
					orchid/o			pexy	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
Named after the inventor of the procedure, Dr. Georgios Papanicolaou					orchid/o			ptosis	
					orchid/o			tomy	
					orch/i			ectomy	
					orch/i		epididym	itis	
					orch/i/o			dynia	
					orch/i/o			pathy	
					orch/i/o			pexy	
					orch/i/o			pexy	
					orch			itis	
					organ/o			genesis	
					orifice				
					ovar			algia	
					ovar/i			algia	
					ovar/i				
					ovar/i/o			centesis	
				ovar/i/o	cyesis				
					ovar/i/o			rrhexis	
					ovar/i/o			stomy	
					ovar			itis	
					ovar			y	
					ov[um]			ula	tion
					ovum				
				oxy	toc[ic]			in	
					Pap[anicolaou]		smear		

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
Literally, "nipple tumor", from Latin papilla, "nipple", diminutive of papula, "swelling" + -oma. The related word "pap" is thought to be from the sound an infant makes when asking for a nipple.						papill[a]		oma	
Latin parere, "bring forth, produce"; para is the feminine form of parere						para			
Also called Müllerian ducts; in males anti-Müllerian hormone stops this development, but the absence of this hormone (or a response to it) results in formation of female genitalia regardless of genetic makeup	para	meso				nephr[os]		ic	
Also called Skene's glands; these are near the urethra, and during sexual arousal they swell with blood and secrete fluid	para					urethr[a]		al	
						parity			
						pelv			
						pelv/i	cephal/o	metry	
						pelv/i		metry	
						penis			
	peri					metr		itis	
	peri					metr		ium	
	peri					nat		al	
	peri					nato		logist	
	peri					nato		logy	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
					perine/o			cele	
					perine/o			plasty	
					perine/o			rrhaphy	
					perine/o			tomy	
Greek perinaion, "space between the anus and scrotum"					perine			um	
Placenta first condition in which placenta is attached near the cervix					placenta		previa		
Early separation of the placenta from the lining of the uterus before the second stage of labor					placental		abruption		
	poly				hydr		amnios		
	poly				men/o			rrhea	
	poly				sperm			y	
	post				nat				
	post				partum				
	pre				eclampsia				
	pre				nat				
French, from Latin præputium, "foreskin"					prepuce				
From ancient Greek minor fertility god named Priapos who is always shown with a permanently erect penis					Priap[os]			ism	
	pro				lapse				
	pro				nucleus				
					prostate				
					prostat			ectomy	
					prostat			itis	
					prostat/o		cyst	itis	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
					prostat/o		lith		
					prostat/o		lith/o	tomy	
					prostat/o			megaly	
					prostat/o			rrhea	
					prostat/o		vesicul	ectomy	
					prostat/o		vesicul	itis	
Latin pseudo- + Greek kúēsis, "pregnancy"	pseudo				cyesis				
	pseudo				hermaphrodite				
				pub	arche				
					pub			ic	
<u>">Latin pubes means "adult, full-grown, manly"; this gives us "os pubis" (pubic bone) which gives us "pubis"</u>					pub			is	
				pub/o	cervic			al	
				py/o	salpinx				
					recto			cele	
					relax			in	
					rete		testis		
					salping			ectomy	
					salping			itis	
					salping/o			cele	
				salping/o	cyesis				
Risk-reducing salpingo-oophorectomy (or oophorosalingectomy) is used to reduce breast cancer risk by about 50% in women that have a dangerous version of the BRCA1/BRCA2 genes.					salping/o		oophor	ectomy	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
Compare etymology of the Carboniferous Period					salping/o		oophor	itis	
					salping/o			pexy	
					salping/o			scope	
					semen				
					semin			al	
					semin			iferous	
					semin			oma	
					som/ato		mammo	trop	in
					sono		hyster/o	graphy	
					speculum				
Latin for "reflector, looking-glass, mirror"					spermatid				
					sperm/at/o			cele	
					sperm/at/o		cyte		
					sperm/at/o			genesis	
					sperm/at/o			genic	
					sperm/at/o			gon	ia
					sperm/at/o			lysis	
					sperm/at/o			zoa	
					sperm/at/o			zoon	
					sperm/i			cide	
">Fibrous connective tissue of an organ, from Latin stroma, "bed covering, mattress; something to sit or lie on"					sperm/o			lytic	
					stroma				
					suckl[e]			ing	
Also called Sertoli cells					sustentacul[um]			ar	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
Derived from a poem from 1500s called "Syphilis, or the French Disease" telling the story of a shepherd named Syphilis who supposedly had the disease						syphilis			
Greek teratos, "belonging to a marvel, sign, wonder, monster"						terat/o		gen	
						terat/o		gen	ic
						terat/o		logy	
						terat		oma	
						testes			
						testicle			
						testic		ul[e]	ar
						testis			
						test		itis	
						test/o	toxic	osis	
Greek tókos, "childbirth"						toco	dyna	graph	
						toco		graph	
						toco		graphy	
						toco		lytic	
">Latin tortio, "torture, torment" from torquere, "to twist or torture"						torsion			
Coined 1909 in French from Greek toxon, "bow, arch" from the organism's shape + plasma						Toxoplasma[a]		osis	
	trans					vagin		al	

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
From parasite <i>Trichomonas vaginalis</i>, a common sexually transmitted disease (STD)					Trichomon[as]			iasis	
	ecto			derm	tropho				
Latin term which means, literally, "white coat", which gives a new meaning to the "white coat ceremony" held by most medical schools					tunica		albuginea		
					urethr			a	
					urethr/o			cele	
					uter			ine	
Latin, "womb, belly" from PIE udero-					uter			us	
					vagin			a	
					vagin			itis	
					vagin/o			dynia	
					vagin/o		myc	osis	
					vagin/o		perine/o	plasty	
					vagin/o		perine/o	rrhaphy	
					vagin/o		perine/o	tomy	
					vagin/o			plasty	
					vagin/o			scope	
					vagin			osis	
					varico		cele		
Latin, "carrying away vessel"					vas		deferens		
					vas			ectomy	
					vaso		vaso	stomy	
					vesico		vagin		

Etymology	Prefix1	Prefix2	"Pre-Root"	"Pre-Root"	Root	Root	"Post-Root"	Suffix1 Word End	Suffix2 Word End
From Greek zygōtos, "yoked"; the sperm and egg are yoked together to form a zygote					vesicul			ectomy	
					vesicul			itis	
					vesicul/o			tomy	
					vulva				
					vulv[a]			itis	
					vulv/o			dynia	
					vulv/o		vagin	itis	
					zygote				

Notes on the Unit 1 Prefixes Table

Notes on the Unit 1 Root Words Table

Words that contain the Greek vowel combination **ai** are often rendered as **æ** or **ae**, especially in Medical British English. For example, *hemorrhage* ([from the Greek **haimo**](#)) is often written *hæmorrhage* in Britain.

Notes on the Unit 1 Suffixes Table

malac means *soft*

add **–ia** (word ending meaning *disease process*)

–malacia is a suffix meaning *softening due to disease*

mega means *large*

add **l** to make it sound good, and **y** (word ending meaning *disease process*)

–megaly is a suffix meaning *enlargement due to disease*

hem is the root word for blood

drop the ‘h’ (you barely said it anyway), now it’s **em**

add **–ia** (word ending meaning “disease process”)

–emia is a suffix meaning *present in blood, indicating disease*

ur is the root word for urine (see root word list)

add **–ia** (word ending meaning disease process)
–uria is a suffix meaning *present in urine, indicating disease*

ec– is a prefix which means *out of* (see prefix list)
–tomy is a suffix which means *cutting*
–ectomy is a suffix which means *cutting out of*

hyper– is a prefix that means *too much* (see prefix list)
–trophy is a suffix that means *nourishment, growth*
–hypertrophy is a suffix which means *too much nourishment* or *too much growth*

Notes on the Unit 2 Prefixes Table

Notes on the Unit 2 Root Words Table

Notes on the Unit 2 Suffixes Table

Notes on the Unit 3 Prefixes Table

di– is a prefix that can be confusing because it seems to have [two disparate \[pun intended\] meanings](#)

dia– means *across, through*

dis– means *double, two*

They both mean *separate, apart*.

When **dia–** or **dis–** are shortened to **di–**, it's hard to separate [pun again] the true meaning.

Basically, we just have to “suck it up, buttercup” and figure out the meaning from the context and the etymology if we care to look it up.

Notes on the Unit 3 Root Words Table

The chemical grouping acetyl is found in vinegar (5% acetic acid, to chemists). The Italian name for vinegar is *aceto*. The same Latin root is also the source of the term *acid*, the next entry in the root word table.

The term **blast** is now used to refer to a rapidly dividing or immature cell, especially in descriptions of cancer (e.g. *glioblastoma*). Its original meaning, from the Greek *blastós*, means *bud* (like on a growing plant) from which it became *rapidly growing* and then *rapidly dividing, out of control cell*.

The Greek word *ballein* has given us the English *ball* and *ballistic*. In clinical medicine, *ballistic* movements are those that cannot be controlled after they are started. In this context, we are throwing molecules around in metab**ol**ism, which is comprised of cata**bol**ism and ana**bol**ism.

The roots **chondr/o** (as in cartilage) and **chondrion** (as in mitochondrion) are related. Cartilage is a grainy tissue, and mitochondria are thread-like grains that can be seen inside a cell through the microscope.

The Greek word *pathos* means deep feeling or emotion. In drama, pathos is feeling that is overdone and schmaltzy. In Medical English, **pathology** is the study of disease (presumably because of the pain or feeling caused by disease). Here, it is being used to describe the feelings that molecules have, either for being around water (**hydrophilic**), against being around water (**hydrophobic**), or okay either way (**amphipathic**).

The Greek word *peptos* (cooked, digested) was part of the description of digestion. We still see it in the brand

name Pepto-Bismol or the fancy word dyspeptic. As the first proteins discovered were byproducts of digestion, *peptide* began to be used to refer to a protein. Now, a *polypeptide* is the name for a small protein.

Note the similarity of *plasm* (a substance that can be molded) to *plastic* (a substance that can be molded) or even *plastic surgery* (molding the structure of the human body).

Notes on the Unit 3 Suffixes Table

Notes on the Unit 3 Word List

Chromosomes are numbered from largest (1) to smallest (21). (Chromosome 22 was thought to be the smallest by scientists looking through microscopes, but when the human genome was sequenced in the 1990s, chromosome 21 turned out to be smaller and it was too confusing to renumber them.)

Chromosome 1 is a polymer with 249,000,000 DNA base pair monomers (A-T or C-G); chromosome 21 is a polymer with “only” 45,000,000 “rungs” on its twisted ladder (double helix).

There are also X chromosomes (size between chromosome 7 and 8) and Y chromosomes (third smallest, slightly larger than 21 or 22).

In a **colloid**, particles are big enough to scatter light but too small to be removed by a medical centrifuge.

In a **solution**, particles are too small to scatter light or be removed by a medical centrifuge. Particles in a solution are molecules or collections of molecules; they are generally about 1/100,000 of a millimeter in size.

In a **suspension**, particles are big enough to scatter light and to be removed by a medical centrifuge. Particles in a suspension are cells or cell fragments; they are generally about 1/1,000 of a millimeter or larger.

Notes on the Unit 4 Prefixes Table

Three prefixes (*e-*, *ec-*, *ex-*) come out of the same Greek term: *ek*. So, they all mean the same thing: *out*, *without*, *from*.

Notes on the Unit 4 Root Words Table

Notes on the Unit 4 Suffixes Table

The ending *-ing* [is used in English to form the present participle](#). This indicates a verb tense that is ongoing, including past, present, and/or future:

- I am thinking about it.
 - I have been thinking about it.
 - I will be thinking about it.
 - We are going.
 - I'm having none of this.
-

Notes on the Unit 4 Word List

Albinos are not evil, but [in cheap fiction and movies, they are](#). It's really lazy writing to do this, but it happens all the time.

Even though this is not a Medical English word, it helps you remember *kerato* as a root for "horn" (it's originally trikeratops, since Greek does not have a "c"). It also increases your street cred with any 5 year olds you might know.
